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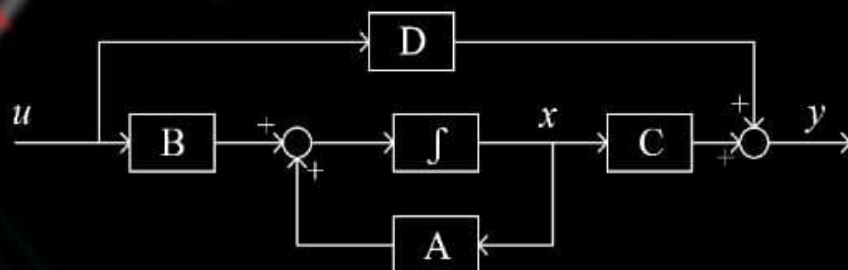
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Modern Control System

Lecture Note

By:- Daniel Abebe Beyene



Preface

This lecture note is a valuable resource for anyone seeking to understand the State Space Representation of Linear Time-Invariant (LTI) Systems. It covers a wide range of topics, including Mathematical Formulation of Control System Problems, Mathematical Modeling and State Space representation of Dynamic Systems, and State Space Representation From Transfer Function.

The lecture note also provides a comprehensive review of Eigenvalues and Eigenvectors, Solution of State Space Equations, Similarity Transformation, Phase Plane, and Stability Analysis of LTI systems. These topics are crucial for understanding the behavior of LTI systems.

In addition, the lecture note covers Controllability and Observability of LTI Systems, Modality Tests, Kalman Decomposition, and Weaker Conditions for LTI Systems. It also discusses Controllability and Observability in the Sense of Transfer Function. These topics are essential for designing control systems that can achieve desired performance.

The lecture notes also include Full-state Feedback Controller Design for LTI Systems, Pole Placement State Feedback Controller Design, Desired State Tracking, Servo-mechanism Controller Design, and Determining Poles of Second Order LTI Systems From Time Domain Specification. These topics are crucial for designing controllers that can achieve desired performance.

Furthermore, the lecture note covers State Estimators/Observers Design for LTI systems, Full Order State Observer Design, Reduced Order State Observer Design, Continuous Time Kalman Filtering, Closed Loop State Feedback Controller Design with Full State Observer, and Closed Loop State Feedback Controller Design with Reduced Order State Observer. These topics are essential for designing state estimators/observers that can estimate the system's state accurately.

The lecture note also provides an Introduction to Optimal Control, Performance Indices, Elements of Calculus of Variation, Solution to Optimal Linear Quadratic Regulator Problem, Optimal Steady State Reference Tracking, and Optimal State Estimation. These topics are crucial for designing optimal control systems that can achieve desired performance.

In addition, the lecture notes cover Introduction to MIMO Systems and Nonlinear Systems, Introduction to Multiple Input Multiple Output Systems, and Introduction to Nonlinear Systems. These topics are essential for understanding the behavior of complex systems.

Finally, the lecture note includes Analog Simulation, Analog Computers and Simulation, Solving Differential Equations with an Analog Computer, Physical Realization of Analog Computer, Programming Differential Equation Using Analog Computer, Magnitude and Time Scaling, Analog Simulation of Transfer Functions, and State Space representation of Systems. These topics are crucial for understanding how analog computers can be used to simulate complex systems.

Overall, this lecture note serves as an essential guide for anyone interested in understanding the State Space Representation of linear time-invariant (LTI) Systems. It provides a comprehensive overview of various topics related to LTI systems and is an excellent resource for students and professionals alike.

In these lectures, the author has taken into consideration students who may not have a strong background in state space representation and analysis for different systems. Therefore, the author has deliberately avoided introducing basic concepts in state space representation and state feedback control for linear systems whenever possible. While some may view this as a drawback, the author hopes that users will not be too critical.

It is important to note that these are only introductory lectures and examples on the topic. Due to time constraints and the availability of other books on the subject, the author has chosen to focus on the most critical aspects and applications of the topic. This decision was made to make the discussion more interesting from both an analytic point of view.

The author hopes that this brief introduction will encourage interested students to explore more about this fascinating mathematical tool for analyzing and designing state feedback controllers for different systems using state space representation. It is important to note that there are many other books available on this topic for further reading, such as "Modern Control Engineering" by Katsuhiko Ogata [1] and "Modern Control Systems" by Richard C. Dorf and Robert H. Bishop [2].

Daniel Abebe, July 30, 2025

Acronym's (Abbreviation)

Abbreviation	Description
DC	Direct Current
BIBO	Bounded Input Bounded Output
FBD	Free Body Diagram
2DOF	2 Degree of Freedom
IPC	Inverted Pendulum on Cart
LQR	Linear Quadratic Regulator
LTI	Linear Time Invariant
MATLAB	Matrix Laboratory
MIMO	Multiple Inputs and Multiple Outputs
SIMO	Single Input and Multiple Outputs

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Chapter 1

State Space Representation of LTI Systems

1.1 Mathematical Modeling of Dynamic Systems

Mathematical modeling is a technique for representing dynamic systems through mathematical expressions, simplifying complex systems to facilitate analysis, prediction, and control of their behavior. These models describe time-dependent processes, aiding in understanding transient responses, such as changes from one steady state to another. They provide insights into system dynamics by characterizing time constants and behaviors. Additionally, mathematical models are instrumental in developing training simulators for operator education and support "what-if" analyses, allowing exploration of scenarios, such as the effects of closing a valve, using tools like MATLAB or LabVIEW. They are essential for optimization and control tasks, including control structure design, controller tuning, and model-based control strategies. Ultimately, models serve as abstractions that encapsulate the behavior of systems and can manifest in various forms, including

- Empirical Models: These models are constructed based on observed data rather than derived from first principles. They often employ statistical methods to fit data, making them effective for predicting system behavior within the range of observed conditions.
- Mathematical Models: These models utilize mathematical language to describe the relationships among system variables. They can be further categorized into:
 - Differential Equations: Applicable to continuous systems, where the state of the system evolves.
 - Difference Equations: Relevant for discrete systems, where changes occur at specific intervals.
- Hybrid Models: These models incorporate both continuous and discrete elements, making them suitable for systems that exhibit both behaviors.

In this course, we will concentrate on continuous-time systems, primarily modeling them through differential equations. The mathematical representation may take the form of a transfer function, which establishes the relationship between the output and input in the Laplace domain as the ratio of their respective Laplace transforms, assuming zero initial conditions. Alternatively, we will explore state space representation, which focuses

on the internal state of the system and employs a set of first-order differential equations. While you have previously covered analysis and controller design using transfer function models in classical control systems, this course will specifically emphasize analysis and controller design within the state space framework.

1.2 Control System

A control system can be viewed informally as a dynamical object (e.g., expressed by an ordinary differential equation) containing a parameter that can be manipulated to influence the behavior of the system to achieve the desired goal.

A system is a thing having parts that can be considered as a single entity. The system must be defined clearly with a particular set of relationships that completely characterize the conceivable states of the system.

The state of the system describes the condition(status) of the system at any time $t \geq 0$ and provides information which (together with the knowledge of the equation describing the system) enables us to calculate/analyze the future behavior of the system under certain control input.

In practice, it is not possible to get all state variables; instead, we get controlled variables (outputs) which depend on somehow on measured states.

1.3 Mathematical Formulation of Control System Problems

We assume that the dynamics of a system are an evolution of its states over time under a given input. The evolution of the state of the system can be viewed as a function of the state, control input, and time. Hence, in the evolution of a system, we have system states, a state mapping function, control inputs, control input mapping function. The state mapping function gives us the effect of the status of the states on the dynamics of a system, while the input mapping function gives us the effect of control input on the dynamics of a system.

Mathematically

$$\dot{\mathbf{x}} = F(\mathbf{x}(t), \mathbf{u}(t), t) \quad (1.3.1)$$

Equation 1.3.1 is called the state equation of a dynamic system. The function $F(\mathbf{x}(t), \mathbf{u}(t), t)$ can be linear or nonlinear, it can be separable or not(i.e it may or may not written as sum of input mapping function and state mapping function explicitly as $F(\mathbf{x}(t), \mathbf{u}(t), t) = f(\mathbf{x}(t), t) + g(\mathbf{u}(t), t)$).

Since all states of the system may not be accessible directly, we define the output (accessible states) of a dynamic system as

$$\mathbf{y}(t) = h(\mathbf{x}(t), \mathbf{u}(t), t) \quad (1.3.2)$$

Equation 1.3.2 is called the output equation of a dynamic system. $\mathbf{y}(t)$ is an output (accessible states/controlled states) of a dynamic system, and $h(\mathbf{x}(t), \mathbf{u}(t), t)$ is an output mapping function. The function $h(\mathbf{x}(t), \mathbf{u}(t), t)$ defines the relation between the output

of the dynamic system, the state of the system, and the control input.

Thus the control system problem is finding admissible (inputs that we can apply to the dynamic system) control input $\mathbf{u}(t)$ that derive the evolution of the dynamic system in the desired way or toward the desired state, by using the relation between the state dynamics and control input as well as the relation between the controlled states (output) of the system and the control input.

In this course, we only deal with linear time-invariant (LTI) dynamic systems. LTI dynamic systems have linear time-invariant state mapping and input mapping functions. This implies that the evolution of LTI dynamic systems is linear concerning the state of the system and control input. Which simplifies the state equation [1.3.1](#)

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t) \quad (1.3.3)$$

Matrices $\mathbf{A}, \mathbf{B}$ are proportionality constant matrices. Matrix $\mathbf{B}$ is called the system matrix, and matrix $\mathbf{B}$ is called the input matrix. Their size is defined based on the number of states and inputs a dynamic system has. For a dynamic system with n states and m control inputs, matrix $\mathbf{A}$ is $n \times n$ matrix, while matrix $\mathbf{B}$ is an $n \times m$ matrix. The output equation [1.3.2](#) is simplified to

$$\mathbf{y}(t) = \mathbf{C}\mathbf{x}(t) + \mathbf{D}\mathbf{u}(t) \quad (1.3.4)$$

for LTI dynamic systems. Matrices $\mathbf{C}, \mathbf{D}$ are proportionality constant matrices. Matrix $\mathbf{C}$ is called an output matrix, and matrix $\mathbf{D}$ is called an input transmission matrix. Their size is defined based on the number of states, outputs, and inputs a dynamic system has. For a dynamic system with n states, p number of outputs, and m number of control inputs, matrix $\mathbf{C}$ is a $p \times n$ matrix, while matrix $\mathbf{D}$ is a $p \times m$ matrix.

Any LTI dynamic system can be represented by a state equation and output equation, which is called the state-space representation of LTI dynamic systems.

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t) \quad (1.3.5)$$

$$\mathbf{y}(t) = \mathbf{C}\mathbf{x}(t) + \mathbf{D}\mathbf{u}(t) \quad (1.3.6)$$

For simplicity of expressions and convenience in the rest of the document, any function of time $f(t)$ is written as f , unless expressed explicitly. With this, the state space representation of LTI dynamic systems is written as:

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} \quad (1.3.7)$$

$$\mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u} \quad (1.3.8)$$

The block diagram of state space representation is shown in [Figure 1.1](#).

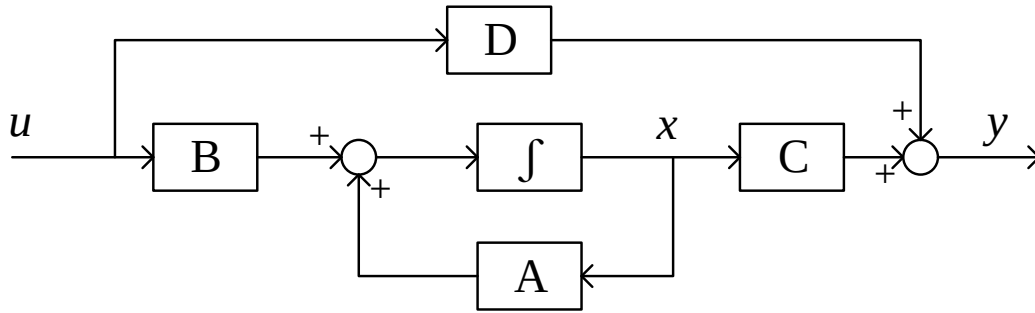


Figure 1.1: *State Space Representation of a Dynamic System in Block Diagram*

1.4 Mathematical Modeling of Electrical Systems

Mathematical modeling of electrical systems often relies on fundamental laws such as Ohm's Law, Kirchhoff's Laws, and the principles of energy conservation.

1.4.1 Basic Laws Used in Modeling Electrical Systems

- **Ohm's Law:** Relates voltage (v_R), current (i_R), and resistance (R) in a linear circuit:

$$v_R = i_R R \quad (1.4.1)$$

- **Kirchhoff's Current Law (KCL):** States that the total current entering a junction equals the total current leaving the junction.
- **Kirchhoff's Voltage Law (KVL):** States that the sum of the electrical potential differences (voltage) around any closed network is zero..
- **Inductor Voltage-Current Relation:** The voltage across an inductor (v_L) is proportional to the rate of change of current (i_L) through it:

$$v_L = L \frac{di_L}{dt} \quad (1.4.2)$$

where L is the inductance of an inductor.

- **Capacitor Voltage-Current Relation:** The current through a capacitor (i_C) is proportional to the rate of change of voltage (v_C) across it:

$$i_C = C \frac{dv_C}{dt} \quad (1.4.3)$$

where C is the capacitance of a capacitor.

1.4.2 Example

RLC Circuit

Consider the series RLC circuit in Figure 1.2, if the concern of the designer is the voltage across the capacitor, From Kirchhoff's current law

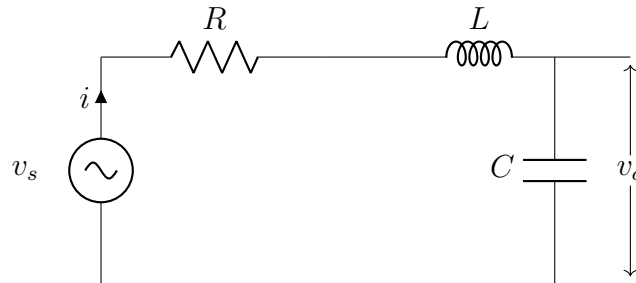


Figure 1.2: *Series RLC Circuit*

$$v_s = iR + L \frac{d}{dt}i + v_c$$

$$\frac{d}{dt}i = -\frac{R}{L}i - \frac{1}{L}v_c + \frac{1}{L}v_s \quad (1.4.4)$$

$$C \frac{d}{dt}v_c = i$$

$$\frac{d}{dt}v_c = \frac{1}{C}i \quad (1.4.5)$$

The state space representation will be

$$\begin{bmatrix} \frac{d}{dt}i \\ \frac{d}{dt}v_c \end{bmatrix} = \begin{bmatrix} -\frac{R}{L} & -\frac{1}{L} \\ \frac{1}{C} & 0 \end{bmatrix} \begin{bmatrix} i \\ v_c \end{bmatrix} + \begin{bmatrix} \frac{1}{L} \\ 0 \end{bmatrix} v_s \quad (1.4.6)$$

$$y = \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} i \\ v_c \end{bmatrix} \quad (1.4.7)$$

1.5 Mathematical Modeling of Magnetic Circuit

Magnetic circuits are analogous to electrical circuits and are used to analyze the behavior of magnetic fields in various materials. The fundamental concepts of magnetic circuit modeling include magnetic flux, magnetomotive force (MMF), and magnetic reluctance.

1.5.1 Magnetic Flux

Magnetic flux (Φ) is defined as the total magnetic field density (B_m) passing through a given area (A). It can be expressed mathematically as:

$$\Phi = B_m \cdot A \quad (1.5.1)$$

where B_m is measured in teslas (T) and A in square meters (m^2). The direction of the magnetic flux is determined by the orientation of the magnetic field lines.

1.5.2 Magnetomotive Force (mmf)

The magnetomotive force (F) is the driving force that produces magnetic flux in a magnetic circuit. It is analogous to voltage in an electrical circuit and is calculated as:

$$F_{mmf} = \mathcal{N} \cdot i \quad (1.5.2)$$

where $\mathcal{N}$ is the number of turns in the coil and I is the current flowing through the coil, measured in amperes (A).

1.5.3 Magnetic Reluctance

Magnetic reluctance (R_m) is the opposition to the flow of magnetic flux in a magnetic circuit, similar to resistance in an electrical circuit. It is defined as:

$$R_m = \frac{F_{mmf}}{\Phi} \quad (1.5.3)$$

The unit of magnetic reluctance is ampere-turns per weber (A/Wb). The relationship between MMF, magnetic flux, and reluctance can be expressed as:

$$F_{mmf} = R_m \cdot \Phi \quad (1.5.4)$$

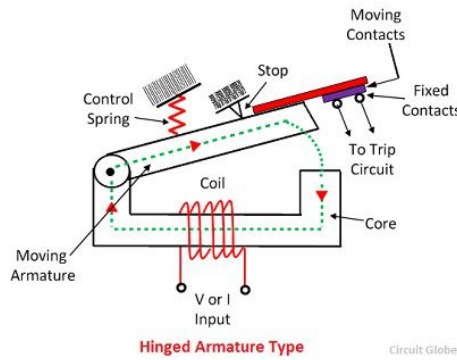
1.5.4 Example

Hinged Armature Type Relay

The hinged armature type relay in Figure 1.3 operates as an electromagnetic circuit consisting of a coil (electromagnet), an armature (movable part), and a core. The strength of this magnetic field is proportional to the current flowing through the coil (I) and the number of turns in the coil (N):

$$B_m = \mu \cdot \frac{N \cdot i}{l}$$

where B_m is the magnetic flux density, μ is the permeability of the core material, and l is the length of the magnetic path.

Figure 1.3: *Hinged armature type relay*

The force (F_{em}) exerted on the armature can be calculated using the magnetic flux and the area of the armature:

$$F_{em} = \frac{B^2 \cdot A}{2\mu_0}$$

where A is the area of the armature and μ_0 is the permeability of free space.

The electromagnetic force is balanced by the spring force. When voltage is applied to the coil, it generates a magnetic field that attracts the pivoted armature, which is held away from the core by a spring. As the armature moves towards the core, it closes the relay contacts, allowing current to flow through the connected circuit and activating the device. When the current is removed, the magnetic field collapses, and the armature returns to its original position by the spring, reopening the contacts and interrupting the current flow.

1.6 Mathematical Modeling of Mechanical Systems

Mathematical modeling of mechanical systems involves the use of mathematical expressions to represent the behavior and dynamics of mechanical components and systems. These models are crucial for understanding, analyzing, and predicting the performance of mechanical systems across various applications in engineering.

1.6.1 Basic Laws Used in Modeling Mechanical Systems

The modeling of mechanical systems is grounded in fundamental physical laws, which provide the framework for understanding how forces and motions interact. The key laws include:

- **Newton's Laws of Motion:** These laws describe the relationship between the motion of an object and the forces acting on it.
- **Conservation of Energy:** This principle states that energy cannot be created or destroyed, only transformed from one form to another. In mechanical systems, this often involves converting potential energy to kinetic energy and vice versa.
- **Principles of Dynamics:** These principles govern the motion of objects under the influence of forces, including concepts such as inertia, momentum, and equilibrium.

1.6.2 Components in Mechanical Systems Modeling

In mechanical modeling, several key components are commonly used to represent the behavior of systems:

Stiffness Elements (Springs)

Springs are mechanical components that store and release energy. The behavior of springs is defined by Hooke's Law, which states that the force or torque exerted by a spring is proportional to its linear displacement (x) or angular displacement (θ) from its equilibrium position:

$$F = -kx \quad (1.6.1)$$

$$\tau = -k\theta \quad (1.6.2)$$

where F and τ are the restoring force and restoring torque, respectively, k is the spring constant (a measure of stiffness), and x and θ are the linear and angular displacements, respectively. This relationship indicates that the force exerted by the spring opposes the

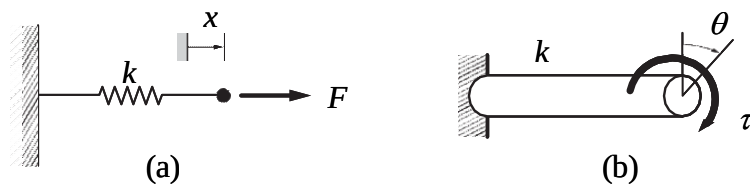


Figure 1.4: *Stiffness elements*

direction of displacement, which is critical in applications such as vibration isolation and energy storage. For example, in automotive suspensions, springs absorb shocks from the road, enhancing ride comfort.

If both ends are free to move or rotate, the equations describing the motion are:

$$F = -k(x_1 - x_2) \quad (1.6.3)$$

$$\tau = -k(\theta_1 - \theta_2) \quad (1.6.4)$$

These equations account for the relative displacements between two points, which is

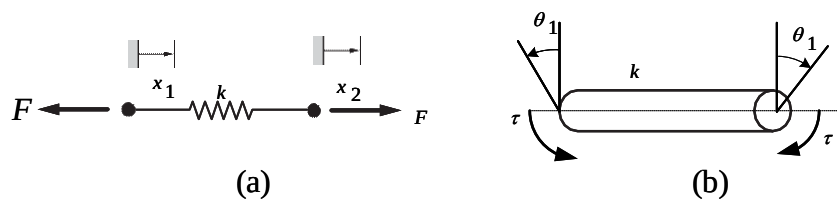


Figure 1.5: *Bi-directional motion stiffness elements*

important in systems where springs may be subjected to forces from multiple directions.

Friction Elements (Dampers)

Dampers are devices that dissipate energy, usually in the form of heat, and are used to reduce oscillations in mechanical systems. The force exerted by a damper is proportional to the velocity (v) of the moving component:

$$F_d = -d\dot{x} \quad (1.6.5)$$

$$\tau_d = -d\dot{\theta} \quad (1.6.6)$$

where F_d and τ_d are the damping force and damping torque, respectively, b is the damping coefficient, and $\dot{x}$ and $\dot{\theta}$ are the linear and angular velocities. This relationship indicates that the damping force opposes the motion, providing a stabilizing effect in dynamic systems. Dampers are essential for controlling vibrations and ensuring system stability. If both ends are free to move or rotate, the equations describing the motion are:

$$F_d = -d(\dot{x}_1 - \dot{x}_2) \quad (1.6.7)$$

$$\tau_d = -d(\dot{\theta}_1 - \dot{\theta}_2) \quad (1.6.8)$$

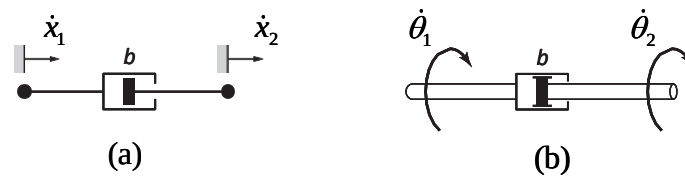


Figure 1.6: *Dampers: A schematic representation of damping mechanisms in mechanical systems.*

These equations illustrate how dampers respond to relative velocities, which is crucial in applications such as automotive shock absorbers, where they mitigate the effects of road irregularities on vehicle stability and comfort.

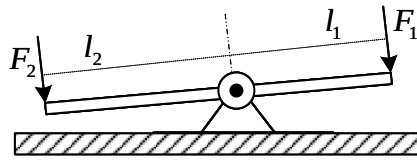
Mechanical Transformers

Mechanical transformers, such as levers and gears, are devices that change the direction or magnitude of forces and motion. These components are fundamental in mechanical systems, allowing for the efficient transfer and modification of energy.

- **Levers:** A lever, as shown in Figure 1.7, amplifies an input force to lift a load. The relationship between the input force (F_1), output force (F_2), input distance (l_1), and output distance (l_2) can be described by the law of the lever:

$$F_1 l_1 = F_2 l_2 \quad (1.6.9)$$

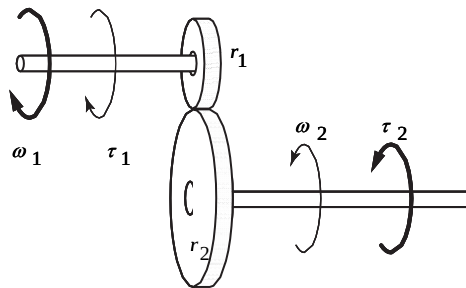
This equation indicates that the moments about the fulcrum are equal, allowing for force amplification depending on the distances from the fulcrum. The longer the input arm (l_1), the smaller the input force needed to lift a given load. This principle is utilized in various applications, such as crowbars and seesaws, demonstrating the mechanical advantage achieved through lever systems.

Figure 1.7: *Lever system*

- **Gears:** Gears, as shown in Figure 1.8, transmit torque and change the speed of rotation. The relationship between the input and output angular velocities (ω_1, ω_2) and torques (τ_1, τ_2) can be expressed as:

$$\frac{\omega_1}{\omega_2} = \frac{r_2}{r_1} \quad \text{and} \quad \tau_1 \omega_1 = \tau_2 \omega_2 \quad (1.6.10)$$

where r_1 and r_2 are the number of teeth on the input and output gears (gear radius ratio), respectively.

Figure 1.8: *Gear system*

This relationship shows how gears can change speed and torque, allowing for mechanical advantage in various applications. For instance, in a bicycle, the gear system enables different pedaling speeds and forces, optimizing performance based on terrain.

1.6.3 Examples

Two Degree of Freedom Quarter Car Suspension System

A vehicle suspension system is a complex vibration system having multiple degrees of freedom. The purpose of the suspension system is to isolate the vehicle body from the road imperfections. Suspension systems serve a dual purpose

- Contributing to the vehicle's road holding/handling and braking for good active safety and driving pleasure, and
- Keeping vehicle occupants comfortable and reasonably well isolated from road noise, bumps, and vibrations.

The system may be

- Passive suspension system, uses springs and dampers only, or
- Semi-active suspension systems include devices such as air springs and switchable shock absorbers, various self-leveling solutions, as well as systems like hydro pneumatic, hydro elastic, and hydro-gas suspensions or
- Fully Active suspension system uses electronic monitoring of vehicle conditions, coupled with the means to impact vehicle suspension and behavior in real time to directly control the motion of the car.

Consider a fully active suspension system shown in Figure 1.9, the free body diagram of the system is shown in Figure 1.10.

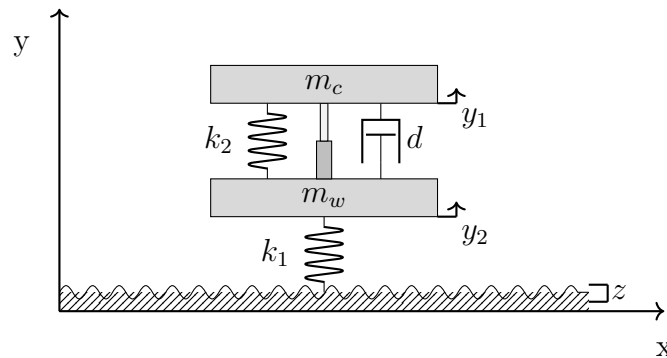


Figure 1.9: 2-DOF Quarter Car Suspension Model

Using Newton's second law of motion, the force on the body of the car is

$$\begin{aligned}
 \Sigma F_y &= m_c \ddot{y}_1 \\
 f_{k_2} + f_d + u &= m_c \ddot{y}_1 \\
 k_2(y_2 - y_1) + d(\dot{y}_2 - \dot{y}_1) + u &= m_c \ddot{y}_1 \\
 \ddot{y}_1 &= -\frac{k_2}{m_c} y_1 + \frac{k_2}{m_c} y_2 - \frac{d}{m_c} \dot{y}_1 + \frac{d}{m_c} \dot{y}_2 + \frac{1}{m_c} u
 \end{aligned} \tag{1.6.11}$$

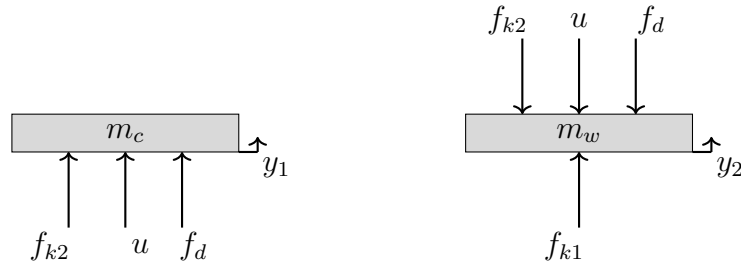


Figure 1.10: FBD of 2-DOF Quarter Car Suspension Model

In the same way, the force on the wheel is

$$\begin{aligned}
 \Sigma F_y &= m_w \ddot{y}_2 \\
 -f_{k2} - f_d - u + f_{k1} &= m_w \ddot{y}_2 \\
 -k_2(y_2 - y_1) - d(\dot{y}_2 - \dot{y}_1) - u + k_1(z - y_2) &= m_w \ddot{y}_2 \\
 \ddot{y}_2 &= \frac{k_2}{m_w} y_1 - \frac{k_1 + k_2}{m_w} y_2 + \frac{d}{m_w} \dot{y}_1 - \frac{d}{m_w} \dot{y}_2 + \frac{k_1}{m_w} z - \frac{1}{m_w} u
 \end{aligned} \tag{1.6.12}$$

Let $y_1 = x_1$, $\dot{y}_1 = x_2$, $y_2 = x_3$ and $\dot{y}_2 = x_4$, the equations 1.6.11 and 1.6.12 simplify

$$\begin{aligned}
 \dot{x}_1 &= x_2 \\
 \dot{x}_2 &= -\frac{k_2}{m_c} x_1 + \frac{k_2}{m_c} x_3 - \frac{d}{m_c} x_2 + \frac{d}{m_c} x_4 + \frac{1}{m_c} u \\
 \dot{x}_3 &= x_4 \\
 \dot{x}_4 &= \frac{k_2}{m_w} x_1 - \frac{k_1 + k_2}{m_w} x_3 + \frac{d}{m_w} x_2 - \frac{d}{m_w} x_4 + \frac{k_1}{m_w} z - \frac{1}{m_w} u
 \end{aligned} \tag{1.6.13}$$

The state space equation of the system is then

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -\frac{k_2}{m_c} & -\frac{d}{m_c} & \frac{k_2}{m_c} & \frac{d}{m_c} \\ 0 & 0 & 0 & 1 \\ \frac{k_2}{m_w} & \frac{d}{m_w} & -\frac{k_1 + k_2}{m_w} & -\frac{d}{m_w} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{m_c} \\ 0 \\ -\frac{1}{m_w} \end{bmatrix} u + \begin{bmatrix} 0 \\ 0 \\ 0 \\ \frac{k_1}{m_w} \end{bmatrix} z \tag{1.6.14}$$

If the output h is the position of the body of the car concerning the road x_1 , the output equation is

$$h = \begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \tag{1.6.15}$$

The goal of the designer is to design a controller that keeps the car body in a predefined range, irrespective of the roughness of the road.

Wind Turbine Generator System

Wind turbine generator systems convert kinetic energy from the wind into electrical energy through a series of mechanical and electrical processes. Understanding the dynamics of these systems is crucial for optimizing performance, predicting energy output, and ensuring reliable operation. The mathematical modeling of a wind turbine involves several key components, including the aerodynamic characteristics of the turbine blades, the mechanical dynamics of the rotor, and the electrical characteristics of the generator.

The relationship between the angular speeds and torques of the turbine and generator is essential for assessing the efficiency and overall performance of the system.

In a typical configuration, the turbine's angular speed ($\dot{\theta}_1$) is associated with the aerodynamic torque (τ_{aero}) generated by the wind acting on the blades. This torque can be expressed as a function of wind speed, blade characteristics, and the turbine's design parameters. The turbine is generally coupled to the generator through a gearbox, which adjusts the speed and torque to enhance generator performance.

The generator, in turn, possesses its angular speed ($\dot{\theta}_2$) and produces a generator torque (τ_g). The gearbox ratio ($N = \frac{r_2}{r_1}$) is critical in relating these parameters, where r_1 and r_2 are the radii of the turbine and generator gears, respectively.

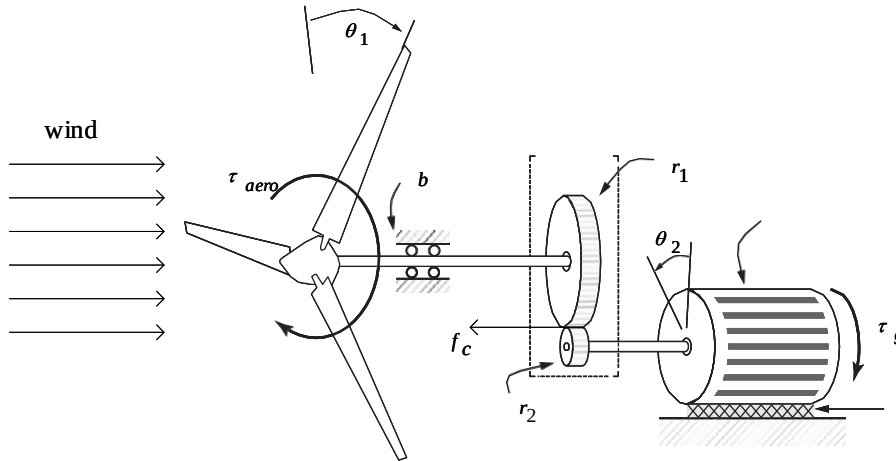


Figure 1.11: Wind Turbine Generator System

The mathematical model of the system can be divided into two main components: the turbine system and the generator system. The mathematical model for the turbine system is given by:

$$\tau_{aero} - b\dot{\theta}_1 - f_c r_1 = J_1 \ddot{\theta}_1 \quad (1.6.16)$$

where f_c represents the contact force between the gears, b_1 is a damping coefficient, and J_1 is the moment of inertia of the turbine system.

Similarly, the mathematical model for the generator system is expressed as:

$$f_c r_2 - b_2 \dot{\theta}_2 - \tau_g = J_2 \ddot{\theta}_2 \quad (1.6.17)$$

where J_2 is the moment of inertia of the generator system, b_2 is a damping coefficient and τ_g is the torque produced by the generator.

To solve for f_c from equation 1.6.17, we can substitute f_c into equation 1.6.16:

$$\tau_{aero} - b_1\dot{\theta}_1 - \frac{1}{N} \left(\tau_g + b_2\dot{\theta}_2 + J_2\ddot{\theta}_2 \right) = J_1\ddot{\theta}_1 \quad (1.6.18)$$

Using the relationship between the angular speeds, we have $\dot{\theta}_1 = N\dot{\theta}_2$, where N is the gearbox ratio. The final expression for τ_{aero} in terms of $\dot{\theta}_2$ is:

$$\begin{aligned} \tau_{aero} &= \left(b_1N + \frac{b_2}{N} \right) \dot{\theta}_2 + \frac{1}{N} \tau_g + \left(\frac{J_2}{N} + J_1N \right) \ddot{\theta}_2 \\ \ddot{\theta}_2 &= \left(\frac{N}{J_2 + J_1N^2} \right) \tau_{aero} - \left(\frac{1}{J_2 + J_1N^2} \right) \tau_g - \left(\frac{b_2 + b_1N^2}{J_2 + J_1N^2} \right) \dot{\theta}_2 \end{aligned} \quad (1.6.19)$$

Let $\gamma_1 = \left(\frac{N}{J_2 + J_1N^2} \right)$, $\gamma_2 = \left(\frac{1}{J_2 + J_1N^2} \right)$, $\gamma_3 = \left(\frac{b_2 + b_1N^2}{J_2 + J_1N^2} \right)$ and the states of system be $x_1 = \theta_2$ and $x_2 = \dot{\theta}_2$ the position and velocity of the mass of the spool, then the state space representation will be

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= -\gamma_3 x_2 + \gamma_1 \tau_{aero} - \gamma_2 \tau_g \end{aligned} \quad (1.6.20)$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & -\gamma_3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ \gamma_1 & -\gamma_2 \end{bmatrix} \begin{bmatrix} \tau_{aero} \\ \tau_g \end{bmatrix} \quad (1.6.21)$$

$$y = \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad (1.6.22)$$

Inverted Pendulum on a Cart (IPC) System

An inverted pendulum on a cart is a mechanical system having a freely rotating inverted pendulum with mass m_p and length l mounted on a cart with mass m_c . The main goal of the control engineer is to determine the control signal u that drives the cart to the desired location and, at the same time, keeps the inverted pendulum from falling. The system is shown in Figure 1.12a.

Let us denote the distance of the cart from the origin by (x, y) and its velocity by $(\dot{x}, \dot{y})$. For simplicity, assume the cart moves along the x axis only; then the velocity and position along the y axis are zero. If the pendulum deviation from the vertical axis is θ , then the position of the pendulum (x_p, y_p) with respect to the origin is given as

$$\begin{bmatrix} x_p \\ y_p \end{bmatrix} = \begin{bmatrix} x + l \sin \theta \\ l \cos \theta \end{bmatrix} \quad (1.6.23)$$

With velocity

$$\begin{bmatrix} \dot{x}_p \\ \dot{y}_p \end{bmatrix} = \begin{bmatrix} \dot{x} + \dot{\theta} \cos \theta \\ -l\dot{\theta} \sin \theta \end{bmatrix} \quad (1.6.24)$$

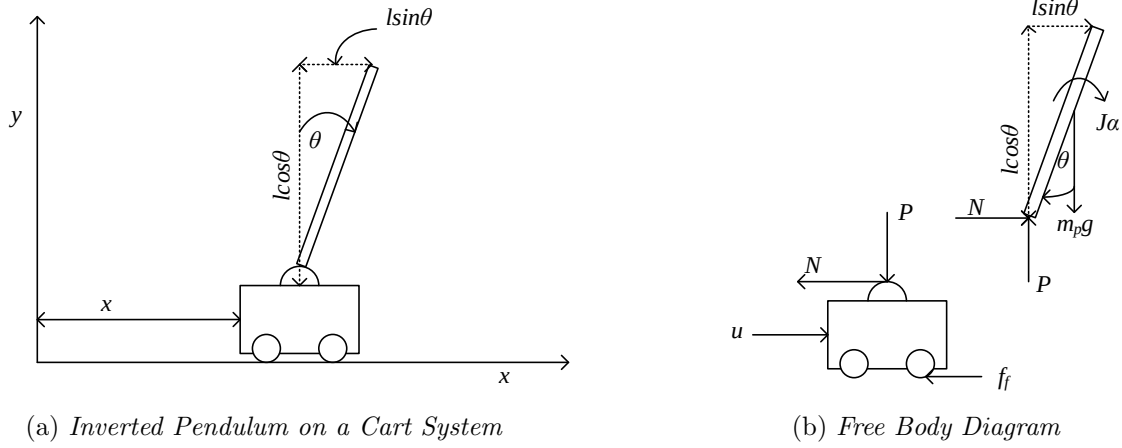


Figure 1.12: Inverted Pendulum on a Cart (IPC) System

The free-body diagram of the IPC is shown in Figure 1.12b. If we denote the coefficient of friction between the cart and the rail by b , the normal force exerted by the cart on the pendulum by P , and the horizontal force exerted by the cart on the pendulum by N .

Using Newton's second law of motion on a pendulum free body diagram, we have,

$$\begin{aligned}\Sigma F_{px} &= m_p \ddot{x}_p \\ N &= m_p \ddot{x}_p = m_p (\ddot{x} + \ddot{\theta} l \cos \theta - \dot{\theta}^2 l \sin \theta)\end{aligned}\quad (1.6.25)$$

$$\begin{aligned}\Sigma F_{py} &= m_p \ddot{y}_p \\ P &= m_p g - m_p \ddot{\theta} l \sin \theta - m_p l \dot{\theta}^2 \cos \theta\end{aligned}\quad (1.6.26)$$

Using Newton's second law of motion forces exerted on a cart are

$$\begin{aligned}\Sigma F_{cx} &= m_c \ddot{x} \\ u &= m_c \ddot{x} + b \dot{x} + N\end{aligned}\quad (1.6.27)$$

$$\Sigma F_{cy} = m_c \ddot{y} = 0 \quad (1.6.28)$$

Substituting for N in equation 1.6.27 we have,

$$u = (m_c + m_p) \ddot{x} + b \dot{x} + m_p l \ddot{\theta} \cos \theta - m_p l \dot{\theta}^2 \sin \theta \quad (1.6.29)$$

Assuming the friction between the pendulum and the cart is negligible, the momentum of the pendulum rod about the pivot is described by,

$$J \alpha = J \ddot{\theta} = P l \sin \theta - N l \cos \theta \quad (1.6.30)$$

Combining equation 1.6.25, equation 1.6.26 and equation 1.6.30 we have

$$(J + m_p l^2) \ddot{\theta} - m_p l g \sin \theta = -m_p l \ddot{x} \cos \theta \quad (1.6.31)$$

Solving for $\ddot{x}$ and $\ddot{\theta}$ in-terms of $\dot{x}$, x , $\dot{\theta}$, θ , we get the two governing dynamic equations for IPC ,

$$\ddot{x} = \frac{(m_p l^2 + I) m_p l \dot{\theta}^2 \sin \theta - (m_p l)^2 g \sin \theta \cos \theta - b(m_p l^2 + J) \dot{x} + (m_p l^2 + I) u}{(m_p + m_c)(m_p l^2 + I) - (m_p l \cos \theta)^2} \quad (1.6.32)$$

$$\ddot{\theta} = \frac{-(m_p l)^2 \dot{\theta}^2 g \sin \theta \cos \theta + b \dot{x} m_p l \cos \theta + (m_p + m_c) m_p l g \sin \theta - m_p l \cos \theta u}{(m_p + m_c)(m_p l^2 + I) - (m_p l \cos \theta)^2} \quad (1.6.33)$$

Using small angle approximation about $\theta = 0$ deg, we know that for small angle $\cos \theta \approx 1$, $\sin \theta \approx \theta$ and $\dot{\theta} \approx 0$. Replacing $\cos \theta = 1$, $\sin \theta = \theta$ and $\dot{\theta} = 0$ in equation 1.6.32 and equation 1.6.33 we have

$$\ddot{x} = \left(\frac{m_p l^2 + J}{q} \right) u - \left(\frac{b(m_p l^2 + J)}{q} \right) \dot{x} - \left(\frac{(m_p l)^2 g}{q} \right) \theta \quad (1.6.34)$$

$$\ddot{\theta} = \left(\frac{b m_p l}{q} \right) \dot{x} + \left(\frac{m_p l g (m_p + m_c)}{q} \right) \dot{\theta} - \left(\frac{m_p l}{q} \right) u \quad (1.6.35)$$

where $q = (m_c + m_p)(m_p l^2 + J) - (m_p l)^2$.

To represent equation 1.6.34 and equation 1.6.35 in state space let $x_1 = x$, $x_2 = \dot{x}$, $x_3 = \theta$ and $x_4 = \dot{\theta}$. The state space representation of IPC is;

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & -\left(\frac{b(m_p l^2 + J)}{q} \right) & -\left(\frac{(m_p l)^2 g}{q} \right) & 0 \\ 0 & 0 & 0 & 1 \\ 0 & \left(\frac{b m_p l}{q} \right) & \left(\frac{m_p l g (m_p + m_c)}{q} \right) & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + \begin{bmatrix} 0 \\ \left(\frac{m_p l^2 + J}{q} \right) \\ 0 \\ -\left(\frac{m_p l}{q} \right) \end{bmatrix} u \quad (1.6.36)$$

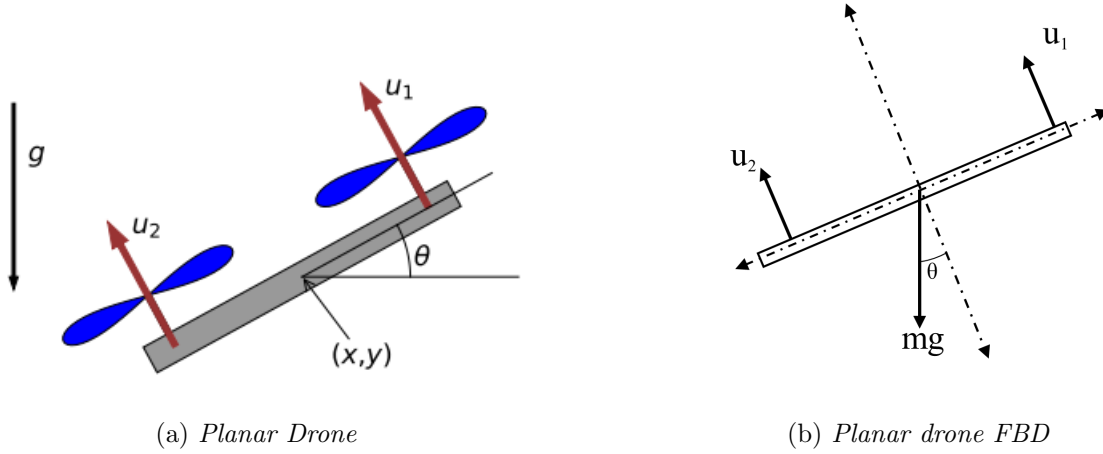
$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \quad (1.6.37)$$

Planar Drone

The planar quadrotor is a simplified model of a quadrotor drone that operates within a two-dimensional plane, making it a valuable tool in control theory and robotics for studying dynamics, control strategies, and navigation algorithms. This model features a frame with four rotors positioned at the corners, each generating lift and capable of independent control. The quadrotor possesses two degrees of freedom in position, allowing movement along the x and y axes, and one degree of freedom in orientation, enabling rotation about the vertical axis (yaw).

Control inputs consist of thrust produced by each rotor, with differential thrust utilized to manage rotation and directional movement. The dynamics of the planar quadrotor can be described by equations derived from Newton's laws or Lagrangian mechanics (see appendix A), incorporating forces from thrust and moments from rotor speeds. These equations govern its position, velocity, orientation, and angular velocity, accounting for factors such as mass, acceleration due to gravity, and moment of inertia.

The planar quadrotor finds applications in various fields, including aerial robotics research, flight dynamics simulation, and the development of control algorithms. Its simplicity allows for easier analysis and the creation of control strategies that can be adapted

Figure 1.13: *Planar Drone*

for more complex three-dimensional systems, making it an essential model for understanding the principles of flight control in aerial vehicles.

The equations of motion are almost trivial, since it is only a single rigid body, and certainly fit into our standard manipulator equations:

$$m\ddot{x} = -mg \sin \theta \quad (1.6.38)$$

$$m\ddot{y} = (u_1 + u_2) - mg \cos \theta \quad (1.6.39)$$

$$I\ddot{\theta} = r(u_1 - u_2) \quad (1.6.40)$$

To express the given equations in state-space representation let $x_1 = x$, $x_2 = \dot{x}$, $x_3 = y$, $x_4 = \dot{y}$, $x_5 = \theta$, $x_6 = \dot{\theta}$ and let $F = u_1 + u_2$ the lift force and $\tau = u_1 - u_2$ the torque, that cause the drone to pitch.

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= -g \sin x_5 \\ \dot{x}_3 &= x_4 \\ \dot{x}_4 &= \frac{F}{m} - g \cos x_5 \\ \dot{x}_5 &= x_6 \\ \dot{x}_6 &= \frac{r}{I} \tau \end{aligned} \quad (1.6.41)$$

Linearizing the given state-space representation using the small angle approximation, we assume that the angle θ or x_5 is small enough such that: $\cos x_5 \approx 1$ and $\sin x_5 \approx x_5$. Using these approximations, we can rewrite the equations as follows:

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= -gx_5 \\ \dot{x}_3 &= x_4 \\ \dot{x}_4 &= \frac{F}{m} - g \\ \dot{x}_5 &= x_6 \\ \dot{x}_6 &= \frac{r}{I} \tau \end{aligned} \quad (1.6.42)$$

The state-space equation is:

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -g & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ \frac{1}{m} & 0 \\ 0 & 0 \\ 0 & \frac{r}{I} \end{bmatrix} \mathbf{u} - \begin{bmatrix} 0 \\ 0 \\ 0 \\ g \\ 0 \\ 0 \end{bmatrix} \quad (1.6.43)$$

The output equation is given by:

$$\mathbf{y} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \mathbf{u} \quad (1.6.44)$$

Ball and Beam System

The ball and beam system is a classic control theory problem that involves balancing a ball on a beam, which can tilt to control the ball's position. This system is widely used in control engineering to study dynamics, stability, and control strategies. The system consists of a beam that can rotate about its center and a ball that rolls along the beam. The goal is to control the beam's angle to regulate the ball's position. The system has two primary degrees of freedom: the position of the ball along the beam and the angular position of the beam.

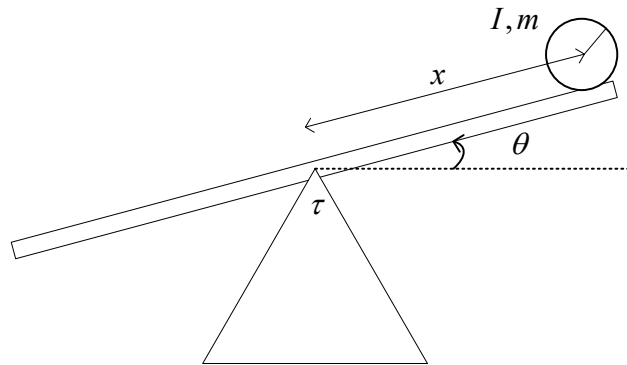


Figure 1.14: *Ball and Beam System*

Let I_1 , m , and I_2 denote the inertia of the beam, the mass of the ball, and the inertia of the ball, respectively. The position and velocity of the center of mass of the ball are

$$x_c = (x \cos \theta, x \sin \theta) \quad (1.6.45)$$

$$\dot{x}_c = (\dot{x} \cos \theta - x \dot{\theta} \sin \theta, \dot{x} \sin \theta + x \dot{\theta} \cos \theta) \quad (1.6.46)$$

The translational part of the kinetic energy is given by

$$K_{\text{trans}} = \frac{1}{2}m\dot{x}_c^2 = \frac{1}{2}m(\dot{x}^2 + x^2\dot{\theta}^2) \quad (1.6.47)$$

The rotational kinetic energy of the beam-and-ball system is

$$K_{\text{rot}} = \frac{1}{2}I_1\dot{\theta}^2 + \frac{1}{2}I_2\left(\frac{\dot{x}}{r}\right)^2 \quad (1.6.48)$$

The total kinetic energy of the system is

$$K = K_{\text{trans}} + K_{\text{rot}} = \frac{1}{2}(I_1 + mx^2)\dot{\theta}^2 + \frac{1}{2}m\left(1 + \frac{I_2}{mr^2}\right)\dot{x}^2 \quad (1.6.49)$$

Let $\lambda = 1 + \frac{I_2}{mr^2}$ is a constant.

The potential energy of the system is

$$V(x, \theta) = mgx \sin \theta \quad (1.6.50)$$

The Lagrangian of the system is

$$\begin{aligned} L &= K(x, \theta) - V(x, \theta) \\ &= \frac{1}{2}(I_1 + mx^2)\dot{\theta}^2 + \frac{1}{2}m\left(1 + \frac{I_2}{mr^2}\right)\dot{x}^2 - mgx \sin \theta \end{aligned} \quad (1.6.51)$$

The only variable that can be actuated is the angle of the beam; the position of the ball is an unactuated variable. Thus the Euler-Lagrangian equation (see appendix A) of the systems is

$$\frac{d}{dt} \left[\frac{\partial L}{\partial \dot{q}} \right] - \frac{\partial L}{\partial q} = \mathbf{Q}_i \quad (1.6.52)$$

where Q_i represents the generalized forces or torques, including friction and external inputs. Assuming that the pivot friction is negligible, the only external force applied to

the system is the torque at the pivot joint τ . Therefore, we have $\mathbf{Q}_i = \begin{bmatrix} \tau \\ 0 \end{bmatrix}$. By solving

the Euler-Lagrange equation, we obtain:

$$(I_1 + mx^2)\ddot{\theta} + 2mx\dot{x}\dot{\theta} + mgx \cos \theta = \tau \quad (1.6.53)$$

$$\lambda\ddot{x} - x\dot{\theta}^2 + g \sin \theta = 0 \quad (1.6.54)$$

With state variables $x_1 = x, x_2 = \dot{x}, x_3 = \theta, x_4 = \dot{\theta}$, the dynamics of the ball and beam system can be described by the following set of nonlinear differential equations:

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= \frac{mx_4^2x_1 - g \sin x_3}{\lambda} \\ \dot{x}_3 &= x_4 \\ \dot{x}_4 &= \frac{\tau - 2mx_1x_2x_3 - mgx_1 \cos x_3}{I + mx_1^2} \end{aligned} \quad (1.6.55)$$

Using small angle approximation ($x_3 = 0$) and the ball is at the center of the beam ($x_1 = 0$), which means ($\sin x_3 \approx x_3$ and $\cos x_3 \approx 1$), the linearized state-space equations become:

$$\begin{aligned}\dot{x}_1 &= x_2 \\ \dot{x}_2 &= \frac{-gx_3}{\lambda} \\ \dot{x}_3 &= x_4 \\ \dot{x}_4 &= \frac{\tau - mgx_1}{I}\end{aligned}\tag{1.6.56}$$

The state-space representation of the system is:

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & \frac{-g}{\lambda} & 0 \\ 0 & 0 & 0 & 1 \\ \frac{-mg}{I} & 0 & 0 & 0 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ \frac{1}{I} \end{bmatrix} \tau\tag{1.6.57}$$

The control objective is to regulate the position of the ball; thus, the output must be x_1 , which results in the output equation

$$y = \begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix} \mathbf{x}\tag{1.6.58}$$

The Acrobot

The Acrobot is representative of the primary challenge in underactuated robots. In order to swing up and balance the entire system, the controller must reason about and exploit the state-dependent coupling between the actuated degree of freedom and the unactuated degree of freedom. It is also an important system because, as we will see, it closely resembles one of the simplest models of a walking robot.

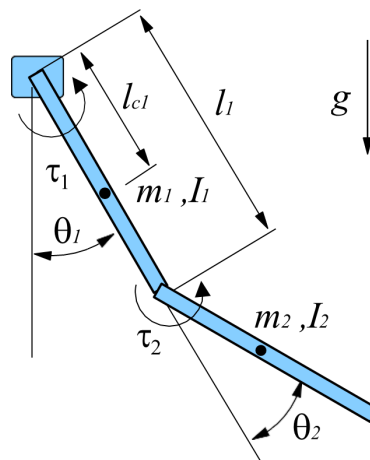


Figure 1.15: Acrobot

Figure 1.15 illustrates the model parameters used in our analysis. θ_1 is the shoulder joint angle, θ_2 is the elbow (relative) joint angle, and we will use $\mathbf{q} = [\theta_1, \theta_2]^\top$, $\mathbf{x} = [q, \dot{q}]^\top$.

The zero configuration is with both links pointed directly down. The moments of inertia, I_1, I_2 , are taken about the pivots. The task is to stabilize the unstable fixed point $\mathbf{x} = [\pi, 0, 0, 0]^\top$.

We will derive the equations of motion for the Acrobot using the method of Lagrange. The locations of the center of mass of each link, p_{c1}, p_{c2} , are given by the kinematics:

$$p_{c1} = \begin{bmatrix} l_{c1}s_1 \\ -l_{c1}c_1 \end{bmatrix}, \quad p_{c2} = \begin{bmatrix} l_1s_1 + l_{c2}s_{1+2} \\ -l_1c_1 - l_{c2}c_{1+2} \end{bmatrix}, \quad (1.6.59)$$

where s_1 is shorthand for $\sin(\theta_1)$, c_{1+2} is shorthand for $\cos(\theta_1 + \theta_2)$, etc. The energy is given by:

$$T = T_1 + T_2, \quad T_1 = \frac{1}{2}I_1\dot{q}_1^2 \quad (1.6.60)$$

$$T_2 = \frac{1}{2}(m_2l_1^2 + I_2 + 2m_2l_1l_{c2}c_2)\dot{q}_1^2 + \frac{1}{2}I_2\dot{q}_2^2 + (I_2 + m_2l_1l_{c2}c_2)\dot{q}_1\dot{q}_2 \quad (1.6.61)$$

$$u = -m_1gl_{c1}c_1 - m_2g(l_1c_1 + l_{c2}c_{1+2}) \quad (1.6.62)$$

Entering these quantities into the Lagrangian yields the equations of motion:

$$(I_1 + I_2 + m_2l_1^2 + 2m_2l_1l_{c2}c_2)\ddot{q}_1 + (I_2 + m_2l_1l_{c2}c_2)\ddot{q}_2 - 2m_2l_1l_{c2}s_2\dot{q}_1\dot{q}_2 \quad (1.6.63)$$

$$-m_2l_1l_{c2}s_2\dot{q}_2^2 + m_1gl_{c1}s_1 + m_2g(l_1s_1 + l_{c2}s_{1+2}) = \tau_1 \quad (1.6.64)$$

$$(I_2 + m_2l_1l_{c2}c_2)\ddot{q}_1 + I_2\ddot{q}_2 + m_2l_1l_{c2}s_2\dot{q}_1^2 + m_2gl_{c2}s_{1+2} = \tau_2 \quad (1.6.65)$$

In standard manipulator equation form:

$$\mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} = \tau_g(\mathbf{q}) + \mathbf{B}u, \quad (1.6.66)$$

using $\mathbf{q} = [\theta_1, \theta_2]^\top$, $u = \tau$, the mass matrix $\mathbf{M}(\mathbf{q})$ is given by:

$$\mathbf{M}(\mathbf{q}) = \begin{bmatrix} I_1 + I_2 + m_2l_1^2 + 2m_2l_1l_{c2}c_2 & I_2 + m_2l_1l_{c2}c_2 \\ I_2 + m_2l_1l_{c2}c_2 & I_2 \end{bmatrix}, \quad (1.6.67)$$

the Coriolis and centrifugal matrix $\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})$ is given by:

$$\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}}) = \begin{bmatrix} -2m_2l_1l_{c2}s_2\dot{q}_2 & -m_2l_1l_{c2}s_2\dot{q}_2 \\ m_2l_1l_{c2}s_2\dot{q}_1 & 0 \end{bmatrix}, \quad (1.6.68)$$

the gravitational torque vector $\tau_g(\mathbf{q})$ is given by:

$$\tau_g(\mathbf{q}) = \begin{bmatrix} -m_1 g l_{c1} s_1 - m_2 g (l_1 s_1 + l_{c2} s_{1+2}) \\ -m_2 g l_{c2} s_{1+2} \end{bmatrix}, \quad (1.6.69)$$

The input matrix $\mathbf{B}$ is given by:

$$\mathbf{B} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}. \quad (1.6.70)$$

The nonlinear equations of motion for the Acrobot are linearized around the equilibrium point $\mathbf{q} = [\pi, 0]^\top$ and $\dot{\mathbf{q}} = [0, 0]^\top$. The linearization is performed using small-angle approximations. Assume small deviations from the equilibrium point:

$$\theta_1 = \pi + \Delta\theta_1, \quad \theta_2 = 0 + \Delta\theta_2, \quad (1.6.71)$$

where $\Delta\theta_1$ and $\Delta\theta_2$ are small angles. For small angles, we use the following approximations:

$$\sin(\theta_1) \approx -\Delta\theta_1, \quad \cos(\theta_1) \approx -1, \quad (1.6.72)$$

$$\sin(\theta_1 + \theta_2) \approx -\Delta\theta_1 + \Delta\theta_2, \quad \cos(\theta_1 + \theta_2) \approx -1. \quad (1.6.73)$$

The mass matrix $\mathbf{M}(\mathbf{q})$ is linearized as:

$$\mathbf{M} = \begin{bmatrix} I_1 + I_2 + m_2 l_1^2 + 2m_2 l_1 l_{c2} & I_2 + m_2 l_1 l_{c2} \\ I_2 + m_2 l_1 l_{c2} & I_2 \end{bmatrix}. \quad (1.6.74)$$

The Coriolis matrix $\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})$ is linearized as:

$$\mathbf{C} \approx \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}. \quad (1.6.75)$$

The linearized gravitational torque vector $\tau_g(\mathbf{q})$ is linearized as:

$$\tau_g = \begin{bmatrix} (m_1 g l_{c1} + m_2 g l_1 - m_2 g l_{c2}) \Delta\theta_1 + m_2 g l_{c2} \Delta\theta_2 \\ -m_2 g l_{c2} \Delta\theta_1 + m_2 g l_{c2} \Delta\theta_2 \end{bmatrix}. \quad (1.6.76)$$

The linearized equations of motion can be written in state-space form:

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}, \quad (1.6.77)$$

Where:

$$\mathbf{x} = \begin{bmatrix} \Delta\theta_1 \\ \Delta\theta_2 \\ \Delta\dot{\theta}_1 \\ \Delta\dot{\theta}_2 \end{bmatrix}, \quad \mathbf{u} = \begin{bmatrix} \tau_1 \\ \tau_2 \end{bmatrix}. \quad (1.6.78)$$

The state matrix $\mathbf{A}$ and input matrix $\mathbf{B}$ are:

$$\mathbf{A} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ a_{31} & a_{32} & 0 & 0 \\ a_{41} & a_{42} & 0 & 0 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ b_{31} & b_{32} \\ b_{41} & b_{42} \end{bmatrix}, \quad (1.6.79)$$

Where:

$$\begin{aligned} a_{31} &= \frac{m_1 g l_{c1} + m_2 g l_1 - m_2 g l_{c2}}{M_{11}}, \\ a_{32} &= \frac{m_2 g l_{c2}}{M_{11}}, \quad a_{41} = \frac{-m_2 g l_{c2}}{M_{22}}, \quad a_{42} = \frac{m_2 g l_{c2}}{M_{22}}, \\ b_{31} &= \frac{1}{M_{11}}, \quad b_{32} = 0, \quad b_{41} = 0, \quad b_{42} = \frac{1}{M_{22}}. \end{aligned}$$

1.7 Mathematical Modeling of Electro-mechanical Systems

Electro-mechanical systems can be represented using differential equations that describe the dynamic behavior of both electrical and mechanical components. A common example of an electro-mechanical system is a solenoid actuator valve and a DC motor.

1.7.1 Example

Solenoid Actuator-Valve System

The 3/2 solenoid actuator valve, depicted in Figure 1.16a, plays a crucial role in regulating fluid flow through three ports: one inlet and two outlets.

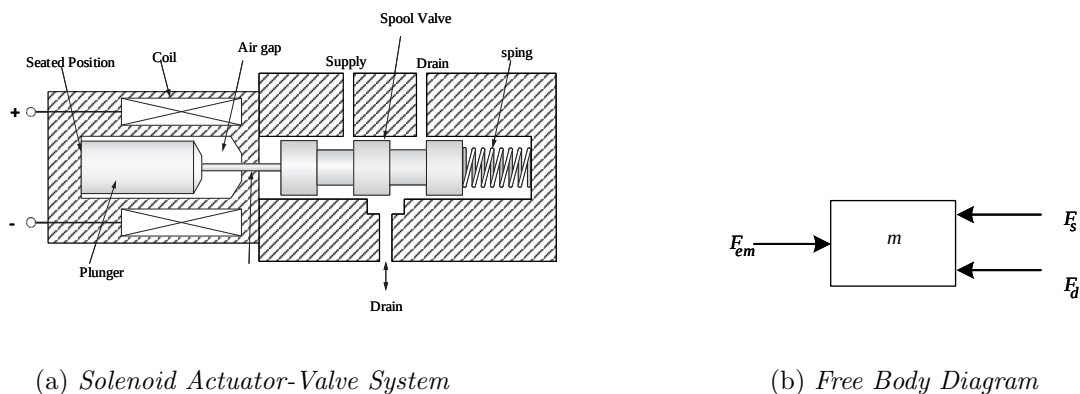


Figure 1.16: *Solenoid Actuator-Valve System*

When an electrical current is applied to the solenoid, it generates a magnetic field that actuates the internal mechanism of the valve. This movement facilitates fluid flow from the inlet to one outlet while simultaneously blocking the other. Upon deactivation of the current, a spring mechanism restores the valve to its original position, thereby redirecting the flow to the second outlet. This efficient operation enables precise control

over actuators and cylinders in various pneumatic and hydraulic applications, establishing the solenoid actuator valve as a vital component in automation and fluid management systems.

Since the spool valve operates within a fluid medium, which introduces viscous friction that influences its dynamic behavior. The free body diagram of the system, illustrated in Figure 1.16b, elucidates the forces acting on the valve and provides insights into its dynamics, particularly regarding the impact of fluid resistance on its operation.

Applying Newton's second law of motion, the force acting on the spool valve can be expressed as:

$$\begin{aligned}\Sigma F &= F_{em} - F_s - F_d = m\ddot{x} \\ Ni - kx - d\dot{x} &= m\ddot{x}\end{aligned}\tag{1.7.1}$$

where m is mass of the spool, x is displacement of the spool from its equilibrium position, N is number of turns in the solenoid, i is the current flowing through the solenoid, k is spring constant and d is damping coefficient.

Let the states of the system be $x_1 = x$ and $x_2 = \dot{x}$, the position and velocity of the mass of the spool. The state space representation will be

$$\begin{aligned}\dot{x}_1 &= x_2 \\ \dot{x}_2 &= -\frac{k}{m}x_1 - \frac{d}{m}x_2 + \frac{N}{m}i\end{aligned}\tag{1.7.2}$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{d}{m} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{N}{m} \end{bmatrix} i\tag{1.7.3}$$

$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\tag{1.7.4}$$

DC Motor System

Consider an armature voltage-controlled DC motor with armature resistance R_a and armature inductance L_a used to drive a mechanical load with load torque τ_L having rotational inertia J through a coupling gear with DC motor shaft gear to load side gear ratio $\frac{r_m}{r_L}$ at a speed ω_L . Let the mechanical friction coefficient between the rotor and the load be b , as shown by the equivalent circuit in Figure 1.17. The control input to the DC motor is armature voltage V_a , and the output is the speed of the load.

The desired goal is to find V_a that can derive the load at the required speed. To determine V_a , we have to know the relation between the speed of the DC motor ω_m and armature voltage V_a , which, of course, needs the mathematical model of the DC motor.

From the electrical characteristics of the DC motor, the torque produced by the motor τ_m is proportional to armature current i_a , while the back emf E_b of the motor is proportional

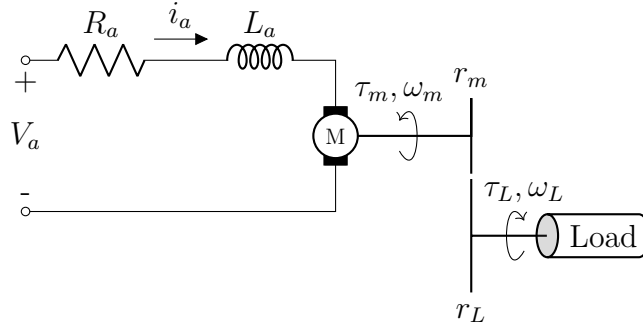


Figure 1.17: Armature Voltage Controlled DC Motor with Mechanical Load

to the speed of the motor ω_m , so we have

$$\tau_m = k_t i_a \quad (1.7.5)$$

$$E_b = k_b \omega_m \quad (1.7.6)$$

where k_t and k_b are torque and back emf proportionality constants, respectively. Using Kirchhoff's current law

$$V_a = L_a \frac{d}{dt} i_a + R_a i_a + E_b \quad (1.7.7)$$

and from the mechanical analysis of the load side

$$\tau_m = J \frac{d}{dt} \omega_L + b \omega_L + \tau_L \quad (1.7.8)$$

Replacing for E_b and τ_m in equation 1.7.7 and equation 1.7.8 respectively and rearranging;

$$\frac{d}{dt} i_a = -\frac{R_a}{L_a} i_a - \frac{k_b}{L_a} \omega_m + \frac{1}{L_a} V_a \quad (1.7.9)$$

$$\frac{d}{dt} \omega_L = \frac{k_t}{J} i_a - \frac{b}{J} \omega_L - \frac{1}{J} \tau_L \quad (1.7.10)$$

From the relation between gear radius ratio and angular speed of the gears, we know that

$$\omega_m = \frac{r_L}{r_m} \omega_L \quad (1.7.11)$$

replacing for ω_m in equation 1.7.9, we have

$$\frac{d}{dt} i_a = -\frac{R_a}{L_a} i_a - \frac{k_b r_L}{L_a r_m} \omega_L + \frac{1}{L_a} V_a \quad (1.7.12)$$

$$\frac{d}{dt} \omega_L = \frac{k_t}{J} i_a - \frac{b}{J} \omega_L - \frac{1}{J} \tau_L \quad (1.7.13)$$

If we consider load torque τ_L as an uncontrolled (disturbance) input, the state space representation of DC motor dynamics is

$$\begin{bmatrix} \dot{i}_a \\ \dot{\omega} \end{bmatrix} = \begin{bmatrix} -\frac{R_a}{L_a} & -\frac{k_b r_L}{L_a r_m} \\ \frac{k_t}{J} & -\frac{b}{J} \end{bmatrix} \begin{bmatrix} i_a \\ \omega_L \end{bmatrix} + \begin{bmatrix} \frac{1}{L_a} \\ 0 \end{bmatrix} V_a + \begin{bmatrix} 0 \\ -\frac{1}{J} \end{bmatrix} \tau_L \quad (1.7.14)$$

$$y = \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} i_a \\ \omega_L \end{bmatrix} \quad (1.7.15)$$

As we can see from equation 1.7.14 and equation 1.7.15, the dc motor dynamics has two states, namely i_a and ω_L .

If the position of the DC motor is required for control purposes, like in servo motors, we add one state $\dot{\theta} = \omega_m$. In this case, the state space representation becomes

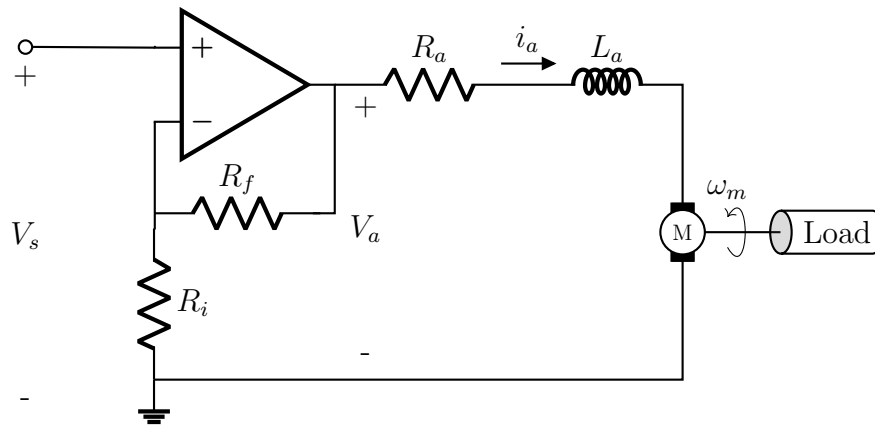
$$\begin{bmatrix} \dot{i}_a \\ \dot{\omega}_L \\ \dot{\theta}_L \end{bmatrix} = \begin{bmatrix} -\frac{R_a}{L_a} & -\frac{k_b r_L}{L_a r_m} & 0 \\ \frac{k_t}{J} & -\frac{J}{r_L} & 0 \\ 0 & \frac{r_L}{r_m} & 0 \end{bmatrix} \begin{bmatrix} i_a \\ \omega_L \\ \theta_L \end{bmatrix} + \begin{bmatrix} \frac{1}{L_a} \\ 0 \\ 0 \end{bmatrix} V_a + \begin{bmatrix} 0 \\ -\frac{1}{J} \\ 0 \end{bmatrix} \tau_L \quad (1.7.16)$$

$$y = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} i_a \\ \omega_L \\ \theta_L \end{bmatrix} \quad (1.7.17)$$

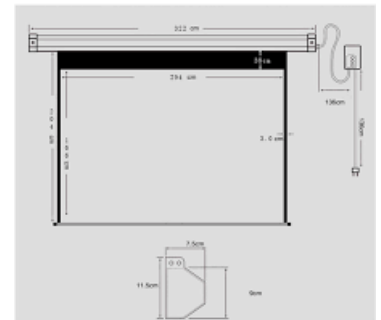
Projector Screen Positioning System

The DC motor position control system for the remote-controlled projector screen incorporates a DC motor paired with a non-inverting pre-amplifier to accurately manage the screen's position. This configuration boosts the input armature voltage supplied to the DC motor, enhancing its performance and responsiveness.

The system setup incorporates a DC motor paired with a pre-amplifier to accurately



(a) Armature Voltage Controlled DC Motor With Pre-amplifier



(b) Remote Controlled Projector Screen

Figure 1.18: Projector Screen Positioning System

control the position of a projector screen. The pre-amplifier boosts the input armature voltage supplied to the DC motor, enhancing its performance and responsiveness. The

position of the leading edge of the projector screen is proportional to the angular position of the coupling gear. Mathematically, this relationship is expressed as:

$$y = \theta r_L \quad (1.7.18)$$

where: y is the position of the leading edge of the screen, θ is the angular position of the coupling gear, r_L is the radius of the coupling gear.

The state-space model describes the dynamics of the system, including how the armature current, angular velocity, and screen position evolve in response to the input voltage and load torque. The output equation specifies that the position of the leading edge of the screen $y(t)$ is the measured output of the system.

$$\begin{bmatrix} \dot{i}_a \\ \dot{\omega}_L \\ \dot{y} \end{bmatrix} = \begin{bmatrix} -\frac{R_a}{L_a} & -\frac{k_b r_L}{L_a r_m} & 0 \\ \frac{k_t}{J} & -\frac{b}{J} & 0 \\ 0 & r_L \frac{r_L}{r_m} & 0 \end{bmatrix} \begin{bmatrix} i_a \\ \omega_L \\ y \end{bmatrix} + \begin{bmatrix} \frac{A_v}{L_a} \\ 0 \\ 0 \end{bmatrix} V_a + \begin{bmatrix} 0 \\ -\frac{1}{J} \\ 0 \end{bmatrix} \tau_L \quad (1.7.19)$$

$$y(t) = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} i_a \\ \omega_L \\ y \end{bmatrix} \quad (1.7.20)$$

1.8 Mathematical Modeling of Fluid Systems

Mathematical modeling of fluid systems involves using mathematical expressions to describe the behavior and dynamics of fluids, whether they are in motion or at rest. These models are essential for understanding and predicting fluid behavior in various engineering applications, such as hydraulics, aerodynamics, and process engineering.

Fluid systems typically consist of a flow source or pressure source, energy-storing elements, and energy-dissipating components, similar to electrical systems.

1.8.1 Basic Laws Used in Fluid Systems Modeling

The modeling of fluid systems is based on fundamental physical laws, including the conservation of mass, momentum, and energy. Key concepts include:

- **Continuity Equation:** This principle states that mass cannot be created or destroyed in a closed system. For incompressible fluids, the continuity equation can be expressed as:

$$\nabla \cdot \mathbf{v} = 0 \quad (1.8.1)$$

where $\mathbf{v}$ is the fluid velocity vector. This equation ensures that the mass flow rate remains constant throughout a streamline.

- **Bernoulli's Equation:** This equation relates the pressure, velocity, and elevation in a moving fluid and is derived from the conservation of energy. For an incompressible, non-viscous fluid, it can be expressed as:

$$P + \frac{1}{2}\rho v^2 + \rho gh = K \quad (1.8.2)$$

where P is the pressure, ρ is the fluid density, v is the flow velocity, g is the acceleration due to gravity, and h is the height above a reference level.

- **Navier-Stokes Equations:** These equations describe the motion of fluid substances and are derived from Newton's second law. They account for the forces acting on the fluid, including viscous forces, pressure forces, and external forces:

$$\rho \left(\frac{\partial \mathbf{v}}{\partial t} + \mathbf{v} \cdot \nabla \mathbf{v} \right) = -\nabla P + \mu \nabla^2 \mathbf{v} + \mathbf{f} \quad (1.8.3)$$

where ρ is the fluid density, p is the pressure, μ is the dynamic viscosity, and $\mathbf{f}$ represents body forces (e.g., gravity).

- **Law of Conservation of Mass:** This law states that the mass of a closed system must remain constant over time. In fluid dynamics, this is often represented by the continuity equation mentioned above.

1.8.2 Key Fluid Properties

Several key properties are used to characterize fluid behavior in mathematical models:

Fluid Inertia

Fluid inertia refers to the tendency of a fluid to resist changes in its state of motion. It is quantified by the mass of the fluid and can be represented in terms of fluid pressure (P) and volumetric flow rate ($\dot{Q}$):

$$L_f = \frac{\Delta P}{\Delta \dot{Q}} \quad (1.8.4)$$

where L_f is the fluid inertia. This property is crucial in analyzing transient flows and dynamic responses in fluid systems.

Fluid Capacitance

Fluid capacitance describes the ability of a fluid system to store volume in response to pressure changes. It is defined as the ratio of the change in volume (ΔV) to the change in pressure (ΔP):

$$C = \frac{\Delta V}{\Delta P} \quad (1.8.5)$$

where C is the fluid capacitance. This property is important in systems like hydraulic accumulators, where fluid storage plays a significant role in performance.

Fluid Bulk Modulus

A fluid is classified as incompressible if its density remains constant, whereas it is considered compressible if its density varies with pressure. Liquids can typically be treated as incompressible under relatively low pressures, while hydraulic fluids at high pressures are regarded as compressible. The bulk modulus (β) is a measure of a fluid's resistance to compression. It is defined as the ratio of the change in pressure to the relative change in volume:

$$\beta = \rho_o \frac{\Delta P}{\Delta \rho} \quad (1.8.6)$$

where ρ_o is a reference fluid density at a nominal pressure and temperature. A high bulk modulus indicates that a fluid is incompressible and can resist changes in volume under pressure, which is a key assumption in many fluid dynamics problems.

1.8.3 Flow Characteristics

Fluid flow can be categorized into laminar and turbulent flow, each exhibiting different characteristics and resistances.

Laminar Flow

Laminar flow occurs when a fluid flows in parallel layers, with minimal disruption between them. This type of flow is characterized by smooth, orderly motion and is typically observed when the source pressure and outlet pressure are the same, or when the diameter of the pipe is larger than the length of the pipe. The laminar flow resistance is proportional to the ratio of pressure difference to volumetric flow rate:

$$R_L = \frac{\Delta P}{Q} \quad (1.8.7)$$

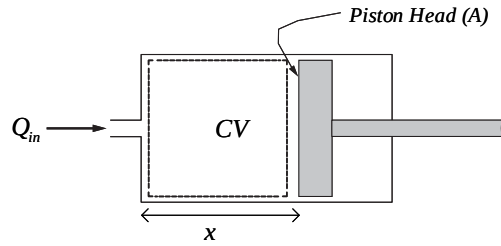
Turbulent Flow

Turbulent flow is characterized by chaotic, irregular fluid motion. This type of flow involves mixing and eddies, leading to increased energy dissipation and resistance. The transition from laminar to turbulent flow is influenced by factors such as flow velocity, fluid viscosity, and pipe diameter. The Turbulent flow resistance is proportional to the ratio of the pressure difference to the square of the volumetric flow rate:

$$R_T = \frac{\Delta P}{Q^2} \quad (1.8.8)$$

1.8.4 Examples*Hydraulic Piston and Cylinder System*

Hydraulic systems are essential in various industrial applications, facilitating the efficient transfer of power through the controlled movement of fluids. Central to many hydraulic systems is the hydraulic piston and cylinder arrangement, which converts hydraulic energy into mechanical work. This system operates on the principles of fluid mechanics and adheres to fundamental laws, including Pascal's principle, which asserts that pressure

Figure 1.19: *Hydraulic Piston and Cylinder*

applied to a confined fluid is transmitted undiminished in all directions. As illustrated in Figure 1.19, a hydraulic piston is positioned within a cylindrical chamber filled with hydraulic fluid. When pressure is applied to the fluid, it generates a force on the piston, prompting its movement. This movement can be utilized to perform a range of tasks, including lifting heavy loads, actuating machinery, and controlling the motion of other components within a system.

Using the law of conservation of mass, we can express the rate of accumulation of mass as the difference between the mass inflow rate and the mass outflow rate. In this case, since there is no mass leaving the system (no outflow), the dynamic equation governing the motion of the piston can be formulated as:

$$\dot{m} = \rho Q_{\text{in}} \quad (1.8.9)$$

Here, $\dot{m}$ represents the rate of change of mass, ρ is the fluid density, and Q_{in} is the volumetric flow rate into the system.

Next, we can substitute the mass m with the product of density and volume, $m = \rho V$. This leads to the following relationship:

$$\rho \dot{V} + V \dot{\rho} = \rho Q_{\text{in}} \quad (1.8.10)$$

In this equation, $\dot{V}$ denotes the rate of change of volume, and V is the volume of the fluid within the cylinder.

To further refine our model, we can replace the density ρ using the relationship derived from the bulk modulus of elasticity, which is given by:

$$\rho = \frac{\rho_o}{\beta} \frac{dP}{dt} \quad (1.8.11)$$

where ρ_o is the reference density, β is the bulk modulus, and $\frac{dP}{dt}$ represents the rate of change of pressure. Additionally, we can express the rate of change of volume $\dot{V}$ in terms of the piston head area A and the displacement of the piston head x :

$$\dot{V} = A \dot{x} \quad (1.8.12)$$

Substituting these expressions into our earlier equation, we arrive at:

$$\begin{aligned}\frac{\rho}{\beta} \dot{P} Ax + \rho A \dot{x} &= \rho Q_{in} \\ \dot{P} &= \frac{\beta Ax}{\rho} (Q_{in} - A \dot{x})\end{aligned}\quad (1.8.13)$$

In this final equation, $\dot{P}$ represents the rate of change of pressure, A is the cross-sectional area of the piston, and $\dot{x}$ is the velocity of the piston head. This equation captures the dynamic relationship between pressure changes, fluid density, and the movement of the piston, allowing us to analyze the behavior of the hydraulic system under varying conditions.

By integrating these principles, we can effectively model the hydraulic piston and cylinder system, facilitating predictions of its performance in practical applications.

Hydraulic Actuator System

The simple hydraulic actuator shown in Figure 1.20a is connected to a load and accompanied by a restoring spring. As the piston moves, it drives the load while compressing the spring. This interaction is crucial for understanding the dynamics of the actuator.

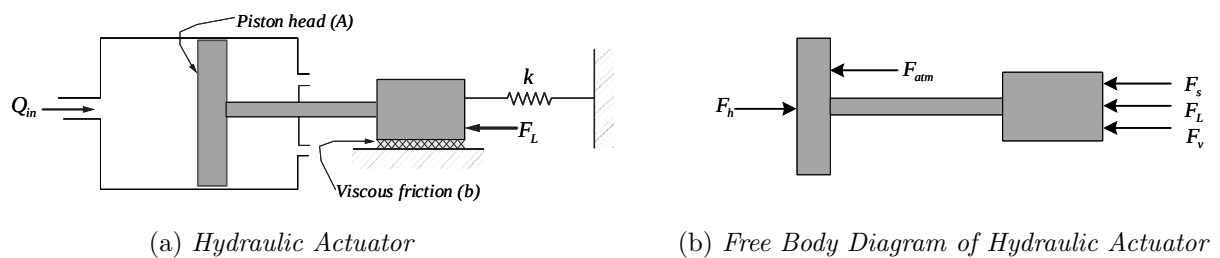


Figure 1.20: Hydraulic Actuator System

The free body diagram of the system is illustrated in Figure 1.20b. This diagram helps visualize the forces acting on the actuator, which include the hydraulic force, atmospheric pressure force, viscous force, spring force, and the load force.

The mathematical model describing the motion of the actuator is derived using Newton's first law of motion, which states that the sum of forces acting on an object is equal to the mass of the object multiplied by its acceleration. This can be expressed as:

$$\Sigma F = F_h - F_{atm} - F_v - F_s - F_L = m\ddot{x} \quad (1.8.14)$$

Where F_h is the hydraulic force acting on the piston, F_{atm} is the force due to atmospheric pressure, F_v is the viscous force, F_s is the spring force, F_L is the load force, m is the mass of the piston, and $\ddot{x}$ is the acceleration of the piston.

By replacing F_h with PA (where P is the pressure and A is the piston head area), and substituting $F_{atm} = P_{atm}A$ for the atmospheric pressure force, the equation becomes:

$$(F - F_{atm})A - F_L = m\ddot{x} + b\dot{x} + kx \quad (1.8.15)$$

Selecting the states of the system $x_1 = x$, $x_2 = \dot{x}$, the state space representation will be

$$\begin{aligned}\dot{x}_1 &= x_2 \\ \dot{x}_2 &= -\frac{k}{m}x_1 - \frac{b}{m}x_2 + \frac{PA}{m} - \frac{1}{m}F_L - \frac{P_{atm}A}{m}\end{aligned}\quad (1.8.16)$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{b}{m} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ \frac{A}{m} & -\frac{1}{m} \end{bmatrix} \begin{bmatrix} P \\ F_L \end{bmatrix} - \begin{bmatrix} 0 \\ \frac{A}{m} \end{bmatrix} P_{atm}\quad (1.8.17)$$

$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\quad (1.8.18)$$

To find the pressure required to hold the piston at its nominal position, we can set the velocity $\dot{x} = 0$ and assume $x = x_0$ (the equilibrium position). This simplifies our equation to:

$$P = P_{atm} + \frac{F_L}{A} + \frac{k}{A}x_0\quad (1.8.19)$$

This equation indicates that the pressure P must balance the load force per unit area, plus the atmospheric pressure acting on the piston. Understanding this relationship is essential for designing hydraulic systems that effectively manage loads and maintain desired positions.

1.9 Mathematical Modeling of Chemical Process

The dynamic model of chemical systems is derived from the balance equations for total mass, energy, and component mass (mole). These equations are crucial for understanding and predicting the behavior of chemical processes. Let's delve into the concepts and equations involved in the mathematical modeling of chemical processes.

The dynamic model of chemical systems is derived from the balance equations for total mass, energy, and component mass (mole). For a system with a well-defined boundary,

$$Change\ Inventory = (IN - OUT)_{TZB} + (Generated - Loss)_{IZB}\quad (1.9.1)$$

Note:- TZB (through the boundary of the system), IZB (In the boundary of the system)
To formulate the dynamic equation, we have to decide the control boundary and which balance we have to use. Once the boundary is set, if the interest is,

- Mass, volume or pressure $\rightarrow$ mass balance,
- Concentration $\rightarrow$ component balance,
- Temperature $\rightarrow$ energy balance, and
- Interaction between flow and pressure $\rightarrow$ Mechanical energy balance (= momentum balance = Bernoulli = Newton's second law)

1.9.1 Fundamental Laws for modeling chemical processes

Mass balance

The rate of change of mass accumulation in a closed boundary is,

$$\frac{dm}{dt} = w_{in} - w_{out} \quad (1.9.2)$$

w_{in} and w_{out} are mass flow rates for the entering and exiting streams (bulk flow).
Introducing density,

$$\rho \frac{dV}{dt} = \rho_{in} q_{in} - \rho_{out} q_{out} \quad (1.9.3)$$

q_{in} and q_{out} are volumetric¹ flow rates.
For constant density

$$\frac{dV}{dt} = q_{in} - q_{out} \quad (1.9.4)$$

In most dynamic processes, volume is not conserved; often the interest is height of the liquid, thus the equation becomes

$$\frac{dV}{dt} = A \frac{dh}{dt} + A \frac{dA}{dh} \frac{dh}{dt} = q_{in} - q_{out} \quad (1.9.5)$$

In most cases, the area is conserved. hence

$$A \frac{dh}{dt} = q_{in} - q_{out} \quad (1.9.6)$$

Dynamic Component Balance

The dynamic component balance can, for an arbitrary component A, be written

$$\frac{dn_A}{dt} = F_{A,in} - F_{A,out} + G_A \quad (1.9.7)$$

Here, n_A [mol A] is the inventory (amount) of component A inside the system's boundary, $F_{A,in} - F_{A,out}$ [mol A/s] are the molar flow rates of A in the streams (bulk flow) and G_A [mol A/s] is net generated in the chemical reactions.

With concentrations $\bar{c}_A = n_A/V$, $c_{A,in} = F_{A,in}/q_{in}$, $c_{A,out} = F_{A,out}/q_{out}$ and the generation of component as average reaction rate times the volume $G_A = \bar{r}_A V$, the equation will be

$$\frac{d(\bar{c}_A V)}{dt} = c_{A,in} q_{in} - c_{A,out} q_{out} + \bar{r}_A V [\text{mol A/s}] \quad (1.9.8)$$

¹Volumetric flow rate is the product of flow rate and cross-sectional area perpendicular to the flow rate

Dynamic Energy Balance

The general energy balance over a small duration of time is

$$\frac{dH}{dt} = \left[H_{in} - H_{out} + Q + W_s - (p_{ex} - p) \frac{dV}{dt} + V \frac{dp}{dt} \right] [J/s] \quad (1.9.9)$$

where $H[J]$ is enthalpy of the system ($H = \mathcal{U}(\text{internal energy}) + pV$), $H_{in} - H_{out}$ is the sum of internal energy in the streams plus the flow work that the streams perform on the system as they are "pushed" in or out of the system. $Q [J/s]$ is supplied heat (through the system's wall) and $W_s [J/s]$ is supplied useful mechanical work (usually shaft work, for example, from a compressor, pump, or turbine).

The term $\left[-(p_{ex} - p) \frac{dV}{dt} + V \frac{dp}{dt} \right]$ occurs if the volume and pressure in the boundary of control changes, which for many processes it is negligible compared to the other parameters.

Using specific enthalpy $\bar{h} = \frac{H}{m}$, we can combine mass balance and energy balance equations, since

$$\frac{dH}{dt} = m \frac{d\bar{h}}{dt} + \bar{h} \frac{dm}{dt} \quad (1.9.10)$$

by replacing for $\frac{dm}{dt} = w_{in} - w_{out}$

$$m \frac{d\bar{h}}{dt} = \left[w_{in}(\bar{h}_{in} - \bar{h}) - w_{out}(\bar{h}_{out} - \bar{h}) + Q + W_s - (p_{ex} - p) \frac{dV}{dt} + V \frac{dp}{dt} \right] \quad (1.9.11)$$

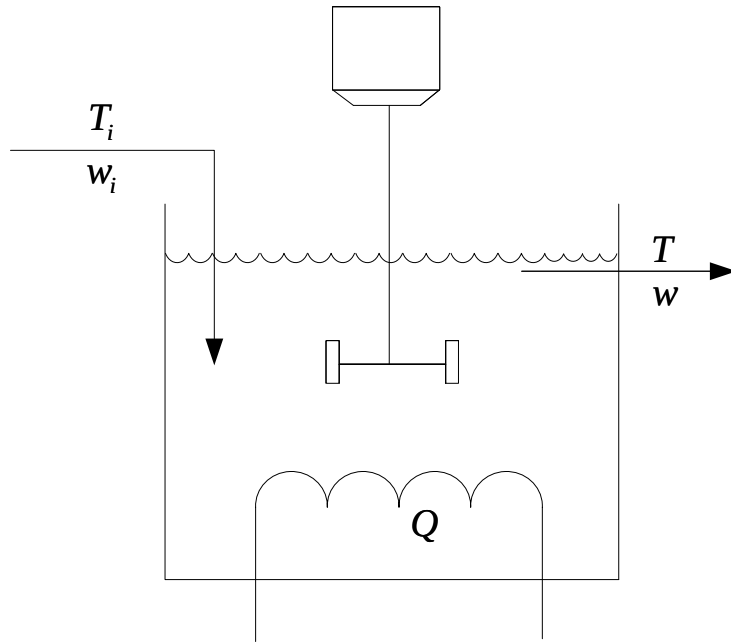
Note:- Enthalpy $\bar{h}(T, p, f, n_j)$ [J/kg] is generally a function of temperature T , pressure p , phase distribution f (where f is fraction of light phase) and composition (n_j). The time derivative of the enthalpy should be treated correctly (i.e., apply chain rule carefully).

1.9.2 Examples

Stirred-Tank Heating Process: Constant Holdup process

The liquid inlet stream consists of a single component with a mass flow rate w_i and an inlet temperature T_i . The tank contents are agitated and heated using an electrical heater that provides a heating rate, Q . The modeling assumptions

- Perfect mixing; thus, the exit temperature T is also the temperature of the tank contents.
- The inlet and outlet flow rates are equal; thus, $w_i = w$ and the liquid holdup V is constant.
- The density ρ and heat capacity C of the liquid are assumed to be constant. Thus, their temperature dependence is neglected.
- Heat losses are negligible

Figure 1.21: *Stirred-Tank Heating Process with Constant Hold-up*

The enthalpy (total energy of the system) is the sum of internal energy and the product of volume and pressure $\bar{h} = \bar{U}_{int} + pV$, but for low pressure system, it can be approximated to $\bar{h} \approx \bar{U}_{int}$, internal energy per unit mass.

The system enthalpy can be considered as a function of only temperature and pressure, $\bar{h}(T, p)$, hence

$$\frac{d\bar{h}}{dt} = f(p) \frac{\partial \bar{h}}{\partial T} \frac{dT}{dt} + f(T) \frac{\partial \bar{h}}{\partial p} \frac{dp}{dt} \quad (1.9.12)$$

For the fluid and solids $\frac{\partial \bar{h}}{\partial p} \approx 0$, and $f(p) \frac{\partial \bar{h}}{\partial T} = C_p$ (heat capacity at constant pressure),
For the total volume we have

$$\frac{dH}{dt} = m \frac{d\bar{h}}{dt} = \rho V C_p \frac{dT}{dt} \quad (1.9.13)$$

Solving for H by integrating we have $H - H_{ref} = \rho V C_p (T - T_{ref})$, in specific enthalpy, $\bar{h} - \bar{h}_{ref} = C_p (T - T_{ref})$, for inlet stream $\bar{h}_i - \bar{h}_{ref} = C_p (T_i - T_{ref})$ and $\bar{h} - \bar{h}_{ref} = C_p (T - T_{ref})$ outlet stream.

Replacing the terms in equation 1.9.11,

$$\begin{aligned}
 m \frac{d\bar{h}}{dt} &= w_{in}(\bar{h}_{in} - \bar{h}) - w_{out}(\bar{h}_{out} - \bar{h}) + Q \\
 m \frac{d\bar{h}}{dt} &= wC_p(T_i - T_{ref}) + wC_p(T - T_{ref}) + Q \\
 m \frac{d\bar{h}}{dt} &= wC_p(T_i - T) + Q \implies \rho V \frac{d\bar{h}}{dt} = wC_p(T_i - T) + Q \\
 \dot{T} &= \left(\frac{w}{\rho V} \right) T_i - \left(\frac{w}{\rho V} \right) T + \left(\frac{1}{\rho V C_p} \right) Q
 \end{aligned} \tag{1.9.14}$$

Stirred-Tank Heating Process: Variable Holdup

In this case, the only difference is the volume changes; hence, there will be a rate of change of mass, and we should consider mass balance.

$$\frac{dm}{dt} = w_i - w \implies \rho \frac{dV}{dt} = w_i - w \tag{1.9.15}$$

We know that the rate of change of the enthalpy of the system is

$$\begin{aligned}
 \frac{dH}{dt} &= H_{in} - H_{out} + Q \\
 \frac{d(m\bar{h})}{dt} &= w_i(\bar{h}_i - \bar{h}_{ref}) - w_{out}(\bar{h}_{out} - \bar{h}_{ref}) + Q \\
 \rho \frac{d(V\bar{h})}{dt} &= w_i(\bar{h}_i - \bar{h}_{ref}) - w_{out}(\bar{h}_{out} - \bar{h}_{ref}) + Q
 \end{aligned} \tag{1.9.16}$$

Since the volume is variable, we have to apply the chain rule to $\frac{d(V\bar{h})}{dt}$, hence

$$\rho \frac{d(V\bar{h})}{dt} = \rho \bar{h} \frac{dV}{dt} + \rho V \frac{d\bar{h}}{dt} \tag{1.9.17}$$

We know that $\bar{h} - \bar{h}_{ref} = C_p(T - T_{ref})$, without loss of generality, we can set $\bar{h}_{ref} = 0$, hence $\bar{h} = C_p(T - T_{ref})$, replacing this values in equation 1.9.11 and $\frac{d\bar{h}}{dt} = C_p \frac{dT}{dt}$,

$$\rho C_p(T - T_{ref}) \frac{dV}{dt} + \rho V C_p \frac{dT}{dt} = w_i C_p(T_i - T_{ref}) - w C_p(T - T_{ref}) + Q \tag{1.9.18}$$

replacing for $\rho \frac{dV}{dt}$ with $w_i - w$ from equation 1.9.15 and rearranging

$$\dot{T} = \frac{1}{\rho V C_p} w_i T_i - \frac{1}{\rho V C_p} T + \frac{1}{\rho V C_p} Q \tag{1.9.19}$$

$$\dot{V} = \frac{1}{\rho} w_i - \frac{1}{\rho} w \tag{1.9.20}$$

Electrically Heated Stirred Tank

In a stirred-tank heating system with constant holdup, the metal heating element has significant thermal capacitance. The electrical heating rate directly impacts the element's

temperature, not the liquid contents. Temperature gradients in the heating element are neglected, assuming it has a uniform temperature (T_e), which represents the average temperature for the heating. Based on this new assumption, and the previous assumptions, the unsteady-state energy balances for the tank and the heating element can be written as

$$mC_p \frac{dT}{dt} = wC_p(T_i - T) + h_e A_e(T_e - T) \quad (1.9.21)$$

$$m_e C_e \frac{dT_e}{dt} = Q - h_e A_e(T_e - T) \quad (1.9.22)$$

Where $m = \rho V$ and $m_e C_e$ is the product of the mass of metal in the heating element and its specific heat. The term $h_e A_e$ is the product of the heat transfer coefficient and area available for heat transfer.

Note that mC and $m_e C_e$ are the thermal capacitance of the tank contents and the heating element, respectively. Q is an input variable, the thermal equivalent of the instantaneous electrical power dissipation in the heating element.

The Continuous Stirred-Tank Reactor (CSTR)

The Continuous Stirred-Tank Reactor (CSTR) is a type of reactor used for continuous mixing of reactants within a vessel. The reactants are continuously fed into the reactor, while the products are continuously removed. The contents of the reactor are well mixed, and the operating conditions, such as temperature, pressure, and concentration, are uniform throughout the reactor. It is the commonly used reactor in industrial processes.

To simplify the analysis, some common assumptions must be made. These assumptions include:

- It is assumed that the contents of the reactor are perfectly mixed, meaning that the concentration of reactants and products is uniform throughout the reactor at any given time.
- The reactor operates under steady-state conditions, where the operating parameters such as flow rates, temperatures, and concentrations do not change with time.
- The volume of the reactor remains constant during the reaction, and there are no significant volume changes due to the reaction.
- Instantaneous reaction: It is assumed that the reaction occurs instantaneously once the reactants are mixed, without any time delay or reaction kinetics effects.
- The reactor is assumed to behave ideally, with no heat losses, no mass transfer limitations, and no non-ideal mixing effects.

Consider a simple liquid-phase, irreversible chemical reaction where chemical species A reacts to form species B . The reaction can be written as $A \rightarrow B$. We assume that the rate of reaction is first-order concerning component A ,

$$r = kC_A \quad (1.9.23)$$

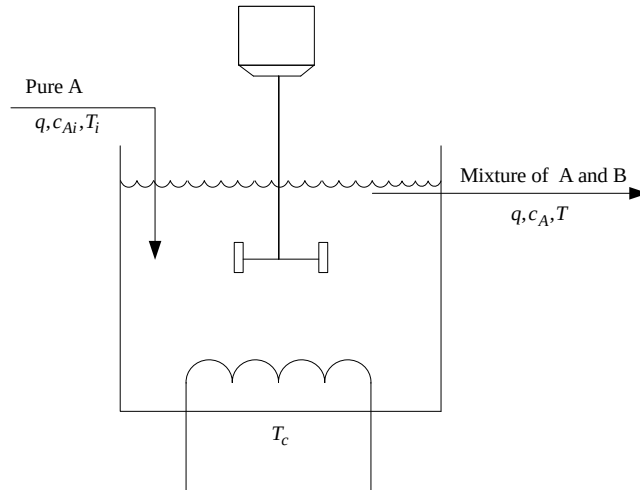


Figure 1.22: *Non-isothermal Continuous Stirred-Tank Reactor*

where r is the reaction rate of A per unit volume, k is the reaction constant, and C_A is the concentration of A. For single-phase reactions, the rate constant is typically a strong function of reaction temperature given by the Arrhenius relation,

$$k = k_o e^{\frac{-E}{RT}} \quad (1.9.24)$$

where k_o is the frequency factor and E is the activation energy, usually determined by fitting experimental data, and R is the gas constant.

The inlet stream consists of pure component A with molar concentration, C_{Ai} . A cooling coil is used to maintain the reaction mixture at the desired operating temperature by removing heat that is released in the exothermic reaction.

Our initial CSTR model development is based on three assumptions:

- The CSTR is perfectly mixed.
- The mass densities of the feed and product streams are equal and constant. They are denoted by ρ .
- The liquid volume V in the reactor is kept constant by an overflow line.

For these assumptions, the unsteady-state mass balance for the CSTR is

$$\frac{d(\rho V)}{dt} = \rho q_i - \rho q \Rightarrow q = q_i \text{ (Since } V \text{ is constant)} \quad (1.9.25)$$

Even though the inlet and outlet flow rates may change due to upstream or downstream conditions, this equation must be satisfied at all times.

Note:- both flow rates are denoted by the symbol q in the Figure.

For the stated assumptions, the unsteady-state component balances for species A (in molar concentration units) are

$$V \frac{dC_A}{dt} = q(C_{Ai} - C_A) - V k C_A \quad (1.9.26)$$

This balance is a special case of the general component balance.

For the unsteady-state energy balance for the CSTR, we need an additional five assumptions:

- The thermal capacitance of the coolant and the cooling coil wall is negligible compared to the thermal capacitance of the liquid in the tank.
- All of the coolant is at a uniform temperature, T_c . (That is, the increase in coolant temperature as the coolant passes through the coil is neglected.)
- The rate of heat transfer from the reactor contents to the coolant is given by

$$Q = UA(T_c - T)$$

where U is the overall heat transfer coefficient and A is the heat transfer area. Both of these model parameters are assumed to be constant.

- The enthalpy change associated with the mixing of the feed and the liquid in the tank is negligible compared with the enthalpy change for the chemical reaction.
- Shaft work and heat losses to the ambient can be neglected.

With the use of the new assumptions and inclusion of energy balance, the dynamic model of CSTR is given by

$$V \rho C \frac{dT}{dt} = w C (T_i - T) + (-\Delta H_R) V k C_A + UA(T_c - T) \quad (1.9.27)$$

This model is nonlinear as a result of the many product terms and the exponential temperature dependence of k ; hence, it should be solved by numerical integration. The model will be more complex if

- More complicated rate expressions are considered
- Additional species or chemical reactions are involved

Reactions involving multiple species are described by high-order, highly coupled, nonlinear reaction models, because several component balances must be written.

Quadruple Tank System

Consider a quadruple tank process shown in Figure 1.23. The objective is to control the level of the liquid in the tanks by using pump 1 and pump 2. The pumps pump the liquid from the reservoir. The volumetric flow rate² q produced by each pump is proportional to the voltage applied to each pump. Let these proportionality constants be k_1 and k_2 for pump 1 and pump 2, respectively. The position of valve 1 controls the volumetric flow rate to tank 1 and tank 4, while the position of valve 2 controls the volumetric flow rate to tank 2 and tank 3. Let the position of valve 1 be γ_1 and the position of valve 2 be γ_2 .

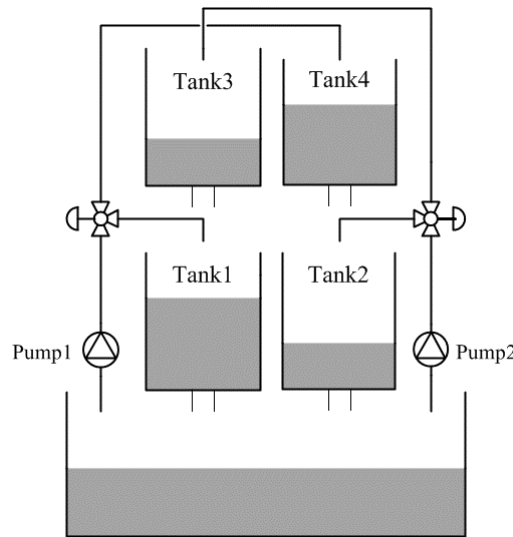


Figure 1.23: Quadruple Tank Process

So we have

$$\begin{aligned} q_{1in} &= \gamma_1 k_1 V_1 & q_{2in} &= \gamma_2 k_2 V_2 \\ q_{3in} &= (1 - \gamma_2) k_2 V_2 & q_{4in} &= (1 - \gamma_1) k_1 V_1 \end{aligned}$$

From fluid dynamics, the rate of mass accumulation in each tank is equal to the difference between mass inflow and mass outflow.

$$\dot{m} = \rho(q_{in} - q_{out}) \quad (1.9.28)$$

where q_{in} is volumetric inflow rate and q_{out} is volumetric outflow rate. The volumetric outflow rate is determined by considering each tank as an isolated system. Using the Bernoulli equation

$$P_1 + gh_1 + \frac{1}{2}v_1^2 = P_2 + gh_2 + \frac{1}{2}v_2^2 \quad (1.9.29)$$

²Volumetric flow rate q is equal to the product of the flow rate of the fluid (ν) times the area (a) perpendicular to the flow. ($q = \nu \times a$)

If we take the top as point 2 and the bottom of the tank as point 1 ($h_1 = 0$), both of them are exposed to atmospheric pressure, so $P_1 = P_2$. Similarly, since there is no inflow from the top, $v_2 = 0$. The fluid flows out of each tank due to gravity only. with these assumption's equation 1.9.29 simplifies to

$$v_1 = \sqrt{2gh_2} \quad (1.9.30)$$

Let the outlet have a cross-sectional area a , the outflow rate will be

$$q_{out} = av = \sqrt{2gh} \quad (1.9.31)$$

from basic physics $\rho = \frac{m}{V} = \frac{m}{Ah}$, rate of accumulation of mass $\dot{m}$ is

$$\dot{m} = \rho A \frac{dh}{dt} \quad (1.9.32)$$

Combining equation 1.9.28 and equation 1.9.32, the dynamic equation for each tank will be

$$A\dot{h} = q_{in} - q_{out} \quad (1.9.33)$$

The dynamic equation of the quadruple tank system is then

$$\begin{aligned} A_1\dot{h}_1 &= q_{1in} + q_{3out} - q_{1out} = \gamma_1 k_1 V_1 + a_3 \sqrt{2gh_3} - a_1 \sqrt{2gh_1} \\ A_2\dot{h}_2 &= q_{2in} + q_{4out} - q_{2out} = \gamma_2 k_2 V_2 + a_4 \sqrt{2gh_4} - a_2 \sqrt{2gh_2} \\ A_3\dot{h}_3 &= q_{3in} - q_{3out} = (1 - \gamma_2) k_2 V_2 - a_3 \sqrt{2gh_3} \\ A_4\dot{h}_4 &= q_{4in} - q_{4out} = (1 - \gamma_1) k_1 V_1 - a_4 \sqrt{2gh_4} \end{aligned}$$

$$\begin{bmatrix} \dot{h}_1 \\ \dot{h}_2 \\ \dot{h}_3 \\ \dot{h}_4 \end{bmatrix} = \begin{bmatrix} \frac{a_3 \sqrt{2gh_3}}{A_1} - \frac{a_1 \sqrt{2gh_1}}{A_1} \\ \frac{a_4 \sqrt{2gh_4}}{A_2} - \frac{a_2 \sqrt{2gh_2}}{A_2} \\ -\frac{a_3 \sqrt{2gh_3}}{A_3} \\ -\frac{a_4 \sqrt{2gh_4}}{A_4} \end{bmatrix} + \begin{bmatrix} \frac{\gamma_1 k_1}{A_1} & 0 \\ 0 & \frac{\gamma_2 k_2}{A_2} \\ 0 & \frac{(1 - \gamma_2) k_2}{A_3} \\ \frac{(1 - \gamma_1) k_1}{A_4} & 0 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} \quad (1.9.34)$$

If the levels of the liquid in tank 1 and tank 2 are required to be controlled, the output equation will be

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} h_1 \\ h_2 \\ h_3 \\ h_4 \end{bmatrix} \quad (1.9.35)$$

1.10 State Space Representation From Transfer Function

When a transfer function is converted to state space representation, the state space commonly has four normal forms (canonical forms). The forms depend on the characteristics polynomial of the transfer function or some other conditions (conditions like controllability and observability, which are explained later).

1. Controllable canonical form

Consider a transfer function

$$\frac{Y(s)}{U(s)} = \frac{N(s)}{D(s)} = \frac{b_n s^n + b_{n-1} s^{n-1} + b_{n-2} s^{n-2} \dots + b_1 s + b_0}{s^n + a_{n-1} s^{n-1} + a_{n-2} s^{n-2} \dots + a_1 s + a_0} \quad (1.10.1)$$

If we take an intermediate function $X(s)$ (state of the system) then we have,

$$\frac{Y(s)}{X(s)} \cdot \frac{X(s)}{U(s)} = \frac{b_n s^n + b_{n-1} s^{n-1} + b_{n-2} s^{n-2} \dots + b_1 s + b_0}{s^n + a_{n-1} s^{n-1} + a_{n-2} s^{n-2} \dots + a_1 s + a_0} \quad (1.10.2)$$

which can be written as

$$\frac{X(s)}{U(s)} = \frac{1}{s^n + a_{n-1} s^{n-1} + a_{n-2} s^{n-2} \dots + a_1 s + a_0} \quad (1.10.3)$$

$$\frac{Y(s)}{X(s)} = b_n s^n + b_{n-1} s^{n-1} + b_{n-2} s^{n-2} \dots + b_1 s + b_0 \quad (1.10.4)$$

From equation 1.10.3

$$U(s) = (s^n + a_{n-1} s^{n-1} + a_{n-2} s^{n-2} \dots + a_1 s + a_0) X(s) \quad (1.10.5)$$

Letting $X_1 = X(s)$, $X_2 = sX(s)$, $X_3 = s^2 X(s)$, $\dots$, $X_n = s^{n-1} X(s)$, we have

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= x_3 \\ &\vdots \\ \dot{x}_{n-1} &= x_n \\ \dot{x}_n &= -a_{n-1}x_n - a_{n-2}x_{n-1} - \dots - a_1x_2 - a_0x_1 + u \end{aligned} \quad (1.10.6)$$

which leads to a state equation

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \vdots \\ \dot{x}_{n-1} \\ \dot{x}_n \end{bmatrix} = \begin{bmatrix} 0 & 1 & \dots & 0 \\ 0 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \\ -a_0 & -a_1 & \dots & -a_{n-1} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{n-1} \\ x_n \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix} u \quad (1.10.7)$$

From equation 1.10.4

$$\begin{aligned} Y(s) &= (b_n s^n + b_{n-1} s^{n-1} + b_{n-2} s^{n-2} + \dots + b_1 s + b_0) X(s) \\ y &= b_n \dot{x}_n + b_{n-1} x_n + b_{n-2} x_{n-1} + \dots + b_1 x_2 + b_0 x_1 \\ y &= (b_0 - b_n a_0) x_1 + (b_1 - b_n a_1) x_2 + \dots + (b_{n-1} - b_n a_{n-1}) x_{n-1} + b_n u \end{aligned} \quad (1.10.8)$$

Which gives us an output equation

$$y = \begin{bmatrix} (b_0 - b_n a_0) & (b_1 - b_n a_1) & \dots & (b_{n-1} - b_n a_{n-1}) \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{n-1} \\ x_n \end{bmatrix} + b_n u \quad (1.10.9)$$

2. Observable canonical form

Factoring s^n and rearranging equation 1.10.1

$$Y(s) = b_n U(s) + s^{-1}(b_{n-1}U(s) - a_{n-1}Y(s) + s^{-1}(b_{n-2}U(s) - a_{n-2}Y(s) + \dots + s^{-1}(b_1U(s) - a_1Y(s) + s^{-1}(b_0U(s) - a_0Y(s)))) \quad (1.10.10)$$

Let

$$\begin{aligned} X_1(s) &= s^{-1}(b_0U(s) - a_0Y(s)) \\ X_2(s) &= s^{-1}(b_1U(s) - a_1Y(s) + X_1(s)) \\ &\vdots \\ X_n(s) &= s^{-1}(b_{n-1}U(s) - a_{n-1}U(s) + X_{n-1}(s)) \end{aligned} \quad (1.10.11)$$

With this equation 1.10.10 becomes,

$$Y(s) = b_n U(s) + X_n(s) \quad (1.10.12)$$

Solving for $Y(s)$ from equation 1.10.12 and replace for $Y(s)$ in equation 1.10.11 and rearranging we have

$$\begin{aligned} sX_1(s) &= ((b_0 - b_n a_0)U(s) - a_0X_n(s)) \\ sX_2(s) &= ((b_1 - b_n a_1)U(s) - a_1X_n(s) + X_1(s)) \\ &\vdots \\ sX_n(s) &= ((b_{n-1} - b_n a_{n-1})U(s) - a_{n-1}X_n(s) + X_{n-1}(s)) \end{aligned} \quad (1.10.13)$$

Which leads to a state equation

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \vdots \\ \dot{x}_{n-1} \\ \dot{x}_n \end{bmatrix} = \begin{bmatrix} 0 & 0 & \dots & 0 & -a_0 \\ 1 & 0 & \dots & 0 & -a_1 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 \dots & 0 & -a_{n-2} \\ 0 & 0 & \dots & 1 & -a_{n-1} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{n-1} \\ x_n \end{bmatrix} + \begin{bmatrix} b_0 - b_n a_0 \\ b_1 - b_n a_1 \\ \vdots \\ b_{n-2} - b_n a_{n-2} \\ b_{n-1} - b_n a_{n-1} \end{bmatrix} u \quad (1.10.14)$$

and output equation

$$y = \begin{bmatrix} 0 & 0 & \dots & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{n-1} \\ x_n \end{bmatrix} + b_n u \quad (1.10.15)$$

As we can see from equation 1.10.7, equation 1.10.9, equation 1.10.14 and equation 1.10.15, the system matrix of observable canonical form is a transpose of the system matrix of controllable canonical form ($\mathbf{A}_o = \mathbf{A}_c^\top$). Input matrix of observable canonical form is equal to the transpose of the output matrix of controllable canonical form ($\mathbf{B}_o = \mathbf{C}_c^\top$). The output matrix of observable canonical form is equal to the transpose of the controllable canonical form input matrix ($\mathbf{C}_o = \mathbf{B}_c^\top$). State transmission matrix of observable canonical form is a transpose of state transmission matrix of controllable canonical form ($\mathbf{D}_o = \mathbf{D}_c^\top$).

3. Diagonal canonical Form

Suppose the transfer function has distinct poles $p_1, p_2, \dots, p_n$ and can be factorized to

$$Y(s) = \frac{c_1}{s + p_1}U(s) + \frac{c_2}{s + p_2}U(s) + \dots + \frac{c_n}{s + p_n}U(s) \quad (1.10.16)$$

Letting $X_1(s) = \frac{1}{s + p_1}U(s), X_2(s) = \frac{1}{s + p_2}U(s), \dots, X_n(s) = \frac{1}{s + p_n}U(s)$ we have

$$\begin{aligned} \dot{x}_1 &= -p_1 x_1 + u \\ \dot{x}_2 &= -p_2 x_2 + u \\ &\vdots \\ \dot{x}_n &= -p_n x_n + u \end{aligned} \quad (1.10.17)$$

Which leads to a state equation

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \vdots \\ \dot{x}_{n-1} \\ \dot{x}_n \end{bmatrix} = \begin{bmatrix} -p_1 & 0 & \dots & 0 \\ 0 & -p_2 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \dots & -p_n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} + \begin{bmatrix} 1 \\ \vdots \\ 1 \\ 1 \end{bmatrix} u \quad (1.10.18)$$

and output equation

$$y = \begin{bmatrix} c_1 & c_2 & \dots & c_n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \quad (1.10.19)$$

4. Jordan Canonical Form

If the transfer function can be factorized and has r multiplicities of pole p_1 , and the rest poles are distinct, the transfer function can be written as

$$Y(s) = \frac{c_1}{s + p_1}U(s) + \frac{c_2}{(s + p_1)^2}U(s) + \dots + \frac{c_r}{(s + p_1)^r}U(s) + \frac{c_{r+1}}{s + p_2}U(s) + \frac{c_{r+2}}{(s + p_3)}U(s) + \dots + \frac{c_n}{(s + p_{(n-r)})}U(s) \quad (1.10.20)$$

Let

$$X_1(s) = \frac{1}{s + p_1}U(s)$$

$$X_2(s) = \frac{1}{(s + p_1)^2}U(s)$$

which can be written as

$$X_2(s) = \frac{1}{s + p_1}X_1(s)$$

in similar way

$$X_3(s) = \frac{1}{s + p_1}X_2(s), \dots, X_r(s) = \frac{1}{s + p_1}X_{(r-1)}(s)$$

Since the rest poles are distinct, the derivation follows the way we drive for diagonal normal form. In the time domain, the states will be

$$\begin{aligned} \dot{x}_1 &= -p_1x_1 + u \\ \dot{x}_2 &= -p_1x_2 + x_1 \\ &\vdots \\ \dot{x}_r &= -p_1x_r + x_{r-1} \\ \dot{x}_{r+1} &= -p_2x_{r+1} + u \\ &\vdots \\ \dot{x}_n &= -p_{n-r}x_{n-r} + u \end{aligned} \quad (1.10.21)$$

Which led to a state equation

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \vdots \\ \dot{x}_r \\ \dot{x}_{r+1} \\ \vdots \\ \dot{x}_n \end{bmatrix} = \begin{bmatrix} -p_1 & 0 & 0 & \dots & 0 & 0 \dots & 0 \\ 1 & -p_1 & 0 & \dots & 0 & 0 \dots & 0 \\ 0 & 1 & -p_1 & \dots & 0 & 0 \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & 1 & -p_1 \dots & 0 \\ 0 & 0 & 0 & \dots & 0 & -p_2 \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & 0 & 0 \dots & -p_{n-r} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_{r-1} \\ x_r \\ \vdots \\ x_n \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \\ 1 \\ \vdots \\ 1 \end{bmatrix} u \quad (1.10.22)$$

The output matrix is similar to the diagonal normal form

$$y = \begin{bmatrix} c_1 & c_2 & \dots & c_n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \quad (1.10.23)$$

1.11 Transfer Function From State Space Representation

Consider a LTI system represented by state space representation on equation 1.3.7 and equation 1.3.8, with initial condition $\mathbf{x}(0) = \mathbf{x}_0$, taking the Laplace transform of the equations, we have

$$\begin{aligned} s\mathbf{X}(s) - \mathbf{x}_0 &= \mathbf{A}\mathbf{X}(s) + \mathbf{B}\mathbf{U}(s) \\ (s\mathbf{I} - \mathbf{A})\mathbf{X}(s) &= \mathbf{x}_0 + \mathbf{B}\mathbf{U}(s) \\ \mathbf{X}(s) &= (s\mathbf{I} - \mathbf{A})^{-1}(\mathbf{x}_0 + \mathbf{B}\mathbf{U}(s)) \end{aligned} \quad (1.11.1)$$

$$\begin{aligned} \mathbf{Y}(s) &= \mathbf{C}\mathbf{X}(s) + \mathbf{D}\mathbf{U}(s) \\ &= \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}(\mathbf{x}_0 + \mathbf{B}\mathbf{U}(s)) + \mathbf{D}\mathbf{U}(s) \\ &= \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{x}_0 + (\mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D})\mathbf{U}(s) \end{aligned} \quad (1.11.2)$$

If the initial condition is zero ($\mathbf{x}_0 = 0$, which is the case for all transfer functions), the transfer function from input to output is

$$\begin{aligned} \frac{\mathbf{Y}(s)}{\mathbf{U}(s)} &= \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D} \\ &= \frac{\mathbf{C} \text{adj}(s\mathbf{I} - \mathbf{A})\mathbf{B} + \det(s\mathbf{I} - \mathbf{A})\mathbf{D}}{\det(s\mathbf{I} - \mathbf{A})} \end{aligned} \quad (1.11.3)$$

The determinant of $(s\mathbf{I} - \mathbf{A})$ is a characteristic polynomial of the transfer function.

1.12 Review on Eigenvalues and Eigenvectors

In linear algebra, given a square matrix $\mathbf{A}$ of size $n \times n$, an eigenvalue λ is a scalar for which there exists a non-zero vector $\mathbf{v}$ (called an eigenvector) such that:

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v} \quad (1.12.1)$$

This equation can be rewritten as:

$$(\mathbf{A} - \lambda\mathbf{I})\mathbf{v} = \mathbf{0} \quad (1.12.2)$$

where $\mathbf{I}$ is the $n \times n$ identity matrix. For a non-trivial solution ($\mathbf{v} \neq \mathbf{0}$) to exist, the matrix $(\mathbf{A} - \lambda\mathbf{I})$ must be singular. This leads to the characteristic equation:

$$\det(\mathbf{A} - \lambda \mathbf{I}) = 0 \quad (1.12.3)$$

Eigenvalues characterize how a linear transformation (represented by a matrix) acts on its eigenvectors. Each eigenvalue λ specifies the scaling factor applied to the corresponding eigenvector while preserving its direction (for real eigenvalues). Eigenvalues may be real numbers, indicating pure scaling, or complex numbers, indicating rotational transformations. In the special case of diagonal matrices, the eigenvalues are simply the entries along the main diagonal.

Any nonzero vectors $\boldsymbol{\nu}_i, \mathbf{q}_i$ which satisfies the matrix equation

$$(\lambda_i \mathbf{I} - \mathbf{A})\mathbf{q}_i = 0$$

$$\boldsymbol{\nu}_i(\lambda_i \mathbf{I} - \mathbf{A}) = 0$$

where λ_i are eigenvalues of the matrix, are called left eigenvectors and right eigenvectors of matrix $\mathbf{A}$ corresponding to eigenvalue λ_i respectively. The matrix formed from right eigenvectors ($\mathbf{q}_i$) of the matrix $\mathbf{A}$ is called the modality matrix of $\mathbf{A}$ and is denoted by $\mathbf{M}$.

$$\mathbf{M} = \begin{bmatrix} q_1 & q_2 & \dots & q_n \end{bmatrix} \quad (1.12.4)$$

Eigenvectors possess a remarkable geometric property: when transformed by the matrix $\mathbf{A}$, they remain within their original span, meaning they don't get rotated but merely get scaled by their corresponding eigenvalue λ . This characteristic makes them fundamental indicators of the matrix's intrinsic stretching behavior. Unlike ordinary vectors that may change direction under transformation, eigenvectors precisely reveal the primary directions along which the matrix:

- Eigenvector stays on its original line ($\text{span}(\mathbf{q}_i)$)
- Transformation only changes the vector's magnitude by a factor λ
- They form natural axes for understanding $\mathbf{A}$'s action

The eigenvalues λ_i quantify exactly how much stretching or compression occurs along each eigenvector direction. Together, the eigenvector-eigenvalue pairs $(\mathbf{q}_i, \lambda_i)$ provide a complete description of the matrix's most fundamental geometric action on the vector space.

Left eigenvectors describe invariant directions under the row transformation induced by $\mathbf{A}$ (as opposed to column transformation for right eigenvectors).

1.12.1 Relationships Between Right and Left Eigenvectors

1. For eigenvectors corresponding to *different* eigenvalues:

$$\mathbf{q}_i \boldsymbol{\nu}_j = 0 \quad \text{if } \lambda_i \neq \lambda_j \quad (1.12.5)$$

This means right and left eigenvectors from distinct eigen spaces are orthogonal.

2. For the *same* eigenvalue λ_i :

$$\mathbf{q}_i \boldsymbol{\nu}_i \neq 0 \quad (1.12.6)$$

The dot product is non-zero, and often normalized to 1.

1.12.2 Application of Eigen structure in Control Systems

The eigenvalues λ_i play a vital role in determining the fundamental dynamic characteristics of the system. The real part $\text{Re}(\lambda_i) < 0$ indicates stability, while the imaginary part determines oscillation characteristics.

The right eigenvectors define mode shapes, which represent specific combinations of state variables that evolve exponentially according to their corresponding eigenvalues. Physically, these manifest as observable vibration patterns.

Left eigenvectors determine the controllability properties of the system through their inner products $\boldsymbol{\nu}_i \mathbf{B}$; when this product is zero, the corresponding mode becomes uncontrollable.

The bi-orthogonality relationship $\mathbf{q}_i \boldsymbol{\nu}_j = \delta_{ij}$ enables modal decomposition, which allows us to express the transfer function as

$$G(s) = \sum_i \frac{(\mathbf{C} \mathbf{q}_i)(\boldsymbol{\nu}_i \mathbf{B})}{s - \lambda_i}. \quad (1.12.7)$$

This eigen-structure underpins key control techniques, including pole placement (eigenvalue assignment), model reduction (mode truncation, approximation of the system with dominant poles), and fault detection (since right eigenvectors project the states to the observable subspace).

1.13 Solution of State Space Equations

Consider the time derivative of $e^{-\mathbf{A}t} \mathbf{x}$, using chain rule we have

$$\begin{aligned} \frac{d}{dt}(e^{-\mathbf{A}t} \mathbf{x}) &= \frac{d}{dt}(e^{-\mathbf{A}t}) \mathbf{x} + e^{-\mathbf{A}t} \frac{d\mathbf{x}}{dt} \\ &= -\mathbf{A}e^{-\mathbf{A}t} \mathbf{x} + e^{-\mathbf{A}t} (\dot{\mathbf{x}}) \\ &= -\mathbf{A}e^{-\mathbf{A}t} \mathbf{x} + e^{-\mathbf{A}t} (\mathbf{A} \mathbf{x} + \mathbf{B} \mathbf{u}) \\ &= e^{-\mathbf{A}t} (-\mathbf{A} \mathbf{x} + \mathbf{A} \mathbf{x}) + e^{-\mathbf{A}t} \mathbf{B} \mathbf{u} \\ &= e^{-\mathbf{A}t} \mathbf{B} \mathbf{u} \end{aligned} \quad (1.13.1)$$

From equation 1.13.1 we can see that $\frac{d}{dt}(e^{-\mathbf{A}t} \mathbf{x}) = e^{-\mathbf{A}t} \mathbf{A} \mathbf{u} = e^{-\mathbf{A}t} (\dot{\mathbf{x}} - \mathbf{A} \mathbf{x})$, using this the solution to state equation is

$$\begin{aligned} e^{-\mathbf{A}t} (\dot{\mathbf{x}} - \mathbf{A} \mathbf{x}) &= e^{-\mathbf{A}t} \mathbf{B} \mathbf{u} \\ \int_{t=0}^t \frac{d}{dt}(e^{-\mathbf{A}t} \mathbf{x}) dt &= \int_{t=0}^t e^{-\mathbf{A}\tau} \mathbf{B} \mathbf{u}(\tau) d\tau \\ e^{-\mathbf{A}t} \mathbf{x} \Big|_{t=0}^{t=t} &= \int_{t=0}^t e^{-\mathbf{A}\tau} \mathbf{B} \mathbf{u}(\tau) d\tau \\ e^{-\mathbf{A}t} (\mathbf{x}(t) - \mathbf{x}(0)) &= \int_{t=0}^t e^{-\mathbf{A}\tau} \mathbf{B} \mathbf{u}(\tau) d\tau \\ \mathbf{x}(t) &= e^{\mathbf{A}t} \mathbf{x}(0) + e^{\mathbf{A}t} \int_{t=0}^t e^{-\mathbf{A}\tau} \mathbf{B} \mathbf{u}(\tau) d\tau \\ \mathbf{x}(t) &= e^{\mathbf{A}t} \mathbf{x}(0) + \int_{t=0}^t e^{\mathbf{A}(t-\tau)} \mathbf{B} \mathbf{u}(\tau) d\tau \end{aligned} \quad (1.13.2)$$

And the output solution will be

$$\begin{aligned}
 \mathbf{y}(t) &= \mathbf{C}\mathbf{x}(t) + \mathbf{D}\mathbf{u}(t) \\
 \mathbf{y}(t) &= \mathbf{C}e^{\mathbf{A}t}\mathbf{x}(0) + \int_{t=0}^t \mathbf{C}e^{\mathbf{A}(t-\tau)}\mathbf{B}\mathbf{u}(\tau)d\tau + \mathbf{D}\mathbf{u}(t) \\
 \mathbf{y}(t) &= \mathbf{C}e^{\mathbf{A}t}\mathbf{x}(0) + \int_{t=0}^t (\mathbf{C}e^{\mathbf{A}(t-\tau)}\mathbf{B} + \mathbf{D})\mathbf{u}(\tau)d\tau
 \end{aligned} \tag{1.13.3}$$

The $\mathbf{C}e^{\mathbf{A}t}\mathbf{x}(0)$ in equation 1.13.3 is called the zero-input response of the dynamic system, while the term $\int_{t=0}^t (\mathbf{C}e^{\mathbf{A}(t-\tau)}\mathbf{B} + \mathbf{D})\mathbf{u}(\tau)d\tau$ is called the zero initial condition response. The frequency domain representation of equation 1.13.3 is equation 1.11.2.

The term $e^{\mathbf{A}t}$ denoted as $\Phi(t)$ is called state transition matrix as it transfer the state of a dynamic system from any initial condition $\mathbf{x}(t_0)$ to a final state $\mathbf{x}(t)$ whether the dynamic system is driven by an input or not. A few properties of the state transition matrix are stated below (It is a property due to the exponential nature of the function).

- $\Phi(0) = I$ since $e^{\mathbf{A}*0} = \mathbf{I}$
- $\Phi(t_2 - t_1)\Phi(t_1 - t_0) = \Phi(t_2 - t_0)$
- $\Phi^{-1}(t) = \Phi(-t)$ since $(e^{\mathbf{A}t})^{-1} = e^{\mathbf{A}(-t)} = \Phi(-t)$
- $\Phi(t_2 - t_1) = \Phi(t_1)\Phi(-t_2)$
- $\Phi(t)^n = \Phi(nt)$

The state transition matrix is determined by one of the following methods,

1. Using Inverse Laplace transform method

$$\Phi(t) = \mathcal{L}^{-1}((s\mathbf{I} - \mathbf{A})^{-1}) \tag{1.13.4}$$

2. Using Series expansion method

$$\Phi(t) = e^{\mathbf{A}t} = \sum_{i=0}^{i=n} \frac{(\mathbf{A}t)^i}{i!} \tag{1.13.5}$$

where n is the dimension of matrix $\mathbf{A}$.

3. Using Cayley Hamilton Theorem

Cayley Hamilton's theorem states that any function of an $n \times n$ square matrix $f(\mathbf{A})$ can be represented by a finite sum of powers of the matrix not exceeding order $n - 1$. Let the eigenvalues of $\mathbf{A}$ be λ_i 's where $i = 1, 2, \dots, n$ and the characteristics polynomial corresponding to each λ_i be $p_i(\lambda)$. So we have n simultaneous equations to solve for the coefficient of the characteristic polynomial. As Cayley Hamilton's theorem states, matrix $\mathbf{A}$ satisfies its characteristic equation, which is why we say

$f(\mathbf{A})$ can be represented by a finite sum of powers of the matrix not exceeding order $n - 1$.

$$f(\mathbf{A}t) = \alpha_0(t)\mathbf{I} + \alpha_1(t)\mathbf{A} + \alpha_2(t)\mathbf{A}^2 + \dots + \alpha_{n-1}(t)\mathbf{A}^{n-1} \quad (1.13.6)$$

The constants are determined from a simultaneous equation resulting from

$$p(\lambda_i t) = \alpha_0(t) + \alpha_1(t)\lambda_i + \alpha_2(t)\lambda_i^2 + \dots + \alpha_{n-1}(t)\lambda_i^{n-1} = f(\lambda_i t) \quad (1.13.7)$$

Where $i = 1, 2, \dots, n$.

If the matrix $\mathbf{A}$ has multiplicity of eigenvalue, the polynomial with the multiplicity eigenvalues is differentiated r times and used as an additional equation, to have an equal number of equations as the number of variables.

4. Using Modality Matrix

Let the diagonal matrix formed from $e^{\lambda_i t}$ ordered from smallest value of λ_i to largest value of λ_i (i.e. $\lambda_1 < \lambda_2 < \dots < \lambda_n$), where λ_i are eigenvalues of $\mathbf{A}$ be denoted as $e^{\mathbf{\Lambda}t}$, then the state transition matrix can be solved using modality matrix $\mathbf{M}$ and $e^{\mathbf{\Lambda}t}$ as

$$\Phi(t) = \mathbf{M}^{-1}e^{\mathbf{\Lambda}t}\mathbf{M} = \mathbf{M}^{-1} \begin{bmatrix} e^{\lambda_1 t} & 0 & \dots & 0 \\ 0 & e^{\lambda_2 t} & \dots & 0 \\ \vdots & \vdots & \dots & \vdots \\ 0 & 0 & \dots & e^{\lambda_n t} \end{bmatrix} \mathbf{M} \quad (1.13.8)$$

If the matrix has r multiplicity of eigenvalues, the determinant of the modality matrix is zero since the eigenvectors are linearly dependent. So, this method will not work in finding a state transition matrix for a matrix with r multiplicity of eigenvalues.

1.14 Similarity Transformation

If there exist a transformation matrix that transform a state $\mathbf{x}$ into new state $\boldsymbol{\xi}$, in such a way that $\mathbf{x} = \mathbf{T}\boldsymbol{\xi}$ which means $\boldsymbol{\xi} = \mathbf{T}^{-1}\mathbf{x}$ and $\dot{\mathbf{x}} = \mathbf{T}\dot{\boldsymbol{\xi}}$ then the state space representation of LTI system on equation 1.3.7 and equation 1.3.8 will be transformed into a dynamics equation with different state coordinates of the form

$$\begin{aligned} \dot{\boldsymbol{\xi}} &= \mathbf{T}^{-1}\mathbf{A}\mathbf{T}\boldsymbol{\xi} + \mathbf{T}^{-1}\mathbf{B}\mathbf{u} \\ \dot{\boldsymbol{\xi}} &= \bar{\mathbf{A}}\boldsymbol{\xi} + \bar{\mathbf{B}}\mathbf{u} \end{aligned} \quad (1.14.1)$$

$$\begin{aligned} \mathbf{y} &= \mathbf{C}\mathbf{T}\boldsymbol{\xi} + \mathbf{D}\mathbf{u} \\ \mathbf{y} &= \bar{\mathbf{C}}\boldsymbol{\xi} + \bar{\mathbf{D}}\mathbf{u} \end{aligned} \quad (1.14.2)$$

where $\bar{\mathbf{A}} = \mathbf{T}^{-1}\mathbf{A}\mathbf{T}$, $\bar{\mathbf{B}} = \mathbf{T}^{-1}\mathbf{B}$, $\bar{\mathbf{C}} = \mathbf{C}\mathbf{T}$ and $\bar{\mathbf{D}} = \mathbf{D}$.

The main purpose of similarity transformation is to simplify the dynamic equation of the system for analysis (converting any state representation of any dynamic system to controllable or observable canonical form will simplify the analysis of the system). Some

dynamic systems are complex in some coordinate systems, but simpler in other coordinate systems.

Similarity transformation only changes the coordinates in which the dynamic system is expressed. It does not change

- Eigenvalues of the system matrix

Proof:- The characteristic equation of the transformed system matrix is

$$\begin{aligned}
 \det(\lambda \mathbf{I} - \bar{\mathbf{A}}) &= 0 \\
 \det(\lambda \mathbf{I} - \mathbf{T}^{-1} \mathbf{A} \mathbf{T}) &= 0 \\
 \det(\mathbf{T}^{-1} \lambda \mathbf{T} - \mathbf{T}^{-1} \mathbf{A} \mathbf{T}) &= 0 \\
 \det(\mathbf{T}^{-1}) \det(\lambda \mathbf{I} - \mathbf{A}) \det(\mathbf{T}) &= 0 \\
 \det(\lambda \mathbf{I} - \mathbf{A}) &= 0
 \end{aligned}$$

- Transfer function of the dynamic equation

Proof:- The transfer function of the new dynamic system is

$$\begin{aligned}
 \frac{\mathbf{Y}(s)}{\mathbf{U}(s)} &= \frac{\bar{\mathbf{C}} \text{adj}(s\mathbf{I} - \bar{\mathbf{A}}) \bar{\mathbf{B}}}{\det(s\mathbf{I} - \bar{\mathbf{A}})} + \bar{\mathbf{D}} \\
 &= \frac{\mathbf{C} \mathbf{T} \text{adj}(s\mathbf{I} - \mathbf{T}^{-1} \mathbf{A} \mathbf{T}) \mathbf{T}^{-1} \mathbf{B}}{\det(s\mathbf{I} - \mathbf{T}^{-1} \mathbf{A} \mathbf{T})} + \mathbf{D} \\
 &= \frac{\mathbf{C} \mathbf{T} \text{adj}(\mathbf{T}^{-1} s \mathbf{T} - \mathbf{T}^{-1} \mathbf{A} \mathbf{T}) \mathbf{T}^{-1} \mathbf{B}}{\det(\mathbf{T}^{-1} s \mathbf{T} - \mathbf{T}^{-1} \mathbf{A} \mathbf{T})} + \mathbf{D} \\
 &= \frac{\mathbf{C} \text{adj}(s\mathbf{I} - \mathbf{A}) \mathbf{B}}{\det(s\mathbf{I} - \mathbf{A})} + \mathbf{D}
 \end{aligned}$$

- Solution of output equation

Proof:- Solution of the new dynamic system have initial condition $\xi_o = \mathbf{T}^{-1} \mathbf{x}_o$, and transfer function same as original dynamic system.

$$\begin{aligned}
 \mathbf{Y}(s) &= \bar{\mathbf{C}}(s\mathbf{I} - \bar{\mathbf{A}})^{-1} \mathbf{T} \mathbf{x}_0 + (\bar{\mathbf{C}}(s\mathbf{I} - \bar{\mathbf{A}})^{-1} \bar{\mathbf{B}} + \bar{\mathbf{D}}) \mathbf{U}(s) \\
 &= \mathbf{C} \mathbf{T} (s\mathbf{I} - \mathbf{T}^{-1} \mathbf{A} \mathbf{T})^{-1} \mathbf{T}^{-1} \mathbf{x}_0 + (\mathbf{C} \mathbf{T} (s\mathbf{I} - \mathbf{T}^{-1} \mathbf{A} \mathbf{T})^{-1} \mathbf{T}^{-1} \mathbf{B} + \mathbf{D}) \mathbf{U}(s) \\
 &= \mathbf{C} (s\mathbf{I} - \mathbf{A})^{-1} \mathbf{x}_0 + (\mathbf{C} (s\mathbf{I} - \mathbf{A})^{-1} \mathbf{B} + \mathbf{D}) \mathbf{U}(s)
 \end{aligned}$$

Note Since the state of the system has changed solution to the state equation is different. But we can derive a solution for the original state equation from the solution of the new state equation by $\mathbf{x}(t) = \mathbf{T} \xi(t)$.

1.15 Phase Plane and Stability Analysis of LTI Systems

1.15.1 Stability analysis of LTI systems

When we say a dynamic system is stable, we mean that the system will stay where it is until it is disturbed by internal or external factors. The point where the system stays forever after it arrives is called a stable equilibrium point. No change in a dynamic system doesn't guarantee us the existence of a stable equilibrium point or even the existence of an equilibrium point at all. But it is a necessary condition for the existence of an equilibrium point. So when we deal with the stability of LTI systems, we try to find whether an LTI system has a stable equilibrium point or whether it can be stabilized at preferred operating points. Stability of the dynamic system may be internal or external. In the next section, we will explain them briefly.

Internal Stability

Internal stability deals with the stability of the states of the dynamic system. It analyzes the behavior of the system state under no external inputs (zero-input response of the states). Consider a dynamic system $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ evolving from an initial condition $\mathbf{x}_0$; the equilibrium point $\mathbf{x}_{eq}(t)$ satisfies the state equation of the dynamic system. This equilibrium point is

- Lyapunov stable if and only if given $\epsilon > 0$ there exist $\sigma > 0$ such that $\|\mathbf{x}_0\| < \sigma$ implies that $\|\mathbf{x}(t)\| < \epsilon$ for all time $t > 0$. This means for every initial condition in a sphere of radius σ , the final state of the system should be in a sphere of radius ϵ . LTI system is Lyapunov stable if and only if $\text{trace}^3(\mathbf{A}) < 0$, $\det(\mathbf{A}) \geq 0$ and $\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{A}^2)$ (i.e. All eigenvalues of system matrix $\mathbf{A}$ should have zero or negative real parts).
- Asymptotically stable if it is stable and if it is possible to choose $\sigma > 0$ such that $\|\mathbf{x}_0\| < \sigma$ implies $\lim_{t \rightarrow \infty} \|\mathbf{x}(t)\| = 0$. If σ can be any real number (anywhere), then the system is globally asymptotically stable. LTI system is asymptotically stable if and only if $\text{trace}(\mathbf{A}) < 0$ and $\det(\mathbf{A}) > 0$ (i.e., all eigenvalues of system matrix $\mathbf{A}$ should have strictly negative real parts).
- Exponentially stable if it is stable and there exist $\sigma > 0, k > 0, \lambda > 0$ such that $\|\mathbf{x}_0\| < \sigma$ implies $\|\mathbf{x}(t)\| < ke^{-\lambda t}\|\mathbf{x}_0\|$ for all $t > 0$. If σ can be any real number (anywhere), then the system is globally exponentially stable.

To determine whether the LTI system is Lyapunov stable or not, Let us take a quadratic function $V = \mathbf{x}^\top \mathbf{P} \mathbf{x}$, where $\mathbf{P}$ are symmetric⁴ positive definite⁵ matrix, then

$$\begin{aligned}\dot{V} &= \mathbf{x}^\top \mathbf{P} \dot{\mathbf{x}} + \dot{\mathbf{x}}^\top \mathbf{P} \mathbf{x} \\ &= \mathbf{x}^\top \mathbf{P} (\mathbf{A} \mathbf{x}) + (\mathbf{A} \mathbf{x})^\top \mathbf{P} \mathbf{x} \\ &= \mathbf{x}^\top (\mathbf{P} \mathbf{A} + \mathbf{A}^\top \mathbf{P}) \mathbf{x}\end{aligned}\tag{1.15.1}$$

³trace($\mathbf{A}$) is the sum of all diagonal elements of $\mathbf{A}$

⁴A matrix $\mathbf{P}$ is called symmetric if and only if $\mathbf{P} = \mathbf{P}^\top$

⁵A matrix $\mathbf{P}$ is positive definite if and only if all eigenvalues of $\mathbf{P}$ are positive or leading principal minors of the are positive

If $(\mathbf{P}\mathbf{A} + \mathbf{A}^\top \mathbf{P}) < 0$ (i.e. negative definite) then $\dot{V}$ decreases monotonically. If we take positive definite symmetric matrix $\mathbf{R}$ such that,

$$\mathbf{P}\mathbf{A} + \mathbf{A}^\top \mathbf{P} + \mathbf{R} = 0 \quad (1.15.2)$$

Equation 1.15.2 is called the Lyapunov equation.

BIBO/External Stability

It is a stability analysis of the response of the dynamic system to bounded input.

Definition:- A dynamic system is bounded-input input bounded-output, and stable if and only if for zero initial condition, if bounded input always invokes bounded output. Consider an LTI dynamic system with zero initial condition response

$$\mathbf{y}(t) = \mathbf{C} \int_0^t e^{\mathbf{A}(t-\tau)} \mathbf{B} \mathbf{u}(\tau) d\tau + \mathbf{D} \mathbf{u}(t) \quad (1.15.3)$$

Since $\mathbf{u}(t)$ is bounded, let $\|\mathbf{u}(t)\| < \alpha$, where $\alpha > 0$ then

$$\begin{aligned} \mathbf{y}(t) &= \int_0^t \mathbf{C} e^{\mathbf{A}(t-\tau)} \mathbf{B} \mathbf{u}(\tau) d\tau + \mathbf{D} \mathbf{u}(t) < \int_0^t \mathbf{C} e^{\mathbf{A}(t-\tau)} \mathbf{B} \alpha d\tau + \mathbf{D} \alpha \\ &< \alpha \int_0^t \mathbf{C} e^{\mathbf{A}(t-\tau)} \mathbf{B} d\tau + \mathbf{D} \end{aligned} \quad (1.15.4)$$

Which implies that for an LTI system to be BIBO stable the weighting matrix pattern must be bounded (i.e. $\int_0^t \mathbf{C} e^{\mathbf{A}(t-\tau)} \mathbf{B} d\tau < \infty$).

For the LTI system, asymptotic stability always implies bounded-input input bounded-output stability. While minimal bounded input and bounded output imply asymptotic stability.

1.15.2 Phase plane analysis of LTI systems

Phase plane analysis is a graphical analysis of a second-order system concerning initial conditions. Consider the following second-order system

$$\begin{aligned} \dot{x}_1 &= f_1(x_1, x_2) \\ \dot{x}_2 &= f_2(x_1, x_2) \end{aligned} \quad (1.15.5)$$

where x_1 and x_2 are states of the system and f_1 and f_2 are linear (for LTI systems) and nonlinear for nonlinear autonomous systems.

Since there is no control input, the phase plane analysis is used to understand the properties of the system under various initial conditions. For linear time invariant systems equation 1.15.5 simplifies to

$$\begin{aligned} \dot{x}_1 &= a_{11}x_1 + a_{12}x_2 \\ \dot{x}_2 &= a_{21}x_1 + a_{22}x_2 \end{aligned} \quad (1.15.6)$$

Geometrically, the state space of this system is a plane having x_1, x_2 as coordinates. This plane is called a phase plane. The solution of equation 1.15.5 or equation 1.15.6 with time varying from zero to infinity can be represented as a curve in the phase plane. Such

a curve is called a phase plane trajectory. A family of phase plane trajectories is called a phase portrait of a system.

Equation 1.15.6 can be written as

$$\begin{aligned}\ddot{x}_1 &= a_{11}\dot{x}_1 + a_{12}\dot{x}_2 = a_{11}\dot{x}_1 + a_{12}(a_{21}x_1 + a_{22}x_2) \\ \ddot{x}_1 &= a_{11}\dot{x}_1 + a_{12}(a_{21}x_1 + a_{22}\frac{\dot{x}_1 - a_{11}x_1}{a_{12}}) \\ \ddot{x}_1 &= \frac{(a_{11}a_{12} + a_{12}a_{22})}{a_{12}}\dot{x}_1 + \frac{(a_{12}a_{21} - a_{12}a_{22}a_{11})}{a_{12}}x_1 \\ \ddot{x}_1 + \xi_1\dot{x}_1 + \xi_2x_1 &= 0\end{aligned}\tag{1.15.7}$$

where $\xi_1 = \frac{(a_{11}a_{12} + a_{12}a_{22})}{a_{12}}$ and $\xi_2 = \frac{(a_{12}a_{21} - a_{12}a_{22}a_{11})}{a_{12}}$.

The solution of equation 1.15.7 has the form

$$x(t) = k_1e^{\lambda_1 t} + k_2e^{\lambda_2 t} \quad \text{for } \lambda_1 \neq \lambda_2 \tag{1.15.8}$$

$$x(t) = k_1e^{\lambda_1 t} + tk_2e^{\lambda_2 t} \quad \text{for } \lambda_1 = \lambda_2 \tag{1.15.9}$$

A point in the phase plane where $\dot{\mathbf{x}} = 0$ is called a singular point or equilibrium point of the system. For a Linear system, we have only one singular/equilibrium point, namely the origin. The trajectories in the vicinity of this singularity point can display quite different characteristics, depending on the values of λ_1 and λ_2 . The following cases can occur

- λ_1, λ_2 are both real and have the same sign (+ or -)
- λ_1, λ_2 are both real and have opposite sign
- λ_1, λ_2 are complex conjugates with non-zero real parts
- λ_1, λ_2 are complex conjugates with real parts equal to 0

The characteristics of the system under these cases are described below

1. The first condition corresponds to a node

- $\lambda_1 < 0, \lambda_2 < 0$ singularity point is called stable node. In this, every phase portrait goes to the singularity point as shown in Figure 1.24, which means the system is asymptotically stable.
- $\lambda_1 > 0, \lambda_2 > 0$ singularity point is called unstable node. In this, every phase portrait goes away from the singularity point as shown in Figure 1.25, which means the system is unstable.

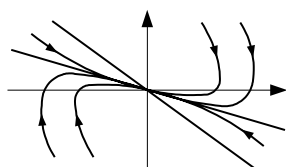


Figure 1.24: *Stable Node*

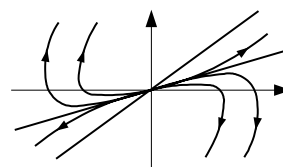
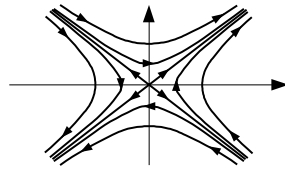


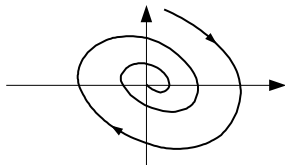
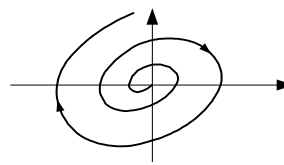
Figure 1.25: *Unstable Node*

- $\lambda_1 < 0 < \lambda_2$, corresponds to a saddle point. Trajectories tend to the equilibrium point (converge to the singular point) at one end and move away from the equilibrium point (diverge to infinity) at the other end as shown in Figure 1.26.

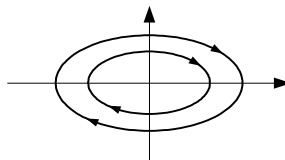
Figure 1.26: *Saddle Point*

3. The third case corresponds to a focus.

- (a) $Re(\lambda_1 < 0, \lambda_2 < 0)$: stable focus, where trajectories circulate the point and finally tend to the point as shown in Figure 1.27. The circulation shows that there is an oscillation in the system response, and the convergence shows that these oscillations will die out and the system becomes stable over time.
- (b) $Re(\lambda_1 > 0, \lambda_2 > 0)$: unstable focus, where trajectories circulate the point and diverge to infinity as shown in Figure 1.28. The circulation shows that there is an oscillation in the system response, and the divergence shows that these oscillations grow large and the system becomes unstable over time.

Figure 1.27: *Stable focus*Figure 1.28: *Unstable Focus*

4. The last condition corresponds to a center point. All trajectories are ellipses, and the singularity point is the center of these ellipses as shown in Figure 1.29.

Figure 1.29: *Center Point*

The stability characteristics of linear systems are uniquely determined by the nature of their singularity points.

1.16 Examples

1. Find the controllable canonical form, observable canonical form, and diagonal normal form or Jordan canonical form of the following transfer function.

$$\begin{aligned}\frac{Y(s)}{U(s)} &= \frac{2s^3 + 5s^2 + 3s + 1}{4s^4 + 2s^2 + 3s + 6} \\ &= \frac{0.5s^3 + 1.25s^2 + 0.75s + 0.25}{s^4 + 0.5s^2 + 0.75s + 1.5}\end{aligned}$$

Solution:

- (a) Controllable canonical form: since the transfer function is of order $n = 4$, we have 4 states, 1 input, and 1 output. The constants are $b_4 = 0, b_3 = 0.5, b_2 = 1.25, b_1 = 0.75, b_0 = 0.25, a_4 = 1, a_3 = 0, a_2 = 0.5, a_1 = 0.75, a_0 = 1.5$

$$\begin{aligned}\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \end{bmatrix} &= \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -1.5 & -0.75 & -0.5 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} u \\ y &= \begin{bmatrix} 0.25 & 0.75 & 1.25 & 0.5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}\end{aligned}$$

- (b) Observable canonical form can be determined from controllable canonical form by $\mathbf{A}_o = \mathbf{A}_c^\top, \mathbf{B}_o = \mathbf{C}_c^\top, \mathbf{C}_o = \mathbf{B}_c^\top$ and $\mathbf{D}_o = \mathbf{D}_c^\top$.

$$\begin{aligned}\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \end{bmatrix} &= \begin{bmatrix} 0 & 0 & 0 & -1.5 \\ 1 & 0 & 0 & -0.75 \\ 0 & 1 & 0 & -0.5 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + \begin{bmatrix} 0.25 \\ 0.75 \\ 1.25 \\ 0.5 \end{bmatrix} u \\ y &= \begin{bmatrix} 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}\end{aligned}$$

- (c) Since the poles of the system are distinct, $p_1 = 0.71 + i, p_2 = 0.71 - i, p_3 = -0.71 + 0.70i$ and $p_4 = -0.71 - 0.70i$. We have a diagonal normal form. The

transfer function is factorized into

$$\frac{Y(s)}{U(s)} = \frac{0.30 - 0.33i}{s + 0.71 + i} + \frac{0.30 + 0.33i}{s + 0.71 - i} + \frac{-0.05 - 0.06i}{s - 0.71 + 0.70i} + \frac{-0.05 + 0.06i}{s - 0.71 - 0.70i}$$

which has the Jordan canonical form of

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \end{bmatrix} = \begin{bmatrix} -0.71 - i & 0 & 0 & 0 \\ 0 & -0.71 + i & 0 & 0 \\ 0 & 0 & 0.71 - 0.70i & 0 \\ 0 & 0 & 0 & 0.71 + 0.70i \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} u$$

$$y = \begin{bmatrix} 0.30 - 0.33i & 0.30 + 0.33i & -0.05 - 0.06i & -0.05 + 0.06i \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$

2. Find the state transition matrix for the following dynamic system

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \mathbf{x}$$

Solution

(a) Method 1:-Using Inverse Laplace transform

$$\begin{aligned} \Phi(t) &= \mathcal{L}^{-1}((s\mathbf{I} - \mathbf{A})^{-1}) = \mathcal{L}^{-1} \left(\left(\begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} - \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \right)^{-1} \right) \\ &= \mathcal{L}^{-1} \left(\begin{bmatrix} s & -1 \\ 2 & s+3 \end{bmatrix}^{-1} \right) \\ &= \mathcal{L}^{-1} \left(\frac{\text{adj} \left(\begin{bmatrix} s & -1 \\ 2 & s+3 \end{bmatrix} \right)}{\det \left(\begin{bmatrix} s & -1 \\ 2 & s+3 \end{bmatrix} \right)} \right) \end{aligned}$$

$$\begin{aligned}
&= \mathcal{L}^{-1} \left(\begin{bmatrix} \frac{s+3}{(s+2)(s+1)} & \frac{1}{(s+2)(s+1)} \\ -\frac{2}{(s+2)(s+1)} & \frac{s}{(s+2)(s+1)} \end{bmatrix} \right) \\
&= \mathcal{L}^{-1} \left(\begin{bmatrix} \frac{-1}{s+2} + \frac{2}{s+1} & \frac{-1}{s+2} + \frac{1}{s+1} \\ \frac{-1}{s+2} + \frac{2}{s+1} & \frac{-1}{s+2} + \frac{1}{s+1} \end{bmatrix} \right) \\
\Phi(t) &= \begin{bmatrix} 2e^{-t} - e^{-2t} & e^{-t} - e^{-2t} \\ 2e^{-2t} - 2e^{-t} & 2e^{-2t} - e^{-t} \end{bmatrix}
\end{aligned}$$

(b) Method 2:-Using Series expansion

$$\begin{aligned}
\Phi(t) &= \sum_{i=0}^{i=2} \frac{(\mathbf{A}t)^i}{i!} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & -1 \\ -2 & -3 \end{bmatrix} t + \begin{bmatrix} 0 & -1 \\ -2 & -3 \end{bmatrix}^2 t^2 \\
&= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & -1 \\ -2 & -3 \end{bmatrix} t + \frac{1}{2} \begin{bmatrix} -2 & -3 \\ 6 & 7 \end{bmatrix} t^2 \\
&= \begin{bmatrix} 1 - t^2 & -t - \frac{3}{2}t^2 \\ -2t + 3t^2 & 1 - 3t + \frac{7}{2}t^2 \end{bmatrix}
\end{aligned}$$

From Taylor expansion of $e^{at} = \sum_{i=0}^{i=2} \frac{(at)^i}{i!}$,

$$\begin{aligned}
e^{-t} &= 1 - t + \frac{1}{2}t^2, & e^{-2t} &= 1 - 2t + \frac{4}{2}t^2 \\
1 - t^2 &= 2e^{-t} - e^{-2t}, & -t + \frac{-3}{2}t^2 &= e^{-t} - e^{-2t} \\
-2t + 3t^2 &= 2e^{-2t} - 2e^{-t}, & 1 - 3t + \frac{7}{2}t^2 &= 2e^{-2t} - e^{-t}
\end{aligned}$$

So,

$$\Phi(t) = \begin{bmatrix} 1 - t^2 & -t + \frac{-3}{2}t^2 \\ -2t + 3t^2 & 1 - 3t + \frac{7}{2}t^2 \end{bmatrix} = \begin{bmatrix} 2e^{-t} - e^{-2t} & e^{-t} - e^{-2t} \\ 2e^{-2t} - 2e^{-t} & 2e^{-2t} - e^{-t} \end{bmatrix}$$

(c) Method 3:-Using Cayley Hamilton Theorem

Determining the eigenvalues

$$\det(\lambda \mathbf{I} - \mathbf{A}) = (\lambda + 2)(\lambda + 1) = 0$$

Which result in eigenvalues of matrix $\mathbf{A}$, $\lambda_1 = -2$ and $\lambda_2 = -1$. Hence the simultaneous equations from $e^{\lambda_i t} = \alpha_0 + \alpha_1 \lambda_i$ are

$$\begin{aligned}
e^{-2t} &= \alpha_0 - 2\alpha_1 \\
e^{-t} &= \alpha_0 - \alpha_1
\end{aligned}$$

Solving for α_0 and α_1 , gives a result

$$\alpha_0 = 2e^{-t} - e^{-2t}, \quad \alpha_1 = e^{-t} - e^{-2t}$$

So by Cayley Hamilton's Theorem

$$\begin{aligned} \Phi(t) &= \alpha_0 \mathbf{I} + \alpha_1 \mathbf{A} \\ &= (2e^{-t} - e^{-2t}) \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + (e^{-t} - e^{-2t}) \begin{bmatrix} 0 & -1 \\ -2 & -3 \end{bmatrix} \\ \Phi(t) &= \begin{bmatrix} 2e^{-t} - e^{-2t} & e^{-t} - e^{-2t} \\ 2e^{-2t} - 2e^{-t} & 2e^{-2t} - e^{-t} \end{bmatrix} \end{aligned}$$

(d) Method 4:-Using Modality Matrix

We have already determined the eigenvalues; the next step is to determine the eigenvector corresponding to each eigenvalue by using $(\lambda_i \mathbf{I} - \mathbf{A})\boldsymbol{\nu}_i = 0$. The eigenvector corresponding to $\lambda_1 = -2$, $\boldsymbol{\nu}_1$ is

$$\begin{aligned} \left[\begin{bmatrix} -2 & 0 \\ 0 & -2 \end{bmatrix} - \begin{bmatrix} 0 & -1 \\ -2 & -3 \end{bmatrix} \right] \begin{bmatrix} \nu_{11} \\ \nu_{12} \end{bmatrix} &= 0 \\ \begin{bmatrix} -2 & 1 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} \nu_{11} \\ \nu_{12} \end{bmatrix} &= 0 \Rightarrow 2\nu_{11} = \nu_{12} \\ \boldsymbol{\nu}_1 &= \begin{bmatrix} \nu_{11} \\ \nu_{12} \end{bmatrix} = m \begin{bmatrix} 1 \\ 2 \end{bmatrix} \end{aligned}$$

The eigenvector corresponding to $\lambda_2 = -1$, $\boldsymbol{\nu}_2$ is

$$\begin{aligned} \left[\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} - \begin{bmatrix} 0 & -1 \\ -2 & -3 \end{bmatrix} \right] \begin{bmatrix} \nu_{21} \\ \nu_{22} \end{bmatrix} &= 0 \\ \begin{bmatrix} -1 & 1 \\ -2 & 2 \end{bmatrix} \begin{bmatrix} \nu_{21} \\ \nu_{22} \end{bmatrix} &= 0 \Rightarrow \nu_{21} = \nu_{22} \\ \Rightarrow \boldsymbol{\nu}_2 &= \begin{bmatrix} \nu_{21} \\ \nu_{22} \end{bmatrix} = m \begin{bmatrix} 1 \\ 1 \end{bmatrix} \end{aligned}$$

Hence the modality $\mathbf{M}$ matrix, it's inverse $\mathbf{M}^{-1}$ and $e^{\mathbf{A}t}$ are

$$\mathbf{M} = \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix}, \quad \mathbf{M}^{-1} = \begin{bmatrix} -1 & 1 \\ 2 & -1 \end{bmatrix}, \quad e^{\mathbf{A}t} = \begin{bmatrix} e^{-2t} & 0 \\ 0 & e^{-t} \end{bmatrix},$$

Finally

$$\begin{aligned}\Phi(t) &= \mathbf{M}^{-1}e^{\mathbf{A}t}\mathbf{M} = \begin{bmatrix} -1 & 1 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} e^{-2t} & 0 \\ 0 & e^{-t} \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 2e^{-t} - e^{-2t} & e^{-t} - e^{-2t} \\ 2e^{-2t} - 2e^{-t} & 2e^{-2t} - e^{-t} \end{bmatrix}\end{aligned}$$

3. Find the transfer function of the following state space representation

$$\begin{aligned}\dot{\mathbf{x}} &= \begin{bmatrix} 1 & 0 & -1 \\ -2 & 2 & -5 \\ 0 & 1 & -1 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} u \\ y &= \begin{bmatrix} 0 & 1 & 1 \end{bmatrix} \mathbf{x} + 2u\end{aligned}$$

Solution

$$\frac{Y(s)}{U(s)} = \frac{\mathbf{C} \text{adj}(s\mathbf{I} - \mathbf{A})\mathbf{B} + \det(s\mathbf{I} - \mathbf{A})\mathbf{D}}{\det(s\mathbf{I} - \mathbf{A})}$$

$$\begin{aligned}&= \frac{\begin{bmatrix} 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} s^2 - s + 3 & -1 & 2 - s \\ -2s - 2 & s^2 - 1 & 7 - 5s \\ -2 & s - 1 & s^2 - 3s + 2 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}}{s^3 - 2s^2 + 4s - 5} + 2 \\ &= \frac{s^2 + s - 2}{s^3 - 2s^2 + 4s - 5} + 2\end{aligned}$$

$$\frac{Y(s)}{U(s)} = \frac{2s^3 - 3s^2 + 9s - 12}{s^3 - 2s^2 + 4s - 5}$$

4. Find the eigenvalues and their corresponding eigenvectors for a matrix

$$\mathbf{A} = \begin{bmatrix} 1 & -1 \\ 0 & -1 \end{bmatrix}$$

Solution

The characteristics equation of the matrix $\mathbf{A}$ determined from $\det(\lambda\mathbf{I} - \mathbf{A}) = 0$

$$\begin{aligned}\det \left(\begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} - \begin{bmatrix} 1 & -1 \\ 0 & -1 \end{bmatrix} \right) &= 0 \\ (\lambda_1 - 1)(\lambda_2 + 1) &= 0 \\ \lambda_1 &= 1, \lambda_2 = -1\end{aligned}$$

The eigenvector corresponding to $\lambda_1 = -1$, $\boldsymbol{\nu}_1$ is

$$\begin{aligned}
 (\lambda_1 \mathbf{I} - \mathbf{A})\boldsymbol{\nu}_1 &= 0 \\
 \left[\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} - \begin{bmatrix} 1 & -1 \\ 0 & -1 \end{bmatrix} \right] \begin{bmatrix} \nu_{11} \\ \nu_{12} \end{bmatrix} &= 0 \\
 \begin{bmatrix} -2 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \nu_{11} \\ \nu_{12} \end{bmatrix} &= 0
 \end{aligned}$$

Which results in

$$-2\nu_{11} + \nu_{12} = 0 \Rightarrow \nu_{12} = 2\nu_{11}$$

Hence $\boldsymbol{\nu}_1$

$$\boldsymbol{\nu}_1 = \begin{bmatrix} \nu_{11} \\ \nu_{12} \end{bmatrix} = m \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

where m is any real number.

The eigenvector corresponding to λ_2 , $\boldsymbol{\nu}_2$ is

$$\begin{aligned}
 (\lambda_2 \mathbf{I} - \mathbf{A})\boldsymbol{\nu}_2 &= 0 \\
 \left[\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 1 & -1 \\ 0 & -1 \end{bmatrix} \right] \begin{bmatrix} \nu_{21} \\ \nu_{22} \end{bmatrix} &= 0 \\
 \begin{bmatrix} 0 & 1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} \nu_{21} \\ \nu_{22} \end{bmatrix} &= 0
 \end{aligned}$$

Which results in

$$\nu_{22} = 0, \quad 2\nu_{22} = 0$$

ν_{21} can be any value and $\nu_{22} = 0$. So $\boldsymbol{\nu}_2$ is

$$\boldsymbol{\nu}_2 = \begin{bmatrix} \nu_{21} \\ \nu_{22} \end{bmatrix} = m \begin{bmatrix} k \\ 0 \end{bmatrix}$$

5. Determine the eigenvalues, right eigenvector, and left eigenvectors of the following matrix

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 3 & 0 \end{bmatrix}$$

Solution

$$\det\left(\begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} - \begin{bmatrix} 1 & 2 \\ 3 & 0 \end{bmatrix}\right) = \det\left(\begin{bmatrix} \lambda - 1 & -2 \\ -3 & \lambda \end{bmatrix}\right) = 0$$

$$\lambda(\lambda - 1) + 5 = 0$$

$$\lambda^2 - \lambda + 6 = 0 \Rightarrow (\lambda - 3)(\lambda + 2) = 0$$

$$\lambda_1 = 3 \quad \lambda_2 = -2$$

the eigenvectors associated with $\lambda_1 = 3$, $\boldsymbol{\nu}_1$ is

$$(\lambda \mathbf{I} - \mathbf{A}) \boldsymbol{\nu}_1 = 0$$

$$\left[\begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} - \begin{bmatrix} 1 & 2 \\ 3 & 0 \end{bmatrix}\right] \begin{bmatrix} \nu_{11} \\ \nu_{12} \end{bmatrix} = 0 \Rightarrow \begin{bmatrix} 2 & -2 \\ -3 & 3 \end{bmatrix} \begin{bmatrix} \nu_{11} \\ \nu_{12} \end{bmatrix} = 0$$

which results in

$$2\nu_{11} - 2\nu_{12} = 0, \quad -3\nu_{11} + 3\nu_{12} = 0$$

$$\nu_{11} = \nu_{12} = 1 \quad \boldsymbol{\nu}_1 \Rightarrow \begin{bmatrix} \nu_{11} \\ \nu_{12} \end{bmatrix} = \nu_{11} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

the eigenvectors associated with $\lambda_2 = -2$, $\boldsymbol{\nu}_2$ is

$$(\lambda \mathbf{I} - \mathbf{A}) \boldsymbol{\nu}_2 = 0$$

$$\left[\begin{bmatrix} -2 & 0 \\ 0 & -2 \end{bmatrix} - \begin{bmatrix} 1 & 2 \\ 3 & 0 \end{bmatrix}\right] \begin{bmatrix} \nu_{21} \\ \nu_{22} \end{bmatrix} = 0 \Rightarrow \begin{bmatrix} -3 & -2 \\ -3 & -2 \end{bmatrix} \begin{bmatrix} \nu_{21} \\ \nu_{22} \end{bmatrix} = 0$$

which results in

$$-3\nu_{21} - 2\nu_{22} = 0, \quad -3\nu_{21} - 2\nu_{22} = 0$$

$$\nu_{21} = -\frac{2}{3}\nu_{22} \quad \boldsymbol{\nu}_2 \Rightarrow \begin{bmatrix} -\frac{2}{3}\nu_{22} \\ \nu_{22} \end{bmatrix} = \frac{\nu_{22}}{3} \begin{bmatrix} -2 \\ 3 \end{bmatrix}$$

the modality matrix $\mathbf{M}$ is

$$\mathbf{M} = [\boldsymbol{\nu}_1 \quad \boldsymbol{\nu}_2] = \begin{bmatrix} 1 & -2 \\ 1 & 3 \end{bmatrix}$$

6. For the following matrix determine the state transition matrix $\Phi(t)$, eigenvectors

$$\mathbf{A} = \begin{bmatrix} -6 & 2 \\ -6 & 1 \end{bmatrix}$$

Solution

$$\det\left(\begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} - \begin{bmatrix} -6 & 2 \\ -6 & 1 \end{bmatrix}\right) = \det\left(\begin{bmatrix} \lambda + 6 & -2 \\ 6 & \lambda - 1 \end{bmatrix}\right) = 0$$

$$(\lambda + 6)(\lambda - 1) + 12 = 0$$

$$\lambda^2 + 5\lambda + 6 = 0 \Rightarrow (\lambda + 3)(\lambda + 2) = 0$$

$$\lambda_1 = -3 \quad \lambda_2 = -2$$

the eigenvectors associated with $\lambda_1 = -3$, $\boldsymbol{\nu}_1$ is

$$(\lambda \mathbf{I} - \mathbf{A}) \boldsymbol{\nu}_1 = 0$$

$$-3 \begin{bmatrix} \nu_{11} \\ \nu_{12} \end{bmatrix} = \begin{bmatrix} -6 & 2 \\ -6 & 1 \end{bmatrix} \begin{bmatrix} \nu_{11} \\ \nu_{12} \end{bmatrix}$$

which results in

$$-6\nu_{11} + 2\nu_{12} = -3\nu_{11}, \quad -3\nu_{11} + 2\nu_{12} = 0$$

$$-6\nu_{11} + \nu_{12} = -3\nu_{12} \quad -6\nu_{11} + 4\nu_{12} = 0$$

$$3\nu_{11} = 2\nu_{12} \Rightarrow \nu_{11} = 2, \quad \nu_{12} = 3$$

$$\Rightarrow \boldsymbol{\nu}_1 = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

the eigenvectors associated with $\lambda_2 = -2$, $\boldsymbol{\nu}_2$ is

$$(\lambda \mathbf{I} - \mathbf{A}) \boldsymbol{\nu}_2 = 0$$

$$[\lambda \mathbf{I} - \mathbf{A}] \boldsymbol{\nu}_1 = 0$$

$$-2 \begin{bmatrix} \nu_{21} \\ \nu_{22} \end{bmatrix} = \begin{bmatrix} -6 & 2 \\ -6 & 1 \end{bmatrix} \begin{bmatrix} \nu_{21} \\ \nu_{22} \end{bmatrix}$$

which results in

$$-6\nu_{21} + 2\nu_{22} = -2\nu_{21}, \quad -3\nu_{21} + 2\nu_{22} = 0$$

$$-6\nu_{21} + \nu_{22} = -2\nu_{22} \quad -6\nu_{21} + 4\nu_{22} = 0$$

$$3\nu_{22} = 2\nu_{21} \Rightarrow \nu_{21} = 1, \quad \nu_{22} = 2$$

$$\Rightarrow \boldsymbol{\nu}_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

the modality matrix $\mathbf{M}$ and its inverse is

$$\mathbf{M} = \begin{bmatrix} \boldsymbol{\nu}_1 & \boldsymbol{\nu}_2 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix} \quad \mathbf{M}^{-1} = \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix}^{-1} = \begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix}$$

The diagonal of matrix $\bar{\mathbf{A}} = \mathbf{M}^{-1}\mathbf{A}\mathbf{M}$

$$\bar{\mathbf{A}} = \begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix} \begin{bmatrix} -6 & 2 \\ -6 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix} = \begin{bmatrix} -3 & 0 \\ 0 & -2 \end{bmatrix}$$

the state transition matrix $\Phi(t)$ is given as

$$\begin{aligned} \Phi(t) &= \mathbf{M}e^{\mathbf{A}t}\mathbf{M}^{-1} = \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} e^{\lambda_1 t} & 0 \\ 0 & e^{\lambda_2 t} \end{bmatrix} \begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix} \\ \Phi(t) &= \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} e^{-3t} & 0 \\ 0 & e^{-2t} \end{bmatrix} \begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix} \\ \Phi(t) &= \begin{bmatrix} 4e^{-3t} - 3e^{-2} & -2e^{-3t} + 2e^{-2t} \\ 6e^{-3t} - 6e^{-2} & -3e^{-3t} + 4e^{-2t} \end{bmatrix} \end{aligned}$$

7. Given matrix $\mathbf{A}$ find $\Phi(t)$

$$\mathbf{A} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 1 & 0 & -1 \end{bmatrix}$$

Solution:

The characteristic equation of the matrix is

$$\begin{aligned} \det \left(\begin{bmatrix} \lambda + 1 & 0 & 0 \\ 0 & \lambda + 1 & 0 \\ -1 & 0 & \lambda + 1 \end{bmatrix} \right) &= 0 \\ (\lambda + 1)(\lambda + 1)(\lambda + 1) &= 0 \\ \Rightarrow \lambda_1 = \lambda_2 = \lambda_3 &= -1 \end{aligned}$$

When any matrix has a multiplicity of eigenvalues, the method we don't rely on is using the modality matrix since the determinant becomes zero. So let us use the

Laplace inverse transform.

$$\begin{aligned}
 \Phi(t) &= \mathcal{L}^{-1}((s\mathbf{I} - \mathbf{A})^{-1}) = \mathcal{L}^{-1}\left(\left[\begin{array}{ccc} s+1 & 0 & 0 \\ 0 & s+1 & 0 \\ -1 & 0 & s+1 \end{array}\right]^{-1}\right) \\
 &= \mathcal{L}^{-1}\left(\left[\begin{array}{ccc} \frac{1}{s+1} & 0 & 0 \\ 0 & \frac{1}{s+1} & 0 \\ -\frac{1}{(s+1)^2} & 0 & \frac{1}{s+1} \end{array}\right]\right) \\
 &= \begin{bmatrix} e^{-t} & 0 & 0 \\ 0 & e^{-t} & 0 \\ te^{-t} & 0 & e^{-t} \end{bmatrix}
 \end{aligned}$$

8. For the following dynamic system, find eigenvalues, zero-input response, and zero-state response to step input for $t > 0$.

$$\dot{\mathbf{x}} = \begin{bmatrix} -1 & -3 \\ 0 & -2 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 5 \\ 6 \end{bmatrix} \mathbf{u}, \quad \mathbf{y} = \begin{bmatrix} 3 & 4 \end{bmatrix} \mathbf{x}$$

with initial condition $\mathbf{x}(0) = \begin{bmatrix} 4 & 5 \end{bmatrix}^\top$.

Solution:

The characteristic equation of the system is,

$$||\lambda\mathbf{I} - \mathbf{A}|| = 0 \Rightarrow (\lambda + 1)(\lambda + 2) = 0 \Rightarrow \lambda_1 = -1, \quad \lambda_2 = -2$$

zero input response y_{zinput} is

$$\begin{aligned}
 y_{zinput} &= \mathcal{L}^{-1}\left(\mathbf{C} \frac{adj(s\mathbf{I} - \mathbf{A})}{det(s\mathbf{I} - \mathbf{A})} \mathbf{x}_0\right) \\
 &= \mathcal{L}^{-1}\left(\mathbf{C} \begin{bmatrix} \frac{1}{s+1} & \frac{-3}{s+1} + \frac{3}{s+2} \\ 0 & \frac{1}{s+2} \end{bmatrix} \begin{bmatrix} 4 \\ 5 \end{bmatrix}\right) \\
 &= \begin{bmatrix} 3 & 4 \end{bmatrix} \begin{bmatrix} e^{-t} & -3e^{-t} + 3e^{-2t} \\ 0 & e^{-2t} \end{bmatrix} \begin{bmatrix} 4 \\ 5 \end{bmatrix} \\
 &= -33e^{-t} + 65e^{-2t}
 \end{aligned}$$

zero state response y_{zs} will be

$$\begin{aligned}
 y_{zs} &= \mathcal{L}^{-1} \left(\mathbf{C} \frac{\text{adj}(s\mathbf{I} - \mathbf{A})}{\det(s\mathbf{I} - \mathbf{A})} \mathbf{B} U(s) \right) \\
 &= \mathcal{L}^{-1} \left(\begin{bmatrix} 3 & 4 \end{bmatrix} \begin{bmatrix} \frac{1}{s+1} & \frac{-3}{s+1} + \frac{3}{s+2} \\ 0 & \frac{1}{s+2} \end{bmatrix} \begin{bmatrix} 5 \\ 6 \end{bmatrix} \frac{1}{s} \right) \\
 &= \mathcal{L}^{-1} \left(\begin{bmatrix} \frac{3}{s+1} & \frac{-9}{s+1} + \frac{9}{s+2} \frac{4}{s+2} \end{bmatrix} \begin{bmatrix} 5 \\ 6 \end{bmatrix} \frac{1}{s} \right) \\
 &= \mathcal{L}^{-1} \left(\begin{bmatrix} \frac{15}{s(s+1)} + \frac{78}{s(s+2)} - \frac{54}{s(s+1)} \end{bmatrix} \right) \\
 &= \mathcal{L}^{-1} \left(\frac{39}{(s+1)(s+2)} \right) \\
 &= \mathcal{L}^{-1} \left(\frac{39}{s+1} - \frac{39}{s+2} \right) \\
 &= 39e^{-t} - 39e^{-2t}
 \end{aligned}$$

the total response of the system is $y = y_{zinput} + y_{zs}$,

$$y = 6e^{-t} - 26e^{-2t}$$

9. Check the stability of the following system matrix

$$\mathbf{A} = \begin{bmatrix} 3 & -2 \\ 1 & -1 \end{bmatrix}$$

Solution:

using Lyapunov stability test: use $\mathbf{R} = \mathbf{I}$

$$\begin{aligned}
 &\mathbf{A}\mathbf{P} + \mathbf{P}\mathbf{A}^\top = -\mathbf{R} \\
 &\begin{bmatrix} 3 & -2 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{bmatrix} + \begin{bmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{bmatrix} \begin{bmatrix} 3 & 1 \\ -2 & -1 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}
 \end{aligned}$$

using the following command on MATLAB:

```

syms p11 p12 p21 p22
A = [3 -1; 1 -1]; R = eye(2); P = [p11 p12; p21 p22];
eqn = A*P + P*A' + R
[p11 p12; p21 p22] = solve(eqn == 0);
P = [p11 p12; p21 p22];

```

The result is

$$\mathbf{P} = \begin{bmatrix} 1 & 7/4 \\ 7/4 & 9/4 \end{bmatrix}$$

$\mathbf{P}$ is symmetric but not positive definite because $\det(\mathbf{P}) = \frac{-13}{16} < 0$. Hence, the $\mathbf{A}$ is not stable. Checking the eigenvalues

$$\|\lambda \mathbf{I} - \mathbf{A}\| = 0 \Rightarrow (\lambda - 3)(\lambda + 1) + 2 = 0 \Rightarrow \lambda_1 = 1 - \sqrt{2}, \quad \lambda_2 = 1 + \sqrt{2}$$

both eigenvalues has strictly positive real parts, so $\mathbf{A}$ is unstable.

10. For dc motor model derive on 1.7.1, check stability with $R_a = 0.25$, $J = 0.0025$, $L_a = 0.003$, $k_b = 0.0067$, $k_t = k_b$.

Solution:

using MATLAB command $[\boldsymbol{\nu}, \mathbf{D}] = \text{eig}(\mathbf{A})$, give us the eigenvalues $\lambda_1 = -83.25$, $\lambda_2 = -8.08$ and corresponding eigenvectors of the dc-motor dynamics

$$\boldsymbol{\nu}_1 = \begin{bmatrix} -1 \\ 0.034 \end{bmatrix} \quad \boldsymbol{\nu}_2 = \begin{bmatrix} 0.03 \\ -1 \end{bmatrix}$$

Since all eigenvalues have strictly negative real parts, the dynamic system is stable.

11. Check stability of IPC dynamic system derived on 1.6.3, with parameters $m_p = 0.5$, $m_c = 0.2$, $b = 0.1$, $l = 0.3$, $j = 0.5m_p l^2$, $g = 9.81$.

Solution:

using MATLAB command $[\boldsymbol{\nu}, \mathbf{D}] = \text{eig}(\mathbf{A})$, give us the eigenvalues $\lambda_1 = 0$, $\lambda_2 = -0.14$, $\lambda_3 = -5.21$, $\lambda_4 = 5.17$ and corresponding eigenvectors of the IPC dynamic system

$$\boldsymbol{\nu}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad \boldsymbol{\nu}_2 = \begin{bmatrix} -0.99 \\ 0.1414 \\ 0.0021 \\ -0.0003 \end{bmatrix} \quad \boldsymbol{\nu}_3 = \begin{bmatrix} -0.01661 \\ 0.0862 \\ -0.1879 \\ 0.9782 \end{bmatrix} \quad \boldsymbol{\nu}_4 = \begin{bmatrix} 0.0158 \\ 0.0816 \\ 0.1891 \\ 0.9784 \end{bmatrix}$$

Since not all eigenvalues doesn't have strictly negative real parts, the dynamic system is unstable.

12. Consider the following simple mass measuring system, with system parameters $k = 20\text{N/m}$, $m_s = 1\text{kg}$, and $d = 9\text{Ns/m}$ shown in Figure C.4b. (Mid-Exam 2021)

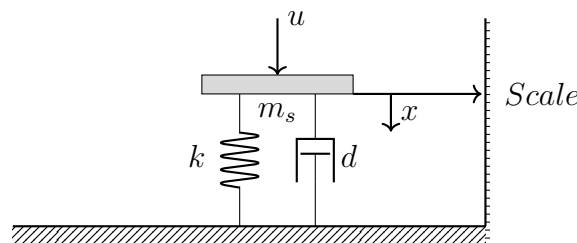


Figure 1.30: Simple Mass Measuring System

The system shows the mass of the object to be measured on the scale. The reading on the scale is proportional to the mass of the object ($y = \alpha m_o$). Where α is a proportionality constant (for this problem, use $\alpha = 1$).

- (a) Derive a state-space representation of the system.
- (b) Find the eigenvalues and eigenvectors of the system.
- (c) Determine whether the system is stable or not.
- (d) Determine the state transition matrix of the system.
- (e) Determine zero input response of the system for initial condition $\mathbf{x}(0) = \begin{bmatrix} 1 & 0 \end{bmatrix}^\top$.
- (f) Determine the zero state response of the system under a step input.
- (g) If the maximum compression of the spring is 10cm , determine the maximum mass the system can handle.
- (h) If you are the designer and you want to increase the maximum mass the system can handle 5 times,
 - What is your best engineering solution?
 - Why do you select the solution?
 - Determine the eigenvalues for your new system.
 - What do you conclude from the result?

Solution:

- (a) From the free body diagram of the system

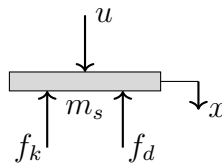


Figure 1.31: *FBD of Simple Mass Measuring System*

$$u - kx - d\dot{x} = m_s\ddot{x}$$

$$\ddot{x} = \frac{-k}{m_s}x - \frac{d}{m_s}\dot{x} + \frac{u}{m}$$

Let $x_1 = x$, $x_2 = \dot{x} = \dot{x}_1$, then the state space representation of the system is:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ \frac{-k}{m_s} & \frac{-d}{m_s} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{m} \end{bmatrix} u$$

The output is $y = \alpha m_o$, but $m_o = \frac{u}{g}$ (since u is the weight of the measured mass), which means $y = \frac{\alpha u}{g}$. The scale shows the result after the transient response dies out (under steady-state conditions). At steady-state conditions,

the only forces on the system are the spring force and the weight of the system. Hence,

$$\begin{aligned}
 u &= mg = kx \\
 y &= \frac{\alpha kx}{g} \Rightarrow y = \left[\frac{\alpha kx_1}{g} \right] \\
 y &= \begin{bmatrix} \frac{\alpha k}{g} & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}
 \end{aligned}$$

with the given parameters

$$\begin{aligned}
 \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} &= \begin{bmatrix} 0 & 1 \\ -20 & -9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u \\
 y &= \begin{bmatrix} 2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}
 \end{aligned}$$

(b) Eigenvalues and vectors of the system are

$$\begin{aligned}
 \|\lambda \mathbf{I} - \mathbf{A}\| &= 0 \Rightarrow \left\| \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} - \begin{bmatrix} 0 & 1 \\ -20 & -9 \end{bmatrix} \right\| = 0 \\
 \lambda(\lambda + 9) + 20 &= 0 \\
 (\lambda + 4)(\lambda + 5) &= 0 \\
 \lambda_1 &= -4 \quad \lambda_2 = -5
 \end{aligned}$$

The eigenvector corresponding to $\lambda_2 = -5$, ($\boldsymbol{\nu}_1$) is

$$\begin{aligned}
 \lambda_1 \begin{bmatrix} \nu_{11} \\ \nu_{12} \end{bmatrix} &= \begin{bmatrix} 0 & 1 \\ -20 & -9 \end{bmatrix} \begin{bmatrix} \nu_{11} \\ \nu_{12} \end{bmatrix} \Rightarrow -5 \begin{bmatrix} \nu_{11} \\ \nu_{12} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -20 & -9 \end{bmatrix} \begin{bmatrix} \nu_{11} \\ \nu_{12} \end{bmatrix} \\
 5\nu_{11} &= -\nu_{12} \quad 20\nu_{11} = -4\nu_{12} \\
 \boldsymbol{\nu}_1 &= \nu_{11} \begin{bmatrix} 1 \\ -5 \end{bmatrix}
 \end{aligned}$$

The eigenvector corresponding to $\lambda_2 = -4$, ($\boldsymbol{\nu}_2$) is

$$\begin{aligned}
 \lambda_2 \begin{bmatrix} \nu_{21} \\ \nu_{22} \end{bmatrix} &= \begin{bmatrix} 0 & 1 \\ -20 & -9 \end{bmatrix} \begin{bmatrix} \nu_{21} \\ \nu_{22} \end{bmatrix} \Rightarrow -4 \begin{bmatrix} \nu_{21} \\ \nu_{22} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -20 & -9 \end{bmatrix} \begin{bmatrix} \nu_{21} \\ \nu_{22} \end{bmatrix} \\
 -4\nu_{21} &= \nu_{22} \quad 20\nu_{21} = -5\nu_{22} \\
 \boldsymbol{\nu}_2 &= \nu_{21} \begin{bmatrix} 1 \\ -4 \end{bmatrix}
 \end{aligned}$$

(c) Stability analysis

Since both eigenvalues are strictly negative, the system is stable.

(d) state transition matrix

$$\begin{aligned}
 \Phi(t) &= \mathcal{L}^{-1}((s\mathbf{I} - \mathbf{A})^{-1}) \\
 &= \mathcal{L}^{-1}\left(\frac{\text{adj}(s\mathbf{I} - \mathbf{A})}{\det(s\mathbf{I} - \mathbf{A})}\right) = \mathcal{L}^{-1}\left(\frac{\begin{bmatrix} s+9 & 1 \\ -20 & s \end{bmatrix}}{(s+5)(s+4)}\right) \\
 &= \mathcal{L}^{-1}\left(\begin{bmatrix} \frac{s+9}{(s+5)(s+4)} & \frac{1}{(s+5)(s+4)} \\ \frac{-20}{(s+5)(s+4)} & \frac{s}{(s+5)(s+4)} \end{bmatrix}\right) \\
 &= \mathcal{L}^{-1}\left(\begin{bmatrix} \frac{20}{s+5} + \frac{5}{s+4} & \frac{-1}{s+5} + \frac{1}{s+4} \\ \frac{-20}{s+5} + \frac{5}{s+4} & \frac{-4}{s+5} + \frac{-4}{s+4} \end{bmatrix}\right) \\
 &= \begin{bmatrix} -4e^{-5t} + 5e^{-4t} & -e^{-5t} + e^{-4t} \\ -20e^{-5t} + 20e^{-4t} & 5e^{-5t} - 4e^{-4t} \end{bmatrix}
 \end{aligned}$$

(e) Zero input response

$$\begin{aligned}
 y_{zi}(t) &= \mathbf{C}\Phi(t)\mathbf{x}_o \\
 &= \begin{bmatrix} 2 & 0 \end{bmatrix} \begin{bmatrix} -4e^{-5t} + 5e^{-4t} & -e^{-5t} + e^{-4t} \\ -20e^{-5t} + 20e^{-4t} & 5e^{-5t} - 4e^{-4t} \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} \\
 &= -8e^{-5t} + 10e^{-4t}
 \end{aligned}$$

(f) Zero state response for step input

$$\begin{aligned}
 y_{zs}(t) &= \mathcal{L}^{-1}\mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B}U(s) \\
 &= \mathcal{L}^{-1}\left(\frac{\mathbf{C}\text{adj}(s\mathbf{I} - \mathbf{A})\mathbf{B}U(s)}{\det(s\mathbf{I} - \mathbf{A})}\right) \\
 &= \mathcal{L}^{-1}\left(\begin{bmatrix} 2 & 0 \end{bmatrix} \begin{bmatrix} \frac{s+9}{(s+5)(s+4)} & \frac{1}{(s+5)(s+4)} \\ \frac{-20}{(s+5)(s+4)} & \frac{s}{(s+5)(s+4)} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \frac{1}{s}\right) \\
 &= \mathcal{L}^{-1}\left(\frac{2}{s(s+5)(s+4)}\right) = \mathcal{L}^{-1}\left(\frac{0.1}{s} + \frac{0.4}{s+5} - \frac{0.5}{s+4}\right) \\
 &= 0.1 + 0.4e^{-5t} - 0.5e^{-4t}
 \end{aligned}$$

(g) The maximum mass of an object that the system can handle is

$$m_{omax} = \frac{kx_{max}}{g} = \frac{20N/m * 0.1m}{10m/s^2} = 0.2kg$$

(h) If we want to increase the maximum mass the system can measure, we have to replace the spring with another spring having a higher spring constant, because the spring force is the only force that supports the weight of the object at a steady state. But this will affect the sensitivity of the system, which has to be calibrated again. In this case, we have to replace the spring with a constant $k_n = 5k = 100N/m$.

The eigenvalues of the new system are

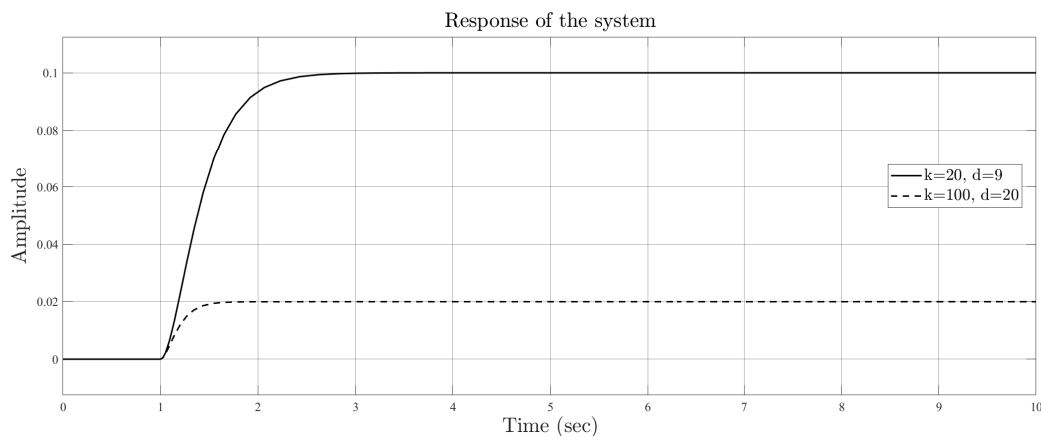


Figure 1.32: *System response with $k = 20, d = 9$ and $k = 100, d = 20$*

$$\begin{aligned} \|\lambda \mathbf{I} - \mathbf{A}\| = 0 &\Rightarrow \left\| \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} - \begin{bmatrix} 0 & 1 \\ -100 & -9 \end{bmatrix} \right\| = 0 \\ \lambda(\lambda + 9) + 100 &= 0 \\ \lambda_1 = -4.5 + j8.93 &\quad \lambda_2 = -4.5 - j8.93 \end{aligned}$$

The system is stable, but there will be an oscillation since the poles are complex conjugates. To solve this problem, we have to use a damper with a higher damping coefficient. If we take a damper with a damping coefficient $d = 20$ (the minimum value for this case). The eigenvalues of the system will be $\lambda_1 = \lambda_2 = -10$. This damper will remove the oscillation.

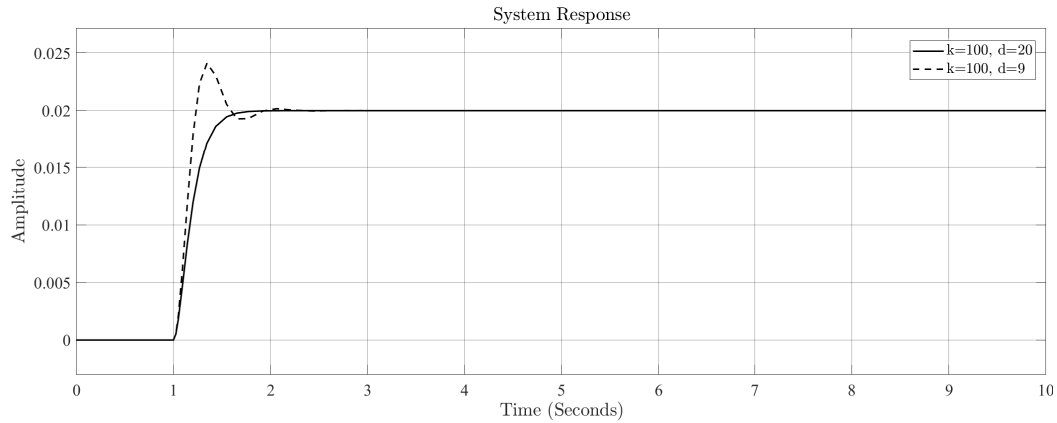


Figure 1.33: *System Response with $k = 100$ $d = 9$ and $d = 20$*

1.17 Exercises

- Find the characteristic polynomial, eigenvalues, and the corresponding eigenvectors for each of the following matrices.

(a) $\mathbf{A} = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$

(b) $\mathbf{C} = \begin{bmatrix} 2 & -2 \\ 2 & 2 \end{bmatrix}$

(c) $\mathbf{D} = \begin{bmatrix} 2 & 2 \\ -1 & -1 \end{bmatrix}$

(d) $\mathbf{E} = \begin{bmatrix} 0 & -1 \\ ab & a+b \end{bmatrix}$

(e) $\mathbf{F} = \begin{bmatrix} 2 & 2 & 0 \\ 1 & 2 & 1 \\ 1 & 2 & 1 \end{bmatrix}$

(f) $\mathbf{H} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$

(g) $\mathbf{I} = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$

- For the following matrix find state transition matrix $\Phi(t)$

(a) $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$

(b) $\mathbf{B} = \begin{bmatrix} 0 & 1 \\ 0 & -0 \end{bmatrix}$

(c) $\mathbf{C} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$

(d) $\mathbf{D} = \begin{bmatrix} -1 & 1 \\ 0 & -1 \end{bmatrix}$

(e) $\mathbf{E} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

(f) $\mathbf{F} = \begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$

(g) $\mathbf{G} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & 7 \end{bmatrix}$

- Find the general solution to the following initialized systems with step input.

(a)

$$\dot{\mathbf{x}} = \begin{bmatrix} -1 & -1 \\ 2 & -4 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u, \quad y = \begin{bmatrix} 0 & 1 \end{bmatrix} \mathbf{x}, \quad \mathbf{x}_0 = \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix}$$

(b)

$$\dot{\mathbf{x}} = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u, \quad y = \begin{bmatrix} 1 & 1 \end{bmatrix} \mathbf{x}, \quad \mathbf{x}_0 = \begin{bmatrix} 0 \\ 0.5 \end{bmatrix}$$

(c)

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u, \quad y = \begin{bmatrix} 1 & 0 \end{bmatrix} \mathbf{x}, \quad \mathbf{x}_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

(d)

$$\dot{\mathbf{x}} = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} u, \quad y = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \mathbf{x}, \quad \mathbf{x}_0 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

4. For a simple harmonic system described by

$$\ddot{z} + \omega z = 0$$

determine the state transition matrix.

5. For a dynamic system described by

$$\begin{aligned} \dot{x}_1 &= x_1 + 4x_2 + u \\ \dot{x}_2 &= 3x_1 + 2x_2 \end{aligned}$$

determine the state transition matrix.

6. For LTI system

$$\dot{\mathbf{x}} = \begin{bmatrix} -1 & -4 \\ -1 & -1 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u, \quad \mathbf{x}_0 = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad \& \quad u = e^{2t}$$

Find the state transition matrix and the solution of the state equation

7. Determine the range of values of k for which the polynomial

$$(3 - k)\lambda^3 + 2\lambda^2 + (5 - 2k)\lambda + 2 = 0$$

is asymptotically stable.

8. Determine the range of values of k , which makes the dynamic system

$$\dot{x}_1 = -kx_1 - 3x_2$$

$$\dot{x}_2 = kx_1 - 2x_2$$

is asymptotically stable.

9. Investigate the stability of

(a)

$$\dot{x}_1 = -x_1 + x_2$$

$$\dot{x}_2 = (x_1 + x_2) \sin k - 3x_2$$

(b)

$$\dot{x}_1 = -kx_1 + x_2$$

$$\dot{x}_2 = -ax_1 - bx_2$$

10. Show that the system described by

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u \quad y = x_1$$

Is BIBO stable but not asymptotically stable?

11. Consider an RLC Circuit in Figure B.3, with system parameters $L = 0.1H$, $C = 0.1F$ and $R = 1\Omega$. If the designer aims to regulate the voltage across the capacitor.

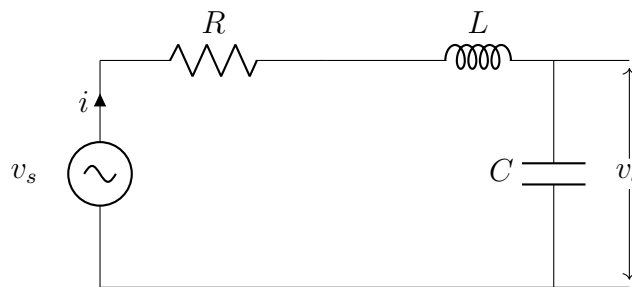


Figure 1.34: *Series RLC circuit*

- Derive a state space representation of the system .
- Find the eigenvalues and eigenvectors of the system.
- Determine whether the system is stable or not.
- Determine the state transition matrix of the system.
- Determine zero input response of the system for initial condition $\mathbf{x}(0) = \begin{bmatrix} 0 & 1 \end{bmatrix}^\top$.

- (f) Determine the zero state response of the system under step $u = e^{-2t}$.
- (g) What is the maximum value of voltage the capacitor can be charged to?
- (h) If the output is the voltage across the inductor, determine the state space representation of the system .
- (i) Compare the result on [11h](#) with the result on [11a](#) and state their difference.
- (j) Determine the zero state response of the state space representation derived in part [11a](#) and part [11h](#) for $u = \cos t$.

Chapter 2

Controllability and Observability of LTI Systems

2.1 Controllability of LTI Systems

In order to do whatever we want with the given dynamic system under control input, the system must be controllable.

Definition:-A linear time invariant system with state space representation on equation 1.3.6 and equation 1.3.7 is completely controllable if for any t_0 any initial state $\mathbf{x}(t_0) = \mathbf{x}_0$ can be transferred to a final state $\mathbf{x}(t_f)$ at t_f using control input u (i.e. if there exist a control signal $\mathbf{u}$ that transfer a state of a dynamic system from any initial condition $\mathbf{x}_0$ to a final state $\mathbf{x}(t_f)$).

Consider A linear time-invariant system on equation 1.3.7 with initial condition $\mathbf{x}_0$ at time $t_0 = 0$. We want to bring the system to the zero final state $\mathbf{x}(t_1) = 0$ at time t_1 . From the solution to the state equation

$$\begin{aligned}\mathbf{x}(t_1) &= e^{\mathbf{A}t_1}\mathbf{x}_0 + \int_0^{t_1} e^{\mathbf{A}(t_1-\tau)}\mathbf{B}\mathbf{u}(\tau)d\tau \\ 0 &= e^{\mathbf{A}t_1}\mathbf{x}_0 + \int_0^{t_1} e^{\mathbf{A}(t_1-\tau)}\mathbf{B}\mathbf{u}(\tau)d\tau \\ \mathbf{x}_0 &= - \int_0^{t_1} e^{-\mathbf{A}\tau}\mathbf{B}\mathbf{u}(\tau)d\tau\end{aligned}\tag{2.1.1}$$

and from Cayley-Hamilton theorem

$$e^{-\mathbf{A}\tau} = \sum_{k=0}^{k=n-1} \alpha(\tau)\mathbf{A}^k\tag{2.1.2}$$

Replacing equation 2.1.2 in equation 2.1.1 an rearranging we have

$$\mathbf{x}_0 = - \sum_{k=0}^{k=n-1} \mathbf{A}^k\mathbf{B} \int_0^{t_1} \alpha(\tau)\mathbf{u}(\tau)d\tau\tag{2.1.3}$$

Letting $\int_0^{t_1} \alpha(\tau) \mathbf{u}(\tau) d\tau = \beta_k$.
we have

$$\mathbf{x}_0 = - \sum_{k=0}^{n-1} \mathbf{A}^k \mathbf{B} \beta_k \quad (2.1.4)$$

$$\mathbf{x}_0 = - \begin{bmatrix} \mathbf{B} & \mathbf{AB} & \mathbf{A}^2\mathbf{B} & \dots & \mathbf{A}^{n-1}\mathbf{B} \end{bmatrix} \begin{bmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_k \end{bmatrix} \quad (2.1.5)$$

If the system is completely controllable, given any initial state $\mathbf{x}_0$, equation 2.1.4 must be satisfied, which requires the matrix

$$\mathbf{C}_o = \begin{bmatrix} \mathbf{B} & \mathbf{AB} & \mathbf{A}^2\mathbf{B} & \dots & \mathbf{A}^{n-1}\mathbf{B} \end{bmatrix} \quad (2.1.6)$$

to have full rank. This imply that the vectors $\mathbf{B}, \mathbf{AB}, \mathbf{A}^2\mathbf{B}, \dots, \mathbf{A}^{n-1}\mathbf{B}$ should be linearly independent. The matrix in equation 2.1.6 is called the controllability matrix. If the controllability matrix has full rank, then the system defined by $(\mathbf{A}, \mathbf{B})$ is completely controllable.

Or if a matrix $\mathbf{Q}$ is right Lyapunov and the solution for $\mathbf{Q}$ of equation 2.1.7

$$\mathbf{A}\mathbf{Q} + \mathbf{Q}\mathbf{A}^\top + \mathbf{B}\mathbf{B}^\top = 0 \quad (2.1.7)$$

is positive definite then the system defined by $(\mathbf{A}, \mathbf{B})$ is completely controllable. $\mathbf{Q}$ is determine from co-variance of $\mathbf{x}(t)$ which is called controllability grammian matrix, defined as

$$\mathbf{Q} = \int_0^\infty \mathbf{x}^\top(t) \mathbf{x}(t) dt = \int_0^\infty (e^{\mathbf{A}^\top t} \mathbf{B}^\top \mathbf{B} e^{\mathbf{A}t}) dt \quad (2.1.8)$$

2.2 Observability of LTI Systems

To see what is going on inside a dynamic system under observation, the system must be completely observable.

Definition:- A linear time invariant system with state space representation on equation 1.3.6 and equation 1.3.7 is completely observable if for any t_0 and any initial state $\mathbf{x}_0$, there exist finite time $t_1 > t_0$ such that the knowledge of $\mathbf{u}(\cdot)$ and $\mathbf{y}(\cdot)$ for $t \in [t_0, t_1]$ suffices to determine $\mathbf{x}_0$ properly.

From the solution to the output equation

$$\mathbf{y}(t) = \mathbf{C}e^{\mathbf{A}t}\mathbf{x}_0 + \int_0^{t_1} \mathbf{C}e^{\mathbf{A}(t-\tau)}\mathbf{B}\mathbf{u}(\tau)d\tau + \mathbf{D}\mathbf{u}(t) \quad (2.2.1)$$

Knowing $u(t)$ and $\mathbf{y}(t)$ will give information about x_0 , if the dynamic system defined by $(\mathbf{A}, \mathbf{C})$ is completely observable. Let the input be zero to derive the required condition. This means the system response is only the response to an initial condition. So,

$$\mathbf{y}(t) = \mathbf{C}e^{\mathbf{A}t}\mathbf{x}_0 \quad (2.2.2)$$

Replacing $e^{\mathbf{A}t} = \sum_{k=0}^{k=n-1} \alpha(t)\mathbf{A}^k$

$$\mathbf{y}(t) = \mathbf{C} \sum_{k=0}^{k=n-1} \alpha(t)\mathbf{A}^k \mathbf{x}_0$$

$$\mathbf{y}(t) = \alpha_0(t)\mathbf{C}\mathbf{x}_0 + \alpha_1(t)\mathbf{C}\mathbf{A}\mathbf{x}_0 + \alpha_2(t)\mathbf{C}\mathbf{A}^2\mathbf{x}_0 + \cdots + \alpha_{n-1}(t)\mathbf{C}\mathbf{A}^{n-1}\mathbf{x}_0$$

$$\mathbf{y}(t) = \begin{bmatrix} \alpha_0(t) & \alpha_1(t) & \cdots & \alpha_{n-1}(t) \end{bmatrix} \begin{bmatrix} \mathbf{C} \\ \mathbf{C}\mathbf{A} \\ \mathbf{C}\mathbf{A}^2 \\ \vdots \\ \mathbf{C}\mathbf{A}^{n-1} \end{bmatrix} \mathbf{x}_0 \quad (2.2.3)$$

Thus, complete observability requires

$$\mathcal{O}_v = \begin{bmatrix} \mathbf{C} \\ \mathbf{C}\mathbf{A} \\ \mathbf{C}\mathbf{A}^2 \\ \vdots \\ \mathbf{C}\mathbf{A}^{n-1} \end{bmatrix} \quad (2.2.4)$$

to have full rank. This implies that the co-vectors¹ $\mathbf{C}, \mathbf{C}\mathbf{A}, \mathbf{C}\mathbf{A}^2 \dots \mathbf{C}\mathbf{A}^{n-1}$ should be linearly independent. The matrix in equation 2.2.4 is called the observability matrix. If

¹A **co-vector** (or *dual vector*) is an element of the dual space V^* of a vector space V . It is a linear functional that maps vectors to scalars.

the observability matrix has full rank, then the system defined by $(\mathbf{A}, \mathbf{C})$ is completely observable.

Or if a matrix $\mathbf{P}$ is left Lyapunov and the solution for $\mathbf{P}$ of the equation 2.2.5

$$\mathbf{A}^\top \mathbf{P} + \mathbf{Q} \mathbf{A} + \mathbf{C}^\top \mathbf{C} = 0 \quad (2.2.5)$$

is positive definite then the system defined by $(\mathbf{A}, \mathbf{C})$ is completely observable. $\mathbf{P}$ is determined from co-variance of $\mathbf{y}(t)$ which is called observability grammian matrix, defined as

$$\mathbf{Q} = \int_0^\infty \mathbf{y}^\top(t) \mathbf{y}(t) dt = \int_0^\infty (e^{\mathbf{A}^\top t} \mathbf{C}^\top \mathbf{C} e^{\mathbf{A} t}) dt \quad (2.2.6)$$

2.3 Modality Tests

A mode of matrix $\mathbf{A}$ is a transformation that converts the matrix into a diagonal form. This transformation is composed of eigenvectors corresponding to the eigenvalues of the matrix. Hence modality test uses this concept to analyze the controllability and observability of a linear time-invariant system.

Consider a linear time invariant system on equation 1.3.6 and equation 1.3.7, if there is a transformation $\mathbf{V}$ that converts a matrix $\mathbf{A}$ to a diagonal matrix $\mathbf{\Lambda}$. Where

$$\mathbf{V}^{-1} \mathbf{\Lambda} \mathbf{V} = \mathbf{A} \quad (2.3.1)$$

Where $\mathbf{\Lambda}$ is a diagonal matrix composed of eigenvalues of matrix $\mathbf{A}$ placed either in decreasing or increasing order.

$$\mathbf{\Lambda} = \begin{bmatrix} \lambda_1 & 0 & 0 & \dots & 0 \\ 0 & \lambda_1 & 0 & \dots & 0 \\ \vdots & & & & \\ 0 & 0 & 0 & \dots & \lambda_1 \end{bmatrix} \quad (2.3.2)$$

The system is completely controllable if and only if no left (row) eigenvectors of $\mathbf{A}$ are orthogonal to $\mathbf{B}$. If $\boldsymbol{\nu}$ is left eigenvector of matrix $\mathbf{A}$ corresponding to any eigenvalue λ , for the system to be completely controllable

$$\boldsymbol{\nu} \cdot \mathbf{B} \neq 0 \quad (2.3.3)$$

The system is completely observable if and only if no right (column) eigenvectors of $\mathbf{A}$ are orthogonal to $\mathbf{C}$. If $\mathbf{q}$ is a right eigenvector of matrix $\mathbf{A}$ corresponding to an eigenvalue λ , for the system to be completely observable

$$\mathbf{C} \cdot \mathbf{q} \neq 0 \quad (2.3.4)$$

These tests are also called PHB (Popov Belvitch Hautus) tests.

2.4 Kalman Decomposition of LTI Systems

If the system is not completely controllable, the system can be decomposed into its controllable states and uncontrollable states using a similarity transformation. Consider LTI system having r controlled states $\mathbf{x}_c$, and $n - r$ uncontrollable states $\mathbf{x}_{\bar{c}}$, with a transformation matrix $\mathbf{T}$, the system can be transformed into

$$\begin{bmatrix} \dot{\mathbf{x}}_c \\ \dot{\mathbf{x}}_{\bar{c}} \end{bmatrix} = \begin{bmatrix} \mathbf{A}_c & \mathbf{A}_{12} \\ \mathbf{0} & \mathbf{A}_{\bar{c}} \end{bmatrix} \begin{bmatrix} \mathbf{x}_c \\ \mathbf{x}_{\bar{c}} \end{bmatrix} + \begin{bmatrix} \mathbf{B}_c \\ \mathbf{0} \end{bmatrix} \mathbf{u} \quad (2.4.1)$$

$$\mathbf{y} = \begin{bmatrix} \mathbf{C}_c & \mathbf{C}_{\bar{c}} \end{bmatrix} \begin{bmatrix} \mathbf{x}_c \\ \mathbf{x}_{\bar{c}} \end{bmatrix} + \mathbf{D}\mathbf{u} \quad (2.4.2)$$

The Block diagram representation of controllable and uncontrollable modes decomposition is shown in Figure 2.1.

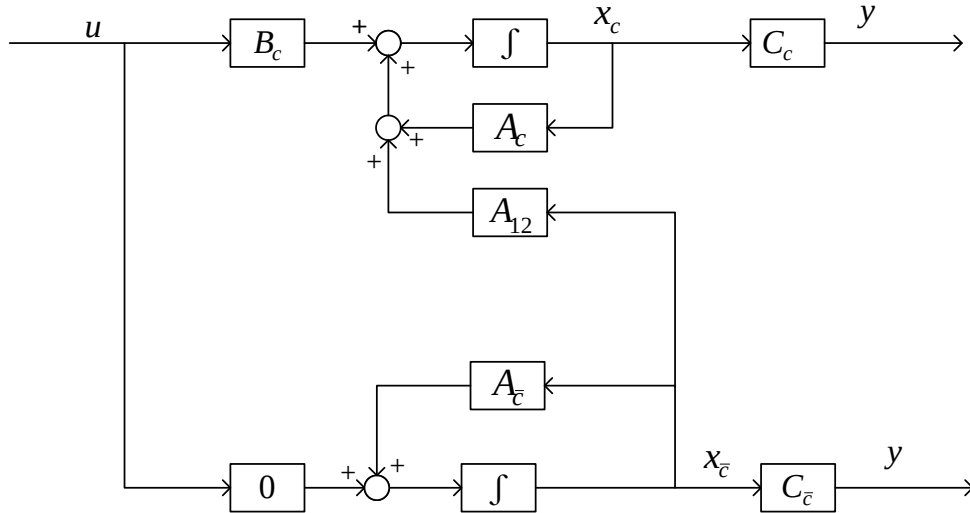


Figure 2.1: Block Diagram Representation of Controllable and Uncontrollable Modes of a System

Similarly, if a system is not completely observable, the system can be decomposed into its observable states and unobservable states using a similarity transformation. Consider LTI system having r observable states $\mathbf{x}_o$, and $n - r$ unobservable states $\mathbf{x}_{\bar{o}}$, with a transformation matrix $\mathbf{T}$, the system can be transformed into

$$\begin{bmatrix} \dot{\mathbf{x}}_o \\ \dot{\mathbf{x}}_{\bar{o}} \end{bmatrix} = \begin{bmatrix} \mathbf{A}_o & \mathbf{0} \\ \mathbf{A}_{21} & \mathbf{A}_{\bar{o}} \end{bmatrix} \begin{bmatrix} \mathbf{x}_o \\ \mathbf{x}_{\bar{o}} \end{bmatrix} + \begin{bmatrix} \mathbf{B}_o \\ \mathbf{B}_{\bar{o}} \end{bmatrix} \mathbf{u} \quad (2.4.3)$$

$$\mathbf{y} = \begin{bmatrix} \mathbf{C}_o & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{x}_o \\ \mathbf{x}_{\bar{o}} \end{bmatrix} + \mathbf{D}\mathbf{u} \quad (2.4.4)$$

The Block diagram representation of observable and unobservable modes decomposition is shown in Figure 2.2.

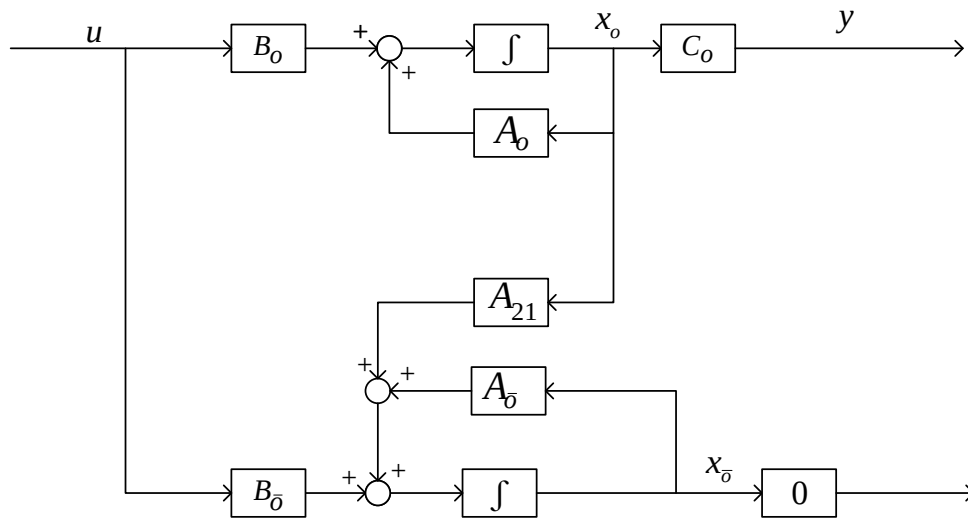


Figure 2.2: *Block Diagram Representation of Observable and Unobservable Modes of a System*

Kalman Decomposition via Eigen-structure

The eigen-structure is utilized to decompose the state space into controllable and uncontrollable subspaces, as well as observable and unobservable subspaces:

1. Controllable and Uncontrollable Subspaces:

- The controllable subspace is spanned by the right eigenvectors $\mathbf{q}_i$ whose corresponding left eigenvectors $\boldsymbol{\nu}_i$ satisfy $\boldsymbol{\nu}_i \mathbf{B} \neq 0$.
- The uncontrollable subspace consists of the modes where $\boldsymbol{\nu}_i \mathbf{B} = 0$. These modes cannot be controlled through any input.

2. Observable and Unobservable Subspaces:

- The observable subspace is defined by the right eigenvectors $\mathbf{q}_i$ for which $\mathbf{C}\mathbf{q}_i \neq 0$. These modes can be detected through the output matrix $\mathbf{C}$.
- The unobservable subspace comprises right eigenvectors $\mathbf{q}_i$ where $\mathbf{C}\mathbf{q}_i = 0$, indicating that these states cannot be inferred from output measurements.

Kalman decomposition is essential for system analysis and design, as it enables us to focus on the controllable and observable parts of the system while understanding the limitations imposed by uncontrollable and unobservable dynamics.

2.4.1 Weaker Conditions

If the uncontrollable mode of the system is stable, or conversely, if the unstable mode is controllable, then the system is said to be stabilizable.

If the unobservable mode of the system is stable, or conversely if the unstable mode is observable, then the system is said to be detectable.

Stabilizability and detectability are called weak conditions since it doesn't impose tight conditions like controllability and observability on system dynamics. If the system state is uncontrollable and stable, there is no need to put additional restrictions since the rest of the states are controllable. Similarly, as long as the unstable states are observable, there is no need to add a restriction on unobservable stable states.

Stability of uncontrollable states can be deduced from the eigenvalues of the uncontrollable subsystem matrix ($\mathbf{A}_{\bar{c}}$, while the stability of unobservable dynamics is deduced from the eigenvalues of the unobservable subsystem matrix $\mathbf{A}_{\bar{o}}$). If the matrices have eigenvalues with a negative real part, then the dynamics are stable.

2.4.2 Transformation Matrix for Kalman Decomposition

To decompose a system into controllable and uncontrollable or observable and unobservable subspaces, we need a transformation matrix $\mathbf{T}$. The transformation matrix $\mathbf{T}$ is composed of the basis vectors that span the controllable and observable space of the system their orthogonal complements. Thus, we need to recap basic concepts in matrix space.

Recap on the Four Fundamental Subspaces of a Matrix

An $m \times n$ matrix operates between two vector spaces: it takes inputs from $\mathbb{R}^n$ (all possible column vectors with n components) and produces outputs in $\mathbb{R}^m$. The matrix's behavior can be completely understood by examining four special subspaces (vector spaces within these larger spaces) that reveal how the matrix transforms vectors.

1. **Column Space (or Range):** The column space, denoted $\text{Col}(\mathbf{A})$, consists of all possible linear combinations of the columns of $\mathbf{A}$. It lives in $\mathbb{R}^m$ and represents all possible outputs the matrix can produce when multiplied by a vector. Its dimension is equal to the rank of $\mathbf{A}$, which tells us how many linearly independent columns $\mathbf{A}$ has. Essentially, it shows us where the matrix can "reach" in the output space.

$$\text{Col}(\mathbf{A}) = \{\mathbf{y} \in \mathbb{R}^m \mid \mathbf{y} = \mathbf{A}\mathbf{x} \text{ for some } \mathbf{x} \in \mathbb{R}^n\}.$$

2. **Null Space (or Kernel):** The null space, denoted $\text{Null}(\mathbf{A})$, contains all vectors $\mathbf{x}$ in $\mathbb{R}^n$ that get mapped to the zero vector when multiplied by $\mathbf{A}$. It shows us which inputs completely disappear when transformed by $\mathbf{A}$. The dimension of the null space is called the nullity and equals $n - \text{rank}(\mathbf{A})$. This tells us how much information gets "lost" when the matrix transforms vectors. The null space consists of **all eigenvectors** for $\lambda = 0$ (plus the zero vector) for a square matrix.

$$\mathbf{A}\mathbf{x} = \mathbf{0}.$$

3. **Row Space:** The row space is the span of all row vectors of $\mathbf{A}$ and lives in $\mathbb{R}^n$. Surprisingly, it always has the same dimension as the column space - both equal to the rank of $\mathbf{A}$. This space represents all possible linear combinations of the matrix's rows, which corresponds to all possible "measurements" or "constraints" that the matrix's rows can represent.

$$\text{Row}(\mathbf{A}) = \{\mathbf{x} \in \mathbb{R}^n \mid \mathbf{x} = \mathbf{A}^\top \mathbf{y} \text{ for some } \mathbf{y} \in \mathbb{R}^m\}.$$

4. **Left Null Space:** This less obvious space, denoted $\text{Null}(\mathbf{A}^\top)$, consists of all vectors $\mathbf{y}$ in $\mathbb{R}^m$ that get mapped to zero when multiplied by $\mathbf{A}^\top$ (the transpose of $\mathbf{A}$). These vectors are orthogonal to all columns of $\mathbf{A}$, and the space has dimension $m - \text{rank}(\mathbf{A})$. It represents all possible "constraints" or "tests" that vectors in the output space must satisfy to be in the column space of $\mathbf{A}$.

$$\mathbf{A}^\top \mathbf{y} = \mathbf{0}.$$

These spaces are connected:

- The row space and null space are perpendicular complements in $\mathbb{R}^n$ ($\text{Null}(\mathbf{A}) \perp \text{Row}(\mathbf{A})$), while the column space and left null space are perpendicular complements in $\mathbb{R}^m$ ($\text{Col}(\mathbf{A}) \perp \text{Null}(\mathbf{A}^\top)$).
- Their dimensions add up perfectly following the Rank-Nullity Theorem: $\dim(\text{Row}(\mathbf{A})) + \dim(\text{Null}(\mathbf{A})) = n$ and $\dim(\text{Col}(\mathbf{A})) + \dim(\text{Null}(\mathbf{A}^\top)) = m$. These relationships form the Fundamental Theorem of Linear Algebra, showing how a matrix connects its input and output spaces through these four fundamental subspaces.

Example

For matrix:

$$\mathbf{B} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} \quad (2.4.5)$$

- **Null Space** $\text{Null}(\mathbf{B})$ is determined by solving:

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad (2.4.6)$$

This gives the system:

$$\begin{aligned} x_1 + x_3 &= 0 \\ x_2 + x_3 &= 0 \end{aligned}$$

The solution is:

$$x_1 = -x_3, \quad x_2 = -x_3, \quad x_3 \text{ is a free variable while } x_1, x_2 \text{ are basic variables.} \quad (2.4.7)$$

Note: In linear algebra, when a matrix is transformed into its reduced row echelon form (RREF), the variables corresponding to pivot columns are called basic variables, while those corresponding to non-pivot columns are free variables. The basic variables can be expressed uniquely in terms of the free variables, which represent the independent degrees of freedom in the solution space.

$$\text{Null}(\mathbf{B}) = \left\{ x_3 \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} \mid x_3 \in \mathbb{R} \right\} \rightarrow \text{span} \left\{ \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} \right\} \quad (2.4.8)$$

- **Column Space** The column space $\text{Col}(\mathbf{B})$ is the span of the column vectors of $\mathbf{B}$. The columns are:

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad (2.4.9)$$

Since the first two columns are linearly independent and the third is their sum, the column space is:

$$\text{Col}(\mathbf{B}) = \text{span} \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\} = \mathbb{R}^2 \quad (2.4.10)$$

- **Row Space** The row space $\text{Row}(\mathbf{B})$ is the span of the row vectors of $\mathbf{B}$. The rows are:

$$\begin{bmatrix} 1 & 0 & 1 \end{bmatrix}, \quad \begin{bmatrix} 0 & 1 & 1 \end{bmatrix} \quad (2.4.11)$$

These rows are linearly independent, so:

$$\text{Row}(\mathbf{B}) = \text{span} \left\{ \begin{bmatrix} 1 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 1 \end{bmatrix} \right\} \quad (2.4.12)$$

- **Left Null Space** The left null space $\text{Null}(\mathbf{B}^\top)$ is the set of solutions to $\mathbf{B}^\top \mathbf{y} = 0$. Compute $\mathbf{B}^\top$:

$$\mathbf{B}^\top = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} \quad (2.4.13)$$

Solve:

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad (2.4.14)$$

This gives:

$$\begin{aligned} y_1 &= 0 \\ y_2 &= 0 \\ y_1 + y_2 &= 0 \end{aligned}$$

The only solution is $y_1 = y_2 = 0$, so:

$$\text{Null}(\mathbf{B}^\top) = \left\{ \begin{bmatrix} 0 \\ 0 \end{bmatrix} \right\} \quad (2.4.15)$$

Application of Matrix Subspace in Control Systems

For a dynamic system described by a state equation:

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} \quad (2.4.16)$$

$$\mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u} \quad (2.4.17)$$

1. Controllability: The controllability matrix $\mathbf{C}_o$ is defined as:

$$\mathbf{C}_o = [\mathbf{B} \quad \mathbf{A}\mathbf{B} \quad \mathbf{A}^2\mathbf{B} \quad \dots \quad \mathbf{A}^{n-1}\mathbf{B}] \quad (2.4.18)$$

- If $\text{Col}(\mathbf{C}_o) = \mathbb{R}^n$ or $\text{rank}(\mathbf{C}_o) = n$, the system is fully controllable.
- If $\text{rank}(\mathbf{C}_o) < n$, there exist $n - \text{rank}(\mathbf{C}_o)$ uncontrollable subspaces.
- The uncontrollable subspaces correspond to $\text{Null}(\mathbf{C}_o^\top)$, as they are orthogonal to $\text{Col}(\mathbf{C}_o)$.

2. Observability: The observability matrix $\mathbf{O}_v$ is defined as:

$$\mathbf{O}_v = [\mathbf{C} \quad \mathbf{C}\mathbf{A} \quad \mathbf{C}\mathbf{A}^2 \quad \dots \quad \mathbf{C}\mathbf{A}^{n-1}]^\top \quad (2.4.19)$$

- If $\text{rank}(\mathbf{O}_v) = n$, the system is completely observable.
- If $\text{rank}(\mathbf{O}_v) < n$, there exist $n - \text{rank}(\mathbf{O}_v)$ unobservable subspaces.
- The unobservable subspaces correspond to $\text{Null}(\mathbf{O}_v)$.

Controllability Decomposition

The concept of controllability fundamentally concerns whether a system can be driven from any initial state $\mathbf{x}(0)$ to any desired final state $\mathbf{x}(t_f)$ in finite time through appropriate control inputs $\mathbf{u}(t)$. This property is characterized by the reachable subspace of the system by control inputs $\mathbf{u}(t)$, which corresponds precisely to the column space of the controllability matrix $\mathbf{C}_o$.

The state space decomposition into controllable and uncontrollable subspaces can be obtained through a similarity transformation matrix $\mathbf{T}_c$ constructed from the basis vectors spanning $\text{Col}(\mathbf{C}_o)$ (the controllable subspace) and the basis vectors spanning $\text{Null}(\mathbf{C}_o^\top)$ (the uncontrollable subspace, being the orthogonal complement).

$$\mathbf{T}_c = [\text{basis}(\text{Col}(\mathbf{C}_o)) \quad \text{basis}(\text{Null}(\mathbf{C}_o^\top))] \quad (2.4.20)$$

Observability Decomposition

The concept of observability fundamentally addresses whether a system's initial state $\mathbf{x}(0)$ can be uniquely determined from knowledge of the input $u(t)$ and output $\mathbf{y}(t)$ over a finite time interval $[0, t_f]$. This property characterizes which state information is visible through the output measurements, defining the observable subspace of the system.

The state space decomposition into observable and unobservable components can be obtained through a similarity transformation matrix $\mathbf{T}_o$ constructed from the basis vectors spanning $\text{Col}(\mathbf{O}_v^\top)$ (the observable subspace, equivalent to the row space of $\mathbf{O}_v$) and the basis vectors spanning $\text{Null}(\mathbf{O}_v)$ (the unobservable subspace).

$$\mathbf{T}_o = \begin{bmatrix} \text{basis}(\text{Col}(\mathbf{O}_v^\top)) & \text{basis}(\text{Null}(\mathbf{O}_v)) \end{bmatrix} \quad (2.4.21)$$

2.5 Controllability and Observability in a Sense of Transfer Function

Consider a transfer function from $U(s)$ to $\mathbf{X}(s)$

$$\frac{X(s)}{U(s)} = \frac{\text{adj}(s\mathbf{I} - \mathbf{A})\mathbf{B}}{\det(s\mathbf{I} - \mathbf{A})} \quad (2.5.1)$$

For a system to be completely controllable, $\text{adj}(s\mathbf{I} - \mathbf{A})\mathbf{B}$ and $\det(s\mathbf{I} - \mathbf{A})$ shouldn't have common factors (i.e, there shouldn't be a pole-zero cancellation in a transfer function from input to state of the system).

Consider a transfer function from $U(s)$ to $Y(s)$

$$\frac{Y(s)}{U(s)} = \mathbf{C} \frac{\text{adj}(s\mathbf{I} - \mathbf{A})\mathbf{B}}{\det(s\mathbf{I} - \mathbf{A})} + \mathbf{D}U(s) \quad (2.5.2)$$

For a system to be completely observable, $\mathbf{C}\text{adj}(s\mathbf{I} - \mathbf{A})\mathbf{B}$ and $\det(s\mathbf{I} - \mathbf{A})$ shouldn't have common factors (i.e, there shouldn't be a pole-zero cancellation in the transfer function from input to output of the system).

2.6 Examples

1. Consider the following dynamic system, check whether the system is completely controllable and completely observable.

$$\dot{\mathbf{x}} = \begin{bmatrix} -1 & -3 \\ 0 & -2 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 5 \\ 6 \end{bmatrix} \mathbf{u}, \quad \mathbf{y} = \begin{bmatrix} 3 & 4 \end{bmatrix} \mathbf{x}$$

Solution:

the controllability matrix $\mathbf{C}_o$ is

$$\mathbf{C}_o = \begin{bmatrix} \mathbf{B} & \mathbf{AB} \end{bmatrix} = \begin{bmatrix} 5 & -23 \\ 6 & -12 \end{bmatrix}$$

checking for linear independence of the columns of $\mathbf{C}_o$

$$\alpha_1 \begin{bmatrix} 5 \\ 6 \end{bmatrix} + \alpha_2 \begin{bmatrix} -23 \\ -12 \end{bmatrix} = 0 \Rightarrow \alpha_1 = \alpha_2 = 0$$

They are linearly independent, hence the controllability matrix has full rank, which means the system is completely controllable.

The observability matrix $\mathbf{O}_v$ is

$$\mathbf{O}_v = \begin{bmatrix} \mathbf{C} \\ \mathbf{CA} \end{bmatrix} = \begin{bmatrix} 3 & 4 \\ -3 & -17 \end{bmatrix}$$

checking for linear independence of the columns of $\mathbf{O}_v$

$$\alpha_1 \begin{bmatrix} 3 \\ -3 \end{bmatrix} + \alpha_2 \begin{bmatrix} 4 \\ -17 \end{bmatrix} = 0 \Rightarrow \alpha_1 = \alpha_2 = 0$$

They are linearly independent, hence the observability matrix has full rank, which means the system is completely observable.

2. Check controllability and observability of the following dynamic system.

$$\dot{\mathbf{x}} = \begin{bmatrix} -2 & -2 & -2 \\ 1 & 0 & 0 \\ 1 & 1 & -1 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} u \quad y = \begin{bmatrix} 1 & 1 & 0 \end{bmatrix} \mathbf{x}$$

The controllability matrix is,

$$\mathbf{C}_o = \begin{bmatrix} 1 & -2 & 0 \\ 0 & 1 & -2 \\ 0 & 1 & -2 \end{bmatrix}$$

The rank $\text{rank}(\mathbf{C}_o) = 2$, hence the dynamic system has an uncontrollable state.

The observability matrix is,

$$\mathbf{O}_v = \begin{bmatrix} 1 & 1 & 0 \\ -1 & -2 & -2 \\ -2 & 0 & 4 \end{bmatrix}$$

having rank $\text{rank}(\mathbf{O}_v) = 2$, hence the dynamic system have unobservable state.

The transformation matrix $\mathbf{T}_c$ that decomposes the system into controllable and uncontrollable subspaces is

$$\mathbf{T}_c = \begin{bmatrix} \text{basis}(\text{Col}(\mathbf{C}_o)) & \text{basis}(\text{Null}(\mathbf{C}_o^\top)) \end{bmatrix}$$

The null space of $(\mathbf{C}_o^\top)$ is determined as

$$\mathbf{C}_o^\top \boldsymbol{\nu} = 0 \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 1 \\ 0 & -2 & -2 \end{bmatrix} \begin{bmatrix} \nu_1 \\ \nu_2 \\ \nu_3 \end{bmatrix} = 0$$

Transforming to Reduced row echelon form,

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \nu_1 \\ \nu_2 \\ \nu_3 \end{bmatrix} = 0 \rightarrow \nu_3 \text{ is the free variable, leading to}$$

$$\boldsymbol{\nu} = \begin{bmatrix} 0 \\ -\nu_3 \\ \nu_3 \end{bmatrix} \quad \text{The Null space of } \mathbf{C}_o^\top \text{ is spanned by } \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}$$

The basis of $(\text{Col}(\mathbf{C}_o))$ is any linearly independent columns of $\mathbf{C}_o$, taking the first two columns and the basis vector that span the null space of $\mathbf{C}_o^\top$, the transformation matrix $\mathbf{T}_c$ becomes,

$$\mathbf{T}_c = \begin{bmatrix} 1 & -2 & 0 \\ 0 & 1 & -1 \\ 0 & 1 & 1 \end{bmatrix}$$

To solve for $\mathbf{T}_c^{-1}$, transform the augmented matrix $[\mathbf{T}_c \mid \mathbf{I}]$ into $[\mathbf{I} \mid \mathbf{F}]$ using the Gauss-Jordan elimination method, where $\mathbf{I}$ is the identity matrix of the same size as $\mathbf{T}_c$. The resulting matrix $\mathbf{F}$ will be the inverse of $\mathbf{T}_c$. Using this method, the inverse of $\mathbf{T}_c$ is

$$\mathbf{T}_c^{-1} = 0.5 \begin{bmatrix} 2 & 2 & 2 \\ 0 & 1 & 1 \\ 0 & -1 & 1 \end{bmatrix}$$

With this transformation matrix, the decomposed system system matrix $\bar{\mathbf{A}}$, input matrix $\bar{\mathbf{B}}$ and $\bar{\mathbf{C}}$ is,

$$\bar{\mathbf{A}}_c = \mathbf{T}^{-1} \mathbf{A} \mathbf{T} = \begin{bmatrix} 0 & -4 & \vdots & -2 \\ 1 & -2 & \vdots & -1 \\ \dots & \dots & \dots & \dots \\ 0 & 0 & \vdots & -1 \end{bmatrix}, \quad \bar{\mathbf{B}}_c = \mathbf{T}^{-1} \mathbf{B} = \begin{bmatrix} 1 \\ 0 \\ \dots \\ 0 \end{bmatrix}$$

$$\bar{\mathbf{C}}_c = \mathbf{C} \mathbf{T} = \begin{bmatrix} 1 & -1 & \vdots & -1 \end{bmatrix}$$

from the decomposition, we see that the uncontrollable state is stable (with eigenvalue -1), hence the system is stabilizable.

The transformation matrix $\mathbf{T}_o$ that decomposes the system into controllable and uncontrollable subspaces is

$$\mathbf{T}_o = \begin{bmatrix} \text{basis}(\text{Col}(\mathbf{O}_v^\top)) & \text{basis}(\text{Null}(\mathbf{O}_v)) \end{bmatrix}$$

The null space of $(\mathbf{O}_v)$ is determined as

$$\mathbf{O}_v \boldsymbol{\nu} = 0 \rightarrow \begin{bmatrix} 1 & 1 & 0 \\ -1 & -2 & -2 \\ -2 & 0 & 4 \end{bmatrix} \begin{bmatrix} \nu_1 \\ \nu_2 \\ \nu_3 \end{bmatrix} = 0$$

Transforming to Reduced row echelon form,

$$\begin{bmatrix} 1 & 0 & -2 \\ 0 & -1 & -2 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \nu_1 \\ \nu_2 \\ \nu_3 \end{bmatrix} = 0 \rightarrow \nu_3 \text{ is the free variable, leading to}$$

$$\boldsymbol{\nu} = \begin{bmatrix} 2\nu_3 \\ -2\nu_3 \\ \nu_3 \end{bmatrix} \quad \text{The Null space of } \mathbf{O}_v^\top \text{ is spanned by } \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix}$$

The basis of $(\text{Col}(\mathbf{O}_o^\top))$ is any linearly independent columns of $\mathbf{O}_v^\top$, taking the first two columns and the basis vector that span the null space of $\mathbf{O}_v$, the transformation matrix $\mathbf{T}_o$ becomes,

$$\mathbf{T}_o = \begin{bmatrix} 1 & 1 & 2 \\ 1 & -2 & -2 \\ 0 & -2 & 1 \end{bmatrix}$$

Solving for $\mathbf{T}_o^{-1}$ result in,

$$\mathbf{T}_o^{-1} = \frac{1}{9} \begin{bmatrix} 6 & 3 & -6 \\ 1 & -1 & -4 \\ 2 & -2 & 1 \end{bmatrix}$$

With this transformation matrix, the decomposed system system matrix $\bar{\mathbf{A}}$, input

matrix $\bar{\mathbf{B}}$ and $\bar{\mathbf{C}}$ is,

$$\bar{\mathbf{A}}_o = \mathbf{T}^{-1}\mathbf{A}\mathbf{T} = \frac{1}{9} \begin{bmatrix} -33 & 63 & \vdots & 0 \\ -13 & 15 & \vdots & 0 \\ \dots & \dots & \dots & \dots \\ -8 & 21 & \vdots & -9 \end{bmatrix}, \quad \bar{\mathbf{B}}_o = \mathbf{T}^{-1}\mathbf{B} = \frac{1}{9} \begin{bmatrix} 6 \\ 1 \\ \dots \\ 2 \end{bmatrix}$$

$$\bar{\mathbf{C}}_o = \mathbf{C}\mathbf{T} = \begin{bmatrix} 2 & -3 & \vdots & 0 \end{bmatrix}$$

From the decomposition, we see that the unobservable state is stable (with eigenvalue -1), hence the system is detectable.

3. Check the stabilizability and detectability of the dynamic system

$$\dot{\mathbf{x}} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & -1 & 1 \\ 0 & 0 & -2 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \mathbf{x}$$

Checking the controllability of the system

$$\mathbf{C}_o = \begin{bmatrix} \mathbf{B} & \mathbf{AB} & \mathbf{A}^2\mathbf{B} \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

The rank of the controllability matrix is 2, hence the system has one uncontrollable dynamics.

Checking the observability of the system

$$\mathbf{O}_v = \begin{bmatrix} \mathbf{C} \\ \mathbf{CA} \\ \mathbf{CA}^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

The rank of the observability matrix is 1; hence, the system has two unobservable dynamics.

The transformation matrix $\mathbf{T}_c$ that decomposes the system into controllable and uncontrollable subspaces is

$$\mathbf{T}_c = \begin{bmatrix} \text{basis}(\text{Col}(\mathbf{C}_o)) & \text{basis}(\text{Null}(\mathbf{C}_o^\top)) \end{bmatrix}$$

The null space of $(\mathbf{C}_o^\top)$ is spanned by

$$\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

The basis of $(\text{Col}(\mathbf{C}_o))$ is any linearly independent columns of $\mathbf{C}_o$, taking the first two column and the basis vector that span the null space of $\mathbf{C}_o^\top$, the transformation matrix $\mathbf{T}_c$ and $\mathbf{T}_c^{-1}$ becomes,

$$\mathbf{T}_c = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \mathbf{T}_c^{-1} = \begin{bmatrix} 0 & -1 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

With this transformation matrix, the decomposed system system matrix $\bar{\mathbf{A}}$, input matrix $\bar{\mathbf{B}}$ and $\bar{\mathbf{C}}$ is,

$$\begin{aligned} \bar{\mathbf{A}}_c = \mathbf{T}^{-1} \mathbf{A} \mathbf{T} &= \begin{bmatrix} 0 & 1 & \vdots & 1 \\ 1 & 0 & \vdots & -1 \\ \dots & \dots & \dots & \dots \\ 0 & 0 & \vdots & -2 \end{bmatrix}, \quad \bar{\mathbf{B}}_c = \mathbf{T}^{-1} \mathbf{B} = \begin{bmatrix} 1 \\ 0 \\ \dots \\ 0 \end{bmatrix} \\ \bar{\mathbf{C}}_c = \mathbf{C} \mathbf{T} &= \begin{bmatrix} 1 & 1 & \vdots & 0 \end{bmatrix} \end{aligned}$$

From the decomposition, we see that the uncontrollable state is stable (with eigenvalue -2), hence the system is stabilizable.

The transformation matrix $\mathbf{T}_o$ that decomposes the system into controllable and uncontrollable subspaces is

$$\mathbf{T}_o = \begin{bmatrix} \text{basis}(\text{Col}(\mathbf{O}_v^\top)) & \text{basis}(\text{Null}(\mathbf{O}_v)) \end{bmatrix}$$

The null space of $(\mathbf{O}_v)$ is spanned by,

$$\begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$$

The basis of $(\text{Col}(\mathbf{O}_o^\top))$ is any linearly independent columns of $\mathbf{O}_v^\top$, taking the first column and the basis vector that span the null space of $\mathbf{O}_v$, the transformation matrix $\mathbf{T}_o$ and $\mathbf{T}_o^{-1}$ becomes,

$$\mathbf{T}_o = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \mathbf{T}_o^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

With this transformation matrix, the decomposed system system matrix $\bar{\mathbf{A}}$, input

matrix $\bar{\mathbf{B}}$ and $\bar{\mathbf{C}}$ is,

$$\bar{\mathbf{A}}_o = \mathbf{T}^{-1}\mathbf{A}\mathbf{T} = \frac{1}{9} \begin{bmatrix} 1 & \vdots & 0 & 0 \\ \cdots & \cdots & \cdots & \cdots \\ 1 & \vdots & -1 & 1 \\ 0 & \vdots & 0 & -2 \end{bmatrix}, \quad \bar{\mathbf{B}}_o = \mathbf{T}^{-1}\mathbf{B} = \begin{bmatrix} 1 \\ \cdots \\ 1 \\ 0 \end{bmatrix}$$

$$\bar{\mathbf{C}}_o = \mathbf{C}\mathbf{T} = \begin{bmatrix} 1 & \vdots & 0 & 0 \end{bmatrix}$$

from the decomposition we see that the unobservable states are stable (with eigenvalues -1 and -2), hence the system is detectable.

4. Check the detectability and stabilizability of the dynamic system

$$\dot{\mathbf{x}} = \begin{bmatrix} -2 & 0 & -3 \\ 0 & 0 & 0 \\ \frac{3}{2} & 0 & \frac{5}{2} \end{bmatrix} \mathbf{x} + \begin{bmatrix} -1 \\ 0 \\ -2 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & \frac{21}{2} & 0 \end{bmatrix} \mathbf{x}$$

The controllability matrix of the system is

$$\mathbf{C}_o = \begin{bmatrix} -1 & 8 & 3.5 \\ 0 & 0 & 0 \\ -2 & -6.5 & -4.25 \end{bmatrix}$$

The rank of the controllability matrix is 2, hence the system has one uncontrollable dynamics.

checking the observability of the dynamic system, the observability matrix $\mathbf{O}_v$,

$$\mathbf{O}_v = \begin{bmatrix} 1 & 10.5 & 0 \\ -2 & 0 & -3 \\ -0.5 & 0 & -1.5 \end{bmatrix}$$

has rank full, hence the dynamic system is completely observable.

The transformation matrix $\mathbf{T}_c$ that decomposes the system into controllable and uncontrollable subspaces is

$$\mathbf{T}_c = \begin{bmatrix} \text{basis}(\text{Col}(\mathbf{C}_o)) & \text{basis}(\text{Null}(\mathbf{C}_o^\top)) \end{bmatrix}$$

The null space of $(\mathbf{C}_o^\top)$ is spanned by

$$\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

The basis of $(\text{Col}(\mathbf{C}_o))$ is any linearly independent columns of $\mathbf{C}_o$, taking the first two column and the basis vector that span the null space of $\mathbf{C}_o^\top$, the transformation matrix $\mathbf{T}_c$ and $\mathbf{T}_c^{-1}$ becomes,

$$\mathbf{T}_c = \begin{bmatrix} -1 & 8 & 0 \\ 0 & 0 & 0 \\ -2 & -6.5 & 1 \end{bmatrix} \quad \mathbf{T}_c^{-1} = \frac{-1}{22.5} \begin{bmatrix} 6.5 & 0 & 8 \\ -2 & 0 & 1 \\ 0 & -22.5 & 0 \end{bmatrix}$$

With this transformation matrix, the decomposed system system matrix $\bar{\mathbf{A}}$, input matrix $\bar{\mathbf{B}}$ and $\bar{\mathbf{C}}$ is,

$$\bar{\mathbf{A}}_c = \mathbf{T}^{-1}\mathbf{A}\mathbf{T} = \frac{-1}{22.5} \begin{bmatrix} 0 & -11.25 & \vdots & 0 \\ -22.5 & -11.25 & \vdots & 0 \\ \dots & \dots & \dots & \dots \\ 0 & 0 & \vdots & 0 \end{bmatrix}, \quad \bar{\mathbf{B}}_c = \mathbf{T}^{-1}\mathbf{B} = \frac{-1}{22.5} \begin{bmatrix} -22.5 \\ 0 \\ \dots \\ 0 \end{bmatrix}$$

$$\bar{\mathbf{C}}_c = \mathbf{C}\mathbf{T} = \begin{bmatrix} -1 & 8 & \vdots & 10.5 \end{bmatrix}$$

From the decomposition, we see that the uncontrollable state is marginally (with eigenvalue 0), hence the system is stabilizable.

5. Check the detectability and stabilizability of the dynamic system

$$\dot{\mathbf{x}} = \begin{bmatrix} -6 & -11 & -6 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} u \quad y = \begin{bmatrix} 0 & 1 & 2 \end{bmatrix} \mathbf{x}$$

The controllability matrix of the system is

$$\mathbf{C}_o = \begin{bmatrix} 1 & -6 & 25 \\ 0 & 1 & -6 \\ 0 & 0 & 1 \end{bmatrix}$$

$\mathbf{C}_o$ has full rank, hence the dynamic system is completely controllable.

The observability matrix of the system is

$$\mathbf{O}_v = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 0 \\ -4 & -11 & -6 \end{bmatrix}$$

which has rank 2, hence the dynamic system has one unobservable mode.

The transformation matrix $\mathbf{T}_o$ that decomposes the system into controllable and uncontrollable subspaces is

$$\mathbf{T}_o = \begin{bmatrix} \text{basis}(\text{Col}(\mathbf{O}_v^\top)) & \text{basis}(\text{Null}(\mathbf{O}_v)) \end{bmatrix}$$

The null space of $(\mathbf{O}_v)$ is spanned by,

$$\begin{bmatrix} 4 \\ -2 \\ 1 \end{bmatrix}$$

The basis of $(\text{Col}(\mathbf{O}_o^\top))$ is any linearly independent columns of $\mathbf{O}_v^\top$, taking the first column and the basis vector that span the null space of $\mathbf{O}_v$, the transformation matrix $\mathbf{T}_o$ and $\mathbf{T}_o^{-1}$ becomes,

$$\mathbf{T}_o = \begin{bmatrix} 0 & 1 & 4 \\ 1 & 2 & -2 \\ 2 & 0 & 4 \end{bmatrix} \quad \mathbf{T}_o^{-1} = \frac{-1}{21} \begin{bmatrix} 2 & -1 & -10 \\ -5 & -8 & 4 \\ -4 & 2 & -1 \end{bmatrix}$$

With this transformation matrix, the decomposed system matrix $\bar{\mathbf{A}}$, input matrix $\bar{\mathbf{B}}$ and $\bar{\mathbf{C}}$ is,

$$\begin{aligned} \bar{\mathbf{A}}_o &= \mathbf{T}^{-1} \mathbf{A} \mathbf{T} = \frac{-1}{21} \begin{bmatrix} -56 & -77 & \vdots & 0 \\ 119 & 140 & \vdots & 0 \\ \dots & \dots & \dots & \dots \\ 91 & 112 & \vdots & 42 \end{bmatrix}, \quad \bar{\mathbf{B}}_o = \mathbf{T}^{-1} \mathbf{B} = \frac{-1}{21} \begin{bmatrix} 2 \\ -5 \\ \dots \\ -4 \end{bmatrix} \\ \bar{\mathbf{C}}_o &= \mathbf{C} \mathbf{T} = \begin{bmatrix} 5 & 2 & \vdots & 0 \end{bmatrix} \end{aligned}$$

From the decomposition, we see that the unobservable state is stable (with eigenvalue -2), hence the system is detectable.

6. Given the following LTI system, determine the value of α for which the system is not c.c.

$$\dot{\mathbf{x}} = \begin{bmatrix} 2 & \alpha - 1 \\ 0 & 2 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 1 \\ \alpha \end{bmatrix} u$$

Solution:

The controllability matrix of the system is

$$\mathbf{C}_o = \begin{bmatrix} 1 & 2 + \alpha(\alpha - 1) \\ \alpha & 2\alpha \end{bmatrix}$$

The determinant of the controllability matrix must be non-zero if the system is c.c. So, to find the value of α that makes the system not c.c.

$$\begin{aligned} \det(\mathbf{C}_o) = 0 &\Rightarrow 2\alpha - \alpha^3 + \alpha^2 - 2\alpha = 0 \\ &\Rightarrow \alpha^2(1 - \alpha) = 0 \Rightarrow \alpha = 0, \text{ or } \alpha = 1 \end{aligned}$$

The same will hold for observability of the system $\det(\mathbf{O}_v) \neq 0$ if the system is c.o.

2.7 Exercises

1. Verify that the following control system is completely controllable

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 & 0 \\ 1 & -1 \end{bmatrix} \mathbf{u}$$

2. For what value of b is the dynamic system below not c.c.

$$\dot{\mathbf{x}} = \begin{bmatrix} -1 & -1 \\ 2 & -4 \end{bmatrix} \mathbf{x} + \begin{bmatrix} b \\ 0 \end{bmatrix} u$$

3. For what value of α is the dynamic system below not c.c.

(a)

$$\dot{\mathbf{x}} = \begin{bmatrix} -1 & -1 & -1 \\ 0 & -1 & \alpha \\ 0 & 1 & 3 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix} u$$

(b)

$$\dot{\mathbf{x}} = \begin{bmatrix} 2 & \alpha - 3 \\ 0 & 2 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 1 & 1 \\ 0 & \alpha^2 - 2 \end{bmatrix} \mathbf{u}$$

4. For a system with output

$$\dot{\mathbf{x}} = \begin{bmatrix} -1 & -1 \\ 2 & -4 \end{bmatrix} \mathbf{x}, \quad y = \begin{bmatrix} 1 & 2 \end{bmatrix} \mathbf{x}$$

find x_0 if the output $y = -20e^{-3t} + 21e^{-2t}$.

5. Show that the control system described by

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= -2x_1 - 3x_2 + u \\ y &= x_1 + x_2 \end{aligned}$$

is not completely observable. Determine the initial state x_0 such that if $u = 0$ for $t > 0$, then $\mathbf{y} = 0$, for $t > 0$.

6. Check the detectability and stabilizability of the dynamic system

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -6 & 0 & 6 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} u \quad y = \begin{bmatrix} 3 & 1 & 0 \end{bmatrix} \mathbf{x}$$

7. Check the detectability and stabilizability of the dynamic system

$$\dot{\mathbf{x}} = \begin{bmatrix} 2 & 0 & -3 \\ 0 & 0 & 0 \\ \frac{3}{2} & 0 & \frac{5}{2} \end{bmatrix} \mathbf{x} + \begin{bmatrix} -1 \\ 0 \\ -2 \end{bmatrix} u \quad y = \begin{bmatrix} 1 & 1 & -1 \end{bmatrix} \mathbf{x}$$

8. Investigate the stabilizability and detectability of the following control systems

(a)

$$\dot{\mathbf{x}} = \begin{bmatrix} 2 & 0 \\ 0 & -1 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u \quad y = \begin{bmatrix} 0 & 1 \end{bmatrix} \mathbf{x}$$

(b)

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 0 \\ 0 & -1 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u \quad y = \begin{bmatrix} 1 & 1 \end{bmatrix} \mathbf{x}$$

(c)

$$\dot{\mathbf{x}} = \begin{bmatrix} 4 & 3 & 5 \\ 1 & -2 & -3 \\ 2 & 1 & 8 \end{bmatrix} \mathbf{x} + \begin{bmatrix} -2 & 1 \\ -1 & 0 \end{bmatrix} \mathbf{u} \quad y = \begin{bmatrix} 0 & 1 & 1 \end{bmatrix} \mathbf{x}$$

(d)

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 & 5 \\ 0 & 0 & 1 \\ -18 & -15 & -2 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u \quad y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \mathbf{x}$$

9. Given the LTI system described by

$$\dot{x}_1 = \alpha x_1 + 2x_2 + u$$

$$\dot{x}_2 = x_1 - x_2$$

$$y = x_1 + \alpha x_2$$

determine for what value of the real parameters α the system is

- (a) completely controllable and completely observable
- (b) completely controllable but not completely observable
- (c) completely observable but not completely controllable

Chapter 3

Full-state Feedback Controller Design for LTI Systems

We know that we can affect the performance and behavior of a dynamic system by changing the location of the closed-loop poles of the system. We can cancel the unstable poles by multiplying the system transfer function by another transfer function having a zero at the location of the unstable pole of the system. This other system is called a controller or compensator.

Consider a completely controllable system on equation 1.3.6 and equation 1.3.7, suppose we apply a linear feedback control law, which means that the control law is a linear combination of state variables,

$$\mathbf{u} = -\mathbf{K}_f \mathbf{x} \quad (3.0.1)$$

where $\mathbf{K}_f$ is $m \times n$ feedback matrix. The closed-loop system under this control law becomes,

$$\begin{aligned} \dot{\mathbf{x}} &= \mathbf{A}\mathbf{x} + \mathbf{B}(-\mathbf{K}_f \mathbf{x}) \\ \dot{\mathbf{x}} &= (\mathbf{A} - \mathbf{B}\mathbf{K}_f)\mathbf{x} \end{aligned} \quad (3.0.2)$$

The block diagram representation of the system with a full state feedback controller is shown in Figure 3.1.

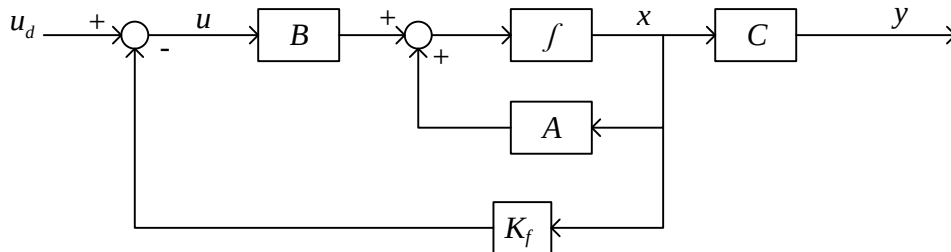


Figure 3.1: Block Diagram Representation of System with Full State Feedback Controller

Pole shifting theorem:- for any controllable system, it is possible to obtain arbitrary eigenvalues for matrix $\mathbf{A} - \mathbf{BK}_f$, using suitable feedback matrix $\mathbf{K}_f$, subjected to the only obvious constraint that complex eigenvalues must appear in pair.

The state feedback controller design can be done either by selecting the location of the poles and determine the feedback gain $\mathbf{K}_f$ that locate the closed-loop poles of the system at the desired location, or determining the gain $\mathbf{K}_f$ that determine the optimal location of the closed-loop poles and that minimize $\mathbf{u}$ at the same time.

3.1 Pole Placement State Feedback Controller Design

In pole placement controller design, given the desired location of the closed-loop system poles (the eigenvalues of $\mathbf{A} - \mathbf{BK}_f$), we are required to determine a feedback gain matrix $\mathbf{K}_f$ such that with a control law $\mathbf{u} = -\mathbf{K}_f\mathbf{x}$ the system has poles at the desired location.

The pole placement or eigenvalue assignment problem can be define as: let $\lambda_1, \lambda_2, \dots, \lambda_n$ be the eigenvalues of the matrix $\mathbf{A}$ of the open loop LTI system and $\tilde{\lambda}_1, \tilde{\lambda}_2, \dots, \tilde{\lambda}_n$ be desired eigenvalues of the matrix $\mathbf{A} - \mathbf{BK}_f$ of the closed loop system. Where all complex eigenvalues exist in complex conjugate pairs. Let $p(s)$ and $\hat{p}(s)$ be respective characteristics polynomial of matrix $\mathbf{A}$ and $\mathbf{A} - \mathbf{BK}_f$ respectively.

$$p(s) = \prod_{i=1}^n (s - \lambda_i) = \|s\mathbf{I} - \mathbf{A}\| = s^n + a_1s^{n-1} + \dots + a_{n-1}s + a_n \quad (3.1.1)$$

$$\hat{p}(s) = \prod_{i=1}^n (s - \tilde{\lambda}_i) = \|s\mathbf{I} - \mathbf{A} + \mathbf{BK}_f\| = s^n + \hat{a}_1s^{n-1} + \dots + \hat{a}_{n-1}s + \hat{a}_n \quad (3.1.2)$$

$\mathbf{K}_f$ is determined if and only if the open loop system is controllable (i.e., the controllability matrix $\mathbf{C}_o$ has full rank).

The feedback matrix $\mathbf{K}_f$ can be determined by one of the following methods.

1. The Base-Gura method:

$$\mathbf{K}_f = (\mathbf{W}^\top \mathbf{C}_o^\top)^{-1}(\hat{\mathbf{a}} - \mathbf{a}) \quad (3.1.3)$$

Where

$$\mathbf{W} = \begin{bmatrix} 1 & a_1 & a_2 & a_3 & \dots & a_{n-1} \\ 0 & 1 & a_1 & a_2 & \dots & a_{n-2} \\ 0 & 0 & 1 & a_1 & \dots & a_{n-3} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \dots & 1 \end{bmatrix}, \hat{\mathbf{a}} = \begin{bmatrix} \hat{a}_1 \\ \hat{a}_2 \\ \hat{a}_3 \\ \vdots \\ \hat{a}_n \end{bmatrix}, \mathbf{a} = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ \vdots \\ a_n \end{bmatrix}$$

2. The phase canonical form formula: Consider the case where $\mathbf{A}$ is in controllable canonical form $\mathbf{A}_c, \mathbf{b}_c$.

$$\mathbf{A}_c = \begin{bmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & 1 \\ -a_0 & -a_1 & -a_2 & \dots & -an_1 \end{bmatrix}, \mathbf{b}_c = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}$$

Which means:

$$\begin{aligned} \mathbf{K}_f &= (\mathbf{W}^\top \mathbf{C}_o^\top)^{-1}(\hat{\mathbf{a}} - \mathbf{a}) = \tilde{I} = (\hat{\mathbf{a}} - \mathbf{a}) \\ \mathbf{K}_f &= (\hat{\mathbf{a}} - \mathbf{a}) \end{aligned} \quad (3.1.4)$$

3. Ackerman's formula:

$$\mathbf{K}_f^\top = \mathbf{e}^\top \mathbf{C}_o^{-1} \hat{p}(\mathbf{A}) \quad (3.1.5)$$

Where

$$\mathbf{e}^\top = [0, 0, \dots, 1],$$

and $\hat{p}(\mathbf{A})$ is obtained from polynomial on equation 3.1.1 by replacing s with matrix A , which means

$$\hat{p}(\mathbf{A}) = \mathbf{A}^n + \hat{a}_1 \mathbf{A}^{n-1} + \dots + \hat{a}_{n-1} \mathbf{A} + \hat{a}_n$$

An LTI system defined by $(\mathbf{A}, \mathbf{B})$ is stabilizable if there exists a matrix $\mathbf{K}_f$ such that $(\mathbf{A} - \mathbf{B}\mathbf{K}_f)$ has a strictly negative real part.

3.2 Desired Input Tracking

If the system is required to track a required reference input trajectory (desired input trajectory ($\mathbf{u}_d(t)$) other than a constant value (which doesn't change with time), such a control problem is called a desired input trajectory tracking problem. If the desired input trajectory is constant, the control problem is called the input regulator problem. Let the desired input trajectory be $\mathbf{u}_d(t)$, hence under state feedback control law

$$\mathbf{u}(t) = \mathbf{u}_d(t) - \mathbf{K}_f \mathbf{x}(t) \quad (3.2.1)$$

Then with this control law, the state equation of the LTI becomes,

$$\begin{aligned} \dot{\mathbf{x}} &= \mathbf{A}\mathbf{x} + \mathbf{B}(\mathbf{u}_d - \mathbf{K}_f \mathbf{x}) \\ \dot{\mathbf{x}} &= (\mathbf{A} - \mathbf{B}\mathbf{K}_f)\mathbf{x} + \mathbf{B}\mathbf{u}_d \end{aligned} \quad (3.2.2)$$

The block diagram representation of a state feedback controller with reference time variant input is shown in Figure 3.2.

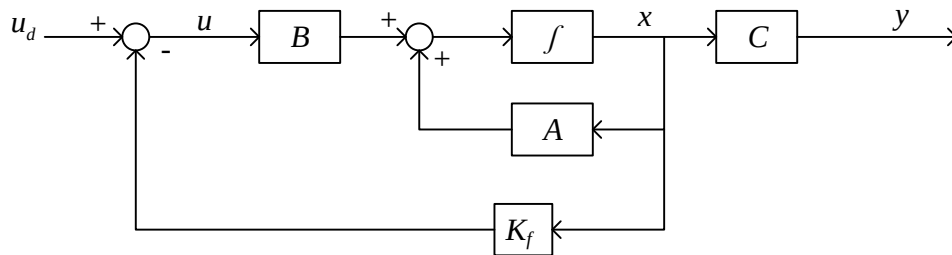


Figure 3.2: Block Diagram Representation of Full State Feedback with Reference State

3.3 Servo-mechanism Controller Design

Suppose we need our system to follow certain input (reference tracking), but due to mathematical approximation, the matrices of the nominal system deviate from the actual system (model uncertainty), and/or there may be parameter variation. In that case, the steady-state performance of the system may be poor. To enhance the performance of the system at a steady state, an integrator error term is added to the dynamics so that the steady-state error tends to zero in finite time. The error dynamics $\dot{\xi}$ is defined as,

$$\dot{\xi}(t) = \mathbf{r}(t) - \mathbf{y}(t) \quad (3.3.1)$$

with the control law,

$$\mathbf{u}(t) = -\mathbf{K}_f \mathbf{x}(t) + \mathbf{k}_1 \xi(t) \quad (3.3.2)$$

The closed-loop dynamic system becomes

$$\begin{aligned} \dot{\mathbf{x}} &= \mathbf{A}\mathbf{x} + \mathbf{B}(-\mathbf{K}_f \mathbf{x} + \mathbf{k}_1 \xi) \\ \dot{\mathbf{x}} &= (\mathbf{A} - \mathbf{B}\mathbf{K}_f)\mathbf{x} + \mathbf{B}\mathbf{k}_1 \xi \end{aligned} \quad (3.3.3)$$

$$\mathbf{y} = \mathbf{C}\mathbf{x} \quad (3.3.4)$$

Which leads to an augmented system

$$\begin{bmatrix} \dot{\mathbf{x}} \\ \dot{\boldsymbol{\xi}} \end{bmatrix} = \begin{bmatrix} \mathbf{A} - \mathbf{BK}_f & \mathbf{Bk}_1 \\ -\mathbf{C} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \boldsymbol{\xi} \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} r \quad (3.3.5)$$

The block diagram representation of a servomechanism is shown in Figure 3.3.

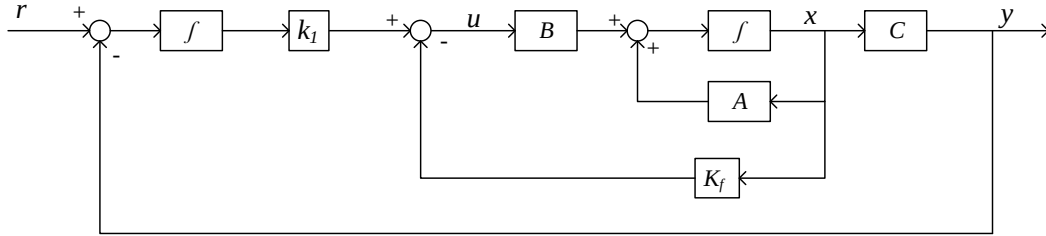


Figure 3.3: Block Diagram Representation of Servomechanism Controller Design

To determine the feedback gain matrix $\mathbf{K}_f$ the matrix

$$\begin{bmatrix} \mathbf{A} - \mathbf{BK}_f & \mathbf{Bk}_1 \\ -\mathbf{C} & \mathbf{0} \end{bmatrix} \Rightarrow \left[\begin{bmatrix} \mathbf{A} & \mathbf{0} \\ -\mathbf{C} & \mathbf{0} \end{bmatrix} - \begin{bmatrix} \mathbf{B} \\ \mathbf{0} \end{bmatrix} \right] \begin{bmatrix} \mathbf{K}_f & -\mathbf{k}_1 \end{bmatrix} \quad (3.3.6)$$

which means

$$\left[\begin{bmatrix} \mathbf{A} & \mathbf{0} \\ -\mathbf{C} & \mathbf{0} \end{bmatrix}, \begin{bmatrix} \mathbf{B} \\ \mathbf{0} \end{bmatrix} \right] \quad (3.3.7)$$

must be completely controllable.

To find the value of $\mathbf{k}_1$, we can analyze the steady state response of the system. At steady state, the output of the system should follow the reference input $\mathbf{y} = \mathbf{r}$, which means that $\dot{\boldsymbol{\xi}} = 0$. This implies that the output of the integrator is a constant value equal to $\mathbf{r}$. Therefore, the states of the system should also follow the desired states of the system $\mathbf{x}_d$, given by $\mathbf{x}_d = \mathbf{k}_1 \mathbf{r}$.

However, at this point, the output of the system is given by $\mathbf{y} = \mathbf{r} = \mathbf{C}\mathbf{x}_d + \mathbf{D}\mathbf{u}$, where $\mathbf{u} = \mathbf{x}_d - \mathbf{K}_f \mathbf{x}_d$. Thus, we can write:

$$\mathbf{r} = \mathbf{C}\mathbf{x}_d + \mathbf{D}(\mathbf{I} - \mathbf{K}_f)\mathbf{x}_d \rightarrow \mathbf{r} = (\mathbf{C} + \mathbf{D}(\mathbf{I} - \mathbf{K}_f))\mathbf{k}_1 \mathbf{r} \quad (3.3.8)$$

Solving for $\mathbf{k}_1$, we get:

$$\mathbf{k}_1 = (\mathbf{C} + \mathbf{D}(\mathbf{I} - \mathbf{K}_f))^{-1} \quad (3.3.9)$$

3.4 Determining Poles of Second Order LTI Systems From Time Domain Specification

Any second-order system can be written as

$$G(s) = \frac{\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2} \quad (3.4.1)$$

where ξ is damping factor and ω_n is undamped natural frequency. Damped natural frequency is $\omega_d = \omega_n \sqrt{1 - \xi^2}$. The time domain specification of the second-order system is shown in Table B.1. where $\theta = \cos^{-1} \xi$.

Table 3.1: *Time Domain Specification of Second Order Systems*

Time domain specification	Formula
Delay time	$t_d = \frac{1 + 0.7\xi}{\omega_n}$
Rise time	$t_r = \frac{\pi - \theta}{\omega_d}$
Peak time	$t_p = \frac{\pi}{\omega_d}$
Percentage Peak overshoot	$M_p = e^{-\left(\frac{\xi\pi}{\sqrt{1-\xi^2}}\right)}$
Settling time with 2% tolerance	$t_s = \frac{4}{\xi\omega_n}$

Since any second-order system can be written in this form, we can determine the desired poles of the system from the time domain specification of the system.

The controllable canonical form of equation 3.4.1 is

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\omega_n^2 & -2\xi\omega_n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u \quad (3.4.2)$$

$$y = \begin{bmatrix} \omega_n^2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad (3.4.3)$$

Note:- The goal of the designer is to find $\mathbf{K}_f$ such that a closed loop time domain specification of the given second order system meets the requirement.

3.5 Examples

1. Given a continuous-time system

$$\begin{aligned}\dot{\mathbf{x}} &= \begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u \\ y &= \begin{bmatrix} 1 & 3 \end{bmatrix} \mathbf{x}\end{aligned}$$

discuss controllability and observability of the system.

Solution

Check for controllability

$$\mathcal{C}_o = [\mathbf{B} \quad \mathbf{AB}] = \begin{bmatrix} 0 & 2 \\ 1 & 1 \end{bmatrix}$$

which has rank 2, hence it is completely controllable.

Check for observability

$$\mathcal{O}_v = \begin{bmatrix} \mathbf{C} \\ \mathbf{CA} \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 7 & 10 \end{bmatrix}$$

which has rank 2, hence it is completely observable.

If we use control law $u = [3 \ 1]\mathbf{x} + \nu$, check controllability and observability of the closed loop system.

$$\dot{\mathbf{x}} = (\mathbf{A} - \mathbf{BK})\mathbf{x} + \mathbf{B}\nu \Rightarrow \dot{\mathbf{x}} = \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} \nu$$

heck for controllability

$$\mathcal{C}_o = [\mathbf{B} \quad \mathbf{AB}] = \begin{bmatrix} 0 & 2 \\ 1 & 0 \end{bmatrix}$$

which has rank 2, hence it is completely controllable.

Check for observability

$$\mathcal{O}_v = \begin{bmatrix} \mathbf{C} \\ \mathbf{CA} \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 1 & 2 \end{bmatrix}$$

the rows of $\mathcal{O}_v$ are not linearly independent, hence rank of $\mathcal{O}_v$ is 1, hence it is not completely observable.

2. Given a continuous-time system

$$H(s) = \frac{(s-1)(s+2)}{(s+1)(s-2)(s+3)}$$

is that possible to convert $H(s)$ to

$$G(s) = \frac{s-1}{(s+2)(s+3)}$$

if possible determine state feedback gain matrix $\mathbf{K}_f$.

Solution

We have to cancel zero at $z = -2$ by a pole at $p = -2$, cancel the pole at $p = -1$ by a zero at $z = -1$, cancel the pole at $p = 2$ by a zero at $z = 2$, and add one pole at $p = -2$ to get $G(s)$ from $H(s)$.

$$\begin{aligned} G(s) &= \frac{(s+1)(s-2)}{(s+2)^2} H(s) = \frac{(s+1)(s-2)}{(s+2)^2} \left(\frac{(s-1)(s+2)}{(s+1)(s-2)(s+3)} \right) \\ &= \frac{(s-1)(s+2)}{(s+2)^2(s+3)} \end{aligned}$$

Since the degree of the numerator is less than the degree of the denominator, it is possible to have such a kind of compensator. Hence it is possible to get $G(s)$ from $H(s)$.

The state-space representation of $H(s)$ is

$$\begin{aligned} \dot{\mathbf{x}} &= \begin{bmatrix} -2 & 5 & 6 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} u \\ y &= \begin{bmatrix} 1 & 1 & -2 \end{bmatrix} \mathbf{x} \end{aligned}$$

We need a state feedback control law $\mathbf{u} = -\mathbf{K}_f \mathbf{x}$ that converts the system $H(s)$ to $G(s)$, hence the desired location of poles of the closed-loop system is at $p = -2, p = -2, p = -3$. Using the Base-Gura method,

$$\det(s\mathbf{I} - \mathbf{A} + \mathbf{BK}_f) = (s+2)^2(s+3)$$

$$\det \left(\begin{bmatrix} s & 0 & 0 \\ 0 & s & 0 \\ 0 & 0 & s \end{bmatrix} - \begin{bmatrix} -2 & 5 & 6 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} k_1 & k_2 & k_3 \end{bmatrix} \right) = s^3 + 7s^2 + 16s + 12$$

$$\begin{aligned} \det \left(\begin{bmatrix} s+2+k_1 & -5+k_2 & -6+k_3 \\ -1 & s & 0 \\ 0 & -1 & s \end{bmatrix} \right) &= s^3 + 7s^2 + 16s + 12 \\ s^3 + (2+k_1)s^2 + (-5+k_2)s + (-6+k_3) &= s^3 + 7s^2 + 16s + 12 \end{aligned}$$

$$2 + k_1 = 7 \Rightarrow k_1 = 5, \quad -5 + 6 = 16 \Rightarrow k_2 = 21, \quad -6 + k_3 = 12 \Rightarrow k_3 = 18$$

$$\mathbf{K}_f = \begin{bmatrix} k_1 & k_2 & k_3 \end{bmatrix} = \begin{bmatrix} 5 & 21 & 18 \end{bmatrix}$$

The closed loop dynamics of the system with full state feedback gain matrix $\mathbf{K}_f$ are

$$\begin{aligned} \dot{\mathbf{x}} &= (\mathbf{A} - \mathbf{BK}_f)\mathbf{x} \\ \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} &= \begin{bmatrix} -2 & 5 & 6 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} - \begin{bmatrix} 5 & 21 & 8 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \\ \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} &= \begin{bmatrix} -7 & -17 & -2 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \end{aligned}$$

3. For IPC dynamic system on 1.6.3 with parameters $m_c = 0.5$, $m_p = 0.2$, $b = 0.1$, $l = 0.3$, $I = 0.5m_pl^2$, $g = 9.81$, Find a feedback gain matrix that place the poles of the closed loop system at $p_1 = -1$, $p_2 = -2$, $p_3 = -1 - i$, $p_4 = -1 + i$.

Solution

To check for complete controllability of IPC dynamic system, use $\mathcal{C}_o = \text{ctrb}(\mathbf{A}, \mathbf{B})$, and $\text{rank}(\mathcal{C}_o)$ commands on MATLAB to check controllability and the rank of controllability matrix, respectively. With the given parameters

$$\begin{aligned} \dot{\mathbf{x}} &= \begin{bmatrix} 0 & 1.00 & 0 & 0 \\ 0 & -0.17 & 2.31 & 0 \\ 0 & 0 & 0 & 1.00 \\ 0 & -0.39 & 26.93 & 0 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 1.75 \\ 0 \\ 3.92 \end{bmatrix} u \\ y &= \begin{bmatrix} 1 & 0 & 1 & 0 \end{bmatrix} \mathbf{x} \end{aligned}$$

which result in controllability matrix $\mathcal{C}_o$, with rank 4.

$$\mathcal{C}_o = \begin{bmatrix} 0 & 1.76 & -0.31 & 9.11 \\ 1.76 & -0.31 & 9.11 & -3.2 \\ 0 & 3.92 & -0.69 & 105.73 \\ 3.92 & -0.69 & 105.73 & -22.21 \end{bmatrix}$$

To design $\mathbf{K}_f$ at the desired locations use MATLAB command $\mathbf{K}_f = \text{place}(\mathbf{A}, \mathbf{B}, \mathbf{p})$, where $\mathbf{p} = \begin{bmatrix} -1 & -2 & -1 - i & -1 + i \end{bmatrix}$. Which results in a feedback gain matrix

$$\mathbf{K}_f = \begin{bmatrix} -0.10 & -0.36 & 9.46 & 1.39 \end{bmatrix}$$

The closed loop dynamics of the system with full state feedback gain matrix $\mathbf{K}_f$ are

$$\dot{\mathbf{x}} = (\mathbf{A} - \mathbf{BK}_f)\mathbf{x}$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \end{bmatrix} = \begin{bmatrix} 0 & 1.00 & 0 & 0 \\ 0 & -0.17 & 2.31 & 0 \\ 0 & 0 & 0 & 1.00 \\ 0 & -0.39 & 26.93 & 0 \end{bmatrix} - \begin{bmatrix} 0 & 0 & 0 & 0 \\ -0.18 & -0.64 & 16.7 & 2.46 \\ 0 & 0 & 0 & 0 \\ -0.41 & -1.41 & 37.11 & 5.46 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0.18 & 0.46 & -14.39 & -2.46 \\ 0 & 0 & 0 & 1 \\ 0.41 & 1.02 & -10.18 & -5.46 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$

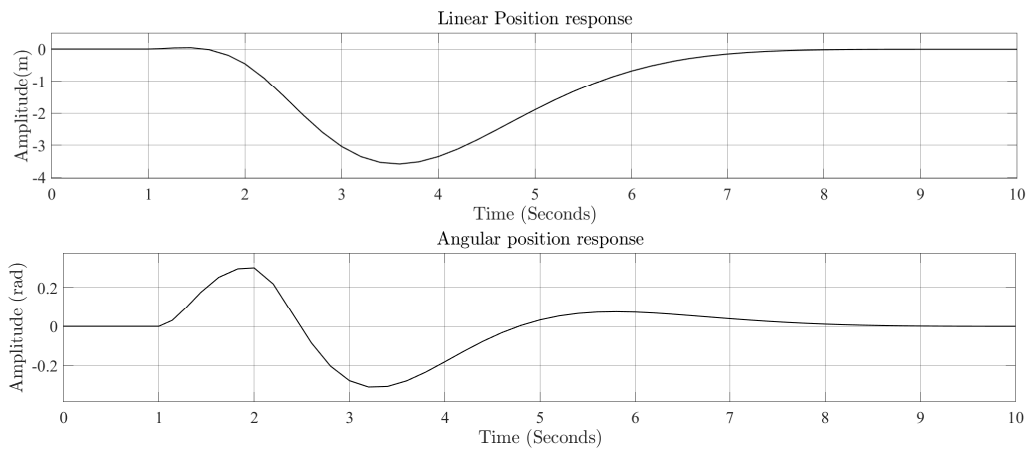


Figure 3.4: *Closed Loop Response of IPC for Impulse Input with $\mathbf{K}_f$*

4. For the following system, determine a feedback gain matrix that places the closed-loop poles of the system at $p = -2 \pm 2i$.

$$\dot{\mathbf{x}} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & -1 & 1 \\ 0 & 0 & -2 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \mathbf{x}$$

Solution

As shown in chapter two example 3, the system is not completely controllable. So we have to decompose the system into controllable and uncontrollable modes as shown in chapter two example 3. So we can design a state feedback controller for

controllable mode $(\mathbf{A}_{11}, \mathbf{B}_1)$.

$$\begin{aligned} \|s\mathbf{I} - \mathbf{A}_{11} + \mathbf{B}_1\mathbf{K}_f\| &= (s + 2 + 2i)(s + 2 - 2i) \\ \left\| \begin{bmatrix} s-1 & 0 \\ -1 & s+1 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} \begin{bmatrix} k_1 & k_2 \end{bmatrix} \right\| &= s^2 + 4s + 8 \\ \left\| \begin{bmatrix} s-1+k_1 & k_2 \\ -1+k_1 & s+1+k_2 \end{bmatrix} \right\| &= s^2 + 4s + 8 \end{aligned}$$

$$\begin{aligned} (s-1+k_1)(s+1+k_2) - k_2(-1+k_1) &= s^2 + 4s + 8 \\ s^2 + (k_1+k_2)s - 1 + k_1 &= s^2 + 4s + 8 \\ k_1 - 1 = 8 \quad \Rightarrow k_1 = 9 \quad k_1 + k_2 = 4 \quad \Rightarrow k_2 = -5 \\ \mathbf{K}_f &= \begin{bmatrix} 9 & -5 \end{bmatrix} \end{aligned}$$

What happens if we want to design a full state feedback controller for as a whole? Let us assume the third pole is at $p = \alpha$, then

$$\begin{aligned} \|s\mathbf{I} - \mathbf{A} + \mathbf{B}\mathbf{K}_f\| &= (s + \alpha)(s^2 + 4s + 8) \\ \left\| \begin{bmatrix} s & 0 & 0 \\ 0 & s & 0 \\ 0 & 0 & s \end{bmatrix} - \begin{bmatrix} 1 & 0 & 0 \\ 1 & -1 & 1 \\ 0 & 0 & -2 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \begin{bmatrix} k_1 & k_2 & k_3 \end{bmatrix} \right\| &= (s + \alpha)(s^2 + 4s + 8) \\ \left\| \begin{bmatrix} s-1 & 0 & 0 \\ -1 & s+1 & -1 \\ 0 & 0 & s+2 \end{bmatrix} + \begin{bmatrix} k_1 & k_2 & k_3 \\ k_1 & k_2 & k_3 \\ 0 & 0 & 0 \end{bmatrix} \right\| &= (s + \alpha)(s^2 + 4s + 8) \end{aligned}$$

which results in

$$(s+2)(s-1+k_1)(s+1+k_2) - k_2(1+k_1) = (s+\alpha)(s^2+4s+8)$$

Hence, the system requires the stability of the uncontrollable mode only. k_3 does not affect the system. If the uncontrollable mode is stable, the feedback gain corresponding to the uncontrollable mode does not affect the system closed-loop dynamics. This example proves this scenario.

From Kalman decomposition, we have

$$\begin{bmatrix} \mathbf{A}_{11} - \mathbf{B}_1 k_1 & \mathbf{A}_{12} - \mathbf{B}_1 k_2 \\ \mathbf{0} & \mathbf{A}_{22} \end{bmatrix}$$

The system only requires the stability of uncontrollable mode; no need to design feedback gain $\mathbf{K}_f$ for the uncontrollable mode, even if the uncontrollable mode is not stable. We cannot impose any effect on the uncontrollable mode (that is why

it is said uncontrollable mode).

The closed loop dynamics of the system with full state feedback gain matrix $\mathbf{K}_f$ are

$$\begin{aligned}\dot{\mathbf{x}} &= (\mathbf{A} - \mathbf{BK}_f)\mathbf{x} \\ \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} &= \left[\begin{bmatrix} 1 & 0 & 0 \\ 1 & -1 & 1 \\ 0 & 0 & -2 \end{bmatrix} - \begin{bmatrix} 9 & -5 & k_3 \\ 9 & -5 & k_3 \\ 0 & 0 & 0 \end{bmatrix} \right] \begin{bmatrix} x_1 \\ x_2 \\ \dots \\ x_3 \end{bmatrix} \\ \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} &= \begin{bmatrix} -8 & 5 & \vdots & -k_3 \\ -8 & 4 & \vdots & 1 - k_3 \\ \dots & \dots & \vdots & \dots \\ 0 & 0 & \vdots & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \dots \\ x_3 \end{bmatrix}\end{aligned}$$

5. Determine a full state feedback matrix for the following second-order system such that the closed loop system has settling time $t_s = 4\text{sec}$ and peak time $t_p = 1.813\text{sec}$.

$$\begin{aligned}\dot{\mathbf{x}} &= \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u \\ y &= \begin{bmatrix} 1 & 0 \end{bmatrix} \mathbf{x}\end{aligned}$$

Solution

From given time domain specifications

$$\begin{aligned}\xi\omega_n &= \frac{4}{t_s} = \frac{4}{4} = 1 \\ \omega_d &= \frac{\pi}{t_p} = \frac{3.14}{1.813} = 1.732 \\ \omega_n\sqrt{1 - \xi^2} &= 1.732 \Rightarrow 1 - \xi^2 = 3\xi^2 \\ \Rightarrow \xi &= 0.5 \quad \Rightarrow \omega_n = 2\end{aligned}$$

Thus the closed loop system has characteristics equation of $s^2 + 2\xi\omega_n s + \omega_n^2 = s^2 + 2s + 4 = (s + 1 + \sqrt{3}i)(s + 1 - \sqrt{3}i)$. The state feedback controller gain must locate the closed-loop system poles at $s = -1 \pm i$. We determine $\mathbf{K}_f$ as follows

$$\begin{aligned}\|s\mathbf{I} - \mathbf{A} + \mathbf{BK}_f\| &= s^2 + 2s + 4 \\ \left\| \begin{bmatrix} s-1 & 0 \\ -1 & s-1 \end{bmatrix} + \begin{bmatrix} k_1 & k_2 \\ 0 & 0 \end{bmatrix} \right\| &= s^2 + 2s + 4 \\ \left\| \begin{bmatrix} s-1+k_1 & k_2 \\ -1 & s-1 \end{bmatrix} \right\| &= s^2 + 2s + 4 \\ s^2 + (-2+k_1)s + 1 - k_1 + k_2 &= s^2 + 2s + 4 \\ -2 + k_1 = 2 \Rightarrow k_1 &= 4 \quad 1 - k_1 + k_2 = 4 \Rightarrow k_2 = 7\end{aligned}$$

$$\mathbf{K}_f = \begin{bmatrix} 4 & 7 \end{bmatrix}$$

The closed loop dynamics of the system with full state feedback gain matrix $\mathbf{K}_f$ are

$$\begin{aligned} \dot{\mathbf{x}} &= (\mathbf{A} - \mathbf{BK}_f)\mathbf{x} \\ \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} &= \left[\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} - \begin{bmatrix} 4 & 7 \\ 0 & 0 \end{bmatrix} \right] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \\ \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} &= \begin{bmatrix} -3 & -7 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \end{aligned}$$

3.6 Exercises

1. Consider a system

$$\dot{\mathbf{x}} = \begin{bmatrix} -1 & -2 \\ 2 & -4 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 1 \\ 3 \end{bmatrix} u$$

find a feedback gain matrix $\mathbf{K}_f$ such that the closed loop system has eigenvalues at -4 and -5 .

2. For LTI system

$$\dot{\mathbf{x}} = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} u$$

find state feedback gain matrix $\mathbf{K}_f$ such that the eigenvalues of closed loop system at $-1, -1 \pm 2i$.

3. For the dynamic system

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 6 & -11 & 6 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} \mathbf{u}$$

Find the state feedback gain matrix $\mathbf{K}_f$ such that the eigenvalues of the closed loop system are at 1, 2, and 3, then check for stability of the system with the control law.

4. Determine whether the system described by

$$\dot{\mathbf{x}} = \begin{bmatrix} -1 & 0 & 3 \\ 0 & -3 & 0 \\ 1 & 0 & -3 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} u$$

is c.c. and show that under linear feedback of the form $u = \alpha x_1 + \beta x_3$, the closed loop system has two fixed eigenvalues, one of which is equal to -3 . Determine the second fixed eigenvalues and also values of α and β such that the third closed loop eigenvalue is -4 .

5. Determine the range of values of k , the linear dynamic system

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -k & -1 & -2 \end{bmatrix} \mathbf{x}$$

is asymptotically stable. If $k = -1$ and control term $\begin{bmatrix} 0 & 1 & 0 \end{bmatrix}^\top u$ is added, find linear feedback gain $\mathbf{K}_f$ that makes the closed loop system to have poles at -1 .

6. For the control system described by

$$\begin{aligned} \dot{x}_1 &= x_1 - x_3 + u \\ \dot{x}_2 &= x_1 + 2x_2 + x_3 \\ \dot{x}_3 &= 2x_1 + 2x_2 + 3x_3 + u \end{aligned}$$

find a suitable feedback matrix $\mathbf{K}_f$ such that the closed loop system has eigenvalues -1 and $-1 \pm 2i$.

7. For the control system described by

$$\begin{aligned} \dot{x}_1 &= x_2 + u_1 \\ \dot{x}_2 &= x_3 + u_2 \\ \dot{x}_3 &= 6x_1 - 11x_2 + 6x_3 + u_1 + u_2 \end{aligned}$$

find a suitable feedback matrix $\mathbf{K}_f$ such that the closed loop system has eigenvalues -1 and $-2 \pm i$.

8. For the following system, find a state feedback controller gain matrix so that the closed-loop system has maximum percentage overshoot $m_p = 14\%$ and rise time $t_r = 1.25 \text{ sec}$.

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & -1 \\ -3 & -2 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$$

9. Let $\mathbf{A}$ is $n \times n$ and $\mathbf{B}$ is $n \times 1$, given any arbitrary set $\Lambda = \{\theta_1, \theta_2, \dots, \theta_m\}$ of complex numbers (appearing in conjugate pairs), there exist a feedback matrix $\mathbf{K}_f$ such that the eigenvalues of $\mathbf{A} - \mathbf{BK}_f$ are the set Λ .

10. Find linear feedback control $\mathbf{u} = -\mathbf{K}_f \mathbf{x}$ for

$$\dot{\mathbf{x}} = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} u$$

that place closed loop poles of the system at $-1, -1 \pm 2i$. Does this verify problem number 9?

Chapter 4

State Estimators/Observers Design for LTI systems

To design a full-state feedback control for a given system, all the states of the system must be available as output (must be measured directly). But in many real-world problems, all states of the system are not available as output. If the system is completely observable, the states of the system are related to the output of the system in some way. So we can recover the unavailable states from the output of the system. The method of recovering these states from the output of the system is called the observer or state estimator design problem. If all states are required to be observed or estimated, the problem is a full state observer problem; if we can measure some states directly and the others are to be observed or estimated, then the problem becomes a reduced-order state observer/estimator.

If the system is partially observable, we can design an observer for the observable mode of the system. Since we cannot say anything about the relation between the output of the system and the unobservable mode of the system, we cannot design an observer for the unobservable mode of the system.

4.1 Full Order State Observer Design

Consider a completely observable dynamic system

$$\dot{\mathbf{x}} = \mathbf{Ax} + \mathbf{Bu} \quad (4.1.1)$$

$$\mathbf{y} = \mathbf{Cx} \quad (4.1.2)$$

Assume the state vector $\mathbf{x}$ is given by its estimate $\hat{\mathbf{x}}$

$$\dot{\hat{\mathbf{x}}} = \hat{\mathbf{A}}\hat{\mathbf{x}} + \hat{\mathbf{B}}\mathbf{u} + \mathbf{L}_c\mathbf{y} \quad (4.1.3)$$

The matrices $\hat{\mathbf{A}}, \hat{\mathbf{B}}$ are unknown.

In the full state observer design problem, $\mathbf{x}$ and $\hat{\mathbf{x}}$ have equal dimension. The deviation (estimation error) $\tilde{\mathbf{x}} = \mathbf{x} - \hat{\mathbf{x}}$, will be

$$\begin{aligned} \dot{\tilde{\mathbf{x}}} &= \dot{\mathbf{x}} - \dot{\hat{\mathbf{x}}} \\ \dot{\tilde{\mathbf{x}}} &= \mathbf{Ax} + \mathbf{Bu} - \hat{\mathbf{A}}\hat{\mathbf{x}} - \hat{\mathbf{B}}\mathbf{u} - \mathbf{L}_c\mathbf{y} \\ \dot{\tilde{\mathbf{x}}} &= \mathbf{Ax} + \mathbf{Bu} - \hat{\mathbf{A}}\hat{\mathbf{x}} - \hat{\mathbf{B}}\mathbf{u} - \mathbf{L}_c\mathbf{Cx} \end{aligned} \quad (4.1.4)$$

replacing $\hat{\mathbf{x}}$ by $\mathbf{x} - \tilde{\mathbf{x}}$, which leads to

$$\begin{aligned}\dot{\tilde{\mathbf{x}}} &= \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} - \hat{\mathbf{A}}(\mathbf{x} - \tilde{\mathbf{x}}) - \hat{\mathbf{B}}\mathbf{u} - \mathbf{L}_c\mathbf{C}\mathbf{x} \\ \dot{\tilde{\mathbf{x}}} &= (\mathbf{A} - \hat{\mathbf{A}} - \mathbf{L}_c\mathbf{C})\mathbf{x} + \hat{\mathbf{A}}\tilde{\mathbf{x}} + (\mathbf{B} - \hat{\mathbf{B}})\mathbf{u}\end{aligned}\quad (4.1.5)$$

If we need $\tilde{\mathbf{x}}$ to become zero independent of $\mathbf{x}$ and $\mathbf{u}$, is only possible if and only if $\hat{\mathbf{B}} = \mathbf{B}$, $\hat{\mathbf{A}} = (\mathbf{A} - \mathbf{L}_c\mathbf{C})$ and $\hat{\mathbf{A}}$ is stable. If this condition is fulfilled, the error dynamics $\tilde{\mathbf{x}}$ becomes

$$\dot{\tilde{\mathbf{x}}} = \hat{\mathbf{A}}\tilde{\mathbf{x}} = (\mathbf{A} - \mathbf{L}_c\mathbf{C})\tilde{\mathbf{x}} \quad (4.1.6)$$

Thus, the design problem is finding a suitable $\mathbf{L}_c$ such that the estimation error dynamics go to zero as $t \rightarrow \infty$, or determining a suitable $\mathbf{L}_c$ that places the closed loop eigenvalues of estimation error dynamics $\dot{\tilde{\mathbf{x}}} = (\mathbf{A} - \mathbf{L}_c\mathbf{C})\tilde{\mathbf{x}}$ at the desired location.

Replacing $\hat{\mathbf{A}}$ with $\mathbf{A} - \mathbf{L}_c\mathbf{C}$ in the full state observer equation on 4.1.3 we have

$$\begin{aligned}\dot{\hat{\mathbf{x}}} &= \mathbf{A}\hat{\mathbf{x}} + \mathbf{B}\mathbf{u} + \mathbf{L}_c(\mathbf{C}\mathbf{x} - \hat{\mathbf{x}}) \\ \dot{\hat{\mathbf{x}}} &= \mathbf{A}\hat{\mathbf{x}} + \mathbf{B}\mathbf{u} + \mathbf{L}_c\mathbf{C}\tilde{\mathbf{x}}\end{aligned}\quad (4.1.7)$$

which is called Luenberger observer.

The full state observer state equation (full-order state observer dynamics) is

$$\begin{bmatrix} \dot{\hat{\mathbf{x}}} \\ \dot{\tilde{\mathbf{x}}} \end{bmatrix} = \begin{bmatrix} \mathbf{A} & \mathbf{L}_c\mathbf{C} \\ \mathbf{0} & \mathbf{A} - \mathbf{L}_c\mathbf{C} \end{bmatrix} \begin{bmatrix} \hat{\mathbf{x}} \\ \tilde{\mathbf{x}} \end{bmatrix} + \begin{bmatrix} \mathbf{B} \\ \mathbf{0} \end{bmatrix} \mathbf{u}$$

If the system is represented in observable canonical form,

$$\dot{\hat{\mathbf{x}}}^\top = (\mathbf{A}_o - \mathbf{L}_c\mathbf{C}_o)^\top \mathbf{x}^\top = (\mathbf{A}_o^\top - \mathbf{C}_o^\top \mathbf{L}_c^\top) \mathbf{x}^\top \Leftrightarrow \dot{\hat{\mathbf{x}}}^\top = (\mathbf{A}_c - \mathbf{B}_c\mathbf{K}) \mathbf{x}^\top \quad (4.1.8)$$

Hence, we can use the methods we used to determine $\mathbf{K}_f$,

$$\mathbf{L}_c = \mathbf{K}^\top \quad (4.1.9)$$

This implies that if estimation error dynamics is expressed in an observable canonical form, the feedback matrix $\mathbf{L}_c$ can be determined from full state feedback matrix $\mathbf{K}_f$ for the corresponding controllable canonical form representation of estimation error dynamics just by transposing $\mathbf{K}_f$ and vise-versa.

If the desired poles of $\dot{\tilde{\mathbf{x}}} = (\mathbf{A} - \mathbf{L}_c\mathbf{C})\tilde{\mathbf{x}}$ and $\dot{\mathbf{x}} = (\mathbf{A} - \mathbf{B}\mathbf{K}_f)\mathbf{x}$ are the same. For a system expressed in observable canonical form $\mathbf{L}_c$ equal to the transpose of the full state feedback matrix $\mathbf{K}_f^\top$, that places the closed-loop eigenvalues of the corresponding controllable canonical form representation of the same system at the desired location.

An LTI system defined by $(\mathbf{A}, \mathbf{C})$ is detectable, if there exists a matrix $\mathbf{L}_c$ such that $(\mathbf{A} - \mathbf{L}_c\mathbf{C})$ has a strictly negative real part.

The block diagram representation of a full-order state observer is shown in Figure 4.1.

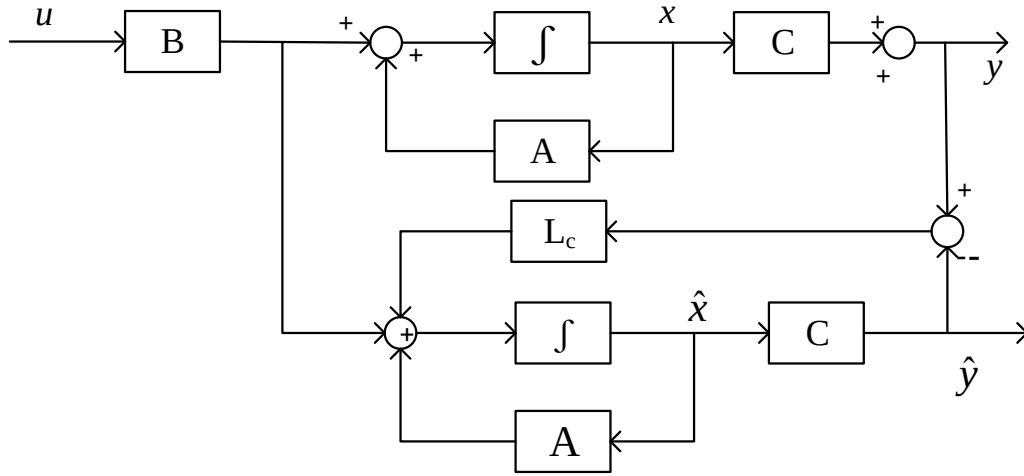


Figure 4.1: *Block Diagram Representation of Full Order State Observer*

4.2 Reduced Order State Observer Design

Consider a completely observable system in which m states are measured directly from the output and $n - m$ states are those that cannot be measured directly. Let $\boldsymbol{\eta}_1$ represent the states that are measured directly, and $\boldsymbol{\eta}_2$ represent the states that cannot be measured directly from outputs.

$$\boldsymbol{\eta}_1 = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_m \end{bmatrix}, \quad \boldsymbol{\eta}_2 = \begin{bmatrix} x_{m+1} \\ x_{m+2} \\ x_{m+3} \\ \vdots \\ x_{n-m} \end{bmatrix}$$

The states to be estimated are related to output so, the output of the state estimator is $\hat{\boldsymbol{\eta}}_2 = \mathbf{L}_c \mathbf{y} + \boldsymbol{\omega}$, where $\boldsymbol{\omega}$ is the state of reduced order dynamics. In terms of $\boldsymbol{\eta}_1$ and $\boldsymbol{\eta}_2$, the dynamic system can be decomposed into directly measured state dynamics and directly unmeasured state dynamics as follows,

$$\begin{aligned} \dot{\boldsymbol{\eta}}_1 &= \mathbf{A}_{11}\boldsymbol{\eta}_1 + \mathbf{A}_{12}\boldsymbol{\eta}_2 + \mathbf{B}_1\mathbf{u} \\ \dot{\boldsymbol{\eta}}_2 &= \mathbf{A}_{21}\boldsymbol{\eta}_1 + \mathbf{A}_{22}\boldsymbol{\eta}_2 + \mathbf{B}_2\mathbf{u} \\ \mathbf{y} &= \boldsymbol{\eta}_1 \end{aligned} \tag{4.2.1}$$

Since $\boldsymbol{\eta}_1$ states are directly measured, the estimated state is equal to the output, which means

$$\hat{\boldsymbol{\eta}}_1 = \boldsymbol{\eta}_1 = \mathbf{y} \tag{4.2.2}$$

But the remaining required states are determined from $\mathbf{y}$

$$\hat{\boldsymbol{\eta}}_2 = \mathbf{L}_c \mathbf{y} + \boldsymbol{\omega} \quad (4.2.3)$$

$$\dot{\hat{\boldsymbol{\eta}}}_2 = \mathbf{L}_c \dot{\mathbf{y}} + \dot{\boldsymbol{\omega}} \quad (4.2.4)$$

Equation 4.2.3 is called state observer output equation.

The estimation error is

$$\begin{bmatrix} \tilde{\boldsymbol{\eta}}_1 \\ \tilde{\boldsymbol{\eta}}_2 \end{bmatrix} = \begin{bmatrix} \boldsymbol{\eta}_1 - \hat{\boldsymbol{\eta}}_1 \\ \boldsymbol{\eta}_2 - \hat{\boldsymbol{\eta}}_2 \end{bmatrix} = \begin{bmatrix} \mathbf{0} \\ \boldsymbol{\eta}_2 - \mathbf{L}_c \boldsymbol{\eta}_1 - \boldsymbol{\omega} \end{bmatrix} \quad (4.2.5)$$

Let $\dot{\boldsymbol{\omega}} = \mathbf{J}\boldsymbol{\omega} + \mathbf{N}\mathbf{y} + \mathbf{M}\mathbf{u}$, which is called minimum order state observer dynamics. The estimation error dynamics become

$$\begin{aligned} \dot{\tilde{\boldsymbol{\eta}}}_2 &= \dot{\boldsymbol{\eta}}_2 - \dot{\hat{\boldsymbol{\eta}}}_2 \\ &= (\mathbf{A}_{21}\boldsymbol{\eta}_1 + \mathbf{A}_{22}\boldsymbol{\eta}_2 + \mathbf{B}_2\mathbf{u}) - (\mathbf{L}_c\dot{\mathbf{y}} + \dot{\boldsymbol{\omega}}) \\ &= (\mathbf{A}_{21}\boldsymbol{\eta}_1 + \mathbf{A}_{22}\boldsymbol{\eta}_2 + \mathbf{B}_2\mathbf{u}) - (\mathbf{L}_c\mathbf{A}_{11}\boldsymbol{\eta}_1 + \mathbf{L}_c\mathbf{A}_{12}\boldsymbol{\eta}_2 + \mathbf{L}_c\mathbf{B}_1\mathbf{u} + \mathbf{J}\boldsymbol{\omega} + \mathbf{N}\boldsymbol{\eta}_1 + \mathbf{M}\mathbf{u}) \end{aligned} \quad (4.2.6)$$

replacing for $\boldsymbol{\omega} = \boldsymbol{\eta}_2 - \mathbf{L}_c\boldsymbol{\eta}_1 - \tilde{\boldsymbol{\eta}}_2$, then we have

$$\begin{aligned} \dot{\tilde{\boldsymbol{\eta}}}_2 &= (\mathbf{A}_{21}\boldsymbol{\eta}_1 + \mathbf{A}_{22}\boldsymbol{\eta}_2 + \mathbf{B}_2\mathbf{u}) - (\mathbf{L}_c\mathbf{A}_{11}\boldsymbol{\eta}_1 + \mathbf{L}_c\mathbf{A}_{12}\boldsymbol{\eta}_2 + \mathbf{L}_c\mathbf{B}_1\mathbf{u}) \\ &\quad - (\mathbf{J}\boldsymbol{\eta}_2 - \mathbf{J}\mathbf{L}_c\boldsymbol{\eta}_1 - \mathbf{J}\tilde{\boldsymbol{\eta}}_2 + \mathbf{N}\boldsymbol{\eta}_1 + \mathbf{M}\mathbf{u}) \\ \dot{\tilde{\boldsymbol{\eta}}}_2 &= (\mathbf{A}_{12} - \mathbf{L}_c\mathbf{A}_{11} + \mathbf{J}\mathbf{L}_c - \mathbf{N})\boldsymbol{\eta}_1 + (\mathbf{A}_{22} - \mathbf{L}_c\mathbf{A}_{12} - \mathbf{J})\boldsymbol{\eta}_2 \\ &\quad + (\mathbf{B}_2 - \mathbf{L}_c\mathbf{B}_1 - \mathbf{M})\mathbf{u} + \mathbf{J}\tilde{\boldsymbol{\eta}}_2 \end{aligned} \quad (4.2.7)$$

For estimation error dynamics to go to zero independent of $\mathbf{x}$ and $\mathbf{u}$, the following conditions should be full-filled

$$\begin{aligned} \mathbf{A}_{12} - \mathbf{L}_c\mathbf{A}_{11} + \mathbf{J}\mathbf{L}_c - \mathbf{N} &= \mathbf{0} \Rightarrow \mathbf{N} = \mathbf{A}_{21} - \mathbf{L}_c\mathbf{A}_{11} + \mathbf{J}\mathbf{L}_c \\ \mathbf{B}_2 - \mathbf{L}_c\mathbf{B}_1 - \mathbf{M} &= \mathbf{0} \Rightarrow \mathbf{M} = \mathbf{B}_2 - \mathbf{L}_c\mathbf{B}_1 \\ \mathbf{A}_{22} - \mathbf{J} - \mathbf{L}_c\mathbf{A}_{12} &= \mathbf{0} \Rightarrow \mathbf{J} = \mathbf{A}_{22} - \mathbf{L}_c\mathbf{A}_{12} \end{aligned}$$

If these conditions are fulfilled, the estimation error dynamics become,

$$\begin{aligned} \dot{\tilde{\boldsymbol{\eta}}}_2 &= \mathbf{J}\tilde{\boldsymbol{\eta}}_2 \\ \dot{\tilde{\boldsymbol{\eta}}}_2 &= (\mathbf{A}_{22} - \mathbf{L}_c\mathbf{A}_{12})\tilde{\boldsymbol{\eta}}_2 \end{aligned} \quad (4.2.8)$$

As we can see, $\mathbf{L}_c$ is the state feedback gain matrix that locates the estimation error dynamics closed-loop poles at the desired location. The reduced order observer state equation will be,

$$\begin{aligned} \dot{\boldsymbol{\omega}} &= \mathbf{J}\boldsymbol{\omega} + \mathbf{N}\boldsymbol{\eta}_1 + \mathbf{M}\mathbf{u} \\ \dot{\boldsymbol{\omega}} &= (\mathbf{A}_{22} - \mathbf{L}_c\mathbf{A}_{12})\boldsymbol{\omega} + (\mathbf{A}_{21} + \mathbf{J}\mathbf{L}_c - \mathbf{L}_c\mathbf{A}_{11})\boldsymbol{\eta}_1 + (\mathbf{B}_2 - \mathbf{L}_c\mathbf{B}_1)\mathbf{u} \end{aligned} \quad (4.2.9)$$

If $\mathbf{J}$ has eigenvalues with a negative real part, the error dynamics go to zero asymptotically. So we determine $\mathbf{L}_c$ such that the closed-loop error dynamics $\mathbf{A}_{22} - \mathbf{L}_c\mathbf{A}_{12}$ has poles at the desired location (i.e., the poles of the reduced-order observer location). In determining $\mathbf{L}_c$, $\mathbf{A}_{22}$ and $\mathbf{A}_{12}$ take a role of $\mathbf{A}$ and $\mathbf{C}$ is full order observer respectively.

Finally, the combined system state equation (state dynamics) with reduced-order observer dynamic equation (state equation) is given as

$$\begin{bmatrix} \dot{\eta}_1 \\ \dot{\eta}_2 \\ \dot{\omega} \end{bmatrix} = \begin{bmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} & \mathbf{0} \\ \mathbf{A}_{21} & \mathbf{A}_{22} & \mathbf{0} \\ [\mathbf{A}_{12} + \mathbf{J}\mathbf{L}_c - \mathbf{L}_c\mathbf{A}_{11}] & \mathbf{0} & [\mathbf{A}_{22} - \mathbf{L}_c\mathbf{A}_{12}] \end{bmatrix} \begin{bmatrix} \eta_1 \\ \eta_2 \\ \omega \end{bmatrix} + \begin{bmatrix} \mathbf{B}_1 \\ \mathbf{B}_2 \\ [\mathbf{B}_2 - \mathbf{L}_c\mathbf{B}_1] \end{bmatrix} \mathbf{u} \quad (4.2.10)$$

The block diagram representation of the reduced-order state observer is shown in Figure 4.2.

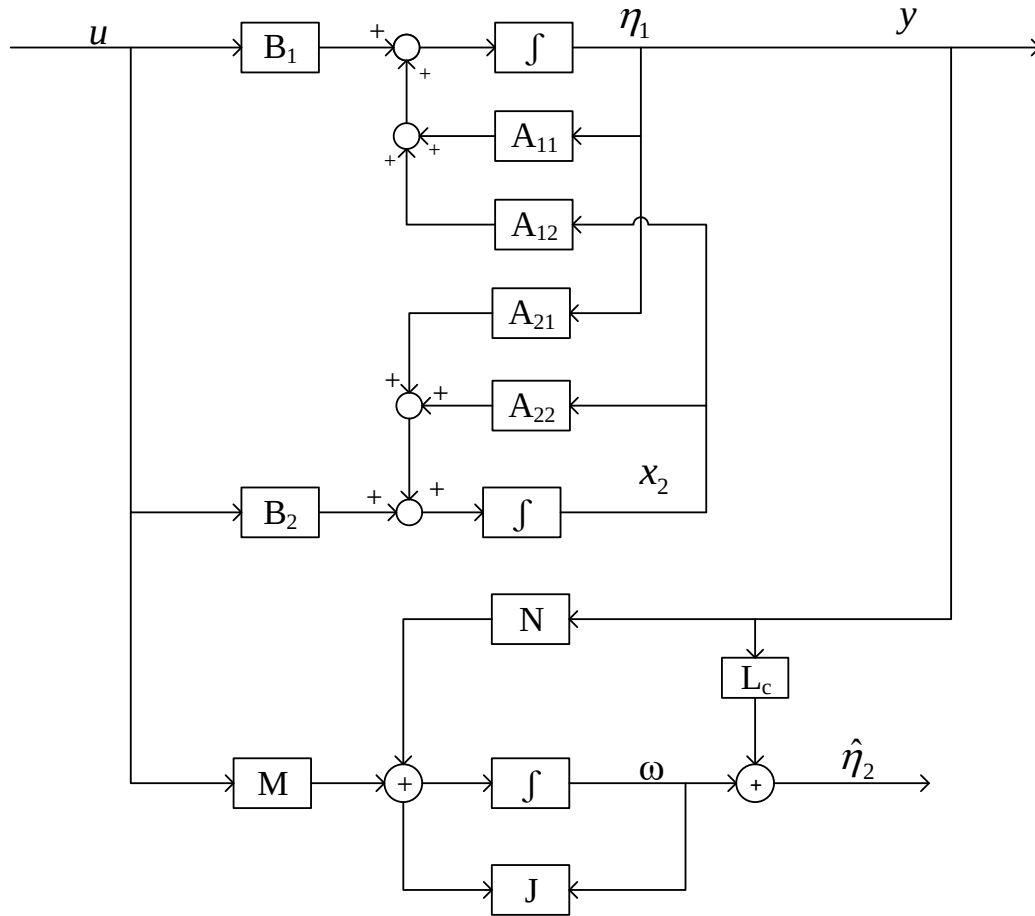


Figure 4.2: Block Diagram Representation of Reduced Order State Observer

4.3 Continuous Time Kalman Filtering

Consider a dynamical system

$$\begin{aligned}\dot{\mathbf{x}} &= \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} + \mathbf{G}\boldsymbol{\omega}(t) \\ \mathbf{y} &= \mathbf{C}\mathbf{x} + \boldsymbol{\nu}(t)\end{aligned}\tag{4.3.1}$$

$\boldsymbol{\omega}(t)$ and $\boldsymbol{\nu}(t)$ are stochastic disturbances, which are estimated to be white noises with zero means and covariance with $\mathbf{x}(0) = \mathbf{N} \sim (0, X_0)$, given as

$$\begin{aligned}E\{\boldsymbol{\omega}(t)\boldsymbol{\omega}(t)^\top\} &= \mathbf{Q}\delta(t - \tau) \\ E\{\boldsymbol{\nu}(t)\boldsymbol{\nu}(t)^\top\} &= \mathbf{R}\delta(t - \tau)\end{aligned}\tag{4.3.2}$$

White noise is an abstraction of the derivation of the Brownian motion process $\mathbf{b}(t)$, which means $\mathbf{b}(t)$ is the integral of $\boldsymbol{\omega}(t)$ white noise. The dynamic system on equation 4.3.1 becomes,

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} + \mathbf{G}\dot{\mathbf{b}}(t)\tag{4.3.3}$$

Using conditional expectation:

$$\hat{\mathbf{x}} = E\{(\mathbf{x}(t) - \hat{\mathbf{x}}(t))^\top (\mathbf{x}(t) - \hat{\mathbf{x}}(t))\}\tag{4.3.4}$$

This coincides with the optimal estimation, which minimizes the sum of the error variances:

$$J = E\{(\mathbf{x}(t) - \hat{\mathbf{x}}(t))^\top (\mathbf{x}(t) - \hat{\mathbf{x}}(t))\}\tag{4.3.5}$$

and the error covariance in the sense of positive definiteness of

$$E\{(\mathbf{x}(t) - \hat{\mathbf{x}}(t))^\top (\mathbf{x}(t) - \hat{\mathbf{x}}(t))\}\tag{4.3.6}$$

The Kalman filter equation dictate the dynamical evolution of $\hat{\mathbf{x}}$ are

$$\begin{aligned}\dot{\hat{\mathbf{x}}} &= \mathbf{A}\hat{\mathbf{x}} + \mathbf{L}_k(\mathbf{y} - \mathbf{C}\hat{\mathbf{x}}) + \mathbf{B}\mathbf{u} \\ &= (\mathbf{A} - \mathbf{L}_k\mathbf{C})\hat{\mathbf{x}} + \mathbf{L}_k\mathbf{y} + \mathbf{B}\mathbf{u}\end{aligned}\tag{4.3.7}$$

where $\mathbf{L}_k = \mathbf{P}\mathbf{C}^\top\mathbf{R}^{-1}$ which result in riccati equation

$$\dot{\mathbf{P}} = \mathbf{A}\mathbf{P} + \mathbf{P}\mathbf{A}^\top + \mathbf{G}^\top\mathbf{Q}\mathbf{G} - \mathbf{P}\mathbf{C}^\top\mathbf{R}^{-1}\mathbf{C}\mathbf{P}\tag{4.3.8}$$

The steady state solution, in case $\mathbf{A}, \mathbf{G}, \mathbf{Q}$ and $\mathbf{R}$ are constants for LTI system and the noise statistics are constant under the assumption that $(\mathbf{A}, \mathbf{C})$ is observable and $(\mathbf{A}, \mathbf{G}\sqrt{\mathbf{Q}})$ is controllable the steady state solution $\mathbf{P}_{ss} = \lim_{t \rightarrow \infty} \mathbf{P}$ of the differential riccati equation is unique. Since $\mathbf{P}$ is constant for LTI systems $\dot{\mathbf{P}} = 0$ which leads to:

$$0 = \mathbf{A}\mathbf{P} + \mathbf{P}\mathbf{A}^\top + \mathbf{G}^\top\mathbf{Q}\mathbf{G} - \mathbf{P}\mathbf{C}^\top\mathbf{R}^{-1}\mathbf{C}\mathbf{P}\tag{4.3.9}$$

further, the estimation error $\tilde{\mathbf{x}} = \mathbf{x} - \hat{\mathbf{x}}$ dynamics is:

$$\dot{\tilde{\mathbf{x}}} = (\mathbf{A} - \mathbf{L}_k\mathbf{C})\tilde{\mathbf{x}}\tag{4.3.10}$$

are stable (if the eigenvalues of $\mathbf{A} - \mathbf{L}_k\mathbf{C}$ has strictly negative real parts).

Note that $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ doesn't need to be stable, and in the filter above

$$\dot{\hat{\mathbf{x}}} = (\mathbf{A} - \mathbf{L}_k\mathbf{C})\hat{\mathbf{x}} + \mathbf{L}_k\mathbf{y} + \mathbf{G}_1\mathbf{u}\tag{4.3.11}$$

If $\mathbf{A}$ has eigenvalues with a positive real part, the state evolution can still be combined and dictated by input. If the input is chosen according to $\mathbf{u} = -\mathbf{K}_f\hat{\mathbf{x}}$, with $\mathbf{K}_f$ chosen so that $(\mathbf{A} - \mathbf{B}\mathbf{K}_f)$ is stable. The closed-loop system is stable with eigenvalues of the union of $(\mathbf{A} - \mathbf{B}\mathbf{K}_f)$ and $(\mathbf{A} - \mathbf{L}_k\mathbf{C})$.

4.4 Closed Loop State Feedback Controller Design with Full State Observer

Consider a completely observable and completely controllable system

$$\begin{aligned}\dot{\mathbf{x}} &= \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} \\ \mathbf{y} &= \mathbf{C}\mathbf{x}\end{aligned}\tag{4.4.1}$$

with state observer

$$\dot{\hat{\mathbf{x}}} = \mathbf{A}\hat{\mathbf{x}} + \mathbf{B}\mathbf{u} + \mathbf{L}_c(\mathbf{y} - \mathbf{C}\hat{\mathbf{x}})\tag{4.4.2}$$

and control law

$$\mathbf{u} = -\mathbf{K}_f\hat{\mathbf{x}}\tag{4.4.3}$$

Replacing $\hat{\mathbf{x}}$ with $\mathbf{x} - \tilde{\mathbf{x}}$ the system becomes

$$\begin{aligned}\dot{\mathbf{x}} &= \mathbf{A}\mathbf{x} - \mathbf{B}\mathbf{K}_f\hat{\mathbf{x}} \\ \dot{\mathbf{x}} &= (\mathbf{A} - \mathbf{B}\mathbf{K}_f)\mathbf{x} + \mathbf{B}\mathbf{K}_f\tilde{\mathbf{x}}\end{aligned}\tag{4.4.4}$$

and

$$\dot{\tilde{\mathbf{x}}} = (\mathbf{A} - \mathbf{L}_c\mathbf{C})\tilde{\mathbf{x}}\tag{4.4.5}$$

This shows that $\mathbf{K}_f$ and $\mathbf{L}_c$ can be designed independently of each other. If $(\mathbf{A}, \mathbf{B})$ is controllable, $\mathbf{K}_f$ can be chosen in a way that the poles of the closed-loop system are at the desired location, and if $(\mathbf{A}, \mathbf{C})$ is observable, $\mathbf{L}_c$ can be chosen so that the error dynamics go to zero rapidly. This property, where the two design problems (observer design problem and state feedback design problem of a system) can be handled independently, is called the separation principle. The dynamics of the closed-loop system of full state feedback with full order state observer is then,

$$\begin{bmatrix} \dot{\mathbf{x}} \\ \dot{\tilde{\mathbf{x}}} \end{bmatrix} = \begin{bmatrix} \mathbf{A} - \mathbf{B}\mathbf{K}_f & \mathbf{B}\mathbf{K}_f \\ \mathbf{0} & \mathbf{A} - \mathbf{L}_c\mathbf{C} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \tilde{\mathbf{x}} \end{bmatrix}$$

The block diagram representation of full state feedback with full order state observer is shown in Figure 4.3.

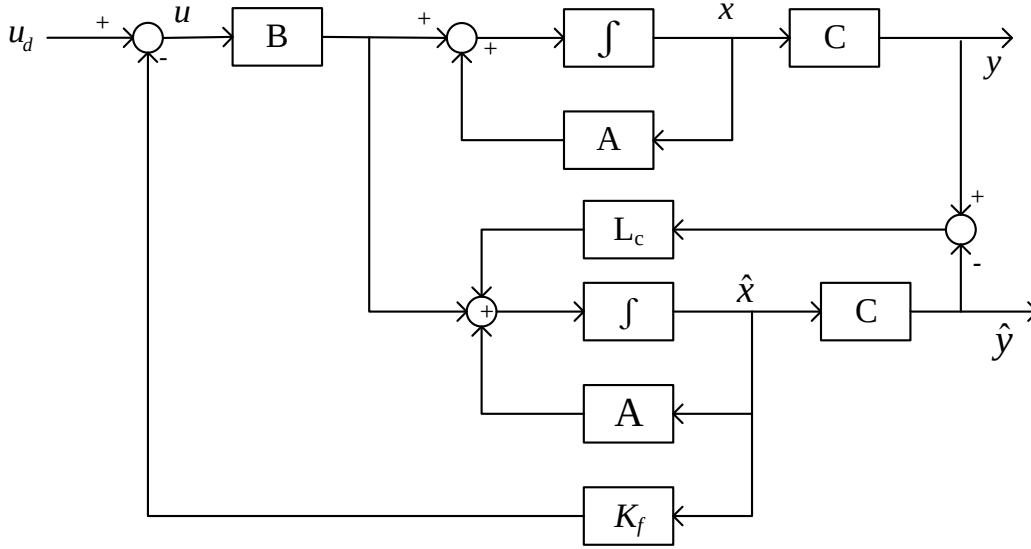


Figure 4.3: Block Diagram Representation of Full State Feedback Controller with Full Order State Observer

4.5 Closed Loop State Feedback Controller Design with Reduced Order State Observer

In this case, we use the same control law as full state observer controller design $\mathbf{u} = -\mathbf{K}\hat{\mathbf{x}}$; the only difference is $\hat{\mathbf{x}}$, which is the reduced-order state observer. The compensator for the states that cannot be measured directly (minimum order state observer dynamics (state equation)) is $\dot{\boldsymbol{\omega}} = \mathbf{J}\boldsymbol{\omega} + \mathbf{N}\mathbf{y} + \mathbf{M}\mathbf{u}$. This means the control law becomes

$$\begin{aligned}\mathbf{u} &= -\mathbf{K}(\mathbf{S}_1\boldsymbol{\eta}_1 + \mathbf{S}_2\hat{\boldsymbol{\eta}}_2) \\ &= -\mathbf{K}(\mathbf{S}_2\boldsymbol{\eta}_1 + \mathbf{S}_2(\boldsymbol{\eta}_2 - \tilde{\boldsymbol{\eta}}_2))\end{aligned}\quad (4.5.1)$$

where $\mathbf{S}_1$ and $\mathbf{S}_2$ are weighing matrices.

With this control law, the state dynamics

$$\begin{aligned}\dot{\boldsymbol{\eta}}_1 &= \mathbf{A}_{11}\boldsymbol{\eta}_1 + \mathbf{A}_{12}\boldsymbol{\eta}_2 + \mathbf{B}_1(-\mathbf{K}\mathbf{S}_1\boldsymbol{\eta}_1 - \mathbf{K}\mathbf{S}_2\hat{\boldsymbol{\eta}}_2) \\ &= \mathbf{A}_{11}\boldsymbol{\eta}_1 + \mathbf{A}_{12}\boldsymbol{\eta}_2 - \mathbf{B}_1\mathbf{K}\mathbf{S}_1\boldsymbol{\eta}_1 - \mathbf{B}_1\mathbf{K}\mathbf{S}_2(\boldsymbol{\eta}_2 - \tilde{\boldsymbol{\eta}}_2) \\ &= (\mathbf{A}_{11} - \mathbf{B}_1\mathbf{K}\mathbf{S}_1)\boldsymbol{\eta}_1 + (\mathbf{A}_{12} - \mathbf{B}_1\mathbf{K}\mathbf{S}_2)\boldsymbol{\eta}_2 + \mathbf{B}_1\mathbf{K}\mathbf{S}_2\tilde{\boldsymbol{\eta}}_2\end{aligned}\quad (4.5.2)$$

$$\begin{aligned}\dot{\boldsymbol{\eta}}_2 &= \mathbf{A}_{21}\boldsymbol{\eta}_1 + \mathbf{A}_{22}\boldsymbol{\eta}_2 + \mathbf{B}_2(-\mathbf{K}\mathbf{S}_1\boldsymbol{\eta}_1 - \mathbf{K}\mathbf{S}_2\hat{\boldsymbol{\eta}}_2) \\ &= \mathbf{A}_{21}\boldsymbol{\eta}_1 + \mathbf{A}_{22}\boldsymbol{\eta}_2 - \mathbf{B}_2\mathbf{K}\mathbf{S}_1\boldsymbol{\eta}_1 - \mathbf{B}_2\mathbf{K}\mathbf{S}_2(\boldsymbol{\eta}_2 - \tilde{\boldsymbol{\eta}}_2) \\ &= (\mathbf{A}_{21} - \mathbf{B}_2\mathbf{K}\mathbf{S}_1)\boldsymbol{\eta}_1 + (\mathbf{A}_{22} - \mathbf{B}_2\mathbf{K}\mathbf{S}_2)\boldsymbol{\eta}_2 + \mathbf{B}_2\mathbf{K}\mathbf{S}_2\tilde{\boldsymbol{\eta}}_2\end{aligned}\quad (4.5.3)$$

and

$$\dot{\tilde{\boldsymbol{\eta}}}_2 = (\mathbf{A}_{22} - \mathbf{L}_c\mathbf{A}_{12})\tilde{\boldsymbol{\eta}}_2\quad (4.5.4)$$

The closed loop dynamics (closed loop state equation) of the system with reduced order observer (reduced order observer state equation) is

$$\begin{bmatrix} \dot{\eta}_1 \\ \dot{\eta}_2 \\ \dot{\tilde{\eta}}_1 \\ \dot{\tilde{\eta}}_2 \end{bmatrix} = \begin{bmatrix} \mathbf{A}_{11} - \mathbf{B}_1 \mathbf{K} \mathbf{S}_1 & \mathbf{A}_{12} - \mathbf{B}_1 \mathbf{K} \mathbf{S}_2 & \mathbf{0} & \mathbf{B}_1 \mathbf{K} \mathbf{S}_2 \\ \mathbf{A}_{21} - \mathbf{B}_2 \mathbf{K} \mathbf{S}_1 & \mathbf{A}_{22} - \mathbf{B}_2 \mathbf{K} \mathbf{S}_2 & \mathbf{0} & \mathbf{B}_2 \mathbf{K} \mathbf{S}_2 \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{A}_{22} - \mathbf{L}_c \mathbf{A}_{12} \end{bmatrix} \begin{bmatrix} \eta_1 \\ \eta_2 \\ \tilde{\eta}_1 \\ \tilde{\eta}_2 \end{bmatrix} \quad (4.5.5)$$

The block diagram representation of full state feedback with reduced order state observer is shown in Figure 4.4.

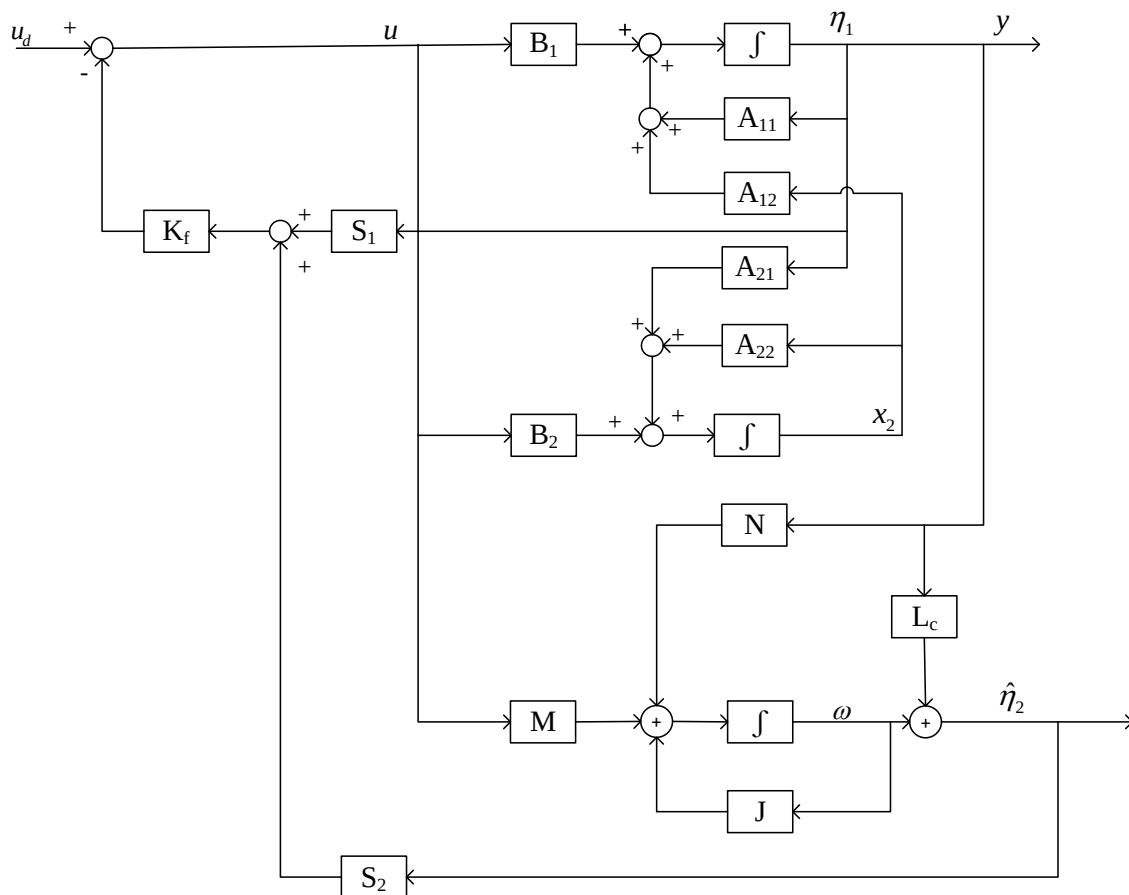


Figure 4.4: *Block Diagram Representation of Full State Feedback Controller with Reduced Order State Observer*

4.6 Examples

1. For a dynamic system

$$\begin{aligned}\dot{\mathbf{x}} &= \begin{bmatrix} 0 & 1 \\ -1 & -2 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u \\ y &= \begin{bmatrix} 0 & 1 \end{bmatrix} \mathbf{x}\end{aligned}$$

Design a full state observer with poles at $p_1 = -0.25, p_2 = -0.5$.

Solution

Check for observability of the system

$$\mathcal{O}_v = \begin{bmatrix} \mathbf{C} \\ \mathbf{CA} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & -2 \end{bmatrix}$$

This has a rank 2, hence the system is observable.

The characteristic of the observer is $(s + 0.25)(s + 0.5) = s^2 + 0.75s + 0.125$. The poles of $(\mathbf{A} - \mathbf{L}_c \mathbf{C})$ must be at the desired poles. so

$$\det(s\mathbf{I} - \mathbf{A} + \mathbf{L}_c \mathbf{C}) = s^2 + 0.75s + 0.125$$

$$\begin{aligned}\det \left(\begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} - \begin{bmatrix} 0 & 1 \\ -1 & -2 \end{bmatrix} + \begin{bmatrix} l_1 \\ l_2 \end{bmatrix} \begin{bmatrix} 0 & 1 \end{bmatrix} \right) &= s^2 + 0.75s + 0.125 \\ \det \left(\begin{bmatrix} s & -1 + l_1 \\ 1 & s + 2 + l_2 \end{bmatrix} \right) &= s^2 + 0.75s + 0.125 \\ s^2 + (2 + l_2)s + (1 - l_1) &= s^2 + 0.75s + 0.125\end{aligned}$$

The state observer gain matrix that puts the poles of the state observer at the desired location is then,

$$\begin{aligned}2 + l_2 = 0.75 &\Rightarrow l_2 = -1.25, \quad 1 - l_1 = 0.125 \Rightarrow l_1 = 0.875 \\ \mathbf{L}_c &= \begin{bmatrix} l_1 \\ l_2 \end{bmatrix} = \begin{bmatrix} 0.875 \\ -1.25 \end{bmatrix}\end{aligned}$$

The full state observer (Luenberger state observer) dynamics is

$$\begin{bmatrix} \dot{\hat{\mathbf{x}}} \\ \dot{\tilde{\mathbf{x}}} \end{bmatrix} = \begin{bmatrix} \mathbf{A} & \mathbf{L}_c \mathbf{C} \\ \mathbf{0} & \mathbf{A} - \mathbf{L}_c \mathbf{C} \end{bmatrix} \begin{bmatrix} \hat{\mathbf{x}} \\ \tilde{\mathbf{x}} \end{bmatrix} + \begin{bmatrix} \mathbf{B} \\ \mathbf{0} \end{bmatrix} \mathbf{u}$$

$$\begin{bmatrix} \dot{\hat{x}}_1 \\ \dot{\hat{x}}_2 \\ \dots \\ \dot{\tilde{x}}_1 \\ \dot{\tilde{x}}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 & \vdots & 0 & 0.875 \\ -1 & -2 & \vdots & 0 & -1.25 \\ \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & \vdots & 0 & 0.125 \\ 0 & 0 & \vdots & -1 & -0.75 \end{bmatrix} \begin{bmatrix} \hat{x}_1 \\ \hat{x}_2 \\ \dots \\ \tilde{x}_1 \\ \tilde{x}_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ \dots \\ 0 \\ 0 \end{bmatrix} u$$

2. For dynamic system

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -4 & -4 & -1 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \mathbf{x}$$

Determine the gain $\mathbf{L}_c$ that result in estimation error dynamics with poles at $p_1 = -1, p_2 = -2, p_3 = -3$ and closed loop dynamics of the system with full state feedback using full state observer, if the feedback gain $\mathbf{K}_f$ required to place the closed loop poles of the system at $s = -2, s = -2 \pm 2i$.

Solution

Check for observability of the system

$$\mathcal{O}_v = \begin{bmatrix} \mathbf{C} \\ \mathbf{CA} \\ \mathbf{CA}^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

which has rank 3, hence the system is observable.

The characteristic of the observer is $(s+1)(s+2)(s+3) = s^3 + 6s^2 + 11s + 6$. The poles of $(\mathbf{A} - \mathbf{L}_c \mathbf{C})$ must be at the desired poles. so

$$\det(s\mathbf{I} - \mathbf{A} + \mathbf{L}_c \mathbf{C}) = s^3 + 6s^2 + 11s + 6$$

$$\det \left(\begin{bmatrix} s & 0 & 0 \\ 0 & s & 0 \\ 0 & 0 & s \end{bmatrix} - \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -4 & -4 & -1 \end{bmatrix} + \begin{bmatrix} l_1 \\ l_2 \\ l_3 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \right) = s^3 + 6s^2 + 11s + 6$$

$$\det \left(\begin{bmatrix} s + l_1 & -1 & 0 \\ l_2 & s & -1 \\ l_3 + 4 & 4 & s + 1 \end{bmatrix} \right) = s^3 + 6s^2 + 11s + 6$$

$$s^3 + (1 + l_1)s^2 + (l_1 + l_2 + 4)s + (4l_1 + l_2 + l_3 + 4) = s^3 + 6s^2 + 11s + 6$$

The state observer gain matrix that puts the poles of the state observer at the desired location is then,

$$1 + l_1 = 6 \Rightarrow l_1 = 5,$$

$$l_1 + l_2 + 4 = 11 \Rightarrow l_2 = 2$$

$$4l_1 + l_2 + l_3 + 4 = 6 \Rightarrow l_3 = -20$$

$$\mathbf{L}_c = \begin{bmatrix} l_1 \\ l_2 \\ l_3 \end{bmatrix} = \begin{bmatrix} 5 \\ 2 \\ -20 \end{bmatrix}$$

The full state observer (Luenberger state observer) dynamics is

$$\begin{bmatrix} \dot{\hat{\mathbf{x}}} \\ \dot{\tilde{\mathbf{x}}} \end{bmatrix} = \begin{bmatrix} \mathbf{A} & \mathbf{L}_c \mathbf{C} \\ \mathbf{0} & \mathbf{A} - \mathbf{L}_c \mathbf{C} \end{bmatrix} \begin{bmatrix} \hat{\mathbf{x}} \\ \tilde{\mathbf{x}} \end{bmatrix} + \begin{bmatrix} \mathbf{B} \\ \mathbf{0} \end{bmatrix} \mathbf{u}$$

$$\begin{bmatrix} \dot{\hat{x}}_1 \\ \dot{\hat{x}}_2 \\ \dot{\hat{x}}_3 \\ \dots \\ \dot{\tilde{x}}_1 \\ \dot{\tilde{x}}_2 \\ \dot{\tilde{x}}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & \vdots & 5 & 0 & 0 \\ 0 & 0 & 1 & \vdots & 2 & 0 & 0 \\ -4 & -4 & -1 & \vdots & -20 & 0 & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \vdots & -5 & 1 & 0 \\ 0 & 0 & 0 & \vdots & -2 & 0 & 1 \\ 0 & 0 & 0 & \vdots & 16 & -4 & -1 \end{bmatrix} \begin{bmatrix} \hat{x}_1 \\ \hat{x}_2 \\ \hat{x}_3 \\ \dots \\ \tilde{x}_1 \\ \tilde{x}_2 \\ \tilde{x}_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \\ \dots \\ 0 \\ 0 \\ 0 \end{bmatrix} u$$

The state feedback gain matrix $\mathbf{K}_f$ can be determined as

$$\begin{aligned} \|s\mathbf{I} - \mathbf{A} + \mathbf{BK}_f\| &= (s+2)(s+2+2i)(s+2-2i) \\ \left\| \begin{bmatrix} s & 0 & 0 \\ 0 & s & 0 \\ 0 & 0 & s \end{bmatrix} - \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -4 & -4 & -1 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ k_1 & k_2 & k_3 \end{bmatrix} \right\| &= (s+2)(s^2+4s+8) \\ \left\| \begin{bmatrix} s & -1 & 0 \\ 0 & s & -1 \\ 4+k_1 & 4+k_2 & s+1+k_3 \end{bmatrix} \right\| &= s^3+6s^2+16s+16 \\ s(s(s+1+k_3)+4+k_2)+4+k_1 &= s^3+6s^2+16s+16 \\ s^3+(1+k_3)s^2+(4+k_2)s+4+k_1 &= s^3+6s^2+16s+16 \\ 1+k_3=6 \Rightarrow k_3=5 & \quad 4+k_2=16 \Rightarrow k_2=12 \quad 4+k_1=16 \Rightarrow k_1=12 \\ \mathbf{K}_f = \begin{bmatrix} k_1 & k_2 & k_3 \end{bmatrix} &= \begin{bmatrix} 12 & 12 & 5 \end{bmatrix} \end{aligned}$$

The closed-loop dynamics with a full-order state observer based on state feedback control are

$$\begin{bmatrix} \dot{\mathbf{x}} \\ \dot{\tilde{\mathbf{x}}} \end{bmatrix} = \begin{bmatrix} \mathbf{A} - \mathbf{BK} & \mathbf{BK}_f \\ \mathbf{0} & \mathbf{A} - \mathbf{L}_c\mathbf{C} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \tilde{\mathbf{x}} \end{bmatrix}$$

Replacing the matrices, we have

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dots \\ \dot{\tilde{x}}_1 \\ \dot{\tilde{x}}_2 \\ \dot{\tilde{x}}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & \vdots & 0 & 0 & 0 \\ 0 & 0 & 1 & \vdots & 0 & 0 & 0 \\ -16 & -16 & -6 & \vdots & 12 & 12 & 5 \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \vdots & -5 & 1 & 0 \\ 0 & 0 & 0 & \vdots & -2 & 0 & 1 \\ 0 & 0 & 0 & \vdots & 16 & -4 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \dots \\ \tilde{x}_1 \\ \tilde{x}_2 \\ \tilde{x}_3 \end{bmatrix}$$

3. For a dynamic system

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -4 & -4 & -1 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Design a reduce order state observer for the system that has poles at $p_1 = -5, p_2 = -5$ and closed loop dynamics of the system with full state feedback using reduced order state observer, if the feedback gain $\mathbf{K}_f$ required to place the closed loop poles of the system at $s = -2, s = -2 \pm 2i$.

Solution

Decomposing the system into a state that can be measured directly from the output and that can't be measured directly from the output,

$$\mathbf{A} = \begin{bmatrix} 0 & \vdots & 1 & 0 \\ \cdots & \cdots & \cdots & \cdots \\ 0 & \vdots & 0 & 1 \\ -4 & \vdots & -4 & -1 \end{bmatrix}, \mathbf{B} = \begin{bmatrix} 1 \\ \cdots \\ 0 \\ 0 \end{bmatrix}$$

$$\mathbf{C} = \begin{bmatrix} 1 & \vdots & 0 & 0 \end{bmatrix}$$

We have

$$\mathbf{A}_{11} = 0, \quad \mathbf{A}_{12} = \begin{bmatrix} 1 & 0 \end{bmatrix} \quad \mathbf{B}_1 = 1 \quad \mathbf{C}_1 = 1$$

$$\mathbf{A}_{21} = \begin{bmatrix} 0 \\ -4 \end{bmatrix} \quad \mathbf{A}_{22} = \begin{bmatrix} 0 & 1 \\ -4 & -1 \end{bmatrix} \quad \mathbf{B}_2 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \mathbf{C}_2 = \begin{bmatrix} 0 & 0 \end{bmatrix}$$

As we can see from the decomposition of the system, x_2 and x_3 can't be measured directly, while x_1 can be measured directly. So the observer is designed for x_2 and x_3 .

The characteristic of the observer is $(s + 5)(s + 5) = s^2 + 10s + 25$. The poles of $(\mathbf{A}_{22} - \mathbf{L}_c \mathbf{A}_{12})$ must be at the desired poles. Checking for the observability of the pair $(\mathbf{A}_{22}, \mathbf{A}_{12})$

$$\mathcal{O}_v = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

which have rank 2, hence it is completely observable.

$$\begin{aligned}
 \det(s\mathbf{I} - \mathbf{A}_{22}\mathbf{A} + \mathbf{L}_c\mathbf{A}_{12}) &= s^2 + 10s + 25 \\
 \det\left(\begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} - \begin{bmatrix} 0 & 1 \\ -4 & -1 \end{bmatrix} + \begin{bmatrix} l_1 \\ l_2 \end{bmatrix} \begin{bmatrix} 1 & 0 \end{bmatrix}\right) &= s^2 + 10s + 25 \\
 \det\left(\begin{bmatrix} s + L_1 & -1 \\ 4 + L_2 & s + 1 \end{bmatrix}\right) &= s^2 + 10s + 25 \\
 s^2 + (1 + l_1)s + (l_1 + l_2 + 4) &= s^2 + 10s + 25
 \end{aligned}$$

The state observer gain matrix that puts the poles of the state observer at the desired location is then,

$$\begin{aligned}
 1 + l_1 &= 10 \Rightarrow L_1 = 9, \quad l_1 + l_2 + 4 = 25 \Rightarrow l_2 = 12 \\
 \mathbf{L}_c &= \begin{bmatrix} l_1 \\ l_2 \end{bmatrix} = \begin{bmatrix} 9 \\ 12 \end{bmatrix}
 \end{aligned}$$

The matrices

$$\begin{aligned}
 \mathbf{J} &= \mathbf{A}_{22} - \mathbf{L}_c\mathbf{A}_{12} = \begin{bmatrix} -9 & 1 \\ -16 & -1 \end{bmatrix} \\
 \mathbf{N} &= \mathbf{A}_{21} + \mathbf{J}\mathbf{L}_c + \mathbf{L}_c\mathbf{A}_{11} = \begin{bmatrix} -69 \\ -160 \end{bmatrix} \\
 \mathbf{M} &= \mathbf{B}_2 - \mathbf{L}_c\mathbf{B}_1 = \begin{bmatrix} 0 \\ 10 \end{bmatrix}
 \end{aligned}$$

The minimum order state observer dynamics is

$$\begin{bmatrix} \dot{\omega}_1 \\ \dot{\omega}_2 \end{bmatrix} = \begin{bmatrix} -9 & 1 \\ -16 & -1 \end{bmatrix} \begin{bmatrix} \omega_1 \\ \omega_2 \end{bmatrix} + \begin{bmatrix} -69 \\ -160 \end{bmatrix} x_1 + \begin{bmatrix} 0 \\ 10 \end{bmatrix} u$$

The reduced-order state observer output equation is then

$$\begin{bmatrix} \hat{x}_2 \\ \hat{x}_3 \end{bmatrix} = \begin{bmatrix} 9y + \omega_1 \\ 12y + \omega_2 \end{bmatrix} = \begin{bmatrix} 9 & 0 & 0 & \vdots & 1 & 0 \\ 12 & 0 & 0 & \vdots & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \dots \\ \omega_1 \\ \omega_2 \end{bmatrix}$$

The combined open-loop system dynamics of the system and reduced-order observer dynamics are given as,

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dots \\ \dot{\omega}_1 \\ \dot{\omega}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & \vdots & 0 & 0 \\ 0 & 0 & 1 & \vdots & 0 & 0 \\ -4 & -4 & -1 & \vdots & 0 & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots \\ -69 & 0 & 0 & \vdots & -9 & 1 \\ -160 & 0 & 0 & \vdots & -16 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \dots \\ \omega_1 \\ \omega_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \\ \dots \\ 0 \\ 1 \end{bmatrix} u$$

Since we have already calculated the feedback gain $\mathbf{K}_f$ in example 2, to determine the closed-loop dynamics of the system with reduced-order observer,

$$\mathbf{A}_{11} - \mathbf{B}_1 \mathbf{K} \mathbf{S}_1 = 0 - \begin{bmatrix} 12 & 12 & 5 \end{bmatrix} \mathbf{S}_1$$

which imply $\mathbf{S}_1$ is 3×1 matrix, Let $\mathbf{S}_1 = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}^\top$, with this

$$\mathbf{A}_{11} - \mathbf{B}_1 \mathbf{K}_f \mathbf{S}_1 = 0 - \begin{bmatrix} 12 & 12 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = -29$$

$$\mathbf{A}_{12} - \mathbf{B}_1 \mathbf{K}_f \mathbf{S}_2 = \begin{bmatrix} 1 & 0 \end{bmatrix} - \begin{bmatrix} 12 & 12 & 5 \end{bmatrix} \mathbf{S}_2$$

which imply $\mathbf{S}_2$ is 3×2 matrix, let $\mathbf{S}_2 = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}^\top$ we have,

$$\mathbf{A}_{12} - \mathbf{B}_1 \mathbf{K} \mathbf{S}_2 = \begin{bmatrix} 1 & 0 \end{bmatrix} - \begin{bmatrix} 12 & 12 & 5 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \end{bmatrix} - \begin{bmatrix} 29 & 29 \end{bmatrix} = \begin{bmatrix} -28 & -29 \end{bmatrix}$$

$$\mathbf{B}_1 \mathbf{K} \mathbf{S}_2 = \begin{bmatrix} 12 & 12 & 5 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 29 & 29 \end{bmatrix}$$

$$\mathbf{A}_{21} - \mathbf{B}_2 \mathbf{K} \mathbf{S}_1 = \begin{bmatrix} 0 \\ -4 \end{bmatrix} - \begin{bmatrix} 0 \\ 0 \end{bmatrix} \begin{bmatrix} 12 & 12 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ -4 \end{bmatrix}$$

$$\mathbf{A}_{22} - \mathbf{B}_2 \mathbf{K} \mathbf{S}_2 = \begin{bmatrix} 0 & 1 \\ -4 & -1 \end{bmatrix} - \begin{bmatrix} 0 \\ 0 \end{bmatrix} \begin{bmatrix} 12 & 12 & 5 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -4 & -1 \end{bmatrix}$$

$$\mathbf{B}_2 \mathbf{K} \mathbf{S}_2 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \begin{bmatrix} 12 & 12 & 5 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\mathbf{A}_{22} - \mathbf{L}_c \mathbf{A}_{12} = \begin{bmatrix} 0 & 1 \\ -4 & -1 \end{bmatrix} - \begin{bmatrix} 9 \\ 12 \end{bmatrix} \begin{bmatrix} 1 & 0 \end{bmatrix} = \begin{bmatrix} -9 & 1 \\ -16 & -1 \end{bmatrix}$$

Resulting in the closed-loop system dynamics with reduced state observer after replacing $\eta_1 = x_1$ (directly measured states) and $\boldsymbol{\eta}_2 = \begin{bmatrix} x_2 & x_3 \end{bmatrix}^\top$ (estimated states) we have,

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dots \\ \dot{\tilde{x}}_1 \\ \dot{\tilde{x}}_2 \\ \dot{\tilde{x}}_3 \end{bmatrix} = \begin{bmatrix} -29 & \vdots & -28 & -29 & \vdots & 0 & \vdots & 29 & 29 \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & \vdots & 0 & 1 & \vdots & 0 & \vdots & 0 & 0 \\ -4 & \vdots & -4 & -1 & \vdots & 0 & \vdots & 0 & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & \vdots & 0 & 0 & \vdots & 0 & \vdots & 0 & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & \vdots & 0 & 0 & \vdots & 0 & \vdots & -9 & 1 \\ 0 & \vdots & 0 & 0 & \vdots & 0 & \vdots & -16 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \dots \\ \tilde{x}_1 \\ \tilde{x}_2 \\ \tilde{x}_3 \end{bmatrix}$$

4. For a dynamic system

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -4 & -4 & -1 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u$$

$$y = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Design a reduce order state observer for the system that has poles at $p_1 = -5, p_2 = -5$ and closed loop dynamics of the system with full state feedback using reduced

order state observer, if the feedback gain $\mathbf{K}_f$ required to place the closed loop poles of the system at $s = -2, s = -2 \pm 2i$.

Solution

Check the controllability of the system

$$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & -1 \\ 1 & -1 & -3 \end{bmatrix}$$

having rank 3 hence the system is completely controllable.

Check the observability of the system

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -4 & -4 & -1 \end{bmatrix}$$

having rank 3 hence the system is completely observable.

Let us arrange the dynamic equation as follows to simplify the problem

$$\begin{bmatrix} \dot{x}_2 \\ \dot{x}_1 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ -4 & -4 & -1 \end{bmatrix} \begin{bmatrix} x_2 \\ x_1 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_2 \\ x_1 \\ x_3 \end{bmatrix}$$

Thus the measurable state is $\eta_1 = x_2$ and unmeasurable state (to be estimated) is $\boldsymbol{\eta}_2 = \begin{bmatrix} x_1 & x_3 \end{bmatrix}^\top$, which is straightforward (similar way as above)

$$\begin{bmatrix} \dot{\eta}_1 \\ \dot{\boldsymbol{\eta}}_2 \end{bmatrix} = \begin{bmatrix} 0 & \vdots & 0 & 1 \\ \dots & \vdots & \dots & \dots \\ 1 & \vdots & 0 & 0 \\ -4 & \vdots & -4 & -1 \end{bmatrix} \begin{bmatrix} \eta_1 \\ \boldsymbol{\eta}_2 \end{bmatrix} + \begin{bmatrix} 0 \\ \dots \\ 0 \\ 1 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & \vdots & 0 & 0 \end{bmatrix} \begin{bmatrix} \eta_1 \\ \boldsymbol{\eta}_2 \end{bmatrix}$$

$$\begin{aligned} \mathbf{A}_{11} &= 0 \quad \mathbf{A}_{12} = \begin{bmatrix} 0 & 1 \end{bmatrix} \quad \mathbf{B}_1 = 0 \quad \mathbf{C}_1 = 1 \\ \mathbf{A}_{21} &= \begin{bmatrix} 1 \\ -4 \end{bmatrix} \quad \mathbf{A}_{22} = \begin{bmatrix} 0 & 0 \\ -4 & -1 \end{bmatrix} \quad \mathbf{B}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad \mathbf{C}_2 = \begin{bmatrix} 0 & 0 \end{bmatrix} \end{aligned}$$

we have

$$\begin{aligned} \mathbf{A}_{22} - \mathbf{L}_c \mathbf{A}_{12} - \mathbf{J} &= 0 \quad \Rightarrow \quad \mathbf{J} = \mathbf{A}_{22} - \mathbf{L}_c \mathbf{A}_{12} \\ \mathbf{A}_{21} - \mathbf{L}_c \mathbf{A}_{11} + \mathbf{J} \mathbf{L}_c - \mathbf{N} &= 0 \quad \Rightarrow \quad \mathbf{N} = \mathbf{A}_{21} - \mathbf{L}_c \mathbf{A}_{11} + \mathbf{J} \mathbf{L}_c \\ \mathbf{B}_2 - \mathbf{L}_c \mathbf{B}_1 - \mathbf{M} &= 0 \quad \Rightarrow \quad \mathbf{M} = \mathbf{B}_2 - \mathbf{L}_c \mathbf{B}_1 \end{aligned}$$

The value of the reduced observer matrix gain $\mathbf{L}_c$ that places the poles of the estimation reduced-order state observer error dynamics at the desired location is,

$$\begin{aligned} \|s\mathbf{I} - \mathbf{J}\| &= s^2 + 10s + 25 \\ \|s\mathbf{I} - \mathbf{A}_{22} + \mathbf{L}_c \mathbf{A}_{12}\| &= s^2 + 10s + 25 \\ \left\| \begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} - \begin{bmatrix} 0 & 0 \\ -4 & -1 \end{bmatrix} + \begin{bmatrix} l_1 \\ l_2 \end{bmatrix} \begin{bmatrix} 0 & 1 \end{bmatrix} \right\| &= s^2 + 10s + 25 \\ \left\| \begin{bmatrix} s & l_1 \\ 4 & s + 1 + l_2 \end{bmatrix} \right\| &= s^2 + 10s + 25 \\ s^2 + (l_2 + 1)s - 4l_1 &= s^2 + 10s + 25 \\ l_2 + 1 = 10 \Rightarrow l_2 &= 9 \quad -4l_1 = 25 \Rightarrow l_1 = -6.25 \quad \mathbf{L}_c = \begin{bmatrix} -6.25 \\ 9 \end{bmatrix} \end{aligned}$$

then

$$\begin{aligned} \mathbf{J} &= \mathbf{A}_{22} - \mathbf{L}_c \mathbf{A}_{12} = \begin{bmatrix} 0 & 6.25 \\ -4 & -10 \end{bmatrix} \\ \mathbf{N} &= \mathbf{A}_{21} - \mathbf{L}_c \mathbf{A}_{11} + \mathbf{J} \mathbf{L}_c = \begin{bmatrix} 57.25 \\ -69 \end{bmatrix} \\ \mathbf{M} &= \mathbf{B}_2 - \mathbf{L}_c \mathbf{B}_1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \end{aligned}$$

The reduced-order state observer output equation is

$$\begin{bmatrix} \hat{x}_1 \\ \hat{x}_3 \end{bmatrix} = \begin{bmatrix} -6.25y + \omega_1 \\ 9y + \omega_2 \end{bmatrix} = \begin{bmatrix} -6.25x_2 + \omega_1 \\ 9x_2 + \omega_2 \end{bmatrix} = \begin{bmatrix} 0 & -6.25 & 0 & \vdots & 1 & 0 \\ 0 & 1 & 0 & \vdots & 0 & 0 \\ 0 & 9 & 0 & \vdots & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \omega_1 \\ \omega_2 \end{bmatrix}$$

The minimum order state observer dynamics is

$$\begin{bmatrix} \dot{\omega}_1 \\ \dot{\omega}_2 \end{bmatrix} = \begin{bmatrix} 0 & 6.25 \\ -4 & -10 \end{bmatrix} \begin{bmatrix} \omega_1 \\ \omega_2 \end{bmatrix} + \begin{bmatrix} 57.25 \\ -69 \end{bmatrix} x_2 + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$$

The open-loop system dynamics combined with minimum state observer dynamics are

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dots \\ \dot{\omega}_1 \\ \dot{\omega}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & \vdots & 0 & 0 \\ 0 & 0 & 1 & \vdots & 0 & 0 \\ -4 & -4 & -1 & \vdots & 0 & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 57.25 & \vdots & 0 & 6.25 \\ 0 & 0 & -69 & \vdots & -4 & -10 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \dots \\ \omega_1 \\ \omega_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ 0 \\ \dots \\ 0 \\ 1 \end{bmatrix} u$$

Since matrix $\mathbf{B}$ is not changed $\mathbf{K}_f = \begin{bmatrix} 12 & 12 & 5 \end{bmatrix}$, the closed loop dynamics of the system with reduced order observer are determined as,

$$\mathbf{A}_{11} - \mathbf{B}_1 \mathbf{K}_f \mathbf{S}_1 = 0 - \begin{bmatrix} 12 & 12 & 5 \end{bmatrix} \mathbf{S}_1$$

which imply $\mathbf{S}_1$ is 3×1 matrix, let $\mathbf{S}_1 = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}^\top$, with this

$$\mathbf{A}_{11} - \mathbf{B}_1 \mathbf{K}_f \mathbf{S}_1 = 0 - [0] \begin{bmatrix} 12 & 12 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = 0$$

$$\mathbf{A}_{12} - \mathbf{B}_1 \mathbf{K}_f \mathbf{S}_2 = \begin{bmatrix} 0 & 1 \end{bmatrix} - [0] \begin{bmatrix} 12 & 12 & 5 \end{bmatrix} \mathbf{S}_2$$

which imply $\mathbf{S}_2$ is 3×2 matrix, let $\mathbf{S}_2 = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}^\top$ we have,

$$\mathbf{A}_{12} - \mathbf{B}_1 \mathbf{K}_f \mathbf{S}_2 = \begin{bmatrix} 0 & 1 \end{bmatrix} - [0] \begin{bmatrix} 12 & 12 & 5 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 \end{bmatrix}$$

$$\mathbf{B}_1 \mathbf{K}_f \mathbf{S}_2 = [0] \begin{bmatrix} 12 & 12 & 5 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \end{bmatrix}$$

$$\mathbf{A}_{21} - \mathbf{B}_2 \mathbf{K}_f \mathbf{S}_1 = \begin{bmatrix} 1 \\ -4 \end{bmatrix} - \begin{bmatrix} 0 \\ 1 \end{bmatrix} \begin{bmatrix} 12 & 12 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -33 \end{bmatrix}$$

$$\mathbf{A}_{22} - \mathbf{B}_2 \mathbf{K}_f \mathbf{S}_2 = \begin{bmatrix} 0 & 0 \\ -4 & -1 \end{bmatrix} - \begin{bmatrix} 0 \\ 1 \end{bmatrix} \begin{bmatrix} 12 & 12 & 5 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ -33 & -30 \end{bmatrix}$$

$$\mathbf{B}_2 \mathbf{K}_f \mathbf{S}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \begin{bmatrix} 12 & 12 & 5 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 29 & 29 \end{bmatrix}$$

$$\mathbf{A}_{22} - \mathbf{L}_c \mathbf{A}_{12} = \begin{bmatrix} 0 & 0 \\ -4 & -1 \end{bmatrix} - \begin{bmatrix} -6.25 \\ 9 \end{bmatrix} \begin{bmatrix} 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 6.25 \\ -4 & -10 \end{bmatrix}$$

Finally, the closed-loop system dynamics with reduced state observer after replacing $\eta_1 = x_2$ (directly measured states) and $\boldsymbol{\eta}_2 = [x_1 \ x_3]^\top$ (unmeasured states) is,

$$\begin{bmatrix} \dot{x}_2 \\ \dot{x}_1 \\ \dot{x}_3 \\ \dots \\ \dot{\tilde{x}}_2 \\ \dot{\tilde{x}}_1 \\ \dot{\tilde{x}}_3 \end{bmatrix} = \begin{bmatrix} 0 & \vdots & 0 & 1 & \vdots & 0 & \vdots & 0 & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ 1 & \vdots & 0 & 0 & \vdots & 0 & \vdots & 0 & 0 \\ -33 & \vdots & -33 & -30 & \vdots & 0 & \vdots & 29 & 29 \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & \vdots & 0 & 0 & \vdots & 0 & \vdots & 0 & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & \vdots & 0 & 0 & \vdots & 0 & \vdots & 0 & 6.25 \\ 0 & \vdots & 0 & 0 & \vdots & 0 & \vdots & -4 & -10 \end{bmatrix} \begin{bmatrix} x_2 \\ x_1 \\ x_3 \\ \dots \\ \tilde{x}_2 \\ \tilde{x}_1 \\ \tilde{x}_3 \end{bmatrix}$$

5. For a dynamic system

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -4 & -4 & -1 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u$$

$$y = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Design a reduce order state observer for the system that has poles at $p_1 = -5, p_2 = -5$ and closed loop dynamics of the system with full state feedback using reduced order state observer, if the feedback gain $\mathbf{K}_f$ required to place the closed loop poles of the system at $s = -2, s = -2 \pm 2i$.

Solution

Check the controllability of the system

$$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & -1 \\ 1 & -1 & -3 \end{bmatrix}$$

having rank 3, hence the system is completely controllable.

Check the observability of the system

$$\begin{bmatrix} 0 & 0 & 1 \\ -4 & -4 & -1 \\ 4 & 0 & -3 \end{bmatrix}$$

having rank 3 hence the system is completely observable.

Let us arrange the dynamic equation as follows to simplify the problem

$$\begin{bmatrix} \dot{x}_3 \\ \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -1 & -4 & -4 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_3 \\ x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_3 \\ x_1 \\ x_2 \end{bmatrix}$$

Thus the measurable state is $\eta_1 = x_3$ and unmeasurable state is $\boldsymbol{\eta}_2 = \begin{bmatrix} x_1 & x_2 \end{bmatrix}^\top$, which is straightforward (similar way as above)

$$\begin{bmatrix} \dot{\eta}_1 \\ \dot{\boldsymbol{\eta}}_2 \end{bmatrix} = \begin{bmatrix} -1 & \vdots & -4 & -4 \\ \dots & \vdots & \dots & \dots \\ 0 & \vdots & 1 & 0 \\ 1 & \vdots & 0 & 0 \end{bmatrix} \begin{bmatrix} \eta_1 \\ \boldsymbol{\eta}_2 \end{bmatrix} + \begin{bmatrix} 1 \\ \dots \\ 0 \\ 0 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & \vdots & 0 & 0 \end{bmatrix} \begin{bmatrix} \eta_1 \\ \boldsymbol{\eta}_2 \end{bmatrix}$$

$$\begin{aligned} \mathbf{A}_{11} &= -1 \quad \mathbf{A}_{12} = \begin{bmatrix} -4 & -4 \end{bmatrix} \quad \mathbf{B}_1 = 1 \quad \mathbf{C}_1 = 1 \\ \mathbf{A}_{21} &= \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad \mathbf{A}_{22} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \quad \mathbf{B}_2 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \mathbf{C}_2 = \begin{bmatrix} 0 & 0 \end{bmatrix} \end{aligned}$$

So we have

$$\begin{aligned} \mathbf{A}_{22} - \mathbf{L}_c \mathbf{A}_{12} - \mathbf{J} &= 0 \quad \Rightarrow \quad \mathbf{J} = \mathbf{A}_{22} - \mathbf{L}_c \mathbf{A}_{12} \\ \mathbf{A}_{21} - \mathbf{L}_c \mathbf{A}_{11} + \mathbf{J} \mathbf{L}_c - \mathbf{N} &= 0 \quad \Rightarrow \quad \mathbf{N} = \mathbf{A}_{21} - \mathbf{L}_c \mathbf{A}_{11} + \mathbf{J} \mathbf{L}_c \\ \mathbf{B}_2 - \mathbf{L}_c \mathbf{B}_1 - \mathbf{M} &= 0 \quad \Rightarrow \quad \mathbf{M} = \mathbf{B}_2 - \mathbf{L}_c \mathbf{B}_1 \end{aligned}$$

The value of the reduced observer matrix gain $\mathbf{L}_c$ that place the poles of estimation reduced order state observer error dynamics at the desired location is

$$\begin{aligned} \|s\mathbf{I} - \mathbf{J}\| &= s^2 + 10s + 25 \\ \|s\mathbf{I} - \mathbf{A}_{22} + \mathbf{L}_c \mathbf{A}_{12}\| &= s^2 + 10s + 25 \\ \left\| \begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} l_1 \\ l_2 \end{bmatrix} \begin{bmatrix} -4 & -4 \end{bmatrix} \right\| &= s^2 + 10s + 25 \\ \left\| \begin{bmatrix} s + 1 - 4l_1 & -4l_1 \\ -4l_2 & s - 4l_2 \end{bmatrix} \right\| &= s^2 + 10s + 25 \\ s^2 + (1 - 4l_1 - 4l_2)s - 4l_2 &= s^2 + 10s + 25 \\ 1 - 4l_1 - 4l_2 = 10 \Rightarrow l_2 = 4 \quad -4l_2 = 25 \Rightarrow l_2 = -6.25 \quad \mathbf{L}_c &= \begin{bmatrix} 4 \\ -6.25 \end{bmatrix} \end{aligned}$$

then

$$\begin{aligned} \mathbf{J} &= \mathbf{A}_{22} - \mathbf{L}_c \mathbf{A}_{12} = \begin{bmatrix} 17 & 16 \\ -25 & -25 \end{bmatrix} \\ \mathbf{N} &= \mathbf{A}_{21} - \mathbf{L}_c \mathbf{A}_{11} + \mathbf{J} \mathbf{L}_c = \begin{bmatrix} -28 \\ 49 \end{bmatrix} \\ \mathbf{M} &= \mathbf{B}_2 - \mathbf{L}_c \mathbf{B}_1 = \begin{bmatrix} -4 \\ 6.25 \end{bmatrix} \end{aligned}$$

The reduced-order state observer output equation is

$$\begin{bmatrix} \hat{x}_1 \\ \hat{x}_2 \end{bmatrix} = \begin{bmatrix} 4y + \omega_1 \\ -6.25y + \omega_2 \end{bmatrix} = \begin{bmatrix} 4x_3 + \omega_1 \\ -6.25x_3 + \omega_2 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 4 & \vdots & 1 & 0 \\ 0 & 0 & -6.25 & \vdots & 0 & 1 \\ 0 & 0 & 1 & \vdots & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \omega_1 \\ \omega_2 \end{bmatrix}$$

The minimum order state observer dynamics is

$$\begin{bmatrix} \dot{\omega}_1 \\ \dot{\omega}_2 \end{bmatrix} = \begin{bmatrix} 17 & 16 \\ -25 & -25 \end{bmatrix} \begin{bmatrix} \omega_1 \\ \omega_2 \end{bmatrix} + \begin{bmatrix} -28 \\ 49 \end{bmatrix} x_3 + \begin{bmatrix} -4 \\ 6.25 \end{bmatrix} u$$

The open-loop system dynamics combined with minimum state observer dynamics are

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dots \\ \dot{\omega}_1 \\ \dot{\omega}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & \vdots & 0 & 0 \\ 0 & 0 & 1 & \vdots & 0 & 0 \\ -4 & -4 & -1 & \vdots & 0 & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & -28 & \vdots & 17 & 16 \\ 0 & 0 & 49 & \vdots & -25 & -25 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \dots \\ \omega_1 \\ \omega_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ 0 \\ \dots \\ 0 \\ 1 \end{bmatrix} u$$

Since matrix $\mathbf{B}$ is not changed $\mathbf{K}_f = \begin{bmatrix} 12 & 12 & 5 \end{bmatrix}$, the closed loop dynamics of the system with reduced order observer are

by letting $\mathbf{S}_1 = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}^\top$, with this

$$\mathbf{A}_{11} - \mathbf{B}_1 \mathbf{K} \mathbf{S}_1 = -30$$

Let $\mathbf{S}_2 = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}^\top$ we have,

$$\mathbf{A}_{12} - \mathbf{B}_1 \mathbf{K} \mathbf{S}_2 = \begin{bmatrix} -33 & -33 \end{bmatrix}$$

$$\mathbf{B}_1 \mathbf{K} \mathbf{S}_2 = \begin{bmatrix} 29 & 29 \end{bmatrix}$$

$$\mathbf{A}_{21} - \mathbf{B}_2 \mathbf{K} \mathbf{S}_1 = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$$

$$\mathbf{A}_{22} - \mathbf{B}_2 \mathbf{K} \mathbf{S}_2 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\mathbf{B}_2 \mathbf{K} \mathbf{S}_2 = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\mathbf{A}_{22} - \mathbf{L}_c \mathbf{A}_{12} = \begin{bmatrix} 17 & 16 \\ -25 & -25 \end{bmatrix}$$

Replacing $\eta_1 = x_3$ (directly measured states) and $\boldsymbol{\eta}_2 = \begin{bmatrix} x_1 & x_2 \end{bmatrix}^\top$ (unmeasured states), the closed-d-loop system with reduced state observer is,

$$\begin{bmatrix} \dot{x}_3 \\ \dot{x}_1 \\ \dot{x}_2 \\ \dots \\ \dot{\tilde{x}}_3 \\ \dot{\tilde{x}}_1 \\ \dot{\tilde{x}}_2 \end{bmatrix} = \begin{bmatrix} -30 & \vdots & -33 & -33 & \vdots & 0 & \vdots & 29 & 29 \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & \vdots & 1 & 0 & \vdots & 0 & \vdots & 0 & 0 \\ -1 & \vdots & 0 & 0 & \vdots & 0 & \vdots & 0 & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & \vdots & 0 & 0 & \vdots & 0 & \vdots & 0 & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & \vdots & 0 & 0 & \vdots & 0 & \vdots & 17 & 16 \\ 0 & \vdots & 0 & 0 & \vdots & 0 & \vdots & -25 & -25 \end{bmatrix} \begin{bmatrix} x_3 \\ x_1 \\ x_2 \\ \dots \\ \tilde{x}_3 \\ \tilde{x}_1 \\ \tilde{x}_2 \end{bmatrix}$$

6. For the system below, design a minimum order state observer.

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 1 \\ -3 \end{bmatrix} u \quad y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Solution

The observability matrix of the system is

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

which has full rank, implying that the system is completely observable. But from the system output matrix, only one state can be directly measured. The two states are not available. Hence, the reduced order observer has order 2. If the observer is required to give a pole at $p = -2 \pm 2i$, then we have

$$\begin{aligned} \dot{\mathbf{x}} &= \begin{bmatrix} 0 & \vdots & 1 & 0 \\ \dots & \dots & \dots & \dots \\ 0 & \vdots & 0 & 1 \\ -6 & \vdots & -11 & -6 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ \dots \\ 1 \\ -3 \end{bmatrix} u \\ \mathbf{y} &= \begin{bmatrix} 1 & \vdots & 0 & 0 \end{bmatrix} \end{aligned}$$

Implying that

$$\mathbf{A}_{22} = \begin{bmatrix} 0 & 1 \\ -11 & -6 \end{bmatrix} \quad \mathbf{A}_{11} = 0 \quad \mathbf{A}_{12} = \begin{bmatrix} 1 & 0 \end{bmatrix} \quad \mathbf{A}_{21} = \begin{bmatrix} 0 \\ -6 \end{bmatrix}$$

$$\mathbf{B}_1 = 0 \quad \mathbf{B}_2 = \begin{bmatrix} 1 \\ -3 \end{bmatrix} \quad \mathbf{C}_1 = 1 \quad \mathbf{C}_2 = \begin{bmatrix} 0 & 0 \end{bmatrix}$$

So $\mathbf{J} = \mathbf{A}_{22} - \mathbf{L}\mathbf{A}_{12}$ is the error dynamics matrix which needs to be stabilized.

$$\left\| \begin{bmatrix} s & -1 \\ 11 & s+6 \end{bmatrix} + \begin{bmatrix} l_1 \\ l_2 \end{bmatrix} \begin{bmatrix} 1 & 0 \end{bmatrix} \right\| = s^2 + 4s + 8$$

$$\left\| \begin{bmatrix} s+l_1 & -1 \\ 11+l_2 & s+6 \end{bmatrix} \right\| = s^2 + 4s + 8$$

$$s^2 + (6+l_1)s + 6l_1 + l_2 + 11 = s^2 + 4s + 8$$

$$6+l_1 = 4 \Rightarrow l_1 = -2 \quad 6l_1 + l_2 + 11 = 8 \Rightarrow l_2 = 9$$

$$\mathbf{L}_c = \begin{bmatrix} -2 \\ 9 \end{bmatrix} \Rightarrow \mathbf{J} = \begin{bmatrix} 2 & 1 \\ -20 & -6 \end{bmatrix}$$

Finally

$$\mathbf{N} = \mathbf{A}_{21} + \mathbf{J}\mathbf{L}_c + \mathbf{L}_c\mathbf{A}_{11} = \begin{bmatrix} 0 \\ -6 \end{bmatrix} + \begin{bmatrix} 13 \\ 4 \end{bmatrix} = \begin{bmatrix} 13 \\ -2 \end{bmatrix}$$

$$\mathbf{M} = \mathbf{B}_2 - \mathbf{L}_c\mathbf{B}_1 = \begin{bmatrix} 1 \\ -3 \end{bmatrix}$$

The minimum order state observer dynamic equation is

$$\begin{bmatrix} \dot{\omega}_1 \\ \dot{\omega}_2 \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ -20 & -6 \end{bmatrix} \begin{bmatrix} \omega_1 \\ \omega_2 \end{bmatrix} + \begin{bmatrix} 13 \\ -2 \end{bmatrix} x_1 + \begin{bmatrix} 1 \\ -3 \end{bmatrix} u$$

The reduced-order state observer output equation is then

$$\begin{bmatrix} \hat{x}_2 \\ \hat{x}_3 \end{bmatrix} = \begin{bmatrix} -2y + \omega_1 \\ 9y + \omega_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & \vdots & 0 & 0 \\ -2 & 0 & 0 & \vdots & 1 & 0 \\ 9 & 0 & 0 & \vdots & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \omega_1 \\ \omega_2 \end{bmatrix}$$

The open-loop dynamics of the system, combined with reduced-order state observer dynamics, are

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dots \\ \dot{\omega}_1 \\ \dot{\omega}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & \vdots & 0 & 0 \\ 0 & 0 & 1 & \vdots & 0 & 0 \\ -6 & -11 & -6 & \vdots & 0 & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots \\ 13 & 0 & 0 & \vdots & -2 & 1 \\ -2 & 0 & 0 & \vdots & -20 & -6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \dots \\ \omega_1 \\ \omega_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ -3 \\ \dots \\ 1 \\ -3 \end{bmatrix} u$$

7. For the system in chapter 2 on example 2 design a minimum order observer.

Solution

As shown in chapter 2 on example 2, the system has one unobservable mode. So we cannot design an observer for the unobservable mode of the system, but we can design for the observable modes of the system. As we can see from the output matrix of the decomposed system, one state can be directly measured, while one cannot. Thus, we design a state observer of order 1. The dynamics look of the observable mode of the system is

$$\begin{aligned} \dot{\mathbf{x}} &= \begin{bmatrix} 2 & \vdots & 1 \\ \dots & \dots & \dots \\ -4 & \vdots & 0 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 1 \\ \dots \\ 1 \end{bmatrix} u \\ \mathbf{y} &= \begin{bmatrix} 1 & \vdots & 0 \end{bmatrix} \mathbf{x} \end{aligned}$$

From this $A_{11} = 2$, $A_{12} = 1$, $A_{21} = -4$, $A_{22} = 0$. If the observer is required to have a pole at $p = -1$, then

$$\begin{aligned} sI - A_{22} + lA_{12} &= s + 1 \\ s - 0 + l &= s + 1 \Rightarrow l = 1 \Rightarrow L_c = l = 1 \\ J &= A_{22} - A_c A_{12} = 0 - 1 = -1 \\ N &= A_{21} + J L_c + L_c A_{11} = -4 - 1 + 2 = -3 \\ M &= B_2 - L_c B_1 = 1 - 1 = 0 \end{aligned}$$

The minimum order state observer dynamic equation is

$$\dot{\omega} = -\omega - 3x_1$$

The reduced-order state observer output equation is then

$$\begin{bmatrix} \hat{x}_2 \end{bmatrix} = \begin{bmatrix} L_c y + \omega \end{bmatrix} = \begin{bmatrix} 1 & 0 & \vdots & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \omega \end{bmatrix}$$

The open-loop system state equation, combined with the reduced-order state observer equation, is

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dots \\ \dot{\omega} \end{bmatrix} = \begin{bmatrix} 2 & 1 & \vdots & 0 \\ -4 & 0 & \vdots & 0 \\ \dots & \dots & \dots & \dots \\ -3 & 0 & \vdots & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \dots \\ \omega \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ \dots \\ 0 \end{bmatrix} u$$

4.7 Exercises

1. Consider a system

$$\dot{\mathbf{x}} = \begin{bmatrix} 0.16 & 2.16 \\ -0.16 & -1.16 \end{bmatrix} \mathbf{x} + \begin{bmatrix} -1 \\ 1 \end{bmatrix} u \quad y = \begin{bmatrix} 1 & 1 \end{bmatrix} \mathbf{x}$$

Find its open-loop eigenvalues, corresponding eigenvectors, and check the stability, controllability, and observability of the system. Determine feedback gain matrix $\mathbf{K}_f$ such that both eigenvalues of the closed-loop system are at -1 , and finally design a minimum order observer for this system that has an eigenvalue at -3 .

2. Design a full state observer for IPC dynamics on [1.6.3](#) that has poles at $-1, -1, -1 \pm i$. Use any valid parameter.
3. Design a full state observer for DC motor dynamics having poles at $-1 \pm 2i$. Use any valid parameters.
4. Design a reduced-order state observer for the dynamics of the DC motor on [1.7.1](#), where the speed is directly measured and the armature current i_a is required to be estimated. The pole of the observer must be at -1 , use any valid parameters.
5. Design a reduced state observer for IPC dynamics on [1.6.3](#), where the position of the cart and position of the pendulum are directly measured and the angular velocity of the pendulum and linear velocity of the cart are not. The poles of the observer are at $-1 \pm 2i$.
6. Design a Kalman-Bucy filter for IPC dynamics and DC motor dynamics. Use any valid parameters.
7. For a dynamic system

$$\begin{aligned} \dot{\mathbf{x}} &= \begin{bmatrix} 0 & 0 \\ 1 & -1 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \boldsymbol{\nu} \\ y &= \begin{bmatrix} 0 & 1 \end{bmatrix} \mathbf{x} + e \end{aligned}$$

Design a Kalman filter.

8. For a system described by

$$\dot{\mathbf{x}} = \begin{bmatrix} -1 & -2 & -2 \\ 0 & -1 & 1 \\ 1 & 0 & -1 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} u \quad y = \begin{bmatrix} 1 & 1 & 0 \end{bmatrix} \mathbf{x}$$

Check controllability, observability, stabilizability, and detectability. Finally, design an observer-based feedback control system that places all closed-loop eigenvalues of the system at -2 . The observer has all poles at -4 .

9. Design a full state observer having poles at $-1, -1 \pm 2i$, for a system described by

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 3 & 0 \\ 0 & 2 & 1 \\ 1 & 1 & 0 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix} u \quad y = \begin{bmatrix} 0 & 2 & 1 \end{bmatrix} \mathbf{x}$$

10. For a dynamic system

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -4 & -4 & -1 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} u$$

$$\mathbf{y} = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix} \mathbf{x}$$

determine minimum order state observer having pole at $-2, -2 \pm 2i$.

11. For a dynamic system

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -4 & -4 & -1 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} u$$

$$y = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix} \mathbf{x}$$

determine minimum order state observer having pole at $-2, -2 \pm 2i$.

12. For a dynamic system

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -4 & -4 & -1 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 & 1 \end{bmatrix} \mathbf{x}$$

determine minimum order state observer having pole at $-2, -2 \pm 2i$.

13. For examples 6 and 7 determine the closed loop dynamics of the system with full state feedback using reduced order state observer, if the feedback gain $\mathbf{K}_f$ required to place the closed loop poles of the system at $s = -1, s = -1 \pm 1i$

Chapter 5

Introduction to Optimal Control

Optimal control deals with the most celebrated problem of modern control theory, "determination of best control strategy, selecting optimum control strategy, selecting optimum control vector $\mathbf{u}(t)$, that minimizes cost function or performance index". Consider an LTI system with initial condition $\mathbf{x}_0$:

$$\begin{aligned}\dot{\mathbf{x}} &= \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} \\ \mathbf{y} &= \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u}\end{aligned}\tag{5.0.1}$$

The objective of optimal control is to determine the control vector $\mathbf{u}$ that forces the behavior of the dynamic system to the required final state by maximizing the required cost function and at the same time fulfilling the physical constraints.

The cost function has a general representation of:

$$J = \theta[\mathbf{x}(t), t] \Big|_{t=t_0}^{t=t_f} + \int_{t_0}^{t_f} \varphi(\mathbf{x}(t), \mathbf{u}(t), t) dt\tag{5.0.2}$$

The first term $\theta[\mathbf{x}(t), t] \Big|_{t=t_0}^{t=t_f}$ is called terminal error optimization problem (terminal cost function) and the second term $\int_{t_0}^{t_f} \varphi(\mathbf{x}(t), \mathbf{u}(t), t) dt$ is called optimal cost function over the entire trajectory (the function to be optimized over the entire trajectory).

5.1 Performance Indices

The cost function (performance index) may be a function that optimizes energy consumed, optimizes the time required to reach the desired state, or minimizes the tracking or terminal value error. Depending on the desired optimization problem, we have five performance indices.

5.1.1 Minimum time control problem

The aim is to determine a control vector that derives the state of the dynamic system from its initial value $\mathbf{x}_0$ to its final value $\mathbf{x}_f$ in minimum time. The best example is car racing, where finishing time is the only required parameter to be minimized. The cost function for minimum time is given as

$$J = \int_{t_0}^{t_f} dt\tag{5.1.1}$$

5.1.2 Minimum terminal error control problem

The aim is to determine a control vector $\mathbf{u}$ that minimizes terminal error. Consider a system with state $\mathbf{x}$, we want to reach $\mathbf{x}_d$ after a certain time t . In a terminal control problem, we need to determine the control signal $\mathbf{u}$ that minimizes $\boldsymbol{\xi} = \mathbf{x}_f - \mathbf{x}_d$. The best example is rocket launching, where the emphasis is put on the target only. The cost function of this control problem is:

$$J = (\mathbf{x}_f - \mathbf{x}_d)^\top \mathbf{S} (\mathbf{x}_f - \mathbf{x}_d) \quad (5.1.2)$$

where $\mathbf{S}$ is $n \times n$, symmetric positive definite matrix.

5.1.3 Minimum effort/energy control problem

The aim is to determine the minimum amount of energy that is required to derive the system from $\mathbf{x}_0$ to $\mathbf{x}_f$. The best example is car braking, where stopping time is required to be achieved with minimum energy. The cost function is:

$$J = \int_{t_0}^{t_f} \mathbf{u}^\top \mathbf{R} \mathbf{u} \, dt \quad (5.1.3)$$

where $\mathbf{R}$ is $m \times m$, symmetric positive definite matrix.

5.1.4 Optimal servo-mechanism or tracking problem

This control problem aims to determine a control vector $\mathbf{u}$ that minimizes the tracking error when driving the system from any initial state $\mathbf{x}_0$ to the desired state (final state). If the desired trajectory is $\mathbf{x}_d$ over the course from t_0 to t_f , the control vector $\mathbf{u}$ is required to minimize $\tilde{\mathbf{x}} = \mathbf{x} - \mathbf{x}_d$. The best example is space shuttle tracking and control, where tracking error, energy consumption, and the difference between the desired terminal point and the shuttle's final point are all required to be minimized. The cost function for this control problem is:

$$J = \int_{t_0}^{t_f} ((\mathbf{x} - \mathbf{x}_d)^\top \mathbf{Q} (\mathbf{x} - \mathbf{x}_d)) dt \quad (5.1.4)$$

where $\mathbf{Q}$ is $n \times n$, symmetric positive definite matrix. If we want to minimize the energy required by the system, in addition to minimizing tracking error, the cost function becomes the sum of tracking error and the minimum effort cost function.

$$\begin{aligned} J &= \int_{t_0}^{t_f} ((\mathbf{x} - \mathbf{x}_d)^\top \mathbf{Q} (\mathbf{x} - \mathbf{x}_d)) dt + \int_{t_0}^{t_f} \mathbf{u}^\top \mathbf{R} \mathbf{u} \, dt \\ &= \int_{t_0}^{t_f} ((\mathbf{x} - \mathbf{x}_d)^\top \mathbf{Q} (\mathbf{x} - \mathbf{x}_d) + \mathbf{u}^\top \mathbf{R} \mathbf{u}) \, dt \end{aligned} \quad (5.1.5)$$

If we add terminal error minimization to the problem in addition to minimizing tracking error and energy optimization, the cost function takes a general form,

$$J = (\mathbf{x}_f - \mathbf{x}_d)^\top \mathbf{S} (\mathbf{x}_f - \mathbf{x}_d) + \int_{t_0}^{t_f} ((\mathbf{x} - \mathbf{x}_d)^\top \mathbf{Q} (\mathbf{x} - \mathbf{x}_d) + \mathbf{u}^\top \mathbf{R} \mathbf{u}) \, dt \quad (5.1.6)$$

5.1.5 Optimal regulator problem

This control problem is a special case of the optimal tracking problem where the desired final state is a point in space, particularly an origin $\mathbf{x}_d = 0$. The best example is restoring the equilibrium point of any dynamic system. The tracking optimal problem cost function is:

$$J = \mathbf{x}_f^\top \mathbf{S} \mathbf{x}_f + \int_{t_0}^{t_f} (\mathbf{x}^\top \mathbf{Q} \mathbf{x} + \mathbf{u}^\top \mathbf{R} \mathbf{u}) dt \quad (5.1.7)$$

5.2 Review on Elements of Calculus of Variation

5.2.1 Increment of a function and a functional

Consider a function $f(t)$, an increment of function $f(t)$ is defined as

$$\Delta f = f(t + \Delta t) - f(t) \quad (5.2.1)$$

Δf depends both on t and Δt , hence it is written as $\Delta f(t, \Delta t)$. In similar way for a functional $J(\mathbf{x}(t), t)$ the incremental of a functional is defined as:

$$\Delta J = J((\mathbf{x}(t) + \delta \mathbf{x}(t)), t) - J(\mathbf{x}(t), t) \quad (5.2.2)$$

$\delta \mathbf{x}(t)$ is called variation of the function $\mathbf{x}(t)$. Since the increment of a functional depends on $\mathbf{x}(t)$ and $\delta \mathbf{x}(t)$ it is written as $J((\mathbf{x}(t), \delta \mathbf{x}(t)))$.

5.2.2 Differential and variation

Differential of a function:- For a function $f(t)$, the Taylor series expansion about a point t^* is given as

$$\Delta f = \left(\frac{\partial f}{\partial t} \right)_{t^*} \Delta t + \frac{1}{2!} \left(\frac{\partial^2 f}{\partial t^2} \right)_{t^*} \Delta t^2 + \dots - f(t^*) \quad (5.2.3)$$

neglecting higher-order terms in Δt ,

$$\begin{aligned} \Delta f &= f(t^*) + \left(\frac{\partial f}{\partial t} \right)_{t^*} \Delta t - f(t^*) \\ \Delta f &= \dot{f}(t^*) \Delta t = df \end{aligned} \quad (5.2.4)$$

df is called the differential of f at the point t^* , and $\dot{f}(t)$ is the slope of f at t^* .

Variation of a functional: Expanding a functional $J(\mathbf{x}(t), t)$ using the Taylor series on a function $\mathbf{x}^*(t)$, gives us:

$$\Delta J = J(\mathbf{x}^*(t)) + \left(\frac{\partial J}{\partial \mathbf{x}} \delta \mathbf{x} \right)_{\mathbf{x}^*(t)} + \left(\frac{1}{2!} \frac{\partial^2 J}{\partial \mathbf{x}^2} \delta \mathbf{x}^2 \right)_{\mathbf{x}^*(t)} + \dots - J(\mathbf{x}^*(t)) \quad (5.2.5)$$

$\delta J = \frac{\partial J}{\partial \mathbf{x}} \delta \mathbf{x}$ is called the first variation, and $\delta^2 J = \frac{1}{2!} \frac{\partial^2 J}{\partial \mathbf{x}^2} \delta \mathbf{x}^2$ is called the second variation of the functional about a function $\mathbf{x}^*(t)$.

Neglecting higher-order terms in $\delta \mathbf{x}$, linear approximation of variation of $J(\mathbf{x}(t), t)$ about a function $\mathbf{x}^*(t)$ is,

$$\Delta J \approx \delta J = \frac{\partial J}{\partial \mathbf{x}} \delta \mathbf{x} \quad (5.2.6)$$

5.2.3 Optimization of a function and unconstrained functional

For a function to have an optimum value at a point, the following two conditions should be met.

- **Necessary Condition:** For a function to have an optimum value at a point t^* , its first derivative must vanish at the point (i.e. $df = 0$ at t^*).
- **Sufficient Condition:** For a function to have an optimum value at a point t^* , either $d^2f > 0$ for the minimum or $d^2f < 0$ for maximum, for all t , such that $(t - t^* < \epsilon, \text{ for } \epsilon > 0)$.
- If $d^2f = 0$, the point is called an inflection point or a stationary point.

Similar conditions hold for functional to have optimum value, the necessary condition is that its first variation must vanish ($\Delta J = 0$) at $\mathbf{x}^*(t)$ and the sufficient condition is that $\Delta^2 J < 0$ for local maximum or $\Delta^2 J > 0$ for local minimum, for all $\mathbf{x}(t)$, such that $(\mathbf{x}(t) - \mathbf{x}^*(t) < \epsilon, \text{ for } \epsilon > 0)$.

A functional may be a function of one or more functions, like a function of several variables $V(\mathbf{x}, \mathbf{y}, \mathbf{z})$. In a similar manner, we have a functional which can be $J(\mathbf{x}(t), \mathbf{y}(t), \mathbf{z}(t))$. The only difference is in function; the variables $\mathbf{x}$, $\mathbf{y}$, and $\mathbf{z}$ are independent variables, while in functional, they are a function of another independent variable.

In a control system, we deal with functions of the form

$$J(\mathbf{x}(t), \dot{\mathbf{x}}(t), \mathbf{u}(t), t) \quad (5.2.7)$$

If the system is time-invariant, the functional will be

$$J(\mathbf{x}(t), \dot{\mathbf{x}}(t), \mathbf{u}(t)) \quad (5.2.8)$$

Consider a functional $J(\mathbf{x}, \dot{\mathbf{x}})$ defined as in equation A.5.1, if we take $\delta \mathbf{x}$ and $\delta \dot{\mathbf{x}}$ as deviation from $\mathbf{x}$ and $\dot{\mathbf{x}}$ respectively,

$$J(\mathbf{x}, \dot{\mathbf{x}}) = \int_{t_0}^{t_f} \varphi(\mathbf{x}, \dot{\mathbf{x}}) dt \quad (5.2.9)$$

The increment of $J(\mathbf{x}, \dot{\mathbf{x}})$ will be

$$\Delta J = \int_{t_0}^{t_f} (\varphi(\mathbf{x} + \delta \mathbf{x}, \dot{\mathbf{x}} + \delta \dot{\mathbf{x}}) - \varphi(\mathbf{x}, \dot{\mathbf{x}})) dt \quad (5.2.10)$$

Using Taylor series expansion:

$$\begin{aligned}
 \Delta J &= \int_{t_0}^{t_f} (\varphi(\mathbf{x} + \delta\mathbf{x}, \dot{\mathbf{x}} + \delta\dot{\mathbf{x}}) - \varphi(\mathbf{x}, \dot{\mathbf{x}})) dt \\
 &= \int_{t_0}^{t_f} \left(\varphi(\mathbf{x}, \dot{\mathbf{x}}) + \frac{\partial\varphi}{\partial\mathbf{x}}\delta\mathbf{x} + \frac{\partial\varphi}{\partial\dot{\mathbf{x}}}\delta\dot{\mathbf{x}} + \dots + hot - \varphi(\mathbf{x}, \dot{\mathbf{x}}) \right) dt \\
 &= \int_{t_0}^{t_f} \left(\frac{\partial\varphi}{\partial\mathbf{x}}\delta\mathbf{x} + \frac{\partial\varphi}{\partial\dot{\mathbf{x}}}\delta\dot{\mathbf{x}} + \dots + hot \right) dt \\
 &= \int_{t_0}^{t_f} \left(\frac{\partial\varphi}{\partial\mathbf{x}}\delta\mathbf{x} + \frac{\partial\varphi}{\partial\dot{\mathbf{x}}}\delta\dot{\mathbf{x}} \right) dt + \int_{t_0}^{t_f} \frac{1}{2!} \left(\frac{\partial^2\varphi}{\partial\mathbf{x}^2}(\delta\mathbf{x})^2 + \frac{\partial^2\varphi}{\partial\dot{\mathbf{x}}^2}(\delta\dot{\mathbf{x}})^2 + 2\frac{\partial^2\varphi}{\partial\dot{\mathbf{x}}\partial\mathbf{x}}\delta\dot{\mathbf{x}}\delta\mathbf{x} \right) dt + \dots + hot \\
 &= \delta J + \delta^2 J + \dots + hot
 \end{aligned} \tag{5.2.11}$$

The linear part of ΔJ with respect to $\delta\mathbf{x}$ and $\delta\dot{\mathbf{x}}$, is

$$\delta J = \int_{t_0}^{t_f} \left(\frac{\partial\varphi}{\partial\mathbf{x}}\delta\mathbf{x} + \frac{\partial\varphi}{\partial\dot{\mathbf{x}}}\delta\dot{\mathbf{x}} \right) dt \tag{5.2.12}$$

For the functional to have optimum value $\Delta J = 0$ (Necessary Condition), so

$$\begin{aligned}
 \int_{t_0}^{t_f} \left(\frac{\partial\varphi}{\partial\mathbf{x}}\delta\mathbf{x} + \frac{\partial\varphi}{\partial\dot{\mathbf{x}}}\delta\dot{\mathbf{x}} \right) dt &= 0 \\
 \int_{t_0}^{t_f} \left(\frac{\partial\varphi}{\partial\mathbf{x}}\delta\mathbf{x} \right) dt + \int_{t_0}^{t_f} \left(\frac{\partial\varphi}{\partial\dot{\mathbf{x}}}\delta\dot{\mathbf{x}} \right) dt &= 0 \implies \int_{t_0}^{t_f} \left(\frac{\partial\varphi}{\partial\mathbf{x}}\delta\mathbf{x} \right) dt + \int_{t_0}^{t_f} \left(\frac{\partial\varphi}{\partial\dot{\mathbf{x}}} \frac{d\delta\mathbf{x}}{dt} \right) dt = 0 \\
 \int_{t_0}^{t_f} \left(\frac{\partial\varphi}{\partial\mathbf{x}}\delta\mathbf{x} \right) dt + \int_{t_0}^{t_f} \left(\frac{\partial\varphi}{\partial\dot{\mathbf{x}}} d(\delta\mathbf{x}) \right) &= 0
 \end{aligned} \tag{5.2.13}$$

Solving the second term in equation 5.2.13 using integration by part

$$\int_a^b u dv = uv|_a^b - \int_a^b v du$$

By letting $u = \frac{\partial\varphi}{\partial\dot{\mathbf{x}}}$ and $dv = d(\delta\mathbf{x})$,

$$\begin{aligned}
 \int_{t_0}^{t_f} \left(\frac{\partial\varphi}{\partial\dot{\mathbf{x}}} d(\delta\mathbf{x}) \right) &= \left(\frac{\partial\varphi}{\partial\dot{\mathbf{x}}}\delta\mathbf{x} \right) \Big|_{t_0}^{t_f} - \int_{t_0}^{t_f} \left(\delta\mathbf{x} \frac{d}{dt} \left(\frac{\partial\varphi}{\partial\dot{\mathbf{x}}} \right) \right) dt \\
 \int_{t_0}^{t_f} \left(\frac{\partial\varphi}{\partial\dot{\mathbf{x}}} d(\delta\mathbf{x}) \right) &= \left(\frac{\partial\varphi}{\partial\dot{\mathbf{x}}}\delta\mathbf{x} \right) \Big|_{t_0}^{t_f} - \int_{t_0}^{t_f} \left(\frac{d}{dt} \left(\frac{\partial\varphi}{\partial\dot{\mathbf{x}}} \right) \delta\mathbf{x} \right) dt
 \end{aligned} \tag{5.2.14}$$

Replacing equation 5.2.14 in equation 5.2.13 for the second part and rearranging we have,

$$\int_{t_0}^{t_f} \left(\left(\frac{\partial\varphi}{\partial\mathbf{x}} - \frac{d}{dt} \left(\frac{\partial\varphi}{\partial\dot{\mathbf{x}}} \right) \right) \delta\mathbf{x} \right) dt + \left(\frac{\partial\varphi}{\partial\dot{\mathbf{x}}}\delta\mathbf{x} \right) \Big|_{t_0}^{t_f} = 0 \tag{5.2.15}$$

Thus, for the functional to have optimum value, both

$$\frac{\partial\varphi}{\partial\mathbf{x}} - \frac{d}{dt} \left(\frac{\partial\varphi}{\partial\dot{\mathbf{x}}} \right) = 0 \tag{5.2.16}$$

over the path and

$$\left(\frac{\partial \varphi}{\partial \dot{\mathbf{x}}} \delta \mathbf{x} \right) = 0 \quad (5.2.17)$$

at the boundaries.

Equation A.4.5 is called the Euler-Lagrange equation, while equation A.4.6 is called the boundary condition equation. For the boundary condition to be zero, either $\frac{\partial \varphi}{\partial \dot{\mathbf{x}}}$ must be zero or $\delta \mathbf{x}$ must be zero at the boundary. This leads to four cases:

1. **Case-I:** $\mathbf{x}$ is fixed at t_0 and t_f (fixed end trajectory).

Here, $\mathbf{x}(t_0) = C_1$ and $\mathbf{x}(t_f) = C_2$, where C_1 and C_2 are constants. Since $\delta \mathbf{x} = 0$ at both t_0 and t_f , there are no restrictions on $\frac{\partial \varphi}{\partial \dot{\mathbf{x}}}$.

2. **Case-II:** $\mathbf{x}$ is fixed at t_0 and free at t_f .

Here, $\mathbf{x}(t_0) = C_1$, and $\frac{\partial \varphi}{\partial \dot{\mathbf{x}}} = 0$ at t_f because $\delta \mathbf{x}$ is unknown at t_f . No restrictions apply to $\frac{\partial \varphi}{\partial \dot{\mathbf{x}}}$ at t_0 .

3. **Case-III:** $\mathbf{x}$ is fixed at t_f and free at t_0 .

Here, $\mathbf{x}(t_f) = C_2$, and $\frac{\partial \varphi}{\partial \dot{\mathbf{x}}} = 0$ at t_0 because $\delta \mathbf{x}$ is unknown at t_0 . No restrictions apply to $\frac{\partial \varphi}{\partial \dot{\mathbf{x}}}$ at t_f .

4. **Case-IV:** $\mathbf{x}$ is free at t_0 and t_f .

Here, $\frac{\partial \varphi}{\partial \dot{\mathbf{x}}} = 0$ at both t_0 and t_f because $\delta \mathbf{x}$ is unknown at both times.

The sufficient condition for the functional to have optimum value is derived using $\delta^2 J$ (second variation),

$$\delta^2 J = \int_{t_0}^{t_f} \frac{1}{2} \left(\frac{\partial^2 \varphi}{\partial \mathbf{x}^2} (\delta \mathbf{x})^2 + \frac{\partial^2 \varphi}{\partial \dot{\mathbf{x}}} (\delta \dot{\mathbf{x}})^2 + 2 \frac{\partial^2 \varphi}{\partial \dot{\mathbf{x}} \partial \mathbf{x}} \delta \dot{\mathbf{x}} \delta \mathbf{x} \right) dt \quad (5.2.18)$$

Solving the last term using integration by part $u = \frac{\partial^2 \varphi}{\partial \mathbf{x}} \delta \dot{\mathbf{x}} \delta \mathbf{x}$ and $dv = \delta \dot{\mathbf{x}}$ and rearranging,

$$\delta^2 J = \frac{1}{2} \int_{t_0}^{t_f} \left[\left(\frac{\partial^2 \varphi}{\partial \mathbf{x}^2} - \frac{d}{dt} \left(\frac{\partial^2 \varphi}{\partial \dot{\mathbf{x}} \partial \mathbf{x}} \right) \right) (\delta \mathbf{x})^2 + \left(\frac{\partial^2 \varphi}{\partial \dot{\mathbf{x}}^2} \right) (\delta \dot{\mathbf{x}})^2 \right] dt \quad (5.2.19)$$

which means

$$\frac{\partial^2 \varphi}{\partial \mathbf{x}^2} - \frac{d}{dt} \left(\frac{\partial^2 \varphi}{\partial \dot{\mathbf{x}} \partial \mathbf{x}} \right) \geq 0 \quad \text{and} \quad \frac{\partial^2 \varphi}{\partial \dot{\mathbf{x}}^2} \geq 0$$

for local minimum and

$$\frac{\partial^2 \varphi}{\partial \mathbf{x}^2} - \frac{d}{dt} \left(\frac{\partial^2 \varphi}{\partial \dot{\mathbf{x}} \partial \mathbf{x}} \right) \leq 0 \quad \text{and} \quad \frac{\partial^2 \varphi}{\partial \dot{\mathbf{x}}^2} \leq 0$$

for local maximum.

5.2.4 Optimum of constrained functional

Real-world problems, especially control problems, are restricted by different natural laws or the limitations of technology. The optimization of such kinds of problems under different constraints is the mainstream of optimal control. For example, the tracking control of a satellite is restricted by the trajectory selected and the maximum input allowed. The objective of such a control problem is to determine a control signal $\mathbf{u}$ that optimizes a cost function subjected to the constraints. If we want to optimize a cost function J at the same time required to fulfill the state dynamics of the system, then the optimal control problem formulation is to determine a control signal $\mathbf{u}$ that minimizes.

$$\int_{t_0}^{t_f} \Upsilon(\mathbf{x}, \dot{\mathbf{x}}, \mathbf{u}) dt \quad (5.2.20)$$

subjected to

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} \quad (5.2.21)$$

Let us define a general function that includes the constraint and the function to be optimized as

$$\varphi(\mathbf{x}, \dot{\mathbf{x}}, \mathbf{u}) = \Upsilon(\mathbf{x}, \dot{\mathbf{x}}, \mathbf{u}) + \boldsymbol{\lambda}^\top (\mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} - \dot{\mathbf{x}}) \quad (5.2.22)$$

The optimization problem is reduced to unconstrained optimization of the form

$$\begin{aligned} & \int_{t_0}^{t_f} (\Upsilon(\mathbf{x}, \dot{\mathbf{x}}, \mathbf{u}) + \boldsymbol{\lambda}^\top (\mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} - \dot{\mathbf{x}})) dt \\ & \int_{t_0}^{t_f} \varphi(\mathbf{x}, \dot{\mathbf{x}}, \mathbf{u}) dt \end{aligned} \quad (5.2.23)$$

By the Euler-Lagrangian equation for the function to have an optimum value

$$\frac{\partial \varphi}{\partial \mathbf{x}} - \frac{d}{dt} \left(\frac{\partial \varphi}{\partial \dot{\mathbf{x}}} \right) = 0 \quad (5.2.24)$$

$$\frac{\partial \varphi}{\partial \mathbf{u}} - \frac{d}{dt} \left(\frac{\partial \varphi}{\partial \dot{\mathbf{u}}} \right) = 0 \quad (5.2.25)$$

$$\frac{\partial \varphi}{\partial \boldsymbol{\lambda}^\top} - \frac{d}{dt} \left(\frac{\partial \varphi}{\partial \dot{\boldsymbol{\lambda}^\top}} \right) = 0 \quad (5.2.26)$$

5.3 Solution to Optimal Linear Quadratic Regulator Problem

The problem is determining control law $\mathbf{u}$ that minimize cost function on equation 5.1.7 subjected to $\dot{\mathbf{x}} = \mathbf{Ax} + \mathbf{Bu}$. Assume we need to bring the final state to zero ($\mathbf{x}_f = 0$).

Which means we need to find $\mathbf{u}$ that minimize

$$J = \int_{t_0}^{t_f} (\mathbf{x}^\top \mathbf{Q} \mathbf{u} + \mathbf{u}^\top \mathbf{R} \mathbf{u} + \boldsymbol{\lambda}^\top (\mathbf{A} \mathbf{u} + \mathbf{B} \mathbf{u} - \dot{\mathbf{x}})) dt \quad (5.3.1)$$

Matrix $\mathbf{Q}$ is a weighing matrix for state error, and $\mathbf{R}$ is a weighing matrix for input. The Choice depended on the optimization problem and design specifications. If state error is required to be minimized matrix $\mathbf{Q}$ should be high, and if input to the system is required to be minimized more than the state tracking error matrix $\mathbf{R}$ should be high. In state feedback controller design, penalizing input is the same as penalizing state, since the control signal is a linear combination of the state of the system.

The possible starting of guessing the values of the matrices can be one of the following

- Simplest choice: $\mathbf{Q} = \mathbf{I}$, $\mathbf{R} = \rho \mathbf{I} \Rightarrow \mathbf{L} = \mathbf{x}^\top \cdot \mathbf{x} + \rho \mathbf{u}^\top \mathbf{u}$ then vary ρ to get something that has good response.
- Diagonal weights

$$\mathbf{Q} = \begin{bmatrix} q_1 & 0 & 0 & 0 \\ 0 & q_2 & 0 & 0 \\ 0 & 0 & q_3 & 0 \\ 0 & 0 & 0 & q_4 \end{bmatrix}$$

Choose each $q_i = \frac{1}{\tilde{x}_i^2}$, where $\tilde{x}_i$ is the tolerable error of each state of the system.

- Output weighting
Let $\mathbf{z} = \mathbf{Hx}$ be the output you want to keep small. Assume $(\mathbf{A}, \mathbf{H})$ observable. Use $\mathbf{Q} = \mathbf{x}^\top \mathbf{x}$, $\mathbf{R} = \rho \mathbf{I} \Rightarrow$ trade off $\mathbf{z}^\top \cdot \mathbf{z}$ vs $\rho \mathbf{u}^2$.
Where $\mathbf{H} = \mathbf{x}^\top \mathbf{Q} \mathbf{x} + \mathbf{u}^\top \mathbf{R} \mathbf{u} + \boldsymbol{\lambda}^\top$ and $\boldsymbol{\lambda}(t) = \mathbf{P}(t) \mathbf{x}(t)$
- Trial and error (on weights)

Most of the time $\mathbf{Q} = \mathbf{C}^\top \mathbf{C}$ and the weight of the matrix is adjusted based on the performance requirement of the system.

Our objective is to determine a control law $\mathbf{u}(t)$ that optimizes the optimal linear quadratic regulator cost function subjected to $\dot{\mathbf{x}} = \mathbf{Ax} + \mathbf{Bu}$. To do so, we use Euler-Lagrangian equations:

$$\frac{\partial \varphi}{\partial \mathbf{x}} - \frac{d}{dt} \left(\frac{\partial \varphi}{\partial \dot{\mathbf{x}}} \right) = 0 \Rightarrow \mathbf{x}^\top \mathbf{Q} + \mathbf{x}^\top \mathbf{Q}^\top + \boldsymbol{\lambda}^\top \mathbf{A} - \dot{\boldsymbol{\lambda}}^\top = 0 \quad (5.3.2)$$

$$\frac{\partial \varphi}{\partial \mathbf{u}} - \frac{d}{dt} \left(\frac{\partial \varphi}{\partial \dot{\mathbf{u}}} \right) = 0 \Rightarrow \mathbf{u}^\top \mathbf{Q} + \mathbf{u}^\top \mathbf{Q}^\top + \boldsymbol{\lambda}^\top \mathbf{B} = 0 \quad (5.3.3)$$

$$\frac{\partial \varphi}{\partial \boldsymbol{\lambda}^\top} - \frac{d}{dt} \left(\frac{\partial \varphi}{\partial \dot{\boldsymbol{\lambda}}^\top} \right) = 0 \Rightarrow \dot{\mathbf{x}} = \mathbf{Ax} + \mathbf{Bu} \quad (5.3.4)$$

Let $\lambda = \mathbf{P}\mathbf{x}$, where $\mathbf{P}$ is positive definite symmetric matrix, So $\lambda^\top = \mathbf{x}^\top \mathbf{P}$ and $\dot{\lambda}^\top = \dot{\mathbf{x}}^\top \mathbf{P} = (\mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u})^\top \mathbf{P} = (\mathbf{x}^\top \mathbf{A}^\top + \mathbf{u}^\top \mathbf{B}^\top) \mathbf{P}$, replacing for $\mathbf{Q} = \frac{\mathbf{Q} + \mathbf{Q}^\top}{2}$, $\mathbf{R} = \frac{\mathbf{R} + \mathbf{R}^\top}{2}$, $\lambda^\top$ and $\dot{\lambda}^\top$, the Euler Lagrangian equation become:

$$\begin{aligned} \mathbf{x}^\top \mathbf{Q} + \mathbf{x}^\top \mathbf{P}\mathbf{A} + (\mathbf{x}^\top \mathbf{A}^\top + \mathbf{u}^\top \mathbf{B}^\top) \mathbf{P} &= 0 \\ \mathbf{x}^\top (\mathbf{Q} + \mathbf{P}\mathbf{A} + \mathbf{A}^\top \mathbf{P}) + \mathbf{u}^\top \mathbf{B}^\top \mathbf{P} &= 0 \end{aligned} \quad (5.3.5)$$

$$\mathbf{u}^\top \mathbf{R} + \mathbf{x}^\top \mathbf{P}\mathbf{B} = 0 \quad (5.3.6)$$

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} \quad (5.3.7)$$

If the control law is design based on state feedback $\mathbf{u} = -\mathbf{K}_f \mathbf{x}$, optimizing $\mathbf{u}(t)$ is same as determining the optimal state feedback gain matrix $\mathbf{K}_f$, hence with state feedback control law, equation 5.3.5 becomes

$$\begin{aligned} \mathbf{x}^\top (\mathbf{Q} + \mathbf{P}\mathbf{A} + \mathbf{A}^\top \mathbf{P}) \mathbf{x} + \mathbf{x}^\top \mathbf{K}_f^\top \mathbf{B}^\top \mathbf{P} &= 0 \\ \mathbf{x}^\top (\mathbf{Q} + \mathbf{P}\mathbf{A} + \mathbf{A}^\top \mathbf{P} - \mathbf{K}_f^\top \mathbf{B}^\top \mathbf{P}) &= 0 \\ \mathbf{Q} + \mathbf{P}\mathbf{A} + \mathbf{A}^\top \mathbf{P} - \mathbf{K}_f^\top \mathbf{B}^\top \mathbf{P} &= 0 \end{aligned} \quad (5.3.8)$$

and from equation 5.3.6.

$$\begin{aligned} -\mathbf{x}^\top \mathbf{K}_f^\top \mathbf{R} + \mathbf{x}^\top \mathbf{P}\mathbf{B} &= 0 \\ \mathbf{x}^\top (-\mathbf{K}_f^\top \mathbf{R} + \mathbf{P}\mathbf{B}) &= 0 \\ \mathbf{K} &= \mathbf{R}^{-1} \mathbf{B}^\top \mathbf{P} \end{aligned} \quad (5.3.9)$$

The control law $\mathbf{u}(t)$ that optimizes the functional (cost function) is,

$$\mathbf{u} = -\mathbf{K}_f \mathbf{x}(t) = -\mathbf{R}^{-1} \mathbf{B}^\top \mathbf{P}\mathbf{x} \quad (5.3.10)$$

$\mathbf{K}$ is the optimal feedback gain matrix that optimizes $\mathbf{u}(t)$, which in turn optimizes our optimal linear quadratic regulator cost function, subjected to $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}$.

Replacing $\mathbf{K}_f$ in equation 5.3.8 results in a Riccati equation in which we solve for the matrix $\mathbf{P}$.

$$\mathbf{Q} + \mathbf{P}\mathbf{A} + \mathbf{A}^\top \mathbf{P} - \mathbf{P}\mathbf{B}\mathbf{R}^{-1} \mathbf{B}^\top \mathbf{P} = 0 \quad (5.3.11)$$

given that $(\mathbf{A}, \mathbf{B})$ is stabilizable.

The other method of finding solution for linear quadratic optimal regulator without using Euler-Lagrangian equations achieved is by letting $\mathbf{x}^\top (\mathbf{Q} + \mathbf{K}_f^\top \mathbf{R} \mathbf{K}_f) \mathbf{x} = -\frac{d}{dt} (\mathbf{x}^\top \mathbf{P} \mathbf{x})$ after replacing for $\mathbf{u}$ in equation 5.1.7 and final state $\mathbf{x}_f = 0$ (same control objective bring the system to zero final state). which results in

$$\begin{aligned} \mathbf{x}^\top (\mathbf{Q} + \mathbf{K}_f^\top \mathbf{R} \mathbf{K}_f) \mathbf{x} &= -\dot{\mathbf{x}}^\top \mathbf{P} \mathbf{x} - \mathbf{x}^\top \mathbf{P} \dot{\mathbf{x}} \\ \mathbf{x}^\top (\mathbf{Q} + \mathbf{K}_f^\top \mathbf{R} \mathbf{K}_f) \mathbf{x} &= -\mathbf{x}^\top (\mathbf{A}^\top - \mathbf{K}_f^\top \mathbf{B}^\top) \mathbf{P} \mathbf{x} - \mathbf{x}^\top \mathbf{P} (\mathbf{A} - \mathbf{B} \mathbf{K}_f) \mathbf{x} \\ \mathbf{A}^\top \mathbf{P} - \mathbf{K}_f^\top \mathbf{B}^\top \mathbf{P} + \mathbf{P}\mathbf{A} - \mathbf{P}\mathbf{B} \mathbf{K}_f &= -(\mathbf{Q} + \mathbf{K}_f^\top \mathbf{R} \mathbf{K}_f) \end{aligned} \quad (5.3.12)$$

Finding $\mathbf{u}$ that minimize J is same as finding $\mathbf{K}_f$ that minimize J , which in-turn minimizes equation 5.3.11. Derivative of equation 5.3.11 with respect to $\mathbf{K}_f$ to get $\mathbf{K}_f$ that minimize the equation result in:

$$\begin{aligned} (\mathbf{B}^\top \mathbf{P})^\top + \mathbf{P}\mathbf{B} - (\mathbf{R}\mathbf{K}_f)^\top + \mathbf{K}_f^\top \mathbf{R} &= 0 \\ \mathbf{K}_f &= \mathbf{R}^{-1} \mathbf{B}^\top \mathbf{P} \end{aligned} \quad (5.3.13)$$

Since $\mathbf{R} = \mathbf{R}^\top$ are $\mathbf{P} = \mathbf{P}^\top$ symmetric positive definite matrices.

So the control signal that minimizes the optimal quadratic regulator function is

$$\mathbf{u} = -\mathbf{K}_f \mathbf{x} = -\mathbf{R}^{-1} \mathbf{B}^\top \mathbf{P} \mathbf{x} \quad (5.3.14)$$

With this control law, the closed-loop system becomes

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} \Rightarrow \dot{\mathbf{x}} = (\mathbf{A} - \mathbf{B}\mathbf{K}_f)\mathbf{x}$$

P is determined from Riccati equation after replacing for $\mathbf{K}_f = \mathbf{R}^{-1} \mathbf{B}^\top \mathbf{P}$ in equation 5.3.12 which is the same as equation 5.3.11. The Block diagram representation of a system with an optimal state feedback controller same as a full state feedback controller, is shown in Figure 5.1.

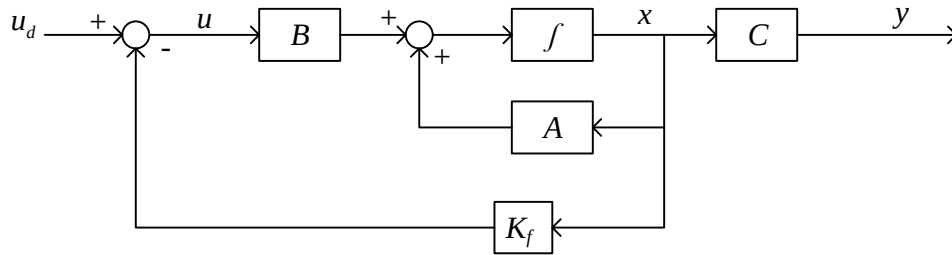


Figure 5.1: Block Diagram Representation of LTI System with Optimal State Feedback Controller

5.4 Optimal Steady State Reference Tracking

In control systems, it is often desired to achieve fast convergence of the controlled output $\mathbf{y}$ to a specific nonzero constant set-point value $\mathbf{r}$. The set-point values $(\mathbf{x}_d, \mathbf{u}_d)$, $\mathbf{x}_d$, represents the desired state of the system and while $\mathbf{u}_d$ represent the desired control signal that the output of system require to follow the reference input.

$$\begin{aligned}\dot{\mathbf{x}} &= \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} \\ \mathbf{y} &= \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u}\end{aligned}\tag{5.4.1}$$

At this point, the rate of change of the state variables will be zero, and the output is expected to be equal to the reference values. Hence for a pair $(\mathbf{x}_d, \mathbf{u}_d)$ equation 5.4.1 will be

$$\begin{aligned}\mathbf{A}\mathbf{x}_d + \mathbf{B}\mathbf{u}_d &= 0 \\ \mathbf{r} &= \mathbf{C}\mathbf{x}_d + \mathbf{D}\mathbf{u}_d\end{aligned}\tag{5.4.2}$$

$$\tag{5.4.3}$$

$$\begin{bmatrix} -\mathbf{A} & \mathbf{B} \\ -\mathbf{C} & \mathbf{D} \end{bmatrix} \begin{bmatrix} -\mathbf{x}_d \\ \mathbf{u}_d \end{bmatrix} = \begin{bmatrix} 0 \\ \mathbf{r} \end{bmatrix}\tag{5.4.4}$$

This corresponds to an LQR criterion of the form

$$J = \int_{t_0}^{t_f} (\tilde{\mathbf{y}}^\top \mathbf{Q} \tilde{\mathbf{y}} + \tilde{\mathbf{u}}^\top \mathbf{R} \tilde{\mathbf{u}}) dt\tag{5.4.5}$$

where $\tilde{\mathbf{y}} = \mathbf{y} - \mathbf{r}$ and $\tilde{\mathbf{u}} = \mathbf{u} - \mathbf{u}_d$. To understand when this is possible, three distinct cases should be considered:

- **Case-I:** When the number of inputs m , is strictly smaller than the number of controlled outputs p , we have an under-actuated system. In this case, the system of equations 5.4.4 generally does not have a solution because it presents more equations than unknowns.
- **Case-II:** When the number of inputs m is equal to the number of controlled outputs p , equation 5.4.4 always has a solution as long as the Rosenbrock's ¹ system matrix

$$P(s) = \begin{bmatrix} s\mathbf{I} - \mathbf{A} & \mathbf{B} \\ -\mathbf{C} & \mathbf{D} \end{bmatrix}\tag{5.4.6}$$

is non-singular for $s = 0$. This means that $s = 0$ should not be an invariant zero of the system and therefore it cannot also be a transmission zero of the transfer function $\mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D}$.

¹Rosenbrock's matrix is obtained by regarding the controller output as the only output of the system.

Intuitively, one should expect problems when $s = 0$ is an invariant zero of the system because when the state converges to an equilibrium point, the control input $\mathbf{u}(t)$ must converge to a constant. By the zero-blocking property, one should then expect the controlled output $\mathbf{y}(t)$ to converge to zero and not to $\mathbf{r}$.

- **Case-III:** When the number of inputs m is strictly larger than the number of controlled outputs p , we have an over-actuated system and equation 5.4.4 generally has multiple solutions.

When $\mathbf{P}(0)$ is full row-rank, i.e., when it has $n + p$ linearly independent rows, the $(n + p) \times (n + p)$ matrix $\mathbf{P}(0)\mathbf{P}(0)^\top$ is non-singular and one solution to equation 5.4.4 is given by

$$\begin{bmatrix} \mathbf{x}_d \\ \mathbf{u}_d \end{bmatrix} = \mathbf{P}(0)^\top (\mathbf{P}(0)\mathbf{P}(0)^\top)^{-1} \begin{bmatrix} \mathbf{0} \\ \mathbf{r} \end{bmatrix} \quad (5.4.7)$$

Also in this case, $s = 0$ should not be an invariant zero of the system because otherwise $\mathbf{P}(0)$ cannot be full rank.

The optimal set-point problem can be reduced to that of optimal regulation by considering an auxiliary system with state $\tilde{\mathbf{x}} = \mathbf{x} - \mathbf{x}_d$, $\tilde{\mathbf{u}} = \mathbf{u} - \mathbf{u}_d$ whose dynamics are:

$$\begin{aligned} \dot{\tilde{\mathbf{x}}} &= \mathbf{A}(\mathbf{x} - \mathbf{x}_d) + \mathbf{B}(\mathbf{u} - \mathbf{u}_d) \\ \dot{\tilde{\mathbf{x}}} &= \mathbf{A}\tilde{\mathbf{x}} + \mathbf{B}\tilde{\mathbf{u}} \end{aligned} \quad (5.4.8)$$

We can then regard equation 5.4.5 and equation 5.4.8 as an optimal regulation problem for which the optimal solution is given by:

$$\tilde{\mathbf{u}} = -\mathbf{K}_f \tilde{\mathbf{x}} \quad (5.4.9)$$

Going back to the original input and state variables $\mathbf{u}$ and $\mathbf{x}$, we conclude that the optimal control for the set-point problem defined by equation 5.4.1 and equation 5.4.5 is given by

$$\mathbf{u} = -\mathbf{K}_f(\mathbf{x} - \mathbf{x}_d) + \mathbf{u}_d \quad (5.4.10)$$

Since the solution to equation 5.4.4 can be written in the form

$$\mathbf{x}_d = \mathbf{F}\mathbf{r} \quad (5.4.11)$$

$$\mathbf{u}_d = \mathbf{N}\mathbf{r} \quad (5.4.12)$$

The values of $\mathbf{F}$ and $\mathbf{N}$ are determined using equation 5.4.2, by replacing for $\mathbf{x}_e = \mathbf{F}\mathbf{r}$ and $\mathbf{u}_d = \mathbf{N}\mathbf{r}$:

$$\mathbf{A}\mathbf{x}_d + \mathbf{B}\mathbf{u}_d = 0 \Rightarrow \mathbf{A}\mathbf{F}\mathbf{r} + \mathbf{B}\mathbf{N}\mathbf{r} = 0 \Rightarrow (\mathbf{A}\mathbf{F} + \mathbf{B}\mathbf{N}) = 0 \quad (5.4.13)$$

$$\mathbf{C}\mathbf{x}_d + \mathbf{D}\mathbf{u}_d = \mathbf{r} \Rightarrow \mathbf{C}\mathbf{F}\mathbf{r} + \mathbf{D}\mathbf{N}\mathbf{r} = \mathbf{r} \Rightarrow (\mathbf{C}\mathbf{F} + \mathbf{D}\mathbf{N}) = \mathbf{I} \quad (5.4.14)$$

Solving for $\mathbf{F}$ from equation 5.4.13 and replacing in equation 5.4.14, the values for matrices $\mathbf{N}$ and $\mathbf{F}$ is:

$$\mathbf{N} = (\mathbf{D} - \mathbf{C}\mathbf{A}^{-1}\mathbf{B})^{-1} \quad (5.4.15)$$

$$\mathbf{F} = -\mathbf{A}^{-1}\mathbf{B}(\mathbf{D} - \mathbf{C}\mathbf{A}^{-1}\mathbf{B})^{-1} \quad (5.4.16)$$

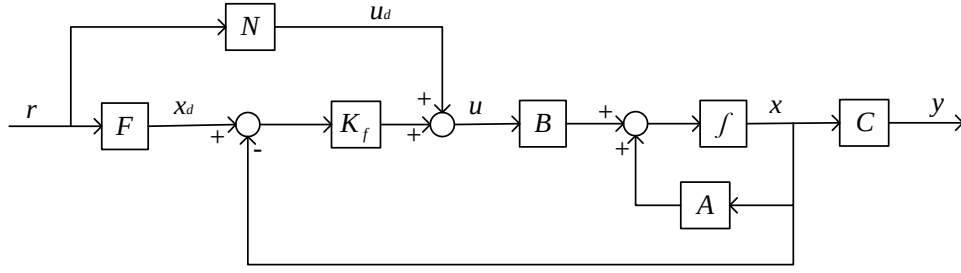


Figure 5.2: Steady State Reference Tracking

The other method is at steady state

$$\mathbf{u} = -\mathbf{K}_f \mathbf{x} + \mathbf{K}_o \mathbf{r} \quad (5.4.17)$$

replacing for $\mathbf{u}$ in the state space equation;

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} - \mathbf{B}\mathbf{u} \Rightarrow \dot{\mathbf{x}} = (\mathbf{A} - \mathbf{B}\mathbf{K}_f)\mathbf{x} + \mathbf{B}\mathbf{K}_o \mathbf{r} \quad (5.4.18)$$

as time goes to infinity ($t \rightarrow \infty$) the state $\mathbf{x}$, goes to the desired state $\mathbf{x}_d$ and the input $\mathbf{u}$, tends toward desired input $\mathbf{u}_d$, having dynamic equation:

$$\dot{\mathbf{x}}_d = \mathbf{A}\mathbf{x}_d + \mathbf{B}\mathbf{u}_d \Rightarrow \dot{\mathbf{x}}_d = (\mathbf{A} - \mathbf{B}\mathbf{K}_f)\mathbf{x}_d + \mathbf{B}\mathbf{K}_o \mathbf{r} \quad (5.4.19)$$

$$\mathbf{u}_d = -\mathbf{K}_f \mathbf{x}_d + \mathbf{K}_o \mathbf{r} \quad (5.4.20)$$

the tracking error $\tilde{\mathbf{x}} = \mathbf{x} - \mathbf{x}_d$ is then

$$\dot{\tilde{\mathbf{x}}} = (\mathbf{A} - \mathbf{B}\mathbf{K}_f)\tilde{\mathbf{x}} \quad (5.4.21)$$

since $\dot{\mathbf{x}}_d$ is zero at $t \rightarrow \infty$, then the dynamic equation 5.4.19, can be written as

$$\mathbf{x}_d = -(\mathbf{A} - \mathbf{B}\mathbf{K}_f)^{-1} \mathbf{B}\mathbf{K}_o \mathbf{r} \quad (5.4.22)$$

replacing equation 5.4.22 in equation 5.4.20, we have

$$\begin{aligned} \mathbf{u}_d &= \mathbf{K}_f(\mathbf{A} - \mathbf{B}\mathbf{K}_f)^{-1} \mathbf{B}\mathbf{K}_o \mathbf{r} + \mathbf{K}_o \mathbf{r} \\ &= (\mathbf{K}_f(\mathbf{A} - \mathbf{B}\mathbf{K}_f)^{-1} \mathbf{B}\mathbf{K}_o + \mathbf{K}_o) \mathbf{r} \end{aligned} \quad (5.4.23)$$

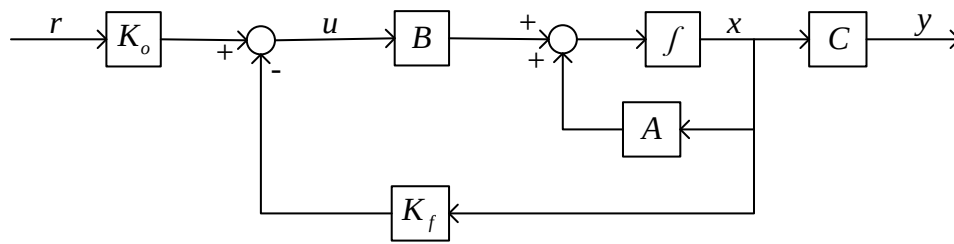
and the output of the dynamic system $\mathbf{y}$, goes to $\mathbf{r}$ as $t \rightarrow \infty$,

$$\mathbf{y} = \mathbf{C}\mathbf{x}_d + \mathbf{D}\mathbf{u}_d = \mathbf{r} \quad (5.4.24)$$

Form equation 5.4.24, by replacing for $\mathbf{u}_d$ and $\mathbf{x}_d$, the value of matrix $\mathbf{K}_o$ is:

$$\mathbf{K}_o = -(\mathbf{C}(\mathbf{A} - \mathbf{B}\mathbf{K}_f)^{-1} \mathbf{B} - \mathbf{D}((\mathbf{K}_f(\mathbf{A} - \mathbf{B}\mathbf{K}_f)^{-1} \mathbf{B} + \mathbf{I}))^{-1} \quad (5.4.25)$$

In cases where the states of the dynamic system are not directly accessible, a state observer can be utilized to estimate the state of the system, thereby allowing the use of the estimated state for control purposes.

Figure 5.3: *Steady State Reference Tracking*

5.5 Optimal State Estimation

The state feedback controller gain matrix design problem involves finding a gain matrix that can stabilize a given linear system by controlling its states. On the other hand, the state estimator gain matrix design problem involves finding a gain matrix that can estimate the states of a given linear system based on its input and output signals. Interestingly, these two problems obey a duality property. This means that the solution to one problem can be used to solve the other problem. In this context, the problem of finding the state feedback controller gain matrix using a linear quadratic regulator (LQR) can be used to design an optimal state estimator gain matrix for linear time-invariant systems.

The LQR approach is a widely used method for designing state feedback controllers. It involves minimizing a quadratic cost function that captures the system's performance and control effort. The optimal state feedback controller gain matrix is obtained by solving the associated algebraic Riccati equation. Once the controller gain matrix is obtained, it can be used to stabilize the system and achieve the desired performance.

On the other hand, the problem of designing an optimal state estimator gain matrix involves minimizing a similar quadratic cost function that captures the difference between the actual state and the estimated state. The optimal estimator gain matrix is obtained by solving another algebraic Riccati equation. Once the estimator gain matrix is obtained, it can be used to estimate the system's state accurately.

The duality property between these two problems arises because the same cost function is used in both cases, but with different weighting matrices. In other words, the solution to one problem can be used to solve the other problem by simply swapping the weighting matrices.

Consider a linear time-invariant system described by the following state-space model:

$$\begin{aligned}\dot{\mathbf{x}} &= \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} \\ \mathbf{y} &= \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u}\end{aligned}\tag{5.5.1}$$

where $\mathbf{x}$ is the state vector, $\mathbf{u}$ is the control input, $\mathbf{y}$ is the output, and $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$, and $\mathbf{D}$ are matrices that describe the system's dynamics.

To design an optimal state feedback controller gain matrix using LQR, we first define the cost function as follows:

$$J = \int_{t_0}^{t_f} ((\hat{\mathbf{x}} - \mathbf{x})^\top \mathbf{Q}(\hat{\mathbf{x}} - \mathbf{x}) + \mathbf{u}^\top \mathbf{R}\mathbf{u}) dt \quad (5.5.2)$$

where $\mathbf{Q}$ and $\mathbf{R}$ are weighting matrices that capture the relative importance of the state and control effort, respectively. The optimal controller gain matrix $\mathbf{K}_f$ is obtained by solving the following algebraic Riccati equation:

$$\mathbf{Q} + \mathbf{P}\mathbf{A} + \mathbf{A}^\top \mathbf{P} - \mathbf{P}\mathbf{B}\mathbf{R}^{-1}\mathbf{B}^\top \mathbf{P} = 0 \quad (5.5.3)$$

where $\mathbf{P}$ is any symmetric positive definite matrix.

Once $\mathbf{K}_f = \mathbf{R}^{-1}\mathbf{B}^\top \mathbf{P}$ is obtained, it can be used to stabilize the system and achieve the desired performance.

To design an optimal state estimator gain matrix using LQR, we define a similar cost function as follows:

$$J = \int_{t_0}^{t_f} ((\hat{\mathbf{x}} - \mathbf{x})^\top \mathbf{Q}(\hat{\mathbf{x}} - \mathbf{x}) + (\mathbf{y}^\top \mathbf{R}_y \mathbf{y})) dt \quad (5.5.4)$$

where $\mathbf{Q}$ and $\mathbf{R}_y$ are weighting matrices that capture the relative importance of the output and state estimation error, respectively. The optimal estimator gain matrix $\mathbf{L}_c$ is obtained by solving the following algebraic Riccati equation:

$$\mathbf{Q} + \mathbf{P}\mathbf{A} + \mathbf{A}^\top \mathbf{P} - \mathbf{P}\mathbf{C}^\top \mathbf{R}_y^{-1} \mathbf{C}\mathbf{P} = 0 \quad (5.5.5)$$

where $\mathbf{P}$ is any symmetric positive definite matrix.

Once $\mathbf{L}_c = \mathbf{P}\mathbf{C}^\top \mathbf{R}_y^{-1}$ is obtained, it can be used to estimate the system's state accurately.

The duality property between the two problems arises from the fact that they share the same mathematical structure. Specifically, the state feedback controller gain matrix design problem can be formulated as a quadratic optimization problem using the linear quadratic regulator (LQR) technique. Similarly, the state estimator gain matrix design problem can be formulated as a quadratic optimization problem using the linear quadratic estimator (LQE) technique.

The LQR and LQE techniques are based on minimizing a cost function that measures the deviation of the system's states from a desired trajectory and the deviation of the estimated states from the actual states, respectively. The cost function is formulated as a quadratic function of the system's states and inputs, and its coefficients are determined by the system's dynamics and performance specifications.

The duality property between the two problems implies that the optimal gain matrices obtained from the LQR and LQE techniques are related to each other. Specifically, the optimal state feedback controller gain matrix is equal to the transpose of the optimal state estimator gain matrix, up to a scaling factor. This means that if we solve the state feedback controller gain matrix design problem using LQR, we can obtain the optimal state estimator gain matrix by transposing the solution and scaling it appropriately.

5.6 Examples

1. Determine the increment of the following functional

$$J = \int_{t_0}^{t_f} (2x^2(t) + 1) dt \quad (5.6.1)$$

Solution

$$\begin{aligned} \Delta J &= J(x(t) + \delta x(t)) - J(x(t)) \\ &= \int_{t_0}^{t_f} (2(x(t) + \delta x(t))^2 + 1) dt - \int_{t_0}^{t_f} (2x^2(t) + 1) dt \\ &= \int_{t_0}^{t_f} (2x^2(t) + 4x(t)\delta x(t) + 2(\delta x(t))^2 + 1 - 2x^2(t) - 1) dt \\ &= \int_{t_0}^{t_f} (4x(t)\delta x(t) + 2(\delta x(t))^2) dt \end{aligned}$$

2. Find the first variation of the following function

$$J = \int_{t_0}^{t_f} (2x^2(t) + 1) dt$$

Solution

$$\begin{aligned} \delta J &= \frac{\partial J}{\partial x} \delta x(t) \\ &= \int_{t_0}^{t_f} \frac{\partial}{\partial x} (4x(t)\delta x(t) + 2(\delta x(t))^2) \delta x(t) dt \\ &= \int_{t_0}^{t_f} (4\delta x(t) + 4\delta x(t)) \delta x(t) dt \\ &= \int_{t_0}^{t_f} 8(\delta x(t))^2 dt \end{aligned}$$

3. Find x that minimizes the following cost function

$$J = \int_0^{\frac{\pi}{2}} (x^2 - \dot{x}^2) dt \quad (5.6.2)$$

with boundary condition $x(0) = 0$, $x(\frac{\pi}{2}) = 1$

Solution:

From the cost function we know that $\varphi(x, \dot{x}) = x^2 - \dot{x}^2$, from Euler Lagrangian equations

$$\frac{\partial \varphi}{\partial x} - \frac{d}{dt} \left(\frac{\partial \varphi}{\partial \dot{x}} \right) = 0$$

which results in a homogeneous second-order system

$$\ddot{x} + x = 0$$

The characteristic equation of this system is

$$\lambda^2 + \lambda = 0 \Rightarrow \lambda_1 = -i, \lambda_2 = i$$

The solution to this system has a form

$$\begin{aligned} x &= C_1 e^{\lambda_1 t} + C_2 e^{\lambda_2 t} \\ &= C_1 e^{-it} + C_2 e^{it} \\ &= (C_1 + C_2) \cos t + i(C_2 - C_1) \sin t \end{aligned}$$

Using boundary condition to solve for the constants C_1 and C_2 , at $t = 0, \mathbf{x} = 0$

$$\begin{aligned} (C_1 + C_2) \cos 0 + i(C_2 - C_1) \sin 0 &= 0 \\ (C_1 + C_2) &= 0 \end{aligned}$$

at $t = \frac{\pi}{2}, x = 1$

$$\begin{aligned} (C_1 + C_2) \cos \frac{\pi}{2} + i(C_2 - C_1) \sin \frac{\pi}{2} &= 1 \\ (C_2 - C_1) &= 1 \end{aligned}$$

Solving the two equations simultaneously results in $C_1 = 0.5$ and $C_2 = -0.5$. Hence, the value of x that minimizes the cost function is

$$x(t) = \sin(t)$$

4. Find u that minimizes the following cost function

$$J = \int_0^1 \frac{1}{2} u^2 dt \quad (5.6.3)$$

subjected to

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$$

with initial and final states $\mathbf{x}(0) = [1 \ 1]^\top$ and $\mathbf{x}(1) = [0 \ 0]^\top$.

Solution:

From constrained optimization we know that $\varphi(\mathbf{x}, \dot{\mathbf{x}}, u) = \frac{1}{2}u^2 + \lambda^\top (\mathbf{A}\mathbf{x} + \mathbf{B}u - \dot{\mathbf{x}})$.
The control law we determine should minimize

$$J = \int_0^1 \left(\frac{1}{2}u^2 + \lambda_1(x_2 - \dot{x}_1) + \lambda_2(u - \dot{x}_2) \right) dt$$

Using the Euler-Lagrange equations

$$\frac{\partial \varphi}{\partial x_1} - \frac{d}{dt} \left(\frac{\partial \varphi}{\partial \dot{x}_1} \right) = 0 \Rightarrow 0 + \dot{\lambda}_1 = 0 \Rightarrow \dot{\lambda}_1 = 0$$

$$\frac{\partial \varphi}{\partial \lambda_1} - \frac{d}{dt} \left(\frac{\partial \varphi}{\partial \dot{\lambda}_1} \right) = 0 \Rightarrow x_2 - \dot{x}_1 = 0 \Rightarrow \dot{x}_1 = x_2$$

result in the state equation.

$$\frac{\partial \varphi}{\partial x_2} - \frac{d}{dt} \left(\frac{\partial \varphi}{\partial \dot{x}_2} \right) = 0 \Rightarrow \lambda_1 + \dot{\lambda}_2 = 0 \Rightarrow \dot{\lambda}_2 = -\lambda_1$$

$$\frac{\partial \varphi}{\partial \lambda_2} - \frac{d}{dt} \left(\frac{\partial \varphi}{\partial \dot{\lambda}_2} \right) = 0 \Rightarrow u - \dot{x}_2 = 0 \Rightarrow \dot{x}_2 = u$$

result in the state equation.

$$\frac{\partial \varphi}{\partial u} - \frac{d}{dt} \left(\frac{\partial \varphi}{\partial \dot{u}} \right) = 0 \Rightarrow u - \lambda_2 = 0 \Rightarrow \lambda_2 = -u$$

Solving these equations

$$\begin{aligned} \dot{\lambda}_1 &= 0 \Rightarrow \lambda_1 = C_1, \\ \dot{\lambda}_2 &= -\lambda_1 \Rightarrow \lambda_2 = -C_1 t + C_2 \\ u(t) &= -\lambda_2 \Rightarrow u(t) = C_1 t + C_2 \end{aligned}$$

from state equations

$$\begin{aligned} \dot{x}_2 &= u \Rightarrow \dot{x}_2 = C_1 t + C_2 \Rightarrow x_2(t) = \frac{C_1}{2} t^2 + C_2 t + C_3 \\ \dot{x}_1 &= x_2 \Rightarrow \dot{x}_1 = \frac{C_1}{2} t^2 + C_2 t + C_3 \Rightarrow x_1(t) = \frac{C_1}{6} t^3 + \frac{C_2}{2} t^2 + C_3 t + C_4 \end{aligned}$$

Using the boundary conditions to solve for the constants C_1, C_2, C_3, C_4 , we have the following equations

$$\begin{aligned} x_2(0) &\Rightarrow C_3 = 1, & x_1(0) &\Rightarrow C_1 - 2C_2 + 2 = 0 \\ x_2(1) &\Rightarrow C_3 = 1, & x_1(1) &\Rightarrow C_1 - 3C_2 + 12 = 0 \end{aligned}$$

solving this equations $C_1 = 18$ and $C_2 = 10$. Hence, the control law that optimizes the constrained cost function is

$$u(t) = 18t - 10$$

5. For the following system, determine an optimal full state feedback gain matrix $\mathbf{K}_f$.

$$\dot{\mathbf{x}} = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} \mathbf{x}$$

Solution

From Riccati equation

$$\mathbf{Q} + \mathbf{P}\mathbf{A} + \mathbf{A}^\top \mathbf{P} - \mathbf{P}\mathbf{B}\mathbf{R}^{-1}\mathbf{B}^\top \mathbf{P} = 0$$

Let

$$\mathbf{P} = \begin{bmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{bmatrix} \quad \mathbf{Q} = \mathbf{C}_1^\top \mathbf{C}_1 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \quad \mathbf{R} = 0.5;$$

$$\begin{aligned} & \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix} + \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{bmatrix} \\ & - \begin{bmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} 2 \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{bmatrix} = 0 \\ & \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} -p_{11} & -2p_{12} \\ -p_{21} & -2p_{22} \end{bmatrix} + \begin{bmatrix} -p_{11} & -p_{12} \\ -2p_{21} & -2p_{22} \end{bmatrix} - 2 \begin{bmatrix} p_{12}p_{21} & p_{12}p_{22} \\ p_{22}p_{21} & p_{22}p_{22} \end{bmatrix} = 0 \\ & \begin{bmatrix} 1 - 2p_{11} - 2p_{12}p_{21} & -3p_{12} - 2p_{12}p_{22} \\ -3p_{12} - 2p_{12}p_{22} & -4p_{22} - 2p_{22}^2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \end{aligned}$$

Since $\mathbf{P}$ must be symmetric $p_{12} = p_{21}$, which imply

$$\begin{bmatrix} 1 - 2p_{11} - 2p_{12}^2 & -3p_{12} - 2p_{12}p_{22} \\ -3p_{12} - 2p_{12}p_{22} & -4p_{22} - 2p_{22}^2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Solving for $\mathbf{P}$

$$\begin{aligned} -4p_{22} - 2p_{22}^2 &= 0 \Rightarrow p_{22} = 0 \text{ or } p_{22} = -2 \\ -3p_{12} - 2p_{12}p_{22} &= 0 \\ 1 - 2p_{11} - 2p_{12}^2 &= 0 \end{aligned}$$

Solving for p_{12} and p_{11}

(a) Case I $p_{22} = 0$

$$\begin{aligned} -3p_{12} - 2p_{12}p_{22} &= 0 \Rightarrow p_{12}(3 + 2p_{22}) = 0 \Rightarrow p_{12} = 0 \\ 1 - 2p_{11} - 2p_{12}^2 &= 0 \Rightarrow 1 - 2p_{11} = 0 \Rightarrow p_{11} = 0.5 \end{aligned}$$

(b) Case II $p_{22} = -2$

$$\begin{aligned} -3p_{12} - 2p_{12}p_{22} &= 0 \Rightarrow p_{12}(3 + 2p_{22}) = 0 \Rightarrow p_{12} = 0 \\ 1 - 2p_{11} - 2p_{12}^2 &= 0 \Rightarrow 1 - 2p_{11} = 0 \Rightarrow p_{11} = 0.5 \end{aligned}$$

So we have

$$\mathbf{P}_1 = \begin{bmatrix} 0.5 & 0 \\ 0 & 0 \end{bmatrix} \quad \mathbf{P}_2 = \begin{bmatrix} 0.5 & 0 \\ 0 & -0.8 \end{bmatrix}$$

Since $\mathbf{P}$ has to be positive definite (all eigenvalues of $\mathbf{P}$ must be greater than or equal to zero), the solution for $\mathbf{P}$ is $\mathbf{P}_1$. The value of feedback gain matrix $\mathbf{K}_f$ is

$$\mathbf{K}_f = \mathbf{R}^{-1}\mathbf{B}^T\mathbf{P} = \begin{bmatrix} 0 & 0 \end{bmatrix}$$

what does $\mathbf{K}_f = 0$ imply?

The control law becomes $\mathbf{u} = -\mathbf{K}_f\mathbf{x} = 0$, with this control law the closed loop system becomes

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} \Rightarrow \dot{\mathbf{x}} = (\mathbf{A} - \mathbf{B}\mathbf{K}_f)\mathbf{x}$$

Eigenvalues of closed loop system $\dot{\mathbf{x}} = (\mathbf{A} - \mathbf{B}\mathbf{K}_f)\mathbf{x}$ is

$$\begin{aligned} \|\lambda\mathbf{I} - (\mathbf{A} - \mathbf{B}\mathbf{K}_f)\| &= 0 \\ \left\| \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} - \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix} \right\| &= 0 \\ \left\| \begin{bmatrix} \lambda + 1 & 0 \\ 0 & \lambda + 2 \end{bmatrix} \right\| &= 0 \\ (\lambda + 1)(\lambda + 2) &= 0 \Rightarrow \lambda_1 = -1 \quad \lambda_2 = -2 \end{aligned}$$

If the error $\tilde{x}_1 = 0.01$ and $\tilde{x}_2 = 0.1$ is tolerable for the system, then $\mathbf{Q}$ will be

$$\mathbf{Q} = \begin{bmatrix} 1000 & 0 \\ 0 & 100 \end{bmatrix}$$

and $\mathbf{R} = 0.5$ with these parameters the value of $\mathbf{P}$ and $\mathbf{K}_f$ is

$$\mathbf{P} = \begin{bmatrix} 500 & 0 \\ 0 & 6.1414 \end{bmatrix} \quad \mathbf{K}_f = \begin{bmatrix} 0 & 12.2829 \end{bmatrix}$$

$u = -\mathbf{K}_f\mathbf{x} \Rightarrow u = -12.2829x_2$, with this control law the eigenvalues of closed loop

system $\dot{\mathbf{x}} = (\mathbf{A} - \mathbf{BK}_f) \mathbf{x}$ is

$$\begin{aligned} \|\lambda \mathbf{I} - (\mathbf{A} - \mathbf{BK}_f)\| &= 0 \\ \left\| \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} - \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 12.2829 \end{bmatrix} \right\| &= 0 \\ \left\| \begin{bmatrix} \lambda + 1 & 0 \\ 0 & \lambda + 14.2829 \end{bmatrix} \right\| &= 0 \\ (\lambda + 1)(\lambda + 14.2829) &= 0 \Rightarrow \lambda_1 = -1 \quad \lambda_2 = -14.2829 \end{aligned}$$

If

$$\mathbf{Q} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

and $\mathbf{R} = 20$ the value of $\mathbf{P}$ and $\mathbf{K}_f$ is

$$\mathbf{P} = \begin{bmatrix} 500 & 0 \\ 0 & 20 \end{bmatrix} \quad \mathbf{K} = \begin{bmatrix} 0 & 1 \end{bmatrix}$$

$u = -\mathbf{K}_f \mathbf{x} \Rightarrow u = -x_2$, with this control law the eigenvalues of closed loop system $\dot{\mathbf{x}} = (\mathbf{A} - \mathbf{BK}_f) \mathbf{x}$ is

$$\begin{aligned} \|\lambda \mathbf{I} - (\mathbf{A} - \mathbf{BK}_f)\| &= 0 \\ \left\| \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} - \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\| &= 0 \\ \left\| \begin{bmatrix} \lambda + 1 & 0 \\ 0 & \lambda + 3 \end{bmatrix} \right\| &= 0 \\ (\lambda + 1)(\lambda + 3) &= 0 \Rightarrow \lambda_1 = -1 \quad \lambda_2 = -3 \end{aligned}$$

6. For the following system, determine an optimal full state feedback gain matrix $\mathbf{K}_f$.

$$\begin{aligned} \dot{\mathbf{x}} &= \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u \\ y &= \begin{bmatrix} 1 & 0 \end{bmatrix} \mathbf{x} \end{aligned}$$

Solution

From Riccati equation

$$\mathbf{Q} + \mathbf{PA} + \mathbf{A}^\top \mathbf{P} - \mathbf{PBR}^{-1} \mathbf{B}^\top \mathbf{P} = 0$$

Let

$$\mathbf{P} = \begin{bmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{bmatrix} \quad \mathbf{Q} = \mathbf{C}_1^\top \mathbf{C}_1 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \quad \mathbf{R} = 1;$$

$$\begin{aligned}
& \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{bmatrix} \\
& - \begin{bmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{bmatrix} = 0 \\
& \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} p_{11} & p_{11} + 2p_{12} \\ p_{21} & p_{21} + 2p_{22} \end{bmatrix} + \begin{bmatrix} p_{11} & p_{12} \\ p_{11} + 2p_{21} & p_{21} + 2p_{22} \end{bmatrix} - \begin{bmatrix} p_{12}p_{21} & p_{12}p_{22} \\ p_{22}p_{21} & p_{22}p_{22} \end{bmatrix} = 0 \\
& \begin{bmatrix} 1 + 2p_{11} - p_{12}p_{21} & p_{11} + 3p_{12} - p_{12}p_{22} \\ p_{11} + 3p_{12} - p_{12}p_{22} & 2(p_{21} + 2p_{22}) - p_{22}^2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}
\end{aligned}$$

Solving for $\mathbf{P}$ results in two values

$$\begin{aligned}
\mathbf{P}_1 &= \begin{bmatrix} -0.3702 & -0.5095 \\ -0.5095 & 3.7265 \end{bmatrix} & \mathbf{P}_2 &= \begin{bmatrix} 19.4317 & 6.3138 \\ 6.3138 & 6.0777 \end{bmatrix} \\
\mathbf{P}_3 &= \begin{bmatrix} 1.4259 & -1.9626 \\ -1.9626 & 2.2735 \end{bmatrix} & \mathbf{P}_4 &= \begin{bmatrix} -0.4875 & 0.1584 \\ 0.1584 & -0.0777 \end{bmatrix}
\end{aligned}$$

Since $\mathbf{P}$ has to be positive definite (all eigenvalues of $\mathbf{P}$ must be greater than or equal to zero). Checking for the positive definiteness

$$\begin{aligned}
eig(\mathbf{P}_1) &= \begin{bmatrix} -0.4326 \\ 3.7890 \end{bmatrix} & eig(\mathbf{P}_2) &= \begin{bmatrix} 3.5653 \\ 21.9442 \end{bmatrix} \\
eig(\mathbf{P}_3) &= \begin{bmatrix} -0.1582 \\ 3.8575 \end{bmatrix} & eig(\mathbf{P}_4) &= \begin{bmatrix} -0.5415 \\ -0.0236 \end{bmatrix}
\end{aligned}$$

Hence $\mathbf{P} = \mathbf{P}_2 = \begin{bmatrix} 19.4317 & 6.3138 \\ 6.3138 & 6.0777 \end{bmatrix}$ with this value of $\mathbf{P}$, the value of feedback gain matrix $\mathbf{K}_f$ is

$$\mathbf{K}_f = \mathbf{R}^{-1}\mathbf{B}^\top \mathbf{P} = \begin{bmatrix} 6.3138 & 6.0777 \end{bmatrix}$$

The control law becomes $u = -\mathbf{K}_f \mathbf{x} = -6.3138x_1 - 6.0777x_2$, with this control law the eigenvalues of closed loop system $\dot{\mathbf{x}} = (\mathbf{A} - \mathbf{BK}_f) \mathbf{x}$ is

$$\begin{aligned}
& \|\lambda \mathbf{I} - (\mathbf{A} - \mathbf{BK}_f)\| = 0 \\
& \left\| \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} - \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 6.3138 & 6.0777 \end{bmatrix} \right\| = 0 \\
& \left\| \begin{bmatrix} \lambda - 1 & -1 \\ 6.3138 & \lambda + 4.0777 \end{bmatrix} \right\| = 0 \\
& (\lambda - 1)(\lambda + 4.0777) + 6.3138 = 0 \Rightarrow \lambda_1 = -2.9990 \quad \lambda_2 = -0.0787
\end{aligned}$$

7. For the following system, determine an optimal full state feedback gain matrix $\mathbf{K}$.

$$\dot{\mathbf{x}} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & -1 & 1 \\ 0 & 0 & -2 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \mathbf{x}$$

Solution

As shown in chapter two example 3, the system is not completely controllable. So, we have to decompose the system into controllable and uncontrollable modes as shown in chapter two example 3. So we can design a state feedback controller for controllable mode $(\mathbf{A}_{11}, \mathbf{B}_1)$. From Riccati equation

$$\mathbf{Q} + \mathbf{P}\mathbf{A} + \mathbf{A}^\top \mathbf{P} - \mathbf{P}\mathbf{B}\mathbf{R}^{-1}\mathbf{B}^\top \mathbf{P} = 0$$

Let

$$\mathbf{P} = \begin{bmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{bmatrix} \quad \mathbf{Q} = \mathbf{C}_1^\top \mathbf{C}_1 = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \quad \mathbf{R} = 2;$$

$$\begin{aligned} & \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} + \begin{bmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{bmatrix} \begin{bmatrix} 2 & -4 \\ 1 & 0 \end{bmatrix} + \begin{bmatrix} 2 & 1 \\ -4 & 0 \end{bmatrix} \begin{bmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{bmatrix} \\ & - \begin{bmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} 2 \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{bmatrix} = 0 \\ & \begin{bmatrix} -\frac{p_{11}^2}{2} + 4p_{11} + 2p_{12} + 1 & 2p_{12} - 4p_{11} + p_{22} - \frac{(p_{11}p_{12})}{2} + 1 \\ 2p_{12} - 4p_{11} + p_{22} - \frac{(p_{11}p_{12})}{2} + 1 & -\frac{p_{12}^2}{2} - 8p_{12} + 1 \end{bmatrix} = 0 \end{aligned}$$

using MATLAB command

```
syms p11 p12 p22
A = [2 -4; 1 0]; B = [1; 0]; C = [1 1]; Q = C' * C; R = 0.5; p21 = p12;
P = [p11 p12; p21 p22];
eqn = Q + P * A + (A') * P - P * B * (1/R) * (B') * P
eqn1 = eqn(1,1); eqn2 = eqn(2,1); eqn3 = eqn(2,2);
vars = [p11, p12, p22]
[p11, p12, p22] = solve(eqn1 == 0, eqn2 == 0, eqn3 == 0, vars);
P1 = double([p11(1,1) p12(1,1); p12(1,1) p22(1,1)])
P2 = double([p11(2,1) p12(2,1); p12(2,1) p22(2,1)])
P3 = double([p11(3,1) p12(3,1); p12(3,1) p22(3,1)])
P4 = double([p11(4,1) p12(4,1); p12(4,1) p22(4,1)])
p1 = eig(P1) p2 = eig(P2) p3 = eig(P3) p4 = eig(P4)
K = (1/R) * (B') * P2
```

P has four the following possible values,

$$\begin{aligned} \mathbf{P}_1 &= \begin{bmatrix} 4.0000 - 6.8188i & -16.1240 + 0.0000i \\ -16.1240 + 0.0000i & 15.0000 + 27.6981i \end{bmatrix} & \mathbf{P}_2 &= \begin{bmatrix} 8.3007 & 0.1240 \\ 0.1240 & 32.4696 \end{bmatrix} \\ \mathbf{P}_3 &= \begin{bmatrix} 4.0000 + 6.8188i & -16.1240 + 0.0000i \\ -16.1240 + 0.0000i & 15.0000 - 27.6981i \end{bmatrix} & \mathbf{P}_4 &= \begin{bmatrix} -0.3007 & 0.1240 \\ 0.1240 & -2.4696 \end{bmatrix} \end{aligned}$$

Since $\mathbf{P}$ has to be a positive definite matrix, the valid answer is (check using all eigenvalues of $\mathbf{P}$ ($\text{eig}(\mathbf{P})$))

$$\mathbf{P}_2 = \begin{bmatrix} 8.3007 & 0.1240 \\ 0.1240 & 32.4696 \end{bmatrix}$$

The corresponding full state feedback gain matrix $\mathbf{K}_f = \mathbf{R}^{-1} \mathbf{B}^\top \mathbf{P}$ is

$$\mathbf{K}_f = \begin{bmatrix} 4.1504 & 0.0620 \end{bmatrix}$$

We can also use MATLAB command $\mathbf{K}_f = \text{lqr}(\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D})$ to determine the optimal feedback gain matrix, which gives the same result.

If we change the input weighting matrix $\mathbf{R} = 0.5$, the value of $\mathbf{P}$ and $\mathbf{K}_f$ becomes

$$\mathbf{P} = \begin{bmatrix} 2.2733 & 0.1213 \\ 0.1213 & 8.4022 \end{bmatrix} \quad \mathbf{K}_f = \begin{bmatrix} 4.5466 & 0.2426 \end{bmatrix}$$

If we change the state weighting matrix

$$\mathbf{Q} = \begin{bmatrix} 10 & 0 \\ 0 & 10 \end{bmatrix}$$

with $\mathbf{R} = 0.5$ the value of $\mathbf{P}$ and $\mathbf{K}_f$ becomes

$$\mathbf{P} = \begin{bmatrix} 3.6458 & 1.0000 \\ 1.0000 & 19.8745 \end{bmatrix} \quad \mathbf{K} = \begin{bmatrix} 7.2915 & 2.0000 \end{bmatrix}$$

$u = -\mathbf{K}_f \mathbf{x} \Rightarrow u = -7.2915x_1 - 2x_2$, with this control law the eigenvalues of closed loop system $\dot{\mathbf{x}} = (\mathbf{A} - \mathbf{BK}_f) \mathbf{x}$ is

$$\begin{aligned} \|\lambda \mathbf{I} - (\mathbf{A} - \mathbf{BK}_f)\| &= 0 \\ \left\| \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} - \begin{bmatrix} 2 & -4 \\ 1 & 0 \end{bmatrix} + \begin{bmatrix} 7.2915 & 2 \\ 0 & 0 \end{bmatrix} \right\| &= 0 \\ \left\| \begin{bmatrix} \lambda + 5.2915 & 6 \\ -1 & \lambda \end{bmatrix} \right\| &= 0 \\ \lambda^2 + 5.2915\lambda + 6 &= 0 \Rightarrow \lambda_1 = -3.6457 \quad \lambda_2 = -1.6458 \end{aligned}$$

5.7 Exercises

1. A system described by

$$\dot{x} = -2x + u$$

and control u is required to minimize a cost function

$$J = \int_0^1 u^2 dt$$

find such u for $x(0) = 1$ and $x(1) = 0$.

2. For a system described by

$$\dot{x} = -2x + 2u$$

find a control law u that minimize the performance index

$$J = \int_0^1 (3x^2 + u^2) dt$$

with $\mathbf{x}(0) = 1$ and $x(1) = 0$.

3. Find a control law that minimizes

$$J = \frac{1}{2}(x_1(2) - 5)^2 + \frac{1}{2}(x_2(2) - 2)^2 + \frac{1}{2} \int_0^2 u^2 dt$$

for a dynamic system

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= -x_2 + u \end{aligned}$$

with $\mathbf{x}(0) = 0$ and $\mathbf{x}(2)$ is on a line $x_1 + 5x_2 = 15$.

4. Find a feedback control law that minimizes

$$J = \int_0^\infty (x_2^2 + \frac{1}{10}u^2) dt$$

subjected to

$$\begin{aligned}\dot{x}_1 &= -x_1 + u \\ \dot{x}_2 &= x_1\end{aligned}$$

5. Use Riccati equation to determine the feedback control law for a system

$$\dot{x} = -x + u$$

which minimize

$$J = \frac{1}{2} \int_0^1 (3x^2 + u^2) dt$$

6. Find an optimal feedback gain $\mathbf{K}_f$ for the following systems with the given initial conditions

(a)

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 \\ -4 & 0 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u, \quad \mathbf{x}_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

that minimize

$$J = \frac{1}{2} \int_0^\infty (\mathbf{x}^\top \mathbf{Q} \mathbf{x} + \mathbf{u}^\top \mathbf{R} \mathbf{u}) dt$$

(b)

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 \\ 2 & -1 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u, \quad \mathbf{x}_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

that minimize

$$J = \frac{1}{2} \int_0^\infty (\mathbf{x}^\top \mathbf{Q} \mathbf{x} + \mathbf{R} u^2) dt$$

(c)

$$\dot{\mathbf{x}} = \begin{bmatrix} -1 & 0 \\ 0 & -4 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u, \quad \mathbf{x}_0 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

that minimize

$$J = \frac{1}{2} \int_0^\infty (\mathbf{x}^\top \mathbf{Q} \mathbf{x} + \mathbf{u}^\top \mathbf{R} \mathbf{u}) dt$$

(d)

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 \\ -6 & -8 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u, \quad \mathbf{x}_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

that minimize

$$J = \frac{1}{2} \int_0^\infty (\mathbf{x}^\top \mathbf{Q} \mathbf{x} + \mathbf{u}^\top \mathbf{R} \mathbf{u}) dt$$

(e)

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 \\ -6 & 0 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u, \quad \mathbf{x}_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

that minimize

$$J = \frac{1}{2} \int_0^\infty (\mathbf{x}^\top \mathbf{Q} \mathbf{x} + \mathbf{u}^\top \mathbf{R} \mathbf{u}) dt$$

7. Design an optimal state feedback matrix $\mathbf{K}_f$ that minimize the cost function

$$J = \frac{1}{2} \int_0^\infty (\mathbf{x}^\top \mathbf{Q} \mathbf{x} + \mathbf{u}^\top \mathbf{R} \mathbf{u}) dt$$

for

- (a) Inverted pendulum dynamics (refer [1.6.3](#))
- (b) DC motor dynamics refer ([1.7.1](#))
- (c) Two degree of freedom car suspension system ([1.6.3](#))

using MATLAB.

Chapter 6

Introduction to Nonlinear Systems and MIMO Systems

6.1 Introduction to Nonlinear Systems

A nonlinear system is a system having an output change that is not proportional to the input. Every nonlinear time-invariant system is represented as

$$\begin{aligned}\dot{\mathbf{x}} &= f(\mathbf{x}, \mathbf{u}) \\ \mathbf{y} &= h(\mathbf{x}, \mathbf{u})\end{aligned}\tag{6.1.1}$$

where $h(\mathbf{x}, \mathbf{u})$ and $f(\mathbf{x}, \mathbf{u})$ are nonlinear functions of state and inputs. The non-linearities may be hard non-linearity and/or soft non-linearity. Hard non-linearity includes backlash, Coulomb's friction, etc., while soft nonlinearities are non-linearities due to state and/or input.

6.1.1 Determining equilibrium points for nonlinear dynamic systems

Nonlinear systems may have a point or set of points or in state space where the dynamics of the system stay there forever after it arrives there. These points are called equilibrium points or limit cycles. These points may be single, multiple, or may not exist for nonlinear systems. At the equilibrium point or limit cycles, $\dot{\mathbf{x}} = 0$ always.

Equilibrium points for IPC dynamics

For IPC dynamics, we derived from 1.6.3, the equilibrium points are

$$\dot{x}_1 = x_2 \tag{6.1.2}$$

$$\dot{x}_2 = \frac{(m_p l^2 + I)m_p l x_4^2 \sin x_3 - (m_p l)^2 g \sin x_3 \cos x_3 - b(m_p l^2 + J)x_2}{(m_p + m_c)(m_p l^2 + I) - (m_p l \cos x_3)^2} \tag{6.1.3}$$

$$\dot{x}_3 = x_4 \tag{6.1.4}$$

$$\dot{x}_4 = \frac{-(m_p l)^2 x_4^2 g \sin x_3 \cos x_3 + b x_2 m_p l \cos x_3 + (m_p + m_c)m_p l g \sin x_3}{(m_p + m_c)(m_p l^2 + I) - (m_p l \cos x_3)^2} \tag{6.1.5}$$

The equilibrium points are determined as

$$\dot{x}_1 = x_2 = 0$$

$$\frac{(m_p l^2 + I)m_p l x_4^2 \sin x_3 - (m_p l)^2 g \sin x_3 \cos x_3}{(m_p + m_c)(m_p l^2 + I) - (m_p l \cos x_3)^2} = 0 \Rightarrow (m_p l)^2 g \sin x_3 \cos x_3 = 0$$

implying that $\sin x_3 = 0$ or $\cos x_3 = 0$,

$$\dot{x}_3 = x_4 = 0$$

$$\frac{-(m_p l)^2 g \sin x_3 \cos x_3 + (m_p + m_c)m_p l g \sin x_3}{(m_p + m_c)(m_p l^2 + I) - (m_p l \cos x_3)^2} = 0 \Rightarrow (m_p + m_c)m_p l g \sin x_3 = 0$$

$$\sin x_3 = 0$$

$$x_3 = n\pi$$

Hence, the equilibrium point for the IPC dynamic system is $(x_1, 0, n\pi, 0)$. We can't draw any conclusion about x_1 since it vanishes during the equilibrium point calculation.

Equilibrium points for quadruple tank process dynamics

Since $\dot{\mathbf{x}} = 0$ at equilibrium points for a quadruple tank process dynamics in equation 1.9.34, the equilibrium points are,

$$\begin{aligned} \gamma_1 k_1 V_{1e} + a_3 \sqrt{2gh_{3e}} &= a_1 \sqrt{2gh_{1e}} \\ \gamma_2 k_2 V_{2e} + a_4 \sqrt{2gh_{4e}} &= a_2 \sqrt{2gh_{2e}} \\ (1 - \gamma_2)k_2 V_{2e} - a_3 \sqrt{2gh_{3e}} &= 0 \implies V_{2e} = \frac{a_3 \sqrt{2gh_{3e}}}{(1 - \gamma_2)k_2} \\ (1 - \gamma_1)k_1 V_{1e} - a_4 \sqrt{2gh_{4e}} &= 0 \implies V_{1e} = \frac{a_4 \sqrt{2gh_{4e}}}{(1 - \gamma_1)k_1} \end{aligned}$$

Since the tanks are coupled, the equilibrium point of each tank cannot be set individually, for instance, to set the tank 4 to a height h_{4e} , a voltage V_{1e} is required which in turn affect the level of tank 1 and tank 2.

$$\begin{aligned} \gamma_1 k_1 V_{1e} + (1 - \gamma_2)k_2 V_{2e} &= a_1 \sqrt{2gh_{1e}} \implies h_{1e} = \frac{1}{2g} \left(\frac{\gamma_1 k_1 V_{1e} + (1 - \gamma_2)k_2 V_{2e}}{a_1} \right) \\ \gamma_2 k_2 V_{2e} + (1 - \gamma_1)k_1 V_{1e} &= a_2 \sqrt{2gh_{2e}} \implies h_{2e} = \frac{1}{2g} \left(\frac{\gamma_2 k_2 V_{2e} + (1 - \gamma_1)k_1 V_{1e}}{a_2} \right) \end{aligned}$$

6.1.2 Linearization of nonlinear systems

A linearization of a nonlinear system is an approximation of the system about some desired point (operating point). Consider a system

$$\dot{\mathbf{x}} = f(\mathbf{x}) \tag{6.1.6}$$

with operating point $\mathbf{x}_e$.

Assume a small deviation $\Delta \mathbf{x}$ from operating point $\mathbf{x}_e$ is given as,

$$\mathbf{x} = \mathbf{x}_e + \Delta \mathbf{x} \tag{6.1.7}$$

The Taylor series expansion of the function about this point $\mathbf{x}_e$ is given as

$$\dot{\mathbf{x}}_e + \Delta\dot{\mathbf{x}} = f(\mathbf{x} + \Delta\mathbf{x}) = f(\mathbf{x}_e) + f'(\mathbf{x}_e)\Delta\mathbf{x} + \frac{1}{2!}f''\mathbf{x}_e(\Delta\mathbf{x})^2 + HOT \quad (6.1.8)$$

Neglecting higher-order terms, the function can be approximated by a linear function as

$$\dot{\mathbf{x}}_e + \Delta\dot{\mathbf{x}} = f(\mathbf{x}_e) + f'(\mathbf{x}_e)\Delta\mathbf{x} \quad (6.1.9)$$

since $\mathbf{x}_e$ satisfies a differential equation $\dot{\mathbf{x}}_e = f(\mathbf{x}_e)$

$$\begin{aligned} f(\mathbf{x}_e) + \Delta\dot{\mathbf{x}} &= f(\mathbf{x}_e) + f'(\mathbf{x}_e)\Delta\mathbf{x} \\ \Delta\dot{\mathbf{x}} &= f'(\mathbf{x}_e)\Delta\mathbf{x} \end{aligned} \quad (6.1.10)$$

Let the deviation $\Delta\mathbf{x}$ be $\mathbf{x}$ then we have

$$\dot{\mathbf{x}} = f'(\mathbf{x}_e)\mathbf{x} \quad (6.1.11)$$

Similarly for multi-variable function $f(\mathbf{x}, \mathbf{u})$ the approximation of the function about operating point $\mathbf{x}_e$ and $\mathbf{u}_e$ given as

$$f(\mathbf{x}, \mathbf{u}) = f(\mathbf{x}_e, \mathbf{u}_e) + \frac{\partial f}{\partial \mathbf{x}}(\mathbf{x} - \mathbf{x}_e) + \frac{\partial f}{\partial \mathbf{u}}(\mathbf{u} - \mathbf{u}_e) \quad (6.1.12)$$

If the operating point is the equilibrium point $\dot{\mathbf{x}} = f(\mathbf{x}_e, \mathbf{u}_e) = 0$, which means

$$\dot{\mathbf{x}} = f(\mathbf{x}, \mathbf{u}) = \frac{\partial f}{\partial \mathbf{x}}\mathbf{x} + \frac{\partial f}{\partial \mathbf{u}}\mathbf{u} \quad (6.1.13)$$

Which implies that $\mathbf{A} = \frac{\partial f}{\partial \mathbf{x}}$ and $\mathbf{B} = \frac{\partial f}{\partial \mathbf{u}}$ evaluated at the equilibrium point $(\mathbf{x}_e, \mathbf{u}_e)$.

We have a nonlinear system with n number of states dynamics $\dot{x}_i = f_i(x_i, u_j)$ $i = 1, 2, \dots, n$, $j = 1, 2, 3, \dots, m$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \vdots \\ \dot{x}_n \end{bmatrix} = \begin{bmatrix} f_1(x_1, x_2, \dots, x_n, u_1, u_2, \dots, u_m) \\ f_2(x_1, x_2, \dots, x_n, u_1, u_2, \dots, u_m) \\ \vdots \\ f_n(x_1, x_2, \dots, x_n, u_1, u_2, \dots, u_m) \end{bmatrix} \quad (6.1.14)$$

Then the approximation of the nonlinear system about an equilibrium point is

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \vdots \\ \dot{x}_n \end{bmatrix} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1}x_1 & \frac{\partial f_1}{\partial x_2}x_2 & \dots & \frac{\partial f_1}{\partial x_n}x_n & \frac{\partial f_1}{\partial u_1}u_1 & \frac{\partial f_1}{\partial u_2}u_2 & \dots & \frac{\partial f_1}{\partial u_m}u_m \\ \frac{\partial f_2}{\partial x_1}x_1 & \frac{\partial f_2}{\partial x_2}x_2 & \dots & \frac{\partial f_2}{\partial x_n}x_n & \frac{\partial f_2}{\partial u_1}u_1 & \frac{\partial f_2}{\partial u_2}u_2 & \dots & \frac{\partial f_2}{\partial u_m}u_m \\ \vdots & \vdots & & \vdots & \vdots & \vdots & & \vdots \\ \frac{\partial f_n}{\partial x_1}x_1 & \frac{\partial f_n}{\partial x_2}x_2 & \dots & \frac{\partial f_n}{\partial x_n}x_n & \frac{\partial f_n}{\partial u_1}u_1 & \frac{\partial f_n}{\partial u_2}u_2 & \dots & \frac{\partial f_n}{\partial u_m}u_m \end{bmatrix} \Big|_{(x_{eq}, u_{eq})} \quad (6.1.15)$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \vdots \\ \dot{x}_n \end{bmatrix} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \cdots & \frac{\partial f_2}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial x_1} & \frac{\partial f_n}{\partial x_2} & \cdots & \frac{\partial f_n}{\partial x_n} \end{bmatrix} \Big|_{(x_e, u_e)} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} + \begin{bmatrix} \frac{\partial f_1}{\partial u_1} & \frac{\partial f_1}{\partial u_2} & \cdots & \frac{\partial f_1}{\partial u_m} \\ \frac{\partial f_2}{\partial u_1} & \frac{\partial f_2}{\partial u_2} & \cdots & \frac{\partial f_2}{\partial u_m} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial u_1} & \frac{\partial f_n}{\partial u_2} & \cdots & \frac{\partial f_n}{\partial u_m} \end{bmatrix} \Big|_{(x_e, u_e)} \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_m \end{bmatrix} \quad (6.1.16)$$

Comparing with $\dot{\mathbf{x}} = \mathbf{Ax} + \mathbf{Bu}$,

$$\mathbf{A} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \cdots & \frac{\partial f_2}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial x_1} & \frac{\partial f_n}{\partial x_2} & \cdots & \frac{\partial f_n}{\partial x_n} \end{bmatrix} \Big|_{(x_{eq}, u_{eq})} \quad (6.1.17)$$

While

$$\mathbf{B} = \begin{bmatrix} \frac{\partial f_1}{\partial u_1} & \frac{\partial f_1}{\partial u_2} & \cdots & \frac{\partial f_1}{\partial u_m} \\ \frac{\partial f_2}{\partial u_1} & \frac{\partial f_2}{\partial u_2} & \cdots & \frac{\partial f_2}{\partial u_m} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial u_1} & \frac{\partial f_n}{\partial u_2} & \cdots & \frac{\partial f_n}{\partial u_m} \end{bmatrix} \Big|_{(x_{eq}, u_{eq})} \quad (6.1.18)$$

In similar way the nonlinear system has p outputs $y_i = h_i(\mathbf{x}, \mathbf{u}), i = 1, 2, \dots, p$ the approximation about equilibrium point will be

$$\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_p \end{bmatrix} = \begin{bmatrix} \frac{\partial h_1}{\partial x_1} x_1 & \frac{\partial h_1}{\partial x_2} x_2 & \cdots & \frac{\partial h_1}{\partial x_n} x_n & \frac{\partial h_1}{\partial u_1} u_1 & \frac{\partial h_1}{\partial u_2} u_2 & \cdots & \frac{\partial h_1}{\partial u_m} u_m \\ \frac{\partial h_2}{\partial x_1} x_1 & \frac{\partial h_2}{\partial x_2} x_2 & \cdots & \frac{\partial h_2}{\partial x_n} x_n & \frac{\partial h_2}{\partial u_1} u_1 & \frac{\partial h_2}{\partial u_2} u_2 & \cdots & \frac{\partial h_2}{\partial u_m} u_m \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ \frac{\partial h_p}{\partial x_1} x_1 & \frac{\partial h_p}{\partial x_2} x_2 & \cdots & \frac{\partial h_p}{\partial x_n} x_n & \frac{\partial h_p}{\partial u_1} u_1 & \frac{\partial h_p}{\partial u_2} u_2 & \cdots & \frac{\partial h_p}{\partial u_m} u_m \end{bmatrix} \Big|_{(x_{eq}, u_{eq})} \quad (6.1.19)$$

$$\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_p \end{bmatrix} = \begin{bmatrix} \frac{\partial h_1}{\partial x_1} & \frac{\partial h_1}{\partial x_2} & \cdots & \frac{\partial h_1}{\partial x_n} \\ \frac{\partial h_2}{\partial x_1} & \frac{\partial h_2}{\partial x_2} & \cdots & \frac{\partial h_2}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial h_p}{\partial x_1} & \frac{\partial h_p}{\partial x_2} & \cdots & \frac{\partial h_p}{\partial x_n} \end{bmatrix} \Big|_{(x_{eq}, u_{eq})} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} + \begin{bmatrix} \frac{\partial h_1}{\partial u_1} & \frac{\partial h_1}{\partial u_2} & \cdots & \frac{\partial h_1}{\partial u_m} \\ \frac{\partial h_2}{\partial u_1} & \frac{\partial h_2}{\partial u_2} & \cdots & \frac{\partial h_2}{\partial u_m} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial h_p}{\partial u_1} & \frac{\partial h_p}{\partial u_2} & \cdots & \frac{\partial h_p}{\partial u_m} \end{bmatrix} \Big|_{(x_{eq}, u_{eq})} \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_m \end{bmatrix} \quad (6.1.20)$$

Comparing with $\mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u}$,

$$\mathbf{C} = \begin{bmatrix} \frac{\partial h_1}{\partial x_1} & \frac{\partial h_1}{\partial x_2} & \cdots & \frac{\partial h_1}{\partial x_n} \\ \frac{\partial h_2}{\partial x_1} & \frac{\partial h_2}{\partial x_2} & \cdots & \frac{\partial h_2}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial h_p}{\partial x_1} & \frac{\partial h_p}{\partial x_2} & \cdots & \frac{\partial h_p}{\partial x_n} \end{bmatrix} \Big|_{(x_{eq}, u_{eq})} \quad (6.1.21)$$

While

$$\mathbf{D} = \begin{bmatrix} \frac{\partial f_1}{\partial u_1} & \frac{\partial f_1}{\partial u_2} & \cdots & \frac{\partial f_1}{\partial u} \\ \frac{\partial f_2}{\partial u_1} & \frac{\partial f_2}{\partial u_2} & \cdots & \frac{\partial f_2}{\partial u_m} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial u_1} & \frac{\partial f_n}{\partial u} & \cdots & \frac{\partial f_n}{\partial u} \end{bmatrix} \Big|_{(x_{eq}, u_{eq})} \quad (6.1.22)$$

Linearizing Inverted Pendulum on a Cart Dynamics

For IPC dynamics derived in Chapter One

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \end{bmatrix} = \begin{bmatrix} f_1(\mathbf{x}, \mathbf{u}) \\ f_2(\mathbf{x}, \mathbf{u}) \\ f_3(\mathbf{x}, \mathbf{u}) \\ f_4(\mathbf{x}, \mathbf{u}) \end{bmatrix} = \begin{bmatrix} x_2 \\ \frac{(m_p l^2 + I)m_p l x_4^2 \sin x_3 - (m_p l)^2 g \sin x_3 \cos x_3 - b(m_p l^2 + J)x_2 + (m_p l^2 + I)u}{(m_p + m_c)(m_p l^2 + I) - (m_p l \cos x_3)^2} \\ x_4 \\ \frac{-(m_p l)^2 x_4^2 g \sin x_3 \cos x_3 + b x_2 m_p l \cos x_3 + (m_p + m_c)m_p l g \sin x_3 - m_p l \cos x_3 u}{(m_p + m_c)(m_p l^2 + I) - (m_p l \cos x_3)^2} \end{bmatrix} \quad (6.1.23)$$

To simplify the equation let

$$f_2 = \frac{\eta_1 + \eta_2}{\eta_3} \quad f_4 = \frac{\xi_1 + \xi_2}{\eta_3}$$

where

$$\begin{aligned} \eta_1 &= (m_p l^2 + I) m_p l x_4^2 \sin x_3 - (m_p l)^2 g \sin x_3 \cos x_3 - b(m_p l^2 + J) x_2 \\ \eta_2 &= (m_p l^2 + I) u \\ \xi_1 &= -(m_p l)^2 x_4^2 g \sin x_3 \cos x_3 + b x_2 m_p l \cos x_3 + (m_p + m_c) m_p l g \sin x_3 \\ \xi_2 &= -m_p l \cos x_3 u \\ \eta_3 &= (m_p + m_c)(m_p l^2 + I) - (m_p l \cos x_3)^2 \end{aligned}$$

The linearization of the functions $f_i(\mathbf{x}, u)$, $i = 1, 2, 3, 4$ about the equilibrium point $(\mathbf{x}_{eq}, u_{eq}) = (x_1, 0, 0, 0, 0)$ is determined using equation 6.1.15.

$$\frac{\partial f_1}{\partial x_1} = \frac{\partial x_2}{\partial x_1} = 0, \quad \frac{\partial f_1}{\partial x_2} = \frac{\partial x_2}{\partial x_2} = 1, \quad \frac{\partial f_1}{\partial x_3} = \frac{\partial x_2}{\partial x_3} = 0, \quad \frac{\partial f_1}{\partial x_4} = \frac{\partial x_2}{\partial x_4} = 0 \quad (6.1.24)$$

$$\frac{\partial f_2}{\partial x_1} = \frac{\partial}{\partial x_1} \left(\frac{\eta_1 - \eta_2}{\eta_3} \right) = 0 \quad (6.1.25)$$

$$\frac{\partial f_2}{\partial x_2} = \frac{\partial}{\partial x_2} \left(\frac{\eta_1 - \eta_2}{\eta_3} \right) \Big|_{(\mathbf{x}_e, u_e)} = \frac{-b(m_p l^2 + J)}{(m_p + m_c)(m_p l^2 + I) - (m_p l)^2} \quad (6.1.26)$$

$$\frac{\partial f_2}{\partial x_3} = \frac{\partial}{\partial x_3} \left(\frac{\eta_1 - \eta_2}{\eta_3} \right) \Big|_{(\mathbf{x}_e, u_e)} = \frac{-(m_p l)^2 g}{(m_p + m_c)(m_p l^2 + I) - (m_p l)^2} \quad (6.1.27)$$

$$\frac{\partial f_2}{\partial x_4} = \frac{\partial}{\partial x_4} \left(\frac{\eta_1 - \eta_2}{\eta_3} \right) = 0 \quad (6.1.28)$$

$$\frac{\partial f_3}{\partial x_1} = \frac{\partial x_3}{\partial x_1} = 0, \quad \frac{\partial f_3}{\partial x_2} = \frac{\partial x_3}{\partial x_2} = 0, \quad \frac{\partial f_3}{\partial x_3} = \frac{\partial x_3}{\partial x_3} = 1, \quad \frac{\partial f_3}{\partial x_4} = \frac{\partial x_3}{\partial x_4} = 0 \quad (6.1.29)$$

$$\frac{\partial f_4}{\partial x_1} = \frac{\partial}{\partial x_1} \left(\frac{\xi_1 - \xi_2}{\eta_3} \right) = 0 \quad (6.1.30)$$

$$\frac{\partial f_4}{\partial x_2} = \frac{\partial}{\partial x_2} \left(\frac{\xi_1 - \xi_2}{\eta_3} \right) \Big|_{(\mathbf{x}_e, u_e)} = \frac{b m_p l}{(m_p + m_c)(m_p l^2 + I) - (m_p l)^2} \quad (6.1.31)$$

$$\frac{\partial f_4}{\partial x_3} = \frac{\partial}{\partial x_3} \left(\frac{\xi_1 - \xi_2}{\eta_3} \right) \Big|_{(\mathbf{x}_e, u_e)} = \frac{m_p l g (m_c + m_p)}{(m_p + m_c)(m_p l^2 + I) - (m_p l)^2} \quad (6.1.32)$$

$$\frac{\partial f_4}{\partial x_4} = \frac{\partial}{\partial x_4} \left(\frac{\xi_1 - \xi_2}{\eta_3} \right) = 0 \quad (6.1.33)$$

$$\frac{\partial f_1}{\partial u} = \frac{\partial x_2}{\partial u} = 0, \quad \frac{\partial f_3}{\partial u} = \frac{\partial x_3}{\partial u} = 0 \quad (6.1.34)$$

$$\frac{\partial f_2}{\partial u} = \frac{\partial}{\partial u} \left(\frac{\xi_1 - \xi_2}{\eta_3} \right) \Big|_{(\mathbf{x}_e, u_e)} = \frac{m_p l^2 + I}{(m_p + m_c)(m_p l^2 + I) - (m_p l)^2} \quad (6.1.35)$$

$$\frac{\partial f_4}{\partial u} = \frac{\partial}{\partial u} \left(\frac{\eta_1 - \eta_2}{\eta_3} \right) \Big|_{(\mathbf{x}_e, u_e)} = \frac{-m_p l g}{(m_p + m_c)(m_p l^2 + I) - (m_p l)^2} \quad (6.1.36)$$

From this, the system matrix is

$$\mathbf{A}_{ij} = \frac{\partial f_i}{\partial x_j} \Big|_{\mathbf{x}_e, \mathbf{u}_e} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & -\left(\frac{b(m_pl^2 + J)}{q}\right) & -\left(\frac{(m_pl)^2 g}{q}\right) & 0 \\ 0 & 0 & 0 & 1 \\ 0 & \left(\frac{bm_pl}{q}\right) & \left(\frac{m_plg(m_p + m_c)}{q}\right) & 0 \end{bmatrix}$$

and the input matrix is

$$\mathbf{B}_{ij} = \frac{\partial f_i}{\partial u_j} \Big|_{\mathbf{x}_e, \mathbf{u}_e} = \begin{bmatrix} 0 \\ \frac{m_pl^2 + J}{q} \\ 0 \\ -\frac{m_pl}{q} \end{bmatrix}$$

where $q = (m_p + m_c)(m_pl^2 + I) - (m_pl)^2$.

with system outputs $y_1 = h_1(\mathbf{x}, u) = x_1$ and $y_2 = h_2(\mathbf{x}, u) = x_3$

$$\frac{\partial h_1}{\partial x_1} = 1, \quad \frac{\partial h_1}{\partial x_2} = 0, \quad \frac{\partial h_1}{\partial x_3} = 0, \quad \frac{\partial h_1}{\partial x_4} = 0 \quad (6.1.37)$$

$$\frac{\partial h_2}{\partial x_1} = 0, \quad \frac{\partial h_2}{\partial x_2} = 0, \quad \frac{\partial h_2}{\partial x_3} = 1, \quad \frac{\partial h_2}{\partial x_4} = 0 \quad (6.1.38)$$

and

$$\frac{\partial h_1}{\partial u} = 0, \quad \frac{\partial h_2}{\partial u} = 0 \quad (6.1.39)$$

The output matrix becomes

$$\mathbf{C}_{ij} = \frac{\partial h_i}{\partial x_j} \Big|_{\mathbf{x}_e, \mathbf{u}_e} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

and input transmission matrix

$$\mathbf{D}_{ij} = \frac{\partial h_i}{\partial u_j} \Big|_{\mathbf{x}_e, \mathbf{u}_e} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Thus the linearized model of IPC about equilibrium point $(\mathbf{x}_e, u_e) = (x_1, 0, 0, 0, 0)$ is given as

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & -\left(\frac{b(m_p l^2 + J)}{q}\right) & -\left(\frac{(m_p l)^2 g}{q}\right) & 0 \\ 0 & 0 & 0 & 1 \\ 0 & \left(\frac{b m_p l}{q}\right) & \left(\frac{m_p l g (m_p + m_c)}{q}\right) & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + \begin{bmatrix} 0 \\ \left(\frac{m_p l^2 + J}{q}\right) \\ 0 \\ -\left(\frac{m_p l}{q}\right) \end{bmatrix} u \quad (6.1.40)$$

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix} u \quad (6.1.41)$$

Which is same as equation 1.6.36 and equation 1.6.37

Linearizing Quadruple Tank Process Dynamics

For quadruple tank process on equation 1.9.34 let $(\mathbf{x}_e, \mathbf{u}_e) = (h_{1_e}, h_{2_e}, h_{3_e}, h_{4_e}, V_{1_e}, V_{2_e})$ be an equilibrium point. We define small deviations:

$$\tilde{\mathbf{x}} = \begin{bmatrix} \tilde{h}_1 \\ \tilde{h}_2 \\ \tilde{h}_3 \\ \tilde{h}_4 \end{bmatrix} = \begin{bmatrix} h_1 - h_{1_e} \\ h_2 - h_{2_e} \\ h_3 - h_{3_e} \\ h_4 - h_{4_e} \end{bmatrix}, \quad \tilde{\mathbf{u}} = \begin{bmatrix} \tilde{V}_1 \\ \tilde{V}_2 \end{bmatrix} = \begin{bmatrix} V_1 - V_{1_e} \\ V_2 - V_{2_e} \end{bmatrix}$$

The linearized system takes the form:

$$\dot{\tilde{\mathbf{x}}} = \mathbf{A} \tilde{\mathbf{x}} + \mathbf{B} \tilde{\mathbf{u}}$$

The matrix $\mathbf{A} \in \mathbb{R}^{4 \times 4}$ is obtained by taking partial derivatives of each equation concerning the states:

$$\mathbf{A}_{ij} = \left. \frac{\partial f_i}{\partial x_j} \right|_{\mathbf{x}_e, \mathbf{u}_e}$$

$$\begin{aligned} \frac{\partial \dot{h}_1}{\partial h_1} &= -\frac{a_1}{A_1} \cdot \frac{g}{\sqrt{2gh_{1_e}}} = -\frac{a_1 \sqrt{g/2h_{1_e}}}{A_1}, & \frac{\partial \dot{h}_1}{\partial h_3} &= \frac{a_3}{A_1} \cdot \frac{g}{\sqrt{2gh_{3_e}}} = \frac{a_3 \sqrt{g/2h_{3_e}}}{A_1} \\ \frac{\partial \dot{h}_2}{\partial h_2} &= -\frac{a_2}{A_2} \cdot \frac{g}{\sqrt{2gh_{2_e}}} = -\frac{a_2 \sqrt{g/2h_{2_e}}}{A_2}, & \frac{\partial \dot{h}_2}{\partial h_4} &= \frac{a_4}{A_2} \cdot \frac{g}{\sqrt{2gh_{4_e}}} = \frac{a_4 \sqrt{g/2h_{4_e}}}{A_2} \\ \frac{\partial \dot{h}_3}{\partial h_3} &= -\frac{a_3}{A_3} \cdot \frac{g}{\sqrt{2gh_{3_e}}} = -\frac{a_3 \sqrt{g/2h_{3_e}}}{A_3}, & \frac{\partial \dot{h}_4}{\partial h_4} &= -\frac{a_4}{A_4} \cdot \frac{g}{\sqrt{2gh_{4_e}}} = -\frac{a_4 \sqrt{g/2h_{4_e}}}{A_4} \end{aligned}$$

Let

$$\alpha_1 = \frac{a_1 \sqrt{g/2h_{1e}}}{A_1}, \quad \alpha_2 = \frac{a_2 \sqrt{g/2h_{2e}}}{A_2}, \quad \alpha_3 = \frac{a_3 \sqrt{g/2h_{3e}}}{A_1}$$

$$\alpha_4 = \frac{a_4 \sqrt{g/2h_{4e}}}{A_2}, \quad \alpha_5 = \frac{a_3 \sqrt{g/2h_{3e}}}{A_3}, \quad \alpha_6 = \frac{a_4 \sqrt{g/2h_{4e}}}{A_4}$$

With this the system matrix $\mathbf{A}$ is then,

$$\mathbf{A} = \begin{bmatrix} -\alpha_1 & 0 & \alpha_3 & 0 \\ 0 & -\alpha_2 & 0 & \alpha_4 \\ 0 & 0 & -\alpha_5 & 0 \\ 0 & 0 & 0 & -\alpha_6 \end{bmatrix} \quad (6.1.42)$$

The matrix $\mathbf{B} \in \mathbb{R}^{4 \times 2}$ is obtained by taking partial derivatives concerning the inputs:

$$\mathbf{B}_{ij} = \left. \frac{\partial f_i}{\partial u_j} \right|_{\mathbf{x}_e, \mathbf{u}_e}$$

$$\frac{\partial \dot{h}_1}{\partial V_1} = \frac{\gamma_1 k_1}{A_1}, \quad \frac{\partial \dot{h}_1}{\partial V_2} = 0, \quad \frac{\partial \dot{h}_2}{\partial V_1} = 0, \quad \frac{\partial \dot{h}_2}{\partial V_2} = \frac{\gamma_2 k_2}{A_2}$$

$$\frac{\partial \dot{h}_3}{\partial V_1} = 0, \quad \frac{\partial \dot{h}_3}{\partial V_2} = \frac{(1 - \gamma_2)k_2}{A_3}, \quad \frac{\partial \dot{h}_4}{\partial V_1} = \frac{(1 - \gamma_1)k_1}{A_4}, \quad \frac{\partial \dot{h}_4}{\partial V_2} = 0$$

Thus:

$$\mathbf{B} = \begin{bmatrix} \frac{\gamma_1 k_1}{A_1} & 0 \\ 0 & \frac{\gamma_2 k_2}{A_2} \\ 0 & \frac{(1 - \gamma_2)k_2}{A_3} \\ \frac{(1 - \gamma_1)k_1}{A_4} & 0 \end{bmatrix} \quad (6.1.43)$$

with system outputs $y_1 = h_1(\mathbf{x}, \mathbf{u}) = h_1$ and $y_2 = h_2(\mathbf{x}, \mathbf{u}) = h_2$, the output matrix becomes

$$\mathbf{C}_{ij} = \left. \frac{\partial h_i}{\partial x_j} \right|_{\mathbf{x}_e, \mathbf{u}_e} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \quad (6.1.44)$$

and input transmission matrix

$$\mathbf{D}_{ij} = \left. \frac{\partial h_i}{\partial u_j} \right|_{\mathbf{x}_e, \mathbf{u}_e} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \quad (6.1.45)$$

6.2 Introduction to Multiple Input Multiple Output Systems

MIMO systems, as their name indicates they have multiple inputs and multiple outputs. Consider a system with n inputs and m outputs; the transfer function of these systems consists of transfer functions from each input to each output.

$$\frac{Y_i(s)}{U_i(s)} = \mathbf{G}_{ij}(s) \quad (6.2.1)$$

where $i = 1, 2, 3, \dots, n$ and $j = 1, 2, 3, \dots, m$.

If we expand equation 6.2.1, we have a set of transfer functions

$$\begin{bmatrix} Y_1(s) \\ Y_2(s) \\ \vdots \\ Y_m(s) \end{bmatrix} = \begin{bmatrix} G_{11}(s) & G_{21}(s) & \dots & G_{n1}(s) \\ G_{21}(s) & G_{22}(s) & \dots & G_{n2}(s) \\ \vdots & \vdots & \ddots & \vdots \\ G_{1m}(s) & G_{2m}(s) & \dots & G_{nm}(s) \end{bmatrix} \begin{bmatrix} U_1(s) \\ U_2(s) \\ \vdots \\ U_n(s) \end{bmatrix} \quad (6.2.2)$$

6.2.1 Poles and zeros of MIMO Systems

The poles of each transfer function are the poles of the MIMO system. If G_{ij} has a pole at $s = p$, then $s = p$ is also a pole of the MIMO system. If one of the transfunctions has poles at frequency r at $s = p_0$, then the MIMO system also has a pole of frequency r at $s = p_0$.

To determine the largest multiplicity pole, determine the largest multiplicity pole among 1×1 minors of the transfer function, then the largest multiplicity pole among 2×2 minors of the transfer function so on. Stop at minors of size $k \times k$ if k is the first size for which the previous multiplicity of the pole is greater than the current minor's pole multiplicity. The largest of all the multiplicities of the minors is the multiplicity of the pole.

Determining zeros of the MIMO system is a tricky problem. To start with, we know that zeros of the system prohibit the control signal from affecting the dynamics of the system. Hence, from this knowledge, a MIMO system has zero at location z_0 if the z_0 causes a reduction in rank of the MIMO system transfer function (i.e., if the z_0 causes zero rows in the transfer function of the MIMO system). Otherwise, the zero of one transfer function will not be considered as the zero of the MIMO system.

Consider IPC dynamics derived in section 1.6.3, the transfer function of IPC will be

$$\frac{Y_1(s)}{U(s)} = -\frac{(m_p l)^2 g}{q s} \left(\frac{m_p l}{(m_p + m_c) m_p l g - q s} \right) + \frac{(m_p l^2 + I)}{s q} \quad (6.2.3)$$

$$\frac{Y_2(s)}{U(s)} = \frac{m_p l}{(m_p + m_c) m_p l g - q s} \quad (6.2.4)$$

$$\begin{bmatrix} Y_1(s) \\ Y_2(s) \end{bmatrix} = \begin{bmatrix} -\frac{(m_p l)^2 g}{qs} \left(\frac{m_p l}{(m_p + m_c)m_p l g - qs} \right) + \frac{(m_p l^2 + I)}{sq} \\ \frac{m_p l}{(m_p + m_c)m_p l g - qs} \end{bmatrix} U(s) \quad (6.2.5)$$

As we can see, the system has only one input; hence, it is a single-input multiple-output system (SIMO). The poles of the system are the poles of equation 6.2.3 and equation 6.2.4. The zero of the system is zero of equation 6.2.3, which will result in one zero row in the system transfer function on equation 6.2.5.

$$\begin{aligned} G_{11}(s) &= -\frac{(m_p l)^3 g}{qs((m_p + m_c)m_p l g - qs)} + \frac{m_p l^2 + I}{sq} \\ &= \frac{-(m_p l)^3 g + (m_p l^2 + I)((m_p + m_c)m_p l g - qs)}{sq((m_p + m_c)m_p l g - qs)} \end{aligned}$$

The numerator becomes:

$$N(s) = m_p l g [-(m_p l)^2 + (m_p l^2 + I)(m_p + m_c)] - (m_p l^2 + I)qs = 0$$

Since $m_p l \neq 0$, we only consider when $N(s) = 0$:

$$s = \frac{m_p l g [-(m_p l)^2 + (m_p l^2 + I)(m_p + m_c)]}{q(m_p l^2 + I)} \quad (6.2.6)$$

For the linearized model of the quadruple tank level control system in section 6.1.2, if height h_1 and h_2 are required to be controlled, we have two outputs and two inputs. The transfer function matrix $\mathbf{G}(s)$ relates inputs to outputs in the Laplace domain:

$$\mathbf{G}(s) = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D} \quad (6.2.7)$$

Given the structure of $\mathbf{A}$:

$$s\mathbf{I} - \mathbf{A} = \begin{bmatrix} s + \alpha_1 & 0 & -\alpha_3 & 0 \\ 0 & s + \alpha_2 & 0 & -\alpha_4 \\ 0 & 0 & s + \alpha_5 & 0 \\ 0 & 0 & 0 & s + \alpha_6 \end{bmatrix} \quad (6.2.8)$$

The matrix is block diagonal, so its inverse is:

$$(s\mathbf{I} - \mathbf{A})^{-1} = \begin{bmatrix} \frac{1}{s + \alpha_1} & 0 & \frac{\alpha_3}{(s + \alpha_1)(s + \alpha_5)} & 0 \\ 0 & \frac{1}{s + \alpha_2} & 0 & \frac{\alpha_4}{(s + \alpha_2)(s + \alpha_6)} \\ 0 & 0 & \frac{1}{s + \alpha_5} & 0 \\ 0 & 0 & 0 & \frac{1}{s + \alpha_6} \end{bmatrix} \quad (6.2.9)$$

$$\mathbf{G}(s) = C(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} = \begin{bmatrix} \frac{\gamma_1 k_1}{A_1(s + \alpha_1)} & \frac{\alpha_3(1 - \gamma_2)k_2}{A_3(s + \alpha_1)(s + \alpha_5)} \\ \frac{\alpha_4(1 - \gamma_1)k_1}{A_4(s + \alpha_2)(s + \alpha_6)} & \frac{\gamma_2 k_2}{A_2(s + \alpha_2)} \end{bmatrix} \quad (6.2.10)$$

The transmission zeros are the values of s that make the transfer function matrix lose rank. We find them by solving:

$$\begin{aligned} \det(\mathbf{G}(s)) = 0 &\implies G_{11}(s)G_{22}(s) - G_{12}(s)G_{21}(s) = 0 \\ &= \left(\frac{\gamma_1 k_1}{A_1(s + \alpha_1)} \right) \left(\frac{\gamma_2 k_2}{A_2(s + \alpha_2)} \right) - \left(\frac{\alpha_3(1 - \gamma_2)k_2}{A_3(s + \alpha_1)(s + \alpha_5)} \right) \left(\frac{\alpha_4(1 - \gamma_1)k_1}{A_4(s + \alpha_2)(s + \alpha_6)} \right) = 0 \end{aligned}$$

Factoring out common terms:

$$\det \mathbf{G}(s) = \frac{\gamma_1 \gamma_2 k_1 k_2}{A_1 A_2 (s + \alpha_1)(s + \alpha_2)} - \frac{\alpha_3 \alpha_4 (1 - \gamma_1)(1 - \gamma_2) k_1 k_2}{A_3 A_4 (s + \alpha_1)(s + \alpha_2)(s + \alpha_5)(s + \alpha_6)} \quad (6.2.11)$$

Combine terms with a common denominator:

$$\det \mathbf{G}(s) = \frac{k_1 k_2 [\gamma_1 \gamma_2 A_3 A_4 (s + \alpha_5)(s + \alpha_6) - \alpha_3 \alpha_4 (1 - \gamma_1)(1 - \gamma_2)]}{A_1 A_2 A_3 A_4 (s + \alpha_1)(s + \alpha_2)(s + \alpha_5)(s + \alpha_6)} \quad (6.2.12)$$

The zeros occur when the numerator polynomial vanishes:

$$\gamma_1 \gamma_2 A_3 A_4 (s + \alpha_5)(s + \alpha_6) - \alpha_3 \alpha_4 (1 - \gamma_1)(1 - \gamma_2) = 0 \quad (6.2.13)$$

Expanding the quadratic:

$$s^2 + (\alpha_5 + \alpha_6)s + \left[\alpha_5 \alpha_6 - \frac{\alpha_3 \alpha_4 (1 - \gamma_1)(1 - \gamma_2)}{\gamma_1 \gamma_2 A_3 A_4} \right] = 0 \quad (6.2.14)$$

The zeros are given by the roots:

$$s = \frac{-(\alpha_5 + \alpha_6) \pm \sqrt{(\alpha_5 + \alpha_6)^2 - 4 \left(\alpha_5 \alpha_6 - \frac{\alpha_3 \alpha_4 (1 - \gamma_1)(1 - \gamma_2)}{\gamma_1 \gamma_2 A_3 A_4} \right)}}{2} \quad (6.2.15)$$

Consider a quadruple tank process with parameters in Table 6.1:

For desired equilibrium heights $h_{1_e} = h_{2_e} = 10$ cm, using

$$\begin{aligned} (1 - \gamma_2)k_2 V_{2_e} &= a_3 \sqrt{2gh_{3_e}}, & (1 - \gamma_1)k_1 V_{1_e} &= a_4 \sqrt{2gh_{4_e}} \\ \gamma_1 k_1 V_{1_e} &= a_1 \sqrt{2gh_{1_e}} - a_3 \sqrt{2gh_{3_e}}, & \gamma_2 k_2 V_{2_e} &= a_2 \sqrt{2gh_{2_e}} - a_4 \sqrt{2gh_{4_e}} \end{aligned}$$

to solve for the rest equilibrium points yields:

$$\begin{aligned} h_{3_e} &= 6.45 \text{ cm}, & h_{4_e} &= 6.45 \text{ cm} \\ V_{1_e} &= 3.15 \text{ V}, & V_{2_e} &= 3.15 \text{ V} \end{aligned}$$

Table 6.1: *Quadruple tank process parameters*

Parameter	Value	Description
A_1, A_2	28 cm ²	Cross-section of lower tanks
A_3, A_4	32 cm ²	Cross-section of upper tanks
a_1, a_2	0.071 cm ²	Outlet holes (lower tanks)
a_3, a_4	0.057 cm ²	Outlet holes (upper tanks)
k_1, k_2	3.33 cm ³ /Vs	Pump constants
γ_1, γ_2	0.7	Valve settings
g	981 cm/s ²	Gravity

which yield the α coefficients:

$$\begin{aligned}\alpha_1 &= \frac{0.071\sqrt{981/(2 \times 10)}}{28} = 0.028 \text{ s}^{-1}, & \alpha_2 &= \alpha_1 = 0.028 \text{ s}^{-1} \\ \alpha_3 &= \frac{0.057\sqrt{981/(2 \times 6.45)}}{28} = 0.023 \text{ s}^{-1}, & \alpha_4 &= \alpha_3 = 0.023 \text{ s}^{-1} \\ \alpha_5 &= \frac{0.057\sqrt{981/(2 \times 6.45)}}{32} = 0.020 \text{ s}^{-1}, & \alpha_6 &= \alpha_5 = 0.020 \text{ s}^{-1}\end{aligned}$$

Thus, the state and input matrices become:

$$\mathbf{A} = \begin{bmatrix} -0.028 & 0 & 0.023 & 0 \\ 0 & -0.028 & 0 & 0.023 \\ 0 & 0 & -0.020 & 0 \\ 0 & 0 & 0 & -0.020 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0.083 & 0 \\ 0 & 0.083 \\ 0 & 0.062 \\ 0.058 & 0 \end{bmatrix}$$

leading to the transfer function of the system,

$$\mathbf{G}(s) = \begin{bmatrix} \frac{0.083}{s + 0.028} & \frac{0.0014}{(s + 0.028)(s + 0.020)} \\ \frac{0.0013}{(s + 0.028)(s + 0.020)} & \frac{0.083}{s + 0.028} \end{bmatrix}$$

solving for the transmission zeros and poles of the system by setting $\det \mathbf{G}(s)$,

$$\begin{aligned}\|\mathbf{G}(s)\| &= \left(\frac{0.0069}{s + 0.028} \right) - \left(\frac{1.8200 \times 10^{-06}}{(s + 0.028)(s + 0.020)} \right) \\ &= \frac{0.006889(s^2 + 0.04s + 0.0001358)}{(s + 0.028)^2(s + 0.020)^2}\end{aligned}$$

The transmission zeros at

$$s^2 + 0.04s + 0.0001358 = 0 \implies s = -0.020 \pm 0.00208j$$

And the poles of the systems are at,

$$(s + 0.028)^2(s + 0.020)^2 = 0 \implies \text{double poles at } s = -0.028 \text{ and } s = -0.02$$

6.3 Examples

1. Find the poles and zeros of the following system

$$\begin{aligned}\dot{x}_1 &= x_2 + u_1 \\ \dot{x}_2 &= x_3 + u_2 \\ \dot{x}_3 &= 6x_1 - 11x_2 + 6x_3 + u_1 + u_2 \\ y_1 &= x_1 \\ y_2 &= x_3\end{aligned}$$

Solution

The state space representation of the system is

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 6 & -11 & 6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

The transfer function of the system is

$$\begin{aligned}\mathbf{G}(s) &= \frac{\mathbf{C} \text{adj}(s\mathbf{I} - \mathbf{A})\mathbf{B}}{\det(s\mathbf{I} - \mathbf{A})} \\ &= \frac{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} s^2 - 6s + 11 & s + 6 & 1 \\ 6 & s^2 - 6s & s \\ 6s & s & s^2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}}{s^3 - 6s^2 + 11s - 6} \\ &= \frac{\begin{bmatrix} s^2 - 6s + 12 & s + 7 \\ s^2 + 6s & s + s^2 \end{bmatrix}}{s^3 - 6s^2 + 11s - 6}\end{aligned}$$

The only number that can cause a zero row in the transfer function matrix is $s = 0$. Hence $s = 0$ is the zero of the system. The poles of the system are zeros of $s^3 - 6s^2 + 11s - 6 = (s + 1)(s + 2)(s + 3) = 0$, hence $s = 1$, $s = 2$ and $s = 3$ are the poles of the system.

2. Find zeros as poles of the following transfer function

$$\mathbf{G}(s) = \begin{bmatrix} \frac{s}{s+1} & \frac{s^2+3}{(s+1)(s+4)} \\ \frac{1}{s^2+1} & \frac{s+3}{s(s+5)} \end{bmatrix}$$

Solution

No number causes zero rows or zero columns in the transfer function matrix, hence the system has no zeros. The poles of the system are

- poles of first minors, $\frac{s}{s+1}$, $\frac{s^2+3}{(s+1)(s+4)}$, $\frac{1}{s^2+1}$ and $\frac{s+3}{s(s+5)}$
- poles of the second minors

$$\left(\frac{s}{s+1}\right)\left(\frac{s+3}{s(s+5)}\right) - \left(\frac{1}{s^2+1}\right)\left(\frac{s^2+3}{(s+1)(s+4)}\right)$$

Thus, the poles of the system are $s = 0$, $s = -1$, $s = -4$, $s = -5$, and $s = \pm i$.

3. Find the equilibrium point/s and linearized model of the system at the equilibrium point/s.

$$\begin{aligned}\dot{x}_1 &= 7x_1 + 2\sin x_2 - x_2^4 + u^3 \\ \dot{x}_2 &= e^{x_1} - 3x_2 - 1 + 5x_1^2 + \sin u \\ y &= e^{x_1+x_2}\end{aligned}$$

Solution

At equilibrium point/s the the state dynamics is zero $\dot{x}_1 = 0$, $\dot{x}_2 = 0$, $u_{eq} = 0$, thus

$$\begin{aligned}f_1(x_1, x_2, u) &= 7x_1 + 2\sin x_2 - x_2^4 + u^3 = 0 \\ f_2(x_1, x_2, u) &= e^{x_1} - 3x_2 - 1 + 5x_1^2 + \sin u = 0\end{aligned}$$

using the following MATLAB code

```
syms x1 x2 u
eqn1 = 7 * x1 + 2 * sin(x2) - x2^4 + u^3;
eqn2 = exp(x1) - 3 * x2 - 1 + 5 * x1^2 + sin(u)
[x1, x2, u] = solve(eqn1 == 0, eqn2 == 0, [x1, x2, u])
```

the only equilibrium point is $(x_{1eq}, x_{2eq}, u_{eq} = 0) = (0, 0, 0)$. The linearized model

about $(0, 0, 0)$ is

$$\begin{aligned}
 \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} &= \begin{bmatrix} \frac{\partial}{\partial x_1} f_1(x_1, x_2, u) & \frac{\partial}{\partial x_2} f_1(x_1, x_2, u) \\ \frac{\partial}{\partial x_1} f_2(x_1, x_2, u) & \frac{\partial}{\partial x_2} f_2(x_1, x_2, u) \end{bmatrix} \Big|_{x_{eq}} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} \frac{\partial}{\partial u} f_1(x_1, x_2, u) \\ \frac{\partial}{\partial u} f_2(x_1, x_2, u) \end{bmatrix} \Big|_{u_{eq}=0} u \\
 &= \begin{bmatrix} 7 & 2 \cos x_2 - 4x_2^3 \\ e^{x_1} + 10x_1 & -3 \end{bmatrix} \Big|_{(x_1=0, x_2=0)} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 3u^2 \\ \cos u \end{bmatrix} \Big|_{u_{eq}=0} u \\
 &= \begin{bmatrix} 7 & 1 \\ 1 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u \\
 y &= \begin{bmatrix} \frac{\partial}{\partial x_1} h(x_1, x_2, u) & \frac{\partial}{\partial x_2} h(x_1, x_2, u) \end{bmatrix} \Big|_{(x_1=0, x_2=0)} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \\
 &= \begin{bmatrix} e^{x_1} & e^{x_2} \end{bmatrix} \Big|_{(x_1=0, x_2=0)} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \\
 &= \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}
 \end{aligned}$$

4. Find the equilibrium point/s of the following system and the linearized model of the system at the point/s.

$$\begin{aligned}
 \dot{x}_1 &= 3x_1 + (x_1^2 + x_2^2) \\
 \dot{x}_2 &= 2x_1x_2(x_1^2 + x_2^2)
 \end{aligned}$$

Solution

At equilibrium point/s the the state dynamics is zero $\dot{x}_1 = 0, \dot{x}_2 = 0$, thus

$$\begin{aligned}
 f_1(x_1, x_2) &= 3x_1 + (x_1^2 - x_2^2) = 0 \\
 f_2(x_1, x_2) &= 2x_1x_2(x_1^2 - x_2^2) = 0
 \end{aligned}$$

The equilibrium points are

$$x_1 = 0, \quad x_2 = 0, \quad x_1^2 - x_2^2 = 0$$

Which means all of the points where $x_1^2x_2^2 = 0 \Rightarrow x_1 = x_2$, or $x_1 = -x_2$ are equilibrium points of the system. This set of points is called a limit cycle of the system.

The linearized model at $(x_1 = x_2)$ is

$$\begin{aligned}
 \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} &= \begin{bmatrix} \frac{\partial}{\partial x_1} f_1(x_1, x_2) & \frac{\partial}{\partial x_2} f_1(x_1, x_2) \\ \frac{\partial}{\partial x_1} f_2(x_1, x_2) & \frac{\partial}{\partial x_2} f_2(x_1, x_2) \end{bmatrix} \bigg|_{x_{eq}} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \\
 &= \begin{bmatrix} 3 + 2x_1 & 2x_2 \\ 4x_1x_2 & 4x_2x_1 \end{bmatrix} \bigg|_{(x_1=x_2)} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \\
 &= \begin{bmatrix} 3 + 2x_1 & 2x_1 \\ 4x_1^2 & 4x_1^2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}
 \end{aligned}$$

or at $(x_1 = -x_2)$ is

$$\begin{aligned}
 \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} &= \begin{bmatrix} \frac{\partial}{\partial x_1} f_1(x_1, x_2) & \frac{\partial}{\partial x_2} f_1(x_1, x_2) \\ \frac{\partial}{\partial x_1} f_2(x_1, x_2) & \frac{\partial}{\partial x_2} f_2(x_1, x_2) \end{bmatrix} \bigg|_{x_{eq}} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \\
 &= \begin{bmatrix} 3 + 2x_1 & 2x_2 \\ 4x_1x_2 & 4x_2x_1 \end{bmatrix} \bigg|_{(x_1=-x_2)} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \\
 &= \begin{bmatrix} 3 - 2x_2 & -2x_2 \\ -4x_2^2 & -4x_2^2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}
 \end{aligned}$$

6.4 Exercises

1. For the following nonlinear system, find the equilibrium point and the linear model at the equilibrium point.

(a) $\dot{x}_1 = -x_1 + x_2$
 $\dot{x}_2 = -x_1 - x_2 - x_1^2$

(b) $\dot{x}_1 = -3x_2$
 $\dot{x}_2 = x_1 - x_2 - 2x_2^3$

(c) $\dot{x}_1 = -x_1^3 - 3x_1 + x_2$
 $\dot{x}_2 = -2x_1$

(d) $\dot{x}_1 = x_1^2 - x_2^2$
 $\dot{x}_2 = -2x_1x_2$

(e) $\dot{x}_1 = -x_1 + x_2$
 $\dot{x}_2 = -x_1 - x_2 - x_1^2$

(f) $\dot{x}_1 = -3x_2$
 $\dot{x}_2 = x_1 - \alpha(x_2^3 - x_2)$

(g) $\dot{x}_1 = -x_1 + x_2 - x_3 + \alpha x_1^3$
 $\dot{x}_2 = -x_1 - x_2$
 $\dot{x}_3 = x_1 - x_2 + \beta x_3 + u$

(h) $\dot{x}_1 = -x_1 + x_2$
 $\dot{x}_2 = (x_1 + x_2) \sin x_1 - 3x_2$

(i) $\ddot{z} + \dot{z} + z^3 + u = 0$

(j) $\ddot{x} - \mu(1 - x^2)\dot{x} + x = 0$

2. For DC motor dynamics on [1.7.1](#), find the transfer function from input voltage to output speed and input torque to output speed.

3. For the following systems, find a transfer function from each input to outputs, zeros, and poles of the system.

(a)

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -4 & -4 & -1 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} \mathbf{u}, \quad \mathbf{y} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \mathbf{x}$$

(b)

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -4 & -4 & -1 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} \mathbf{u}, \quad \mathbf{y} = \begin{bmatrix} 1 & 0 & -2 \\ 0 & 2 & 1 \end{bmatrix} \mathbf{x}$$

(c)

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 1 \\ -3 \end{bmatrix} u, \quad \mathbf{y} = \begin{bmatrix} -3 & 1 & 0 \\ 4 & 0 & 1 \end{bmatrix} \mathbf{x}$$

(d)

$$\dot{\mathbf{x}} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \mathbf{u}, \quad \mathbf{y} = \begin{bmatrix} 1 & 0 & -0.5 \\ -1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \mathbf{x}$$

Chapter 7

Analog Simulation

7.1 Analog Computers and Simulation

An analog computer is a type of computer that uses physical quantities, such as voltage, to represent and solve mathematical problems. It works by creating an electrical circuit that is analogous to the mathematical equation that represents the problem. The computer then measures the voltage at different points in the circuit to obtain the solution to the problem.

Analog computers are particularly well-suited for solving differential equations, which are equations that describe how a quantity changes over time. This is because analog computers can continuously measure and adjust the voltage in the circuit to represent the changing quantity.

Analog computers were widely used in the 20th century to solve problems in engineering, physics, and mathematics. However, they have been largely replaced by digital computers, which are more accurate and versatile. However, analog computers are still used in some specialized applications, such as aircraft flight control systems.

Here is a simplified explanation of how an analog computer works:

- The mathematical model of the problem is converted into an analog block diagram. The block diagram shows how the different parts of the problem are related to each other.
- The analog block diagram is used to create an electrical circuit. The electrical components are connected together in a way that represents the mathematical relationships in the block diagram.
- The electrical circuit is powered up, and the voltages at different points in the circuit are measured. The voltages represent the solutions to the problem.
- The voltages are converted back to the original problem variables. This gives the final solution to the problem.

Analog computers have a number of advantages over digital computers for certain types of problems. For example, analog computers can solve differential equations in real time. This makes them ideal for applications such as aircraft flight control systems, where the computer needs to be able to respond to changes in the environment very quickly.

However, analog computers also have a number of disadvantages. For example, analog computers are less accurate than digital computers. They are also more difficult to program and maintain.

7.2 Solving Differential Equations with an Analog Computer

A typical simulation of a physical system involves a mathematical model consisting of a set of one or more differential equations and initial conditions on the variables. If the system is linear, the differential equations are linear, and the operations required are

1. Summation
2. Sign inversion
3. Multiplication by a constant
4. Integration
5. Differentiation

For practical reasons, the integration operation is easier to implement than the differentiation operation. The reason lies in the fact that computer signals are real voltages and, therefore, are corrupted by noise to some extent. Since integration tends to average out the effects of noise (while differentiation will accentuate it), a more precise solution can be obtained using integration techniques.

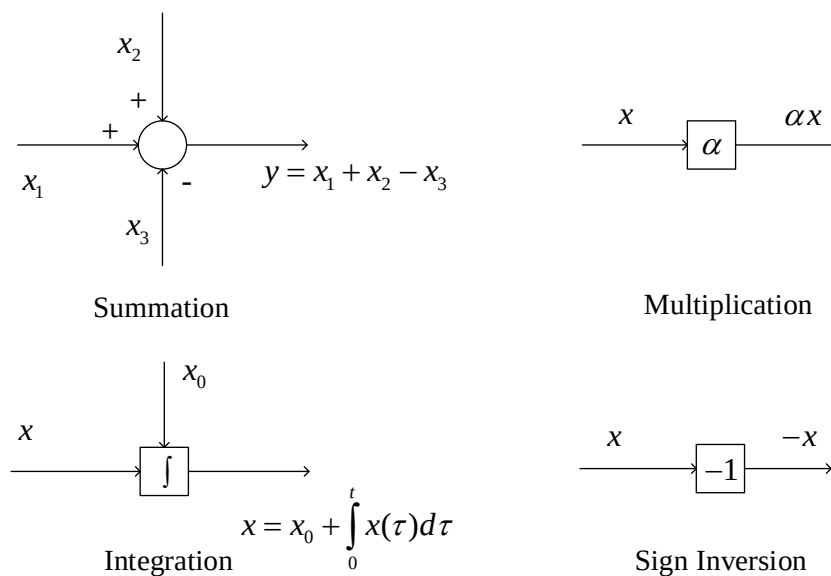


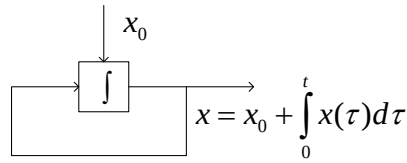
Figure 7.1: *Analog Computing Operations*

Consider an unforced first-order system (zero input signal), with initial condition x_0

$$\dot{x} = x \quad x_0 = 1 \quad (7.2.1)$$

Integrating both sides

$$x = x_0 + \int_0^t x(\tau) d\tau \quad (7.2.2)$$

Figure 7.2: *Unforced First Order System Realization*

As we can understand from Figure 7.2, it is not necessary to know the input of the integrator to solve equation 7.2.1. The only requirement is that the input to the integrator must always be equal to the output. Hence, the concept of feeding the unknown output back to the integrator is the solution to the differential equation in an analog computer.

This is because the solution to a differential equation is general form. An initial or boundary condition is required to determine a specific solution for a particular differential equation.

Since higher-order differential equations can be reduced to cascaded first-order differential equations, the method of realization stays the same. For example, consider a second-order unforced differential equation

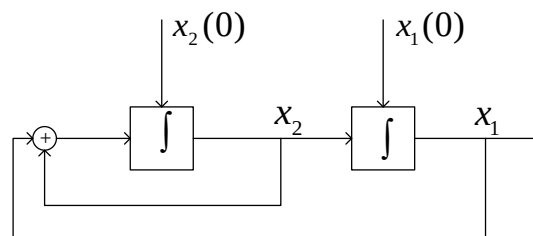
$$\ddot{x} + \dot{x} + x = 0 \quad x_0 = 1 \quad \dot{x}_0 = 0 \quad (7.2.3)$$

This equation can be written as

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= -x_1 - x_2 \end{aligned} \quad (7.2.4)$$

Integration on both sides

$$\begin{aligned} x_1 &= x_1(0) + \int_0^t x_2(\tau) d\tau \\ x_2 &= x_2(0) - \int_0^t [x_1(\tau) + x_2(\tau)] d\tau \end{aligned} \quad (7.2.5)$$

Figure 7.3: *Unforced Second Order System Realization*

7.3 Physical Realization of Analog Computer

1. Summation

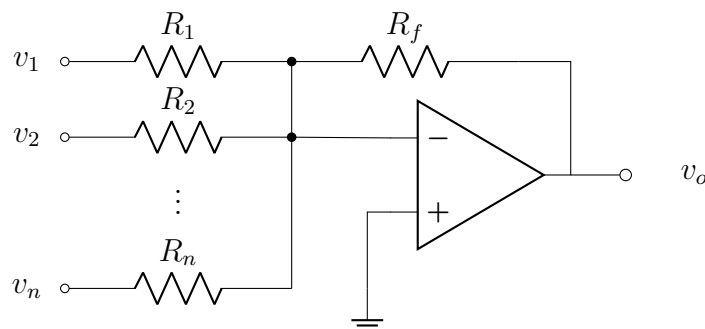


Figure 7.4: *Summer Circuit Diagram Using Op-amp*

From the inverting operational amplifier, the voltage transfer characteristics we have,

$$v_o = -\frac{R_f}{R_1}v_1 - \frac{R_f}{R_2}v_2 - \dots - \frac{R_f}{R_n}v_n \quad (7.3.1)$$

Let $k_1 = \frac{R_f}{R_1}$, $k_2 = \frac{R_f}{R_2}$, ..., and $k_n = \frac{R_f}{R_n}$ we have,

$$v_o = -k_1v_1 - k_2v_2 - \dots - k_nv_n \quad (7.3.2)$$

2. Integration

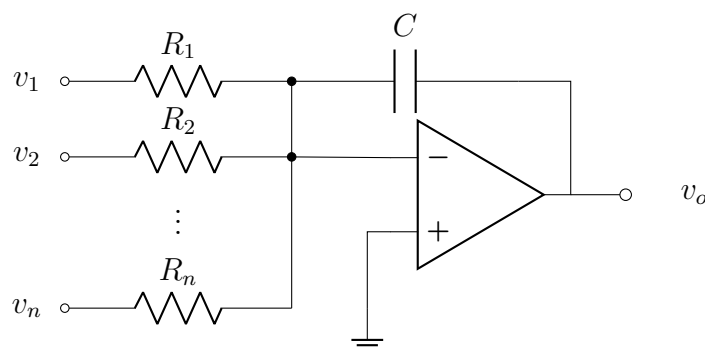


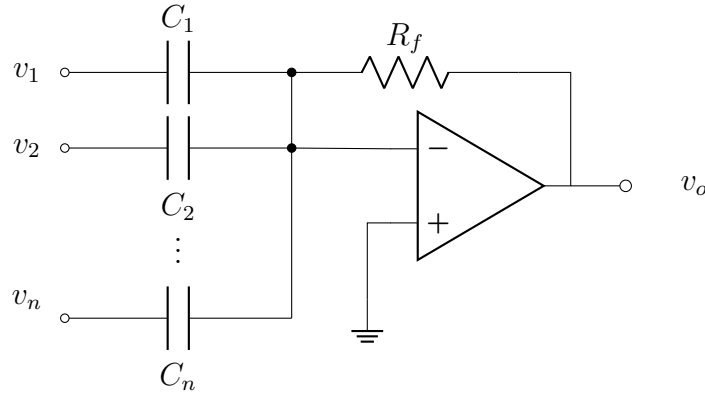
Figure 7.5: *Integrator Circuit Diagram Using Op-amp*

For the integrator circuit, the voltage transfer function is,

$$\begin{aligned} v_o &= v_o(0) - \int_0^t \frac{1}{CR_1}v_1(\tau)d\tau - \int_0^t \frac{1}{CR_2}v_2(\tau)d\tau - \dots - \int_0^t \frac{1}{CR_n}v_n(\tau)d\tau \\ v_o &= v_o(0) - \int_0^t \left[\frac{1}{CR_1}v_1(\tau) + \frac{1}{CR_2}v_2(\tau) + \dots + \frac{1}{CR_n}v_n(\tau) \right] d\tau \end{aligned} \quad (7.3.3)$$

Let $k_1 = \frac{1}{CR_1}$, $k_2 = \frac{1}{CR_2}$, ..., and $k_n = \frac{1}{CR_n}$ we have,

$$v_o = v_o(0) - \int_0^t [k_1v_1(\tau) + k_2v_2(\tau) + \dots + k_nv_n(\tau)] d\tau \quad (7.3.4)$$

Figure 7.6: *Differentiator Circuit Diagram Using Op-amp*

3. Differentiation

For the differentiator circuit, the voltage transfer function is,

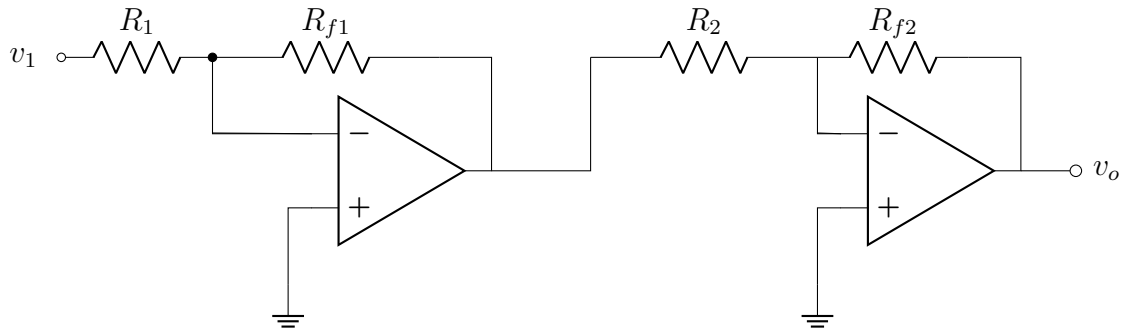
$$v_o = -R_f C_1 \frac{dv_1}{dt} - R_f C_2 \frac{dv_2}{dt} - \dots - R_f C_n \frac{dv_n}{dt}$$

$$v_o = v_o(0) - \int_0^t \left[\frac{1}{C R_1} v_1(\tau) + \frac{1}{C R_2} v_2(\tau) + \dots + \frac{1}{C R_n} v_n(\tau) \right] d\tau \quad (7.3.5)$$

Let $k_1 = R_f C_1$, $k_2 = R_f C_2$, ..., and $k_n = R_f C_n$ we have,

$$v_o = -k_1 \frac{dv_1}{dt} - k_2 \frac{dv_2}{dt} - \dots - k_n \frac{dv_n}{dt} \quad (7.3.6)$$

4. Sign inversion

Figure 7.7: *Sign Invertor Circuit Diagram Using Op-amp*

For the sign inversion circuit, the voltage transfer function is,

$$v_o = \frac{R_{f2} R_{f1}}{R_1 R_2} v_1 \quad (7.3.7)$$

If $\frac{R_{f2} R_{f1}}{R_1 R_2} = 1$, we have $v_o = v_1$.

7.4 Programming Differential Equation Using Analog Computer

To solve differential equations using an analog computer, we can combine the circuits for an n^{th} order integrator and n^{th} order differentiator.

The circuit for an n^{th} order integrator consists of n operational amplifiers (op-amps) connected in series with capacitors and resistors. The output of each op-amp is connected to the input of the next op-amp, and the input signal is applied to the first op-amp. The output of the last op-amp is the integrated signal.

The circuit for an n^{th} order differentiator is similar, but with capacitors and resistors connected in parallel instead of in series. The input signal is applied to the first capacitor, and the output of each op-amp is connected to the input of the next op-amp. The output of the last op-amp is the differentiated signal.

By combining these circuits, we can solve differential equations by converting them into an analog block diagram, creating an electrical circuit based on the block diagram, and measuring the voltages at different points in the circuit to obtain the solution to the problem.

Consider n^{th} order differential equation

$$\frac{d^n}{dt^n}x + \frac{d^{n-1}}{dt^{n-1}}x + \dots + \frac{d}{dt}x + x = 0 \quad (7.4.1)$$

with n initial conditions.

If $\frac{d^n}{dt^n}x$ is known, then all the rest variables can be determined by integrating $\frac{d^n}{dt^n}x$, or if $\frac{d^{n-1}}{dt^{n-1}}x, \dots, \frac{d}{dt}x$ and x are known, we can determine $\frac{d^n}{dt^n}x$. This implies that we are only expected to know the input and output of the integrator. The analog circuit that integrates $\frac{d^n}{dt^n}x$ is shown in Figure 7.8.

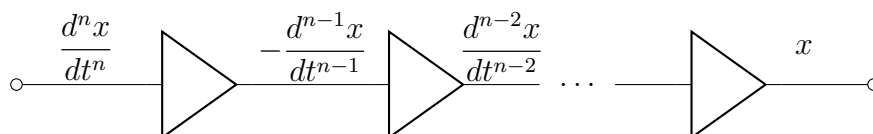


Figure 7.8: Analog Circuit for a n^{th} Order Integrator

The analog circuit that can sum $\frac{d^{n-1}}{dt^{n-1}}x, \dots, \frac{d}{dt}x$ and x to get $\frac{d^n}{dt^n}x$ is shown in Figure 7.8.

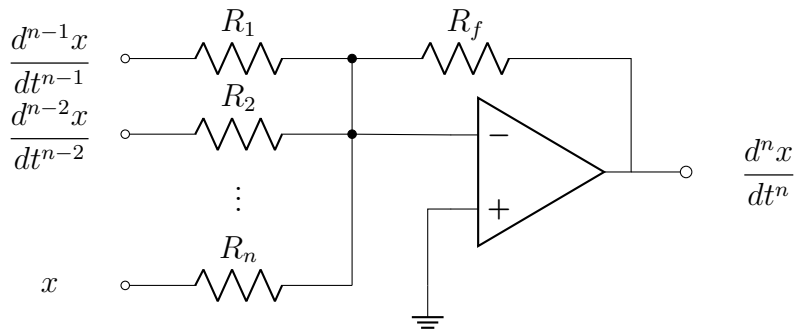


Figure 7.9: *Summer Circuit*

Combining the two analog circuits will give us an analog circuit used to solve n^{th} order ordinary differential equations shown in Figure 7.10.

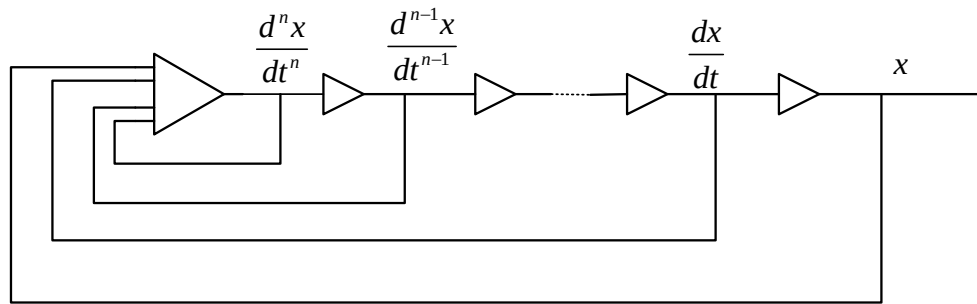


Figure 7.10: n^{th} Order Ordinary Differential Solver

7.5 Magnitude and Time Scaling

In the field of analog computing, equipment constraints often limit the maximum voltage output of each amplifier to 10 volts. This is because exceeding this voltage output can result in amplifier overload and render the solution invalid. When faced with a problem that requires a large voltage output, it becomes necessary to scale the problem down to fit the computer's limitations.

For instance, if we were to solve a problem involving rocket speed on an analog computer, it would be impossible to relate 1 volt on the computer to 1cm/sec in the real problem. However, we could relate 1 volt on the computer to 1km/sec in the problem. This illustrates the need to scale the problem so that it will fit the computer.

Similarly, in the same problem, an acceleration term might have a maximum value of 1cm/sec^2 . Using the basic unit of miles for length as determined for scaling the velocity would give a voltage of approximately 0.002 volts, which corresponds to the acceleration term. However, this signal is in the noise level of the computer and would yield erroneous results. Thus, it is necessary to scale up some parts of a problem and scale down other parts.

Time scaling is also necessary in certain situations. For example, when studying a physical phenomenon that evolves much faster or slower than is convenient for the computer operator, time scaling can be used. If we wish to study planetary motion, waiting for a complete period of revolution about the sun would be impractical. Instead, an actual time of one year may be represented by 10 seconds of computer time.

Equipment constraints and limitations require careful consideration when solving problems on analog computers. Scaling techniques must be employed to ensure that the problem fits within the computer's limitations. Time scaling can also be used to study physical phenomena that evolve at different rates than is convenient for the computer operator.

7.5.1 Magnitude scaling

Magnitude scaling is necessary when a problem variable, existing as an amplifier output, becomes too large and causes amplifier overload or becomes too small, causing the amplifier output noise to dominate the signal.

Scaling is performed based on the maximum voltage supplied to the analog computer and the maximum magnitude that appears in the solution of the differential equation. The input and output of the analog computer are scaled down using the voltage divider rule, while operational amplifiers are used to scale them up. Similarly, the initial conditions can be scaled up or down.

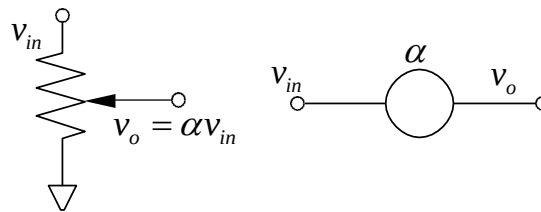


Figure 7.11: *Circuit Diagram of Scale Down Using Potentiometer*

A Systematic Approach to Magnitude Scaling

1. Draw the unscaled analog diagram. Remember that + 10 volts is the maximum reference available on the computer.
2. Estimate the maximum value of each variable appearing as an amplifier output. Only the amplifier outputs need to be considered since all other elements have a gain that is less than unity.
3. Define a scaling relationship between each problem variable and the corresponding computer variable, and prepare a scaling table.
4. Write the scaled analog equations for each amplifier in terms of the computer variables (amplifier outputs) defined in step 3.
5. Draw the scaled analog diagram using the equations obtained in step 5. Label all computer variables on the diagram. The final diagram should have the same form as the unscaled diagram except for different potentiometer values and amplifier gains.

Suppose we need to find an analog simulation that converts m to cm , with the maximum value of m converted being $5m$. Let x be the number of m and y be the number of cm . So we have

$$y = ax$$

Where $a = 100$.

We must scale the problem so that we can represent 500 centimeters, y_{max} , as an analog voltage. We also know

$$V_x = L_x x \quad V_y = L_y y$$

The analog simulation diagram for the system is shown below.

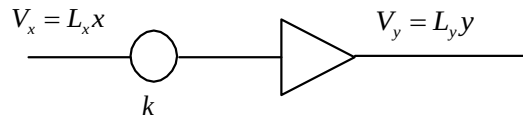


Figure 7.12: *Scaling Diagram*

A potentiometer is set to some value which we denote by 1. Electronically,

$$\begin{aligned}
 V_{out} &= \text{Overall Gain } V_{in} \\
 -L_y y &= -kGL_x x \\
 y &= \frac{kGL_x}{L_y} x
 \end{aligned}$$

Which means $a = \frac{kGL_x}{L_y}$. So the gain of the potentiometer must have a setting

$$k = \frac{aL_y}{L_x}$$

In this example

$$L_x = \frac{10}{5} = 2 \quad L_y = \frac{10}{100} = 0.1 \Rightarrow k = \frac{aL_y}{L_x} = \frac{100 * 0.1}{2} = 5$$

Generally

$$k = \frac{\text{Level Out COEF}}{\text{Level In gain}} \quad (7.5.1)$$

Where

- *Level Out* is the level of the variable immediately following the pot to be set,
- *Level In* the level of the variable preceding the pot,
- *COEF* relates to the equation to be simulated and,
- *gain* is the gain of a summer, inverter, or integrator following the pot.

7.5.2 Time scaling

Analog simulation time can be adjusted to run faster or slower than real time.

$$\tau_c = \beta t \quad (7.5.2)$$

where τ_c computer, t is real time and β scaling factor.

Time scaling can be achieved by multiplying the gains of every integrator by $\frac{1}{\beta}$. Therefore, we can adjust the previous pot setting formula to take care of magnitude and time scaling. We simply use a single pot for each input to the integrators such that:

$$k = \frac{\text{Level Out COEF}}{\text{Level In gain}} \frac{1}{\beta} \quad (7.5.3)$$

Since there is no time variation for initial conditions and non-integrating operational amplifiers, time scaling will not be applied to them.

7.6 Analog Simulation of Transfer Functions and State Space Representation of Systems

To develop a scaled analog simulation diagram of transfer functions, use the following procedures.

1. Convert the transfer function into differential equations using direct or indirect programming or standard programming methods.
2. Write the differential equation to be solved in normal form.
3. Estimate maximum values for variables.
4. Draw integrators needed and put a pot between each pair of integrators.
5. Determine the scaling factors based on the maximum variable occurring in the differential equation, as described in step 3.

To develop a scaled analog simulation diagram of state space representation, use the following procedures.

1. Derive the differential equations from the state space equations.
2. Write the differential equation to be solved in normal form.
3. Estimate maximum values for variables.
4. Draw all integrators needed and put a pot between each pair of integrators.
5. Determine the scaling factors based on the maximum variable occurring in the differential equation, as described in step 3.

7.6.1 Initial conditions

Suppose that we wish to simulate a transfer function, but with initial conditions included. This is not a well-posed problem, as such, but we can work around that. Consider a second-order transfer function

$$G(s) = \frac{a_2 s^2 + a_1 s + a_0}{s^2 + b_1 s + b_0} \quad (7.6.1)$$

In order to incorporate these initial conditions, we need to “translate” them into the initial conditions of the actual integrators, and possibly the input.

$$\begin{aligned} y(0^-) &= a_0 x(0^-) + a_1 \dot{x}(0^-) + a_2 \ddot{x}(0^-) \\ &= a_0 x(0^-) + a_1 \dot{x}(0^-) + a_2(-u(0^-) - b_1 \dot{x}(0^-) - b_0 x(0)) \\ &= (a_0 - a_2 b_0)x(0^-) + (a_1 - a_2 b_1)\dot{x}(0^-) - a_2 u(0^-) \end{aligned} \quad (7.6.2)$$

and also

$$\begin{aligned} \dot{y}(0^-) &= a_0 \dot{x}(0^-) + a_1 \ddot{x}(0^-) + a_2 x^{(3)}(0^-) \\ &= a_0 \dot{x}(0^-) + a_1(-u(0^-) - b_1 \dot{x}(0^-) - b_0 x(0)) + a_2(-\dot{u}(0^-) - b_1 \ddot{x}(0^-) - b_0 \dot{x}(0)) \\ &= -b_0(a_1 - a_2 b_1)x(0^{-1}) + [a_0 - a_2 b_0 - b_1(a_1 - a_2 b_1)]\dot{x}(0^{-1}) - (a_1 - a_2 b_1)u(0^{-1}) \\ &\quad - a_2 \dot{u}(0^{-1}) \end{aligned} \quad (7.6.3)$$

We note that the initial states of the integrators can be rewritten in terms of the initial states of y and $\dot{y}$, and upon the input and its derivative. Note that when we take the Laplace transform of a signal (like $u(t)$), we assume the signal to be zero and all its derivatives to be zero. With is assumptions equations 7.6.2 and 7.6.3 simplifies to

$$y(0^-) = (a_0 - a_2 b_0)x(0^-) + (a_1 - a_2 b_1)\dot{x}(0^-) \quad (7.6.4)$$

$$\dot{y}(0^-) = -b_0(a_1 - a_2 b_1)x(0^{-1}) + [a_0 - a_2 b_0 - b_1(a_1 - a_2 b_1)]\dot{x}(0^{-1}) \quad (7.6.5)$$

Let $a_{11} = a_0 - a_2 b_0$, $a_{12} = a_1 - a_2 b_1$, $a_{21} = -b_0(a_1 - a_2 b_1)$ and $a_{22} = a_0 - a_2 b_0 - b_1(a_1 - a_2 b_1)$,

$$y(0^-) = a_{11}x(0^-) + a_{12}\dot{x}(0^-) \quad (7.6.6)$$

$$\dot{y}(0^-) = a_{21}x(0^{-1}) + a_{22}\dot{x}(0^{-1}) \quad (7.6.7)$$

$$\begin{bmatrix} y(0^-) \\ \dot{y}(0^-) \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} x(0^-) \\ \dot{x}(0^-) \end{bmatrix} \Leftrightarrow \begin{bmatrix} x(0^-) \\ \dot{x}(0^-) \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}^{-1} \begin{bmatrix} y(0^-) \\ \dot{y}(0^-) \end{bmatrix} \quad (7.6.8)$$

thus we obtain

$$x(0^-) = \frac{\alpha_{22}y(0^-) - \alpha_{12}\dot{y}(0^-)}{\alpha_{11}\alpha_{22} - \alpha_{12}\alpha_{21}} \quad (7.6.9)$$

$$\dot{x}(0^-) = \frac{-\alpha_{12}y(0^-) - \alpha_{22}\dot{y}(0^-)}{\alpha_{11}\alpha_{22} - \alpha_{12}\alpha_{21}} \quad (7.6.10)$$

7.7 Examples

1. Determine the circuit to produce the output voltage given by

$$v_o = -v_1 - 10v_2 + v_3 + 10v_4$$

Solution:

The analog circuit diagram that must be patched on the analog computer

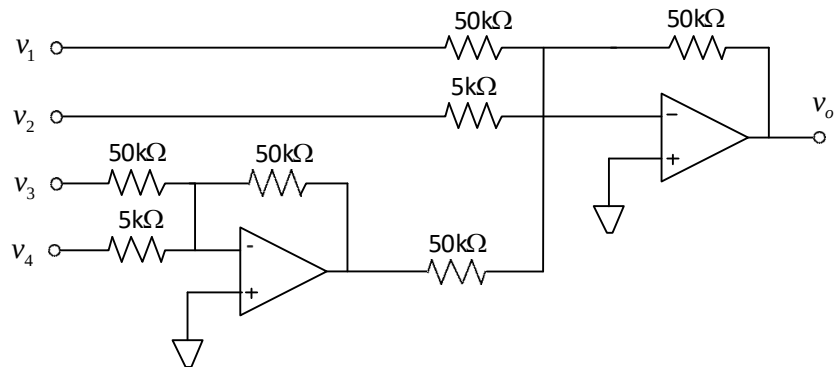


Figure 7.13: *Circuit Diagram for Example 1*

2. Determine the analog diagram and circuit to implement the equation

$$v_o = -0.35v_1 + 5.24v_2 + 2.6$$

Solution:

The analog circuit diagram must be patched on the analog computer

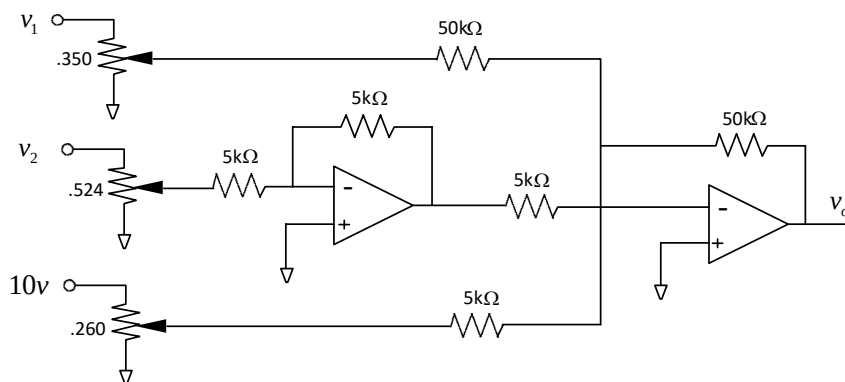


Figure 7.14: *Circuit Diagram for Example 2*

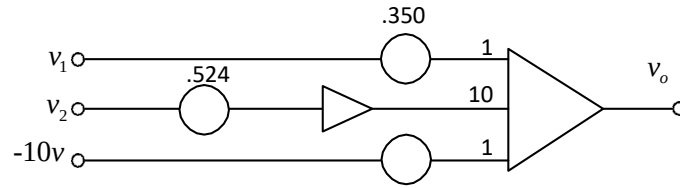


Figure 7.15: Analog Diagram for Example 2

3. For the following second-order linear differential equation and initial conditions, determine the analog circuit diagram.

$$\ddot{x} + 0.5\dot{x} + x = 4 \quad \dot{x}_0 = 1 \quad \text{and} \quad x_0 = 0 \quad (7.7.1)$$

Solution:

The analog circuit diagram that must be patched on the analog computer

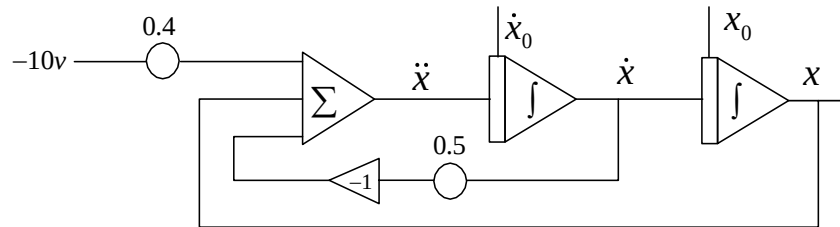


Figure 7.16: Circuit Diagram for Example 3

4. For the following transfer function, determine an analog circuit diagram

$$G(s) = \frac{a_2 s^2 + a_1 s + a_0}{s^2 + b_1 s + b_0} = \frac{N(s)}{D(s)}$$

Solution:

Using direct programming, we have

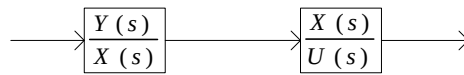
$$\frac{Y(s)}{U(s)} = \frac{a_2 s^2 + a_1 s + a_0}{s^2 + b_1 s + b_0};$$

With an intermediate term $x(s)$ we have

$$\begin{aligned} \frac{Y(s)}{X(s)} &= a_2 s^2 + a_1 s + a_0 \\ Y(s) &= a_2 s^2 X(s) + a_1 s X(s) + a_0 X(s) \end{aligned}$$

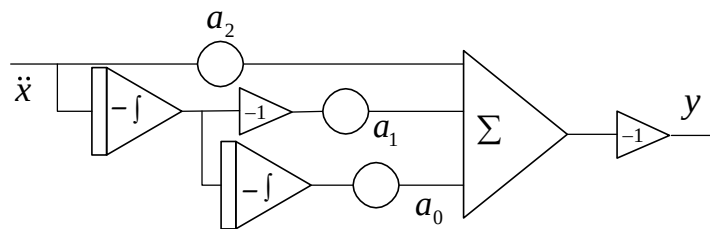
and

$$\begin{aligned} \frac{X(s)}{U(s)} &= \frac{1}{s^2 + b_1 s + b_0} \\ U(s) &= s^2 X(s) + b_1 s X(s) + b_0 X(s) \\ s^2 X(s) - b_1 s X(s) - b_0 X(s) + U(s) &= 0 \end{aligned}$$

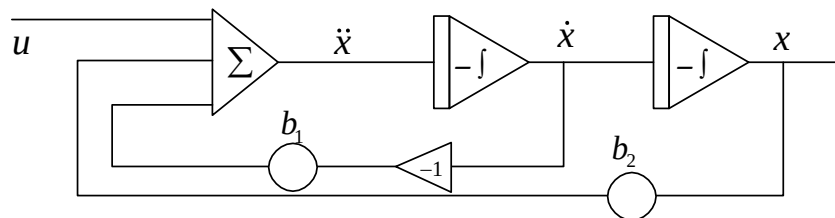
Figure 7.17: *Block diagram Representation of Transfer Function*

This implies that the analog circuit diagram of the two differential equations can be designed same way and combined as follows

The analog circuit diagram of the transfer function from the state to the output is

Figure 7.18: *Analog Circuit Diagram of Transfer Function From State to Input*

The analog circuit diagram of the transfer function from the input to the state is

Figure 7.19: *Analog Circuit Diagram of Transfer Function from Input to State*

Combining the two analog circuit diagrams, the transfer function from input to output is,

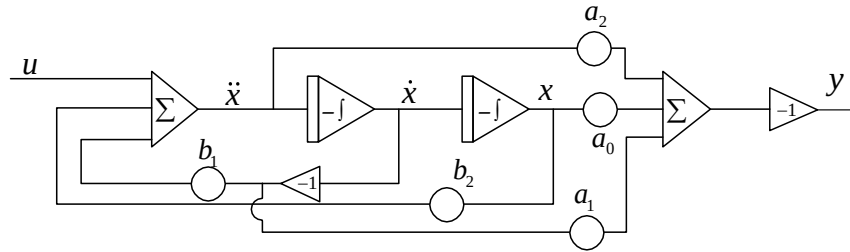


Figure 7.20: Analog Circuit Diagram of Transfer Function from Input to Output

5. Simulate the following system and magnitude scale so that no overload occurs and the maximum amplifier range is used

$$\ddot{x} + 8\dot{x} + 25x = 500, \quad x(0) = 40 \quad \dot{x}(0) = 150$$

with $|x|_{max} = 50$ and $|\dot{x}|_{max} = 250$.

Solution:

- (a) Step 1: Draw an unscaled analog diagram

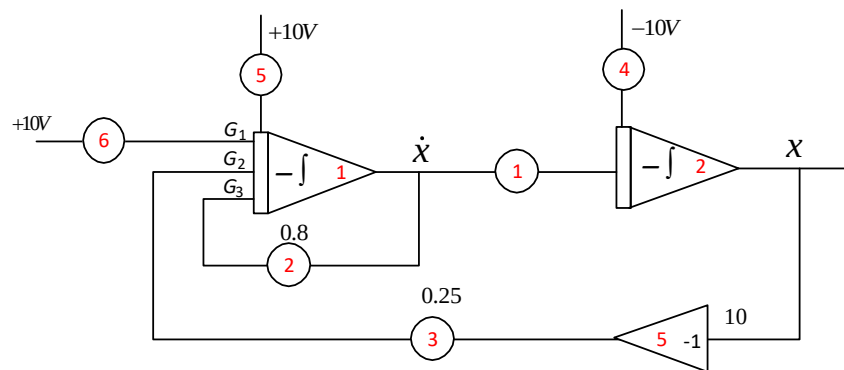


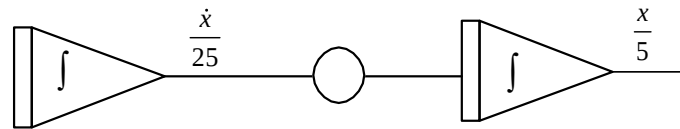
Figure 7.21: Unscaled Analog Circuit Diagram

- (b) Step 2: This step is not necessary since the maximum values are already specified in the problem statement.
- (c) Step 3: Prepare a scaling table
- (d) Step 4: Write the scaled equations for each amplifier.

$k_1 = \frac{1/5}{1/25} \frac{1}{G_1}$, if we select $G_1 = 10$, then $k_1 = 0.5$. Similarly, computing for

Table 7.1: *Scaling table*

Problem variable	Estimated Maximum	Level	Computer variable
$-\dot{x}$	250	1/25	$[-\dot{x}/25]$
x	50	1/5	$[x/5]$
$-2.5x$	125	1/12.5	$[-x/5]$

Figure 7.22: *Establishing Integrator Outputs*

the rest of the circuit, we have

$$k_2 = \frac{1/25}{1/5} \frac{8}{G_1} = 0.8 \quad G_1 = 10$$

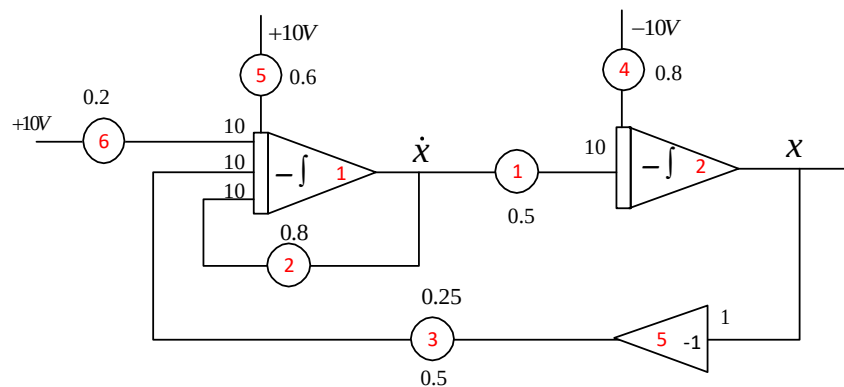
$$k_3 = \frac{1/25}{1/5} \frac{25}{G_2} = 0.5 \quad G_2 = 10$$

$$k_4 = \frac{1/5}{10} \frac{40}{1} = 0.8$$

$$k_5 = \frac{1/25}{10} \frac{150}{1} = 0.6$$

$$k_6 = \frac{1/25}{10} \frac{500}{G_3} = 0.2 \quad G_3 = 10$$

(e) Step 5: Draw the scaled analog computer diagram

Figure 7.23: *Fully Scaled Diagram*

7.8 Exercises

1. For the following system, find an analog circuit diagram

$$G_1(s) = \frac{s^3 + 2s^2 + 1}{s^3 + 5s^2 + 3s + 6} \quad G_2(s) = \frac{2s^3 + 5s^2 + 3s + 1}{4s^4 + 2s^2 + 3s + 6}$$

2. For the following system, find an analog circuit diagram

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & -1 \\ 2 & -4 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 2 \\ 0 \end{bmatrix} u$$

3. For the following system, find an analog circuit diagram

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & -1 \\ 2 & -4 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \mathbf{u}$$

4. Magnitude-scale the following problem so that no overload occurs and maximum amplifier range is used.

$$\ddot{x} + \dot{x} + 25x = 0, \quad x(0) = 20 \quad \dot{x}(0) = 0$$

$$\text{with } |x|_{\max} \leq 20 \text{ and } |\dot{x}|_{\max} \leq 100.$$

5. Magnitude-scale the following problem

$$\ddot{x} + 10\dot{x} + 25x = 20, \quad x(0) = 0 \quad \dot{x}(0) = 0$$

$$\text{with } |x|_{\max} \leq 1.6 \text{ and } |\dot{x}|_{\max} \leq 4.$$

6. Magnitude-scale and time scale the following problem

$$\ddot{x} + 10\dot{x} + 50x = 20, \quad x(0) = 10 \quad \dot{x}(0) = 0$$

$$\text{with } |x|_{\max} \leq 1.6, |\dot{x}|_{\max} \leq 4 \text{ and } \beta = 2.5.$$

Appendix A

Hamilton's Principle and Lagrangian Mechanics

A.1 Hamilton's Principle (Generalized Classical Mechanics)

Hamilton's principle states that the motion of a system between two points in time t_1 and t_2 is such that the **action** S is stationary (usually optimized):

$$J = \int_{t_1}^{t_2} L dt, \tag{A.1.1}$$

where L is the Lagrangian.

A.1.1 What is the Lagrangian?

The Lagrangian L is a function of generalized coordinates $\mathbf{q}$, generalized velocities $\dot{\mathbf{q}}$, and time t (in some cases). It is defined as:

$$L(\mathbf{q}, \dot{\mathbf{q}}, t) = T(\mathbf{q}, \dot{\mathbf{q}}, t) - V(\mathbf{q}, t) \tag{A.1.2}$$

where:

- $T(\mathbf{q}, \dot{\mathbf{q}}, t)$ is the kinetic energy of the system in generalized coordinates.
- $V(\mathbf{q}, t)$ is the potential energy of the system in generalized coordinates.

A.1.2 Properties of the Lagrangian

The Lagrangian of the system:

- Encodes the dynamics of the system in a single function.
- Is a scalar quantity, making it easier to work with than vector quantities like force.
- is particularly useful for systems with constraints or complex geometries.

A.1.3 Connection to Newton's Second Law

Consider a particle with mass m moving in a generalized coordinate system. The kinetic energy of the particle is $T = \frac{1}{2}m\dot{\mathbf{q}}^2$, and its potential energy is $V = V(\mathbf{q})$. The Lagrangian of the system is:

$$L = \frac{1}{2}m\dot{\mathbf{q}}^2 - V(\mathbf{q}) \quad (\text{A.1.3})$$

Applying the Euler-Lagrange equation:

$$-\frac{\partial V}{\partial \mathbf{q}} - \frac{d}{dt}(m\dot{\mathbf{q}}) = 0 \implies \mathbf{F} = m\ddot{\mathbf{q}} \quad (\text{A.1.4})$$

This recovers **Newton's second law**.

A.1.4 Why Use the Euler-Lagrange Method?

The Euler-Lagrange method provides:

- A unified framework for mechanics:
 - Generalizes classical mechanics to more complex systems (e.g., systems with constraints).
 - Works in any coordinate system (non-Cartesian coordinates).
- Extensions to other fields (e.g., electromagnetism, fluid dynamics).
- A foundation for quantum mechanics and modern physics.

A.2 Derivation of Euler-Lagrange Equations

A.2.1 Functional vs Function

A function depends on one or more independent variables:

- Function: $V(\mathbf{x}, \mathbf{y}, \mathbf{z})$

A functional can depend on one or more functions:

- Functional: $J(\mathbf{x}(t), \mathbf{y}(t), \mathbf{z}(t))$

Note: The key difference is that in a functional, $\mathbf{x}$, $\mathbf{y}$, and $\mathbf{z}$ are functions of another independent variable.

A.2.2 Increment of a Function and a Functional

Consider a function $f(t)$. The increment of $f(t)$ is defined as:

$$\Delta f = f(t + \Delta t) - f(t) \quad (\text{A.2.1})$$

Δf depends on both t and Δt , hence it is written as $\Delta f(t, \Delta t)$. Similarly, for a functional $J(\mathbf{x}(t), t)$, the increment is defined as:

$$\Delta J = J((\mathbf{x}(t) + \delta \mathbf{x}(t)), t) - J(\mathbf{x}(t), t) \quad (\text{A.2.2})$$

Here, $\delta \mathbf{x}(t)$ is called the variation of the function $\mathbf{x}(t)$. Since the increment of a functional depends on $\mathbf{x}(t)$ and $\delta \mathbf{x}(t)$, it is written as $J(\mathbf{x}(t), \delta \mathbf{x}(t))$.

A.2.3 Differential of a Function

For a function $f(t)$, the Taylor series expansion about a point t^* is given as:

$$\Delta f = f(t^*) + \left(\frac{\partial f}{\partial t} \right)_{t^*} \Delta t + \frac{1}{2!} \left(\frac{\partial^2 f}{\partial t^2} \right)_{t^*} \Delta t^2 + \dots - f(t^*) \quad (\text{A.2.3})$$

Neglecting higher-order terms in Δt :

$$\begin{aligned} \Delta f &= f(t^*) + \left(\frac{\partial f}{\partial t} \right)_{t^*} \Delta t - f(t^*) \\ \Delta f &= \dot{f}(t^*) \Delta t = df \end{aligned} \quad (\text{A.2.4})$$

Here, df is called the differential of f at point t^* , and $\dot{f}(t)$ is the slope of f at t^* .

A.3 Optimization of Functions and Functionals

A.3.1 Optimum of a Function

For a function to have an optimum value at a point, the following conditions must be met:

- **Necessary Condition:** The first derivative must vanish at the point (i.e., $df = 0$ at t^*).
- **Sufficient Condition:**
 - $d^2f > 0$ for a minimum.
 - $d^2f < 0$ for a maximum.
- If $d^2f = 0$, the point is called an inflection point or stationary point.

A.3.2 Variation of a Functional

Expanding a functional $J(\mathbf{x}(t), t)$ using Taylor series about a function $\mathbf{x}^*(t)$, gives us:

$$\begin{aligned} \Delta J &= J(\mathbf{x}^*(t)) + \left(\frac{\partial J}{\partial \mathbf{x}} \delta \mathbf{x} \right)_{\mathbf{x}^*(t)} + \left(\frac{1}{2!} \frac{\partial^2 J}{\partial \mathbf{x}^2} \delta \mathbf{x}^2 \right)_{\mathbf{x}^*(t)} + \dots - J(\mathbf{x}^*(t)) \\ \Delta J &= \delta J + \delta^2 J + \dots + \text{higher-order terms} \end{aligned} \quad (\text{A.3.1})$$

Here, $\delta J = \frac{\partial J}{\partial \mathbf{x}} \delta \mathbf{x}$ is called the first variation, and $\delta^2 J = \frac{1}{2!} \frac{\partial^2 J}{\partial \mathbf{x}^2} \delta \mathbf{x}^2$ is called the second variation of the functional about a function $\mathbf{x}^*(t)$.

Neglecting higher-order terms in $\delta \mathbf{x}$, the linear approximation of variation of $J(\mathbf{x}(t), t)$ about a function $\mathbf{x}^*(t)$ is:

$$\Delta J = \delta J = \frac{\partial J}{\partial \mathbf{x}} \delta \mathbf{x} \quad (\text{A.3.2})$$

A.4 Optimization of an Unconstrained Functional

For a function to have an optimum value:

- **Necessary Condition:** The first variation must vanish ($\delta J = 0$).
- **Sufficient Condition:**
 - $\delta^2 J > 0$ for a local minimum.
 - $\delta^2 J < 0$ for a local maximum.

A.4.1 Typical Functional in Control Systems

For linear time systems with functional $J(\mathbf{x}, \dot{\mathbf{x}})$:

$$\Delta J = \int_{t_0}^{t_f} (\varphi(\mathbf{x} + \delta \mathbf{x}, \dot{\mathbf{x}} + \delta \dot{\mathbf{x}})) dt \quad (\text{A.4.1})$$

Using Taylor Series Expansions:

$$\Delta J = \int_{t_0}^{t_f} \left(\frac{\partial \varphi}{\partial \mathbf{x}} \delta \mathbf{x} + \frac{\partial \varphi}{\partial \dot{\mathbf{x}}} \delta \dot{\mathbf{x}} + \text{HOT} \right) dt = 0 \quad (\text{A.4.2})$$

The first variation of the functional is:

$$\delta J = \int_{t_0}^{t_f} \left(\frac{\partial \varphi}{\partial \mathbf{x}} \delta \mathbf{x} + \frac{\partial \varphi}{\partial \dot{\mathbf{x}}} \delta \dot{\mathbf{x}} \right) dt = 0 \quad (\text{A.4.3})$$

A.4.2 Necessary Condition for Optimum

For the functional to have an optimum value, $\delta J = 0$. This leads to:

$$\int_{t_0}^{t_f} \left(\frac{\partial \varphi}{\partial \mathbf{x}} \delta \mathbf{x} + \frac{\partial \varphi}{\partial \dot{\mathbf{x}}} \delta \dot{\mathbf{x}} \right) dt = 0 \quad (\text{A.4.4})$$

Using integration by parts for the second term, we derive the Euler-Lagrange equation:

$$\frac{\partial \varphi}{\partial \mathbf{x}} - \frac{d}{dt} \left(\frac{\partial \varphi}{\partial \dot{\mathbf{x}}} \right) = 0 \quad (\text{A.4.5})$$

And the boundary condition:

$$\left(\frac{\partial \varphi}{\partial \dot{\mathbf{x}}} \delta \mathbf{x} \right) = 0 \quad (\text{A.4.6})$$

Boundary Conditions

For the boundary condition to be zero, either $\frac{\partial\varphi}{\partial\dot{\mathbf{x}}}$ or $\delta\mathbf{x}$ must be zero at the boundary. This leads to four cases:

1. **Case-I:** $\mathbf{x}$ is fixed at t_0 and t_f (fixed end trajectory). Here, $\mathbf{x}(t_0) = C_1$ and $\mathbf{x}(t_f) = C_2$, where C_1 and C_2 are constants. Since $\delta\mathbf{x} = 0$ at both t_0 and t_f , there are no restrictions on $\frac{\partial\varphi}{\partial\dot{\mathbf{x}}}$.
2. **Case-II:** $\mathbf{x}$ is fixed at t_0 and free at t_f . Here, $\mathbf{x}(t_0) = C_1$, and $\frac{\partial\varphi}{\partial\dot{\mathbf{x}}} = 0$ at t_f because $\delta\mathbf{x}$ is unknown at t_f . No restrictions apply to $\frac{\partial\varphi}{\partial\dot{\mathbf{x}}}$ at t_0 .
3. **Case-III:** $\mathbf{x}$ is fixed at t_f and free at t_0 . Here, $\mathbf{x}(t_f) = C_2$, and $\frac{\partial\varphi}{\partial\dot{\mathbf{x}}} = 0$ at t_0 because $\delta\mathbf{x}$ is unknown at t_0 . No restrictions apply to $\frac{\partial\varphi}{\partial\dot{\mathbf{x}}}$ at t_f .
4. **Case-IV:** $\mathbf{x}$ is free at t_0 and t_f . Here, $\frac{\partial\varphi}{\partial\dot{\mathbf{x}}} = 0$ at both t_0 and t_f because $\delta\mathbf{x}$ is unknown at both times.

A.4.3 Sufficient Condition for Optimum

The sufficient condition for an optimum is derived from the second variation of the functional J , denoted as $\delta^2 J$. The expression for $\delta^2 J$ is given by:

$$\delta^2 J = \frac{1}{2} \int_{t_0}^{t_f} \left[\left(\frac{\partial^2 \varphi}{\partial \mathbf{x}^2} - \frac{d}{dt} \left(\frac{\partial^2 \varphi}{\partial \mathbf{x} \partial \dot{\mathbf{x}}} \right) \right) (\delta \mathbf{x})^2 + \left(\frac{\partial^2 \varphi}{\partial \dot{\mathbf{x}}^2} \right) (\delta \dot{\mathbf{x}})^2 \right] dt \quad (\text{A.4.7})$$

For a local minimum, the following conditions must be satisfied:

$$\frac{\partial^2 \varphi}{\partial \mathbf{x}^2} - \frac{d}{dt} \left(\frac{\partial^2 \varphi}{\partial \mathbf{x} \partial \dot{\mathbf{x}}} \right) \geq 0 \quad \text{and} \quad \frac{\partial^2 \varphi}{\partial \dot{\mathbf{x}}^2} \geq 0 \quad (\text{A.4.8})$$

For a local maximum, the conditions are:

$$\frac{\partial^2 \varphi}{\partial \mathbf{x}^2} - \frac{d}{dt} \left(\frac{\partial^2 \varphi}{\partial \mathbf{x} \partial \dot{\mathbf{x}}} \right) \leq 0 \quad \text{and} \quad \frac{\partial^2 \varphi}{\partial \dot{\mathbf{x}}^2} \leq 0 \quad (\text{A.4.9})$$

A.5 Typical Functional in Control Systems

In control systems, a typical functional has the form $J(\mathbf{x}, \dot{\mathbf{x}}, \mathbf{u}, \dot{\mathbf{u}}, t)$:

$$J(\mathbf{x}, \dot{\mathbf{x}}, \mathbf{u}, \dot{\mathbf{u}}, t) = \int_{t_0}^{t_f} \varphi(\mathbf{x}, \dot{\mathbf{x}}, \mathbf{u}, \dot{\mathbf{u}}, t) dt \quad (\text{A.5.1})$$

The increment of $J(\mathbf{x}, \dot{\mathbf{x}}, \mathbf{u}, \dot{\mathbf{u}}, t)$ is given by:

$$\Delta J = \int_{t_0}^{t_f} (\varphi(\mathbf{x} + \delta\mathbf{x}, \dot{\mathbf{x}} + \delta\dot{\mathbf{x}}, \mathbf{u} + \delta\mathbf{u}, \dot{\mathbf{u}} + \delta\dot{\mathbf{u}}, t + \delta t) - \varphi(\mathbf{x}, \dot{\mathbf{x}}, \mathbf{u}, \dot{\mathbf{u}}, t)) dt \quad (\text{A.5.2})$$

For linear time-invariant systems, the increment simplifies to:

$$\Delta J = \int_{t_0}^{t_f} (\varphi(\mathbf{x} + \delta\mathbf{x}, \dot{\mathbf{x}} + \delta\dot{\mathbf{x}}, \mathbf{u} + \delta\mathbf{u}, \dot{\mathbf{u}} + \delta\dot{\mathbf{u}}) - \varphi(\mathbf{x}, \dot{\mathbf{x}}, \mathbf{u}, \dot{\mathbf{u}})) dt \quad (\text{A.5.3})$$

Using Taylor Series expansions, the first variation of the functional is:

$$\Delta J = \int_{t_0}^{t_f} \left(\frac{\partial \varphi}{\partial \mathbf{x}} \delta\mathbf{x} + \frac{\partial \varphi}{\partial \dot{\mathbf{x}}} \delta\dot{\mathbf{x}} + \frac{\partial \varphi}{\partial \mathbf{u}} \delta\mathbf{u} + \frac{\partial \varphi}{\partial \dot{\mathbf{u}}} \delta\dot{\mathbf{u}} + \text{hot} \right) dt = 0 \quad (\text{A.5.4})$$

The first variation of the functional is:

$$\delta J = \int_{t_0}^{t_f} \left(\frac{\partial \varphi}{\partial \mathbf{x}} \delta\mathbf{x} + \frac{\partial \varphi}{\partial \dot{\mathbf{x}}} \delta\dot{\mathbf{x}} \right) dt + \int_{t_0}^{t_f} \left(\frac{\partial \varphi}{\partial \mathbf{u}} \delta\mathbf{u} + \frac{\partial \varphi}{\partial \dot{\mathbf{u}}} \delta\dot{\mathbf{u}} \right) dt = 0 \quad (\text{A.5.5})$$

A.6 Optimization of Constrained Functional

In real-world problems, especially in control systems, optimization is often subject to constraints imposed by natural laws or technological limitations. For example, satellite tracking control is restricted by the selected trajectory and the maximum input allowed. The objective is to determine a control signal $\mathbf{u}$ that optimizes a cost function while satisfying these constraints.

A.6.1 Problem Formulation

The problem is to optimize a cost function J while fulfilling the state dynamics of the system. Specifically, we aim to determine a control signal $\mathbf{u}$ that minimizes:

$$\int_{t_0}^{t_f} \Upsilon(\mathbf{x}, \dot{\mathbf{x}}, \mathbf{u}) dt \quad (\text{A.6.1})$$

Subject to the state dynamics:

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} \quad (\text{A.6.2})$$

A.6.2 General Function Definition

To incorporate the constraints into the optimization problem, we define a general function that includes both the cost function and the constraints:

$$\varphi(\mathbf{x}, \dot{\mathbf{x}}, \mathbf{u}) = \Upsilon(\mathbf{x}, \dot{\mathbf{x}}, \mathbf{u}) + \boldsymbol{\lambda}^\top (\mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} - \dot{\mathbf{x}}) \quad (\text{A.6.3})$$

This reduces the constrained optimization problem to an unconstrained one:

$$\begin{aligned} J &= \int_{t_0}^{t_f} (\Upsilon(\mathbf{x}, \dot{\mathbf{x}}, \mathbf{u}) + \boldsymbol{\lambda}^\top (\mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} - \dot{\mathbf{x}})) dt \\ &= \int_{t_0}^{t_f} \varphi(\mathbf{x}, \dot{\mathbf{x}}, \mathbf{u}) dt \end{aligned} \quad (\text{A.6.4})$$

A.6.3 Euler-Lagrangian Equation

For the function to have an optimum value, the Euler-Lagrangian equations must be satisfied:

$$\frac{\partial \varphi}{\partial \mathbf{x}} - \frac{d}{dt} \left(\frac{\partial \varphi}{\partial \dot{\mathbf{x}}} \right) = 0 \quad (\text{A.6.5})$$

$$\frac{\partial \varphi}{\partial \mathbf{u}} - \frac{d}{dt} \left(\frac{\partial \varphi}{\partial \dot{\mathbf{u}}} \right) = 0 \quad (\text{A.6.6})$$

$$\frac{\partial \varphi}{\partial \boldsymbol{\lambda}^\top} - \frac{d}{dt} \left(\frac{\partial \varphi}{\partial \dot{\boldsymbol{\lambda}}^\top} \right) = 0 \quad (\text{A.6.7})$$

A.7 Euler-Lagrangian Equation for systems with Generalized Forces

The first variation of the functional is:

$$\delta J = \int_{t_0}^{t_f} \left(\frac{\partial \varphi}{\partial \mathbf{x}} \delta \mathbf{x} + \frac{\partial \varphi}{\partial \dot{\mathbf{x}}} \delta \dot{\mathbf{x}} \right) dt = F_i \quad (\text{A.7.1})$$

where F_i are the generalized forces, including friction and external input.

Which leads to the Euler-Lagrangian equation:

$$\frac{\partial L}{\partial q} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) = F_i, \quad (\text{A.7.2})$$

A.8 Example:-1 Inverted Pendulum on a Cart System Dynamics

The objective of the inverted pendulum on a cart control system is to maintain the pendulum in an upright position while simultaneously controlling the movement of the cart along a track. This system is a classic example in control theory and is used to study stability and control strategies.

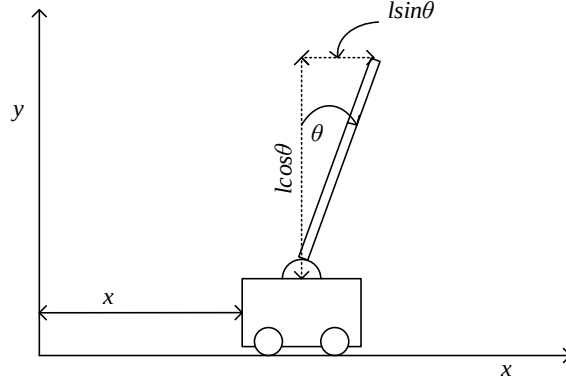


Figure A.1: *Inverted Pendulum on a Cart (IPC) System*

A.8.1 System Description

The system consists of:

- A cart moving along a rail.
- A pendulum attached to the cart.

The generalized coordinates are:

$$q_1 = x \quad (\text{cart position}), \quad q_2 = \theta \quad (\text{pendulum angle}). \quad (\text{A.8.1})$$

Friction forces in the system include:

- $b_c \dot{q}_1$: Friction between the cart and the rail (viscous damping).
- $b_p \dot{q}_2$: Friction at the pendulum joint (viscous damping).

A.8.2 Dynamic Model of the System

The linear velocity of the cart and the angular velocity of the pendulum are given by:

$$\begin{bmatrix} \dot{x}_c \\ \dot{y}_c \end{bmatrix} = \begin{bmatrix} \dot{q}_1 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} \dot{x}_p \\ \dot{y}_p \end{bmatrix} = \begin{bmatrix} \dot{q}_1 + l \dot{q}_2 \cos q_2 \\ -l \dot{q}_2 \sin q_2 \end{bmatrix} \quad (\text{A.8.2})$$

For the pendulum:

$$\begin{bmatrix} x_p \\ y_p \end{bmatrix} = \begin{bmatrix} q_1 + l \sin q_2 \\ l \cos q_2 \end{bmatrix}, \quad \begin{bmatrix} \dot{x}_p \\ \dot{y}_p \end{bmatrix} = \begin{bmatrix} \dot{q}_1 + l \dot{q}_2 \cos q_2 \\ -l \dot{q}_2 \sin q_2 \end{bmatrix} \quad (\text{A.8.3})$$

Kinetic and Potential Energy

The total kinetic energy of the system is:

$$K_e = \frac{1}{2}m_c\dot{x}_c^2 + \frac{1}{2}m_p(\dot{x}_p^2 + \dot{y}_p^2) + \frac{1}{2}I\dot{q}_2^2 \quad (\text{A.8.4})$$

Substituting the velocities:

$$K_e = \frac{1}{2}(m_c + m_p)\dot{q}_1^2 + m_pl\dot{q}_1\dot{q}_2 \cos q_2 + \frac{1}{2}(m_pl^2 + I)\dot{q}_2^2 \quad (\text{A.8.5})$$

The potential energy of the system is:

$$V = m_pgl \cos q_2 \quad (\text{A.8.6})$$

Lagrangian Formulation

The Lagrangian is defined as:

$$L = K_e - V \quad (\text{A.8.7})$$

Substituting K_e and V :

$$L = \frac{1}{2}(m_c + m_p)\dot{q}_1^2 + m_pl\dot{q}_1\dot{q}_2 \cos q_2 + \frac{1}{2}(m_pl^2 + I)\dot{q}_2^2 - m_pgl \cos q_2 \quad (\text{A.8.8})$$

Euler-Lagrange Equations with Friction

The Euler-Lagrange equations are:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = F_i \quad (\text{A.8.9})$$

where F_i are the generalized forces, including friction and external input.

Generalized Forces

- For q_1 (cart motion):

$$F_1 = u - b_c\dot{q}_1 \quad (\text{A.8.10})$$

where $\mathbf{u}$ is the external force applied to the cart, and $b_c\dot{q}_1$ is the friction force.

- For q_2 (pendulum motion):

$$F_2 = -b_p\dot{q}_2 \quad (\text{A.8.11})$$

where $b_p\dot{q}_2$ is the friction torque at the pendulum joint.

Equations of Motion

The equations of motion for the system are derived as follows:

Equation for q_1 (Cart Motion)

$$(m_c + m_p)\ddot{q}_1 + m_pl\ddot{q}_2 \cos q_2 - m_pl\dot{q}_2^2 \sin q_2 = u - b_c\dot{q}_1 \quad (\text{A.8.12})$$

Equation for q_2 (Pendulum Motion)

$$m_pl\ddot{q}_1 \cos q_2 + (m_pl^2 + I)\ddot{q}_2 - m_pgl \sin q_2 = -b_p\dot{q}_2 \quad (\text{A.8.13})$$

State-Space Representation

Let:

$$x_1 = q_1, \quad x_2 = \dot{q}_1, \quad x_3 = q_2, \quad x_4 = \dot{q}_2 \quad (\text{A.8.14})$$

The state-space equations become:

$$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = \frac{1}{\lambda} [(m_pl^2 + I)(u - b_c x_2 + m_pl x_4^2 \sin x_3) - m_pl \cos x_3 (m_pgl \sin x_3 - b_p x_4)] \\ \dot{x}_3 = x_4 \\ \dot{x}_4 = \frac{1}{\lambda} [-(m_pl \cos x_3)(u - b_c x_2 + m_pl x_4^2 \sin x_3) + (m_c + m_p)(m_pgl \sin x_3 - b_p x_4)] \end{cases} \quad (\text{A.8.15})$$

Where:

$$\lambda = (m_c + m_p)(m_pl^2 + I) - (m_pl \cos x_3)^2 \quad (\text{A.8.16})$$

Linearization

Using the small angle approximation about $\theta = 0^\circ$, we assume $\cos \theta \approx 1$, $\sin \theta \approx \theta$, and $\dot{\theta} \approx 0$. Substituting these approximations, the linear model is:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & \left(-\frac{b_c(m_pl^2 + I)}{\lambda}\right) & \left(-\frac{(m_pl)^2 g}{\lambda}\right) & \left(\frac{b_p m_pl}{\lambda}\right) \\ 0 & 0 & 0 & 1 \\ 0 & \left(\frac{b_c m_pl}{\lambda}\right) & \left(\frac{m_pl g (m_p + m_c)}{\lambda}\right) & \left(-\frac{b_p (m_p + m_c)}{\lambda}\right) \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + \begin{bmatrix} 0 \\ \left(\frac{m_pl^2 + I}{\lambda}\right) \\ 0 \\ \left(-\frac{m_pl}{\lambda}\right) \end{bmatrix} u \quad (\text{A.8.17})$$

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \quad (\text{A.8.18})$$

A.9 Example:-2 Ball and Beam System

The ball and beam system is a classic control theory problem that involves balancing a ball on a beam, which can tilt to control the ball's position.

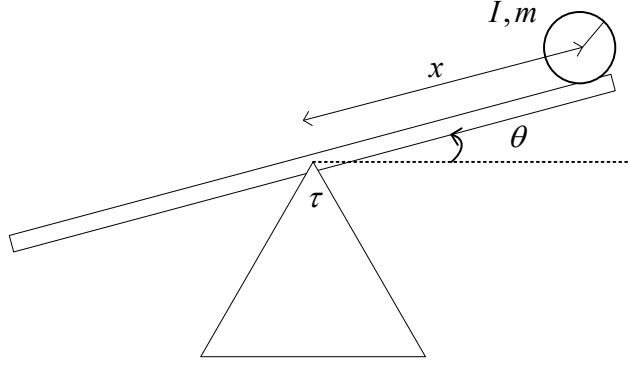


Figure A.2: *Ball and Beam System*

A.9.1 System Description

- The system consists of a beam that can rotate about its center and a ball that rolls along the beam.
- The goal is to control the beam's angle to regulate the ball's position.
- The system has two primary degrees of freedom: the position of the ball along the beam and the angular position of the beam.

A.9.2 Dynamic Model of the System

Let I_1 , m , and I_2 denote the inertia of the beam, the mass of the ball, and the inertia of the ball, respectively.

- Position of the center of mass of the ball:

$$x_c = (x \cos \theta, x \sin \theta) \quad (\text{A.9.1})$$

- Velocity of the center of mass of the ball:

$$\dot{x}_c = (\dot{x} \cos \theta - x \dot{\theta} \sin \theta, \dot{x} \sin \theta + x \dot{\theta} \cos \theta) \quad (\text{A.9.2})$$

- Translational kinetic energy:

$$K_{\text{trans}} = \frac{1}{2} m \dot{x}_c^2 = \frac{1}{2} m (\dot{x}^2 + x^2 \dot{\theta}^2) \quad (\text{A.9.3})$$

- Rotational kinetic energy:

$$K_{\text{rot}} = \frac{1}{2} I_1 \dot{\theta}^2 + \frac{1}{2} I_2 \left(\frac{\dot{x}}{r} \right)^2 \quad (\text{A.9.4})$$

Total Kinetic and Potential Energy

- Total kinetic energy:

$$K = K_{\text{trans}} + K_{\text{rot}} = \frac{1}{2}(I_1 + mx^2)\dot{\theta}^2 + \frac{1}{2}m \left(1 + \frac{I_2}{mr^2}\right) \dot{x}^2 \quad (\text{A.9.5})$$

- Let $\lambda = 1 + \frac{I_2}{mr^2}$ be a constant.
- Potential energy:

$$V(x, \theta) = mgx \sin \theta \quad (\text{A.9.6})$$

- Lagrangian of the system:

$$L = \frac{1}{2}(I_1 + mx^2)\dot{\theta}^2 + \frac{1}{2}m \left(1 + \frac{I_2}{mr^2}\right) \dot{x}^2 - mgx \sin \theta \quad (\text{A.9.7})$$

Euler-Lagrange Equations

The Euler-Lagrange equation of the system is:

$$\frac{d}{dt} \left[\frac{\partial L}{\partial \dot{q}} \right] - \frac{\partial L}{\partial q} = F_i \quad (\text{A.9.8})$$

Where F_i are the generalized forces or torques, including friction and external input. In this case, assuming the pivot friction is negligible, the only external force applied to the system is the torque at the pivot joint τ .

- Resulting equations of motion:

$$(I_1 + mx^2)\ddot{\theta} + 2mx\dot{x}\dot{\theta} + mgx \cos \theta = \tau \quad (\text{A.9.9})$$

$$\lambda \ddot{x} - x\dot{\theta}^2 + g \sin \theta = 0 \quad (\text{A.9.10})$$

State-Space Representation

With state variables $x_1 = x$, $x_2 = \dot{x}$, $x_3 = \theta$, and $x_4 = \dot{\theta}$, the dynamics of the system are described by:

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= \frac{mx_4^2 x_1 - g \sin x_3}{\lambda} \\ \dot{x}_3 &= x_4 \\ \dot{x}_4 &= \frac{\tau - 2mx_1 x_2 x_3 - mgx_1 \cos x_3}{I + mx_1^2} \end{aligned}$$

Linearized Model State-Space Representation

Using small-angle approximations ($\sin x_3 \approx x_3$ and $\cos x_3 \approx 1$) and assuming the ball is at the center ($x_1 = 0$), the linearized state-space equations become:

$$\begin{aligned}\dot{x}_1 &= x_2 \\ \dot{x}_2 &= \frac{-gx_3}{\lambda} \\ \dot{x}_3 &= x_4 \\ \dot{x}_4 &= \frac{\tau - mgx_1}{I}\end{aligned}$$

The state-space representation of the linearized system is:

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & \frac{-g}{\lambda} & 0 \\ 0 & 0 & 0 & 1 \\ \frac{-mg}{I} & 0 & 0 & 0 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ \frac{1}{I} \end{bmatrix} \tau \quad (\text{A.9.11})$$

The control objective is to regulate the position of the ball; thus, the output must be x_1 . The output equation is:

$$y = \begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix} \mathbf{x} \quad (\text{A.9.12})$$

Appendix B

Modern Control System

B.1 Modern Control System Tests

B.1.1 Test 1 (2024 G.C.)

Part I:-Conceptual Question(3 points)

1. Define stabilizability and detectability for a dynamic system and the requirements that the system dynamics should have to be detectable and stabilizable. (3 points)

Part II:- Workout problem(12 points)

1. A cable reel dynamic system has the following overall transfer function,

$$G(s) = \frac{2}{s^2 + \beta s + 21} \quad (\text{B.1.1})$$

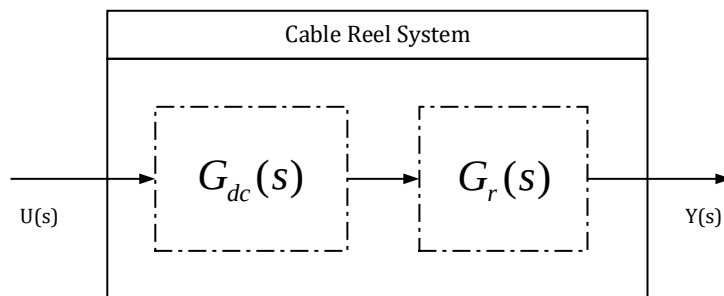


Figure B.1: *Cable Reel System Block Diagram*

For the cable reel system

- (a) Derive a state space representation of the system. (3 points)
- (b) Determine the range of β for which system is stable. (3 points)
- (c) Select a single value of β from the possible β values you have determined in part **1b** and determine the zero initial condition step response of the resulting system. (3 points)
- (d) Is the system completely controllable and completely observable with the value of β you have selected in part **1c**? Why? Justify your answer mathematically. (3 points)

B.1.2 Test 2 (2024 G.C.)

Part I:-Conceptual Question(3 points)

1. Define stabilizability and detectability for a dynamic system and the requirements that the system dynamics should have to be detectable and stabilizable. (3 points)

Part II:- Workout problem(12 points)

1. Consider the following dynamic system,

$$G(s) = \frac{s + \alpha}{s^2 + 5s + 6} \quad (\text{B.1.2})$$

- (a) Derive diagonal normal form state space representation for the dynamic system.(3 points)
- (b) Determine the range of α for which system is completely observable. (3 points)
- (c) Select single value of α from the possible α values you have determined in part **1b** and determine state transition matrix of the system. (3 points)
- (d) Does the value of α affect the stability of the system? Justify your answer mathematically. (3 points)

B.1.3 Test 3 (2020 G.C.)

Part I:-Conceptual Questions(5 points)

1. Show that the solution of the state equation will be changed with a similarity transformation. (2.5 points)
2. Derive an expression for the solution of the state space equation.(2.5 points)

Part II:- Calculation(15 points)

Consider the RLC Circuit, with system parameters $L = 0.1H$, $C = 0.1F$, and $R = 1\Omega$.

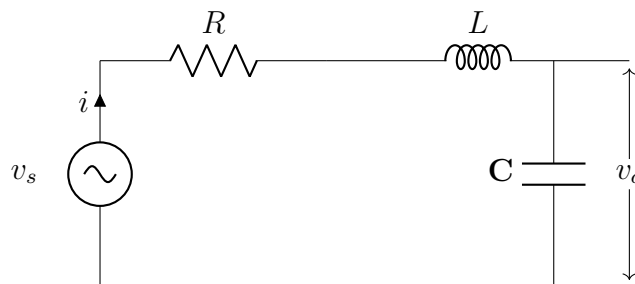


Figure B.2: *Series RLC Circuit*

1. If the designer aims to regulate the voltage across the capacitor.
 - (a) Derive a state space representation of the system. (2.5 points)
 - (b) Find the eigenvalues and eigenvectors of the system. (1.5 points)
 - (c) Determine whether the system is stable or not. (0.5 points)
 - (d) Determine the state transition matrix of the system. (2.5 points)
 - (e) Determine zero input response of the system for initial condition $\mathbf{x}(0) = \begin{bmatrix} 0 & 1 \end{bmatrix}^T$. (2.5 points)
 - (f) Determine the zero state response of the system under step $u = e^{-2t}$. (2.5 points)
 - (g) What is the maximum value of voltage the capacitor can be charged to? (0.5 points)
2. If the output is the voltage across the inductor, determine the output state space representation of the system. (2 points)
3. Compare the result on 2 with the result on 1a and state their difference. (0.5 points)

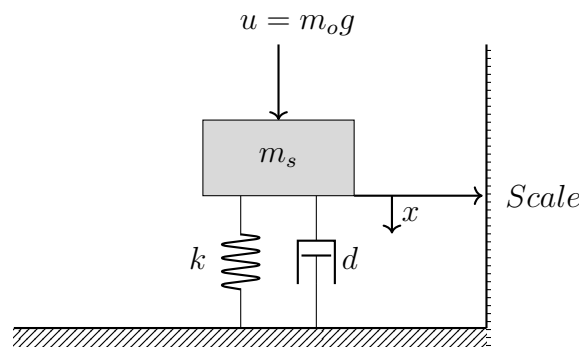
B.1.4 Test 4 (2020 G.C.)

Part I:-Conceptual Question(5 points)

1. Show that the similarity transformation doesn't change the transfer function of the system and the solution of output equation. (2.5 points each)

Part II:- Calculation(15 points)

1. Consider the following simple weight measuring system, with system parameters $k = 20\text{N/m}$, $m_s = 1\text{kg}$, and $d = 9\text{Ns/m}$.



If the output, which is shown on the scale, is the mass of the object to be measured ($m_o = \frac{kx}{g}$). (use $g = 10\text{m/s}^2$)

- (a) Derive a state space representation of the system. (2.5 points)
- (b) Find the eigenvalues and eigenvectors of the system. (1.5 points)
- (c) Determine whether the system is stable or not. (0.5 points)
- (d) Determine the state transition matrix of the system. (2.5 points)

- (e) Determine zero input response of the system for initial condition $\mathbf{x}(0) = \begin{bmatrix} 1 & 0 \end{bmatrix}^T$. (2.5 points)
- (f) Determine the zero state response of the system under a step input. (2.5 points)
2. If the maximum compression of the spring is 10cm , determine the maximum mass the system can handle. (0.5 points)
3. If you are the designer and you want to increase the maximum mass the system can handle 5 times,
- What is your best engineering solution? (0.625 points)
 - Why do you select the solution? (0.625 points)
 - Determine the eigenvalues for your new system. (0.625 points)
 - What do you conclude from the result? (0.625 points)

B.2 Modern Control Systems Final Exams

B.2.1 Final Exam 1 (2020 G.C.)

Part I:-Conceptual Questions(10 points)

1. What does it mean when we say a system is stabilizable or detectable? (5 points)
2. What is optimal control? Give an example of a terminal error minimization control problem. (5 points)

Part II:- Calculation(40 points)

1. Consider the RLC Circuit, with system parameters $L = 0.1\text{H}$, $C = 0.1\text{F}$ and $R = 1\Omega$. If the output of the system is the voltage across the capacitor.

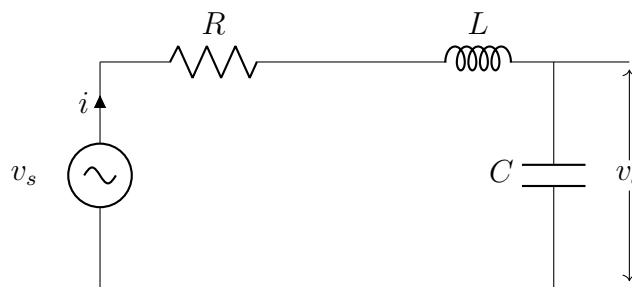


Figure B.3: *Series RLC Circuit*

- (a) Derive a state space representation of the system. (3 points)
- (b) Determine the range of the system parameters (R, L, C) for which the system is completely controllable. (2 points)
- (c) Determine the range of the system parameters (R, L, C) for which the system is completely observable. (2 points)
- (d) Is $R = 0$ a system detectable? If explain why. (2 points)

- (e) If C is very large, check for system observability and controllability and state your conclusion? (2 points)
2. For the following completely controllable and completely observable system, (10 points)

$$\begin{aligned}\dot{\mathbf{x}} &= \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -5 & -10 & -5 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u \\ y &= \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \mathbf{x}\end{aligned}\tag{B.2.1}$$

- (a) Determine the feedback gain $\mathbf{K}_f$ so that the control law is $u = -\mathbf{K}_f \mathbf{x}$, assuming the desired poles of the closed loop system are at $s = -2 \pm j4$ and $s = -2$. (3.5 points)
- (b) Design an full order observer gain $\mathbf{K}_e$ or $\mathbf{L}_c$ such that the eigenvalues of the observer are located at $s = -5 \pm j4$ and $s = -10$. (3.5 points)
- (c) Draw the block diagram implementation of the full state feedback with full state observer. (3 points)
3. For the system shown on equation B.2.1 (9 points)
- (a) Determine the reduced observer gain matrix $\mathbf{K}_e$ or $\mathbf{L}_c$ assume that desired observer poles to be at $s = -5, s = -5$. (4 points)
- (b) Determine the reduced-order state observer dynamics and reduced-order state observer output equations(open loop system dynamics).(3 points)
- (c) Draw the overall block diagram for the above system (for question 3).(2 points)
4. For the dynamic system (10 points)

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u\tag{B.2.2}$$

$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} \mathbf{x}\tag{B.2.3}$$

The performance index J to be optimized is

$$J = \frac{1}{2} \int_0^\infty (\mathbf{x}^\top \mathbf{Q} \mathbf{x} + \mathbf{u}^\top \mathbf{R} \mathbf{u}) dt\tag{B.2.4}$$

where

$$\mathbf{Q} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{R} = 0.25\tag{B.2.5}$$

- (a) Determine the Riccati matrix $\mathbf{P}$. (3 points)

- (b) Design the linear quadratic Regulator (LQR) feedback controller $\mathbf{K}_f$. (2 points)
- (c) The control signal $\mathbf{u}$. (1 points)
- (d) The closed loop eigenvalues of the system. (2 points)
- (e) Draw the block diagram for the LQR controller. (2 points)

B.2.2 Final Exam 2 (2024 G.C.)

Part I:-Conceptual Questions (16 points)

1. Explain briefly the difference between state space representation and transfer function representation of dynamic systems.(4 points)
2. Explain briefly the basic difference between the pole placement controller design technique and the optimal linear quadratic regulator (LQR) controller design technique. (4 points)
3. Explain briefly what optimal control is, and state the generalized cost function/performance index and explain the terms in the cost function, giving an individual example for each term.(8 points)

Part II:- Workout problems (34 points)

Consider an open-loop system with the transfer function:

$$G(s) = \frac{s}{s^2 + 2s + 2} \quad (\text{B.2.6})$$

1. For the system in equation B.2.6. (6 points)
 - (a) Check the stability of the system.
 - (b) Check whether the system is completely controllable or not.
 - (c) Check whether the system is completely observable or not.
2. Identify the values of closed loop poles of the system p_1 , and p_2 for which the system in equation B.2.6 meets the following specifications: $M_p = 0.1630\%$ and $t_s(2\% \text{criterion}) = 1s$. (8 points)
3. Design a state feedback controller based on the poles p_1, p_2 , identified in the previous part for the system in equation B.2.6 . (6 points)
 - (a) Determine the state feedback gain matrix $\mathbf{K}_f$.
 - (b) Determine the state feedback control law $\mathbf{u}$.
 - (c) Draw a block diagram representation of the system with a full state feedback controller.
4. Design a full state observer based on the poles p_1, p_2 identified in the previous part for the system in equation B.2.6. (7 points)
 - (a) Determine the state observer gain matrix $\mathbf{L}_c$.
 - (b) Determine the dynamic equation of the observer.
 - (c) Determine the state feedback control law u with a full state observer.
 - (d) Draw a block diagram representation of the full state feedback controller with full state observer.

5. Design a reduced state observer based on the poles p_1, p_2 identified in the previous part for the system in equation B.2.6. (7 points)
- Determine the reduced state observer gain matrix $\mathbf{L}_c$, that locate the pole of the reduced order observer at $p = p_1 + p_2$.
 - Determine the output equation of the observer.
 - Determine the reduced-order state observer dynamic equation.
 - Determine the state feedback control law $\mathbf{u}$ with full state observer.
 - Draw a block diagram representation of the full state feedback system controller with reduced state observer.

Use the following table and equations if necessary

$$G_{2^{order}}(s) = \frac{\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2} \quad (\text{B.2.7})$$

Table B.1: *Time domain specification of second order systems*

Time domain specification	Formula
Delay time	$t_d = \frac{1 + 0.7\xi}{\omega_n}$
Rise time	$t_r = \frac{\pi - \theta}{\omega_d}$
Peak time	$t_p = \frac{\pi}{\omega_d}$
Percentage Peak overshoot ($\%M_p$)	$M_p = e^{-\left(\frac{\xi\pi}{\sqrt{1-\xi^2}}\right)}$
Settling time with 2% tolerance	$t_s = \frac{4}{\xi\omega_n}$

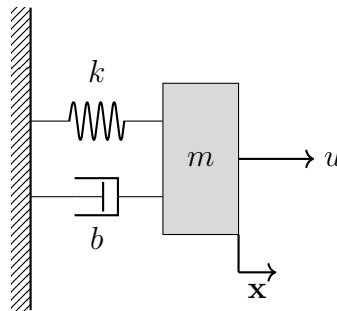
B.2.3 Final Exam 3 (2020 G.C.)

Part I:-Conceptual Questions(10 points)

1. What does it mean when we say a system is controllable or observable? (5 points)
2. What is optimal control? Give an example of a terminal error minimization control problem. (5 points)

Part II:- Calculation(40 points)

1. Consider the following simple mass spring damper system (11 points)



If the output of the system is the velocity of the mass M .

- (a) Derive a state space representation of the system.(3 points)
 - (b) Determine the range of the system parameters (k, b, m) for which the system is completely controllable.(2 points)
 - (c) Determine the range of the system parameters (k, b, m) for which the system is completely observable.(2 points)
 - (d) Assume $k = 0$ and $b \neq 0$ is the system detectable? If explain why.(2 points)
 - (e) If m is very large, check for system observability and controllability and state your conclusion? (2 points)
2. For the following completely controllable and completely observable system, (10 points)

$$\begin{aligned}\dot{\mathbf{x}} &= \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -5 & -10 & -5 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u \\ y &= \begin{bmatrix} 0 & 1 & 0 \end{bmatrix} \mathbf{x}\end{aligned}\tag{B.2.8}$$

- (a) Determine the feedback gain $\mathbf{K}_f$ so that the control law is $\mathbf{u} = -\mathbf{K}_f \mathbf{x}$, assuming the desired pole of the closed loop system is at $s = -2 \pm j4$ and $s = -2$.(3.5 points)
- (b) Design an full order observer gain $\mathbf{K}_e$ or $\mathbf{L}_c$ such that the eigenvalues of the observer are located at $s = -5 \pm j4$ and $s = -10$. (3.5 points)

- (c) Draw the block diagram implementation of the full state feedback with full state observer. (3 points)
3. For the system shown on equation B.2.8 (9 points)
- (a) Determine the reduced observer gain matrix $\mathbf{K}_e$ or $\mathbf{L}_c$ assume that desired observer poles to be at $s = -5, s = -5$. (4 points)
- (b) Determine the reduced-order state observer dynamics and reduced-order state observer output equations(open loop system dynamics).(3 points)
- (c) Draw the overall block diagram for the above system (for question 3).(2 points)
4. Consider the Dynamic System (10 points)

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 \\ 0 & -1 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u \quad (\text{B.2.9})$$

$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} \mathbf{x} \quad (\text{B.2.10})$$

The performance index J to be optimized is

$$J = \frac{1}{2} \int_0^\infty (\mathbf{x}^\top \mathbf{Q} \mathbf{x} + \mathbf{u}^\top \mathbf{R} \mathbf{u}) dt \quad (\text{B.2.11})$$

where

$$\mathbf{Q} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad R = 1 \quad (\text{B.2.12})$$

- (a) Determine the Riccati matrix $\mathbf{P}$. (3 points)
- (b) Design the LQR feedback controller $\mathbf{K}_f$. (2 points)
- (c) The control signal $\mathbf{u}$. (1 points)
- (d) The closed loop eigenvalues of the system. (2 points)
- (e) Draw the block diagram for the LQR controller. (2 points)

B.2.4 Final Exam 4 (2021 G.C.)

Part I:-Conceptual Questions (16 points)

1. What does it mean when we say a system is stabilizable or detectable? (4 points)
2. What does it mean when we say a system is completely controllable or completely observable? (4 points)
3. What is optimal control? List and give an example of the most popular optimal control cost functions (performance indexes). (6 points)
4. What is a state observer/state estimator, and where do we use a state estimator/observer? (2 points)

Part II:- Calculation (34 points)

Consider the following dynamic system

$$\begin{aligned}\dot{x}_1 &= x_2 \\ \dot{x}_2 &= x_2 + u \\ y &= x_1\end{aligned}\tag{B.2.13}$$

1. For the system shown on equation (B.2.13) (6 points)
 - (a) Check whether the system is completely controllable. (3 points)
 - (b) Check whether the system is completely observable. (3 points)
2. For the system shown on equation (B.2.13) (10 points)
 - (a) Determine the feedback gain $\mathbf{K}_f$ so that the control law is $u = -\mathbf{K}_f \mathbf{x}$, assuming the desired poles of the closed loop system are at $s = -4$ and $s = -1$. (3.5 points)
 - (b) Design an full order observer gain $\mathbf{L}_c$ such that the eigenvalues of the observer are located at $s = -4$ and $s = -1$. (3.5 points)
 - (c) Draw the block diagram implementation of the system with full state feedback using estimated states. (3 points)
3. For the system shown on equation (B.2.13) (8 points)
 - (a) Determine the reduced observer gain matrix $\mathbf{L}_c$ assume that desired observer pole is at $s = -5$. (4 points)
 - (b) Determine the reduced-order state observer dynamics and reduced-order state observer output equation (open loop system dynamics). (3 points)
 - (c) Draw the overall block diagram for the system with reduced state observer. (1 points)
4. For the system shown on equation (B.2.13) (10 points) If the designer wants to optimize the performance index J ,

$$J = \frac{1}{2} \int_0^{\infty} (\mathbf{x}^T \mathbf{Q} \mathbf{x} + \mathbf{u}^T \mathbf{R} \mathbf{u}) dt$$

where

$$\mathbf{Q} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad R = 1$$

- (a) Determine the Riccati matrix $\mathbf{P}$. (3 points)
- (b) Design the LQR feedback controller $\mathbf{K}_f$. (2 points)
- (c) The control signal $\mathbf{u}$. (1 points)
- (d) The closed loop eigenvalues of the system. (2 points)
- (e) Draw the block diagram for the system with LQR controller. (2 points)

Appendix C

Control Systems Exams

C.1 Control Systems Exams Test

C.1.1 Test 1 (2025 G.C)

Solenoid Actuator Valve System

The 3/2 solenoid actuator valve, depicted in Figure C.1, plays a crucial role in regulating fluid flow through three ports: one inlet and two outlets. When an electrical current is

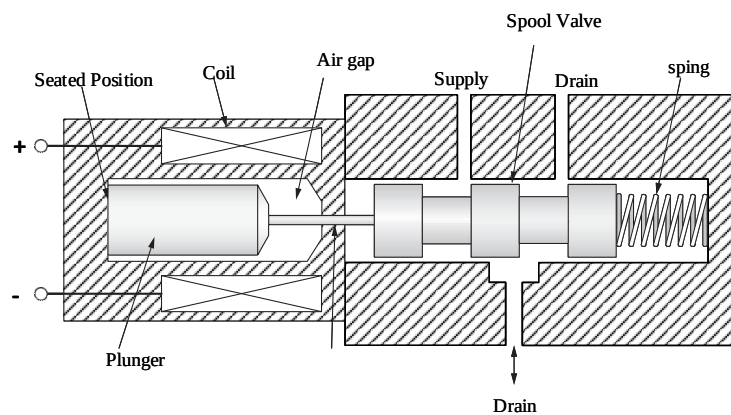


Figure C.1: *Solenoid actuator valve*

applied to the solenoid, it generates a magnetic field that actuates the internal mechanism of the valve. This movement facilitates fluid flow from the inlet to one outlet while simultaneously blocking the other. Upon deactivation of the current, a spring mechanism restores the valve to its original position, thereby redirecting the flow to the second outlet. This efficient operation enables precise control over actuators and cylinders in various pneumatic and hydraulic applications, establishing the solenoid actuator valve as a vital component in automation and fluid management systems.

Since the spool valve operates within a fluid medium, which introduces viscous friction that influences its dynamic behavior. The free body diagram of the system, illustrated in Figure C.2, elucidates the forces acting on the valve and provides insights into its dynamics, particularly regarding the impact of fluid resistance on its operation.

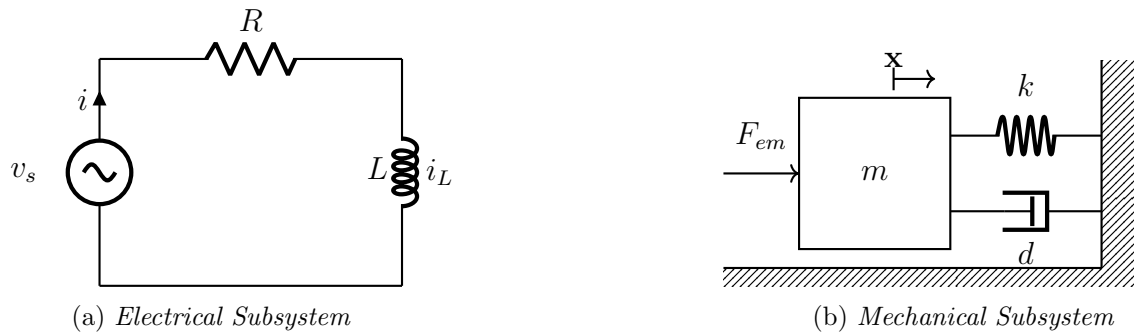


Figure C.2: *Solenoid Actuator Valve System*

1. For Solenoid Actuator Valve System: (12 points)
 - (a) Derive the transfer function of the system from the input (source voltage $V_s(s)$) to the output (Spool Valve position $X(s)$) in zero-pole-gain form (zpk form). (3 points)
 - (b) Derive the transfer function, poles, and zeros of the system. (3 points)
 - (c) Determine the DC gain of the system. Which parameter affects the DC gain of the system? What conclusions can you draw from your result? (2 points)
 - (d) Approximate the system to second order using dominant poles approximation. (**Note:-** The DC gain of the original system transfer function (1a) and reduced order (approximated) system should be equal.) (5 points)
 - (e) Which system parameter affects the steady-state error (e_{ss}) for a step input? (2 points)
2. If you want to reduce the settling time by 20%: (8 points)
 - (a) Which parameter do you adjust and why? (3 points)
 - (b) What would be the effect on the poles of the system with your suggestion? (3 points)
 - (c) What would be the effect on peak time (t_p)? (2 points)
3. If we use unity negative feedback: (5 points)
 - (a) Does it affect the steady-state error of the system for a step input? Compare your results with (1d) and draw a conclusion. (3 points)
 - (b) Does it affect the DC gain of the system? Compare your results with (1c) and draw a conclusion. (2 points)

C.1.2 Test 2 (2025 G.C)

A shock absorber reduces unwanted oscillations in mechanical systems, enhancing stability and performance. It works by absorbing and dissipating energy from vibrations through a combination of hydraulic fluid and mechanical components. When a force is applied, the shock absorber compresses, allowing the fluid to flow through valves, which dampens the motion. The stiffness and damping coefficients can vary based on application, design, and specific requirements, enabling tailored performance for different environments, such as vehicles or industrial machinery.

The 3D model and Mechanical System Model of a typical shock absorber are shown in Figures C.3a and C.3b, respectively.

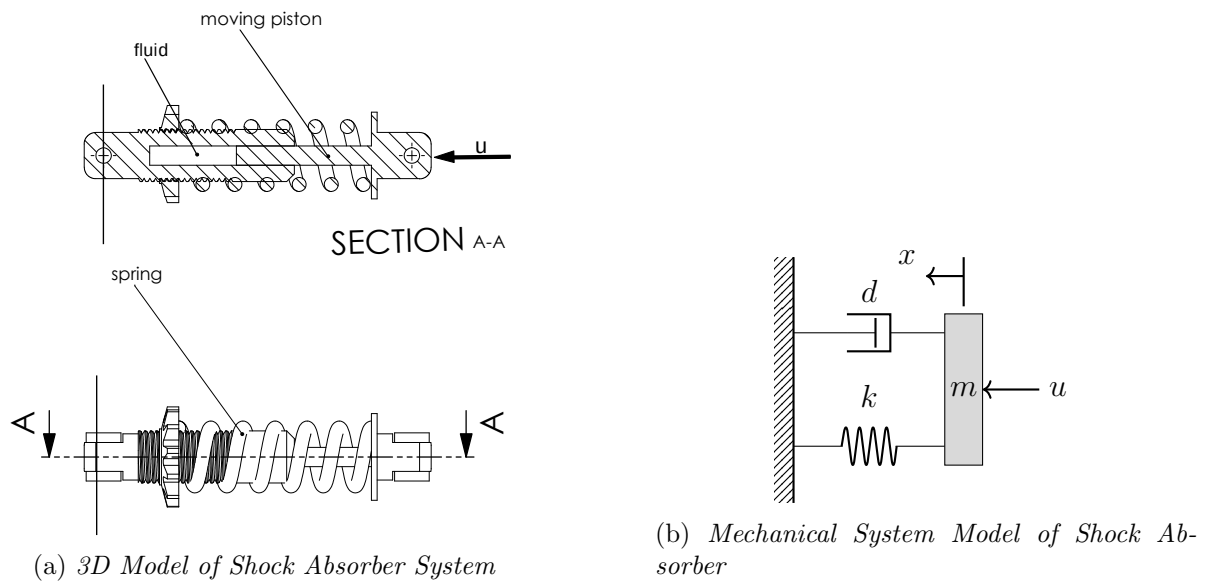


Figure C.3: Shock Absorber System

1. For the Shock absorber: (12 points)
 - (a) Derive the transfer function, poles, and zeros of the system. (3 points)
 - (b) Determine the DC gain of the system. Which parameter affects the DC gain of the system? What conclusions can you draw from your result? (2 points)
 - (c) Determine the conditions for the system to be undamped, under-damped, critically damped, and overdamped. How does each scenario influence the system's poles and its response to inputs? (5 points)
 - (d) Which system parameter affects the steady-state error (e_{ss}) for a step input? (2 points)
2. If you want to reduce the settling time by 20%: (8 points)
 - (a) Which parameter do you adjust and why? (3 points)
 - (b) What would be the effect on the poles of the system with your suggestion? (3 points)

- (c) What would be the effect on peak time (t_p)? (2 points)
- 3. If we use unity negative feedback: (5 points)
 - (a) Does it affect the steady-state error of the system for a step input? Compare your results with (1d) and draw a conclusion. (3 points)
 - (b) Does it affect the DC gain of the system? Compare your results with (1b) and draw a conclusion. (2 points)

C.1.3 Test 3 (2025 G.C)

1. Given a transfer function of a dynamic system,

$$G(s) = \frac{60}{0.5s^2 + 2.5s + 3} \quad (\text{C.1.1})$$

- (a) Draw a block diagram representation of the system.
- (b) Is the system Stable? Why? (2 points)
- (c) Determine the steady-state error (e_{ss}) of the system for a step input. (2 points)
- (d) Determine the settling time, maximum percentage overshoot, and steady-state error of the system with unity feedback for a step input. (2 points)
- (e) For the closed-loop frequency response, determine the resonant frequency, the magnitude of the system at the resonant frequency, and the phase angle at the resonant frequency. What implications does your answer have for the closed-loop time-domain analysis of the system? Relate your answer to part 1d. (4 points)
- (f) Determine the response of the closed loop system for $r(t) = 2\sin(4t)$ input. (4 points)
- (g) Determine the gain crossover frequency (ω_{gc}), phase crossover frequency (ω_{pc}), gain margin (GM), and phase margin (PM) of the open loop transfer function of the system and interpret the implications of the results. (4 points)
- (h) Design a compensator (any compensator, a lead or a lag or PI or PD, or PID) with proper reasoning for your choice. The compensator should ensure that the steady-state error for a step input to the specified system (referenced as C.1.1) is less than 0.02, while achieving a desired phase margin of at least 45° . (10 points)

C.2 Control Systems Exams Final

C.2.1 Final Exam 1 (2025 G.C.)

Solenoid Actuator Valve System

The operational parameters of the solenoid actuator valve system in Figure C.1 are given in Table C.2, including the parameters and voltage gain of the non-inverting operational amplifier configuration.

Table C.1: *System Parameters*

Parameter	Value
Stiffness Coefficient (K)	$100N/m$
Number of Turn (N)	400
Winding Resistance (R)	$1\ \Omega$
Winding Inductance (L)	$0.5H$
Spool valve mass (m)	$1kg$
Damping Friction Coefficient (b)	$25Ns/m$
Electromagnetic Force	$F_{em} = Ni_L$

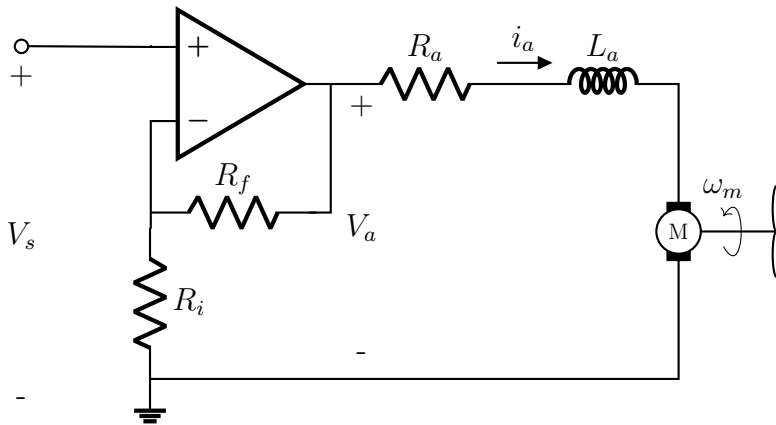
1. For the solenoid actuator valve system shown in Figure C.1; (12 points)
 - (a) Derive the transfer function of the system from the input (source voltage $V_s(s)$) to the output (Spool Valve position $X(s)$) in zero-pole-gain form (zpk form). (8 points)
 - (b) Draw a block diagram representation of the system.
 - (c) Approximate the system to second order using dominant poles approximation. (4 points)
2. For the reduced order system 1c, (18 points)
 - (a) Determine the time constants for the electrical and mechanical subsystems. What conclusions can you draw from your results about part 1a? (5 points)
 - (b) Determine the DC gain of the system. (2 points)
 - (c) Determine the damping factor (ξ) and natural frequency (ω_n) of the system. (5 points)
 - (d) Is the system Stable? Why? (3 points)
 - (e) Determine the steady-state error (e_{ss}) of the system for a step input. (2 points)
 - (f) Determine the rise time, maximum percentage overshoot, and steady-state error of the system with unity feedback for a step input. (5 points)

- (g) For the closed-loop frequency response, determine the resonant frequency, the magnitude of the system at the resonant frequency, and the phase angle at the resonant frequency. What implications does your answer have for the closed-loop time-domain analysis of the system? Relate your answer to part 2f. (6 points)
 - (h) Determine the response of the closed loop system for $r(t) = 2 \cos(4t)$ input. (4 points)
 - (i) Determine the gain crossover frequency (ω_{gc}), phase crossover frequency (ω_{pc}), gain margin (GM), and phase margin (PM) of the open loop transfer function of the system and interpret the implications of the results. (7 points)
3. Design a lead compensator to ensure that the steady-state error for a step input is less than 0.02, while achieving a desired phase margin of at least 45° at 20 rad/s frequency for the system you derived in part 1b. (20 points)

C.2.2 Final Exam 2 (2025 G.C.)

DC fan speed control system

The DC fan speed control system integrates a DC motor with a pre-amplifier configured in a non-inverting manner to precisely regulate the fan speed. This design facilitates an elevation of the input armature voltage supplied to the DC motor, thereby enhancing its operational performance and responsiveness.



(a) Armature Voltage Controlled DC Motor With Pre-amplifier



(b) DC-Fan

Figure C.4: DC Fan Speed Control System

In this configuration, the weight of the fan is considered negligible (**Zero load torque**), which contributes to optimal efficiency. The motor is directly coupled to the fan blades, ensuring smooth and responsive operation. The pre-amplifier is instrumental in stepping up the input voltage, allowing for refined control over the motor's speed and improving the overall effectiveness of the system.

The operational parameters of the DC motor are given in Table C.2, including the parameters and voltage gain of the non-inverting operational amplifier configuration.

Table C.2: DC Motor Parameters

Parameter	Value
Back EMF constant (K_b)	0.001 V-s/rad
Torque constant (K_t)	0.001 V-s/rad
Winding Resistance (R_a)	1 Ω
Winding Inductance (L_a)	0.5 H
Motor Inertia (J_m)	0.001 kg/m ²
Windage friction (b)	0.01 N-s
Input resistance (R_i)	1 k Ω
Feedback Resistance (R_f)	470 k Ω
Non-Inverting Op-amp voltage gain (A_v)	$\left(1 + \frac{R_f}{R_i}\right)$

1. For the DC fan speed control system shown in Figure C.4; (30 points)
 - (a) Derive the transfer function of the system from the input (source voltage $V_s(s)$) to the output (motor speed $\omega_m(s)$) in zero-pole-gain form (zpk form). (8 points)
 - (b) Draw a block diagram representation of the system.
 - (c) Determine the time constants for the electrical and mechanical subsystems. What conclusions can you draw from your results about part 1a? (5 points)
 - (d) Determine the DC gain of the system. (2 points)
 - (e) Determine the damping factor (ξ) and natural frequency (ω_n) of the system. (5 points)
 - (f) Is the system Stable? Why? (3 points)
 - (g) Determine the steady-state error (e_{ss}) of the system for a step input. (2 points)
 - (h) Determine the settling time, maximum percentage overshoot, and steady-state error of the system with unity feedback for a step input. (5 points)
2. Frequency domain analysis,
 - (a) For the closed-loop frequency response, determine the resonant frequency, the magnitude of the system at the resonant frequency, and the phase angle at the resonant frequency. What implications does your answer have for the closed-loop time-domain analysis of the system? Relate your answer to part 2d. (4 points)
 - (b) Determine the response of the closed loop system for $r(t) = 2 \cos(4t)$ input. (4 points)
 - (c) Determine the gain crossover frequency (ω_{gc}), phase crossover frequency (ω_{pc}), gain margin (GM), and phase margin (PM) of the open loop transfer function of the system and interpret the implications of the results. (8 points)

3. Design a lead compensator to ensure that the steady-state error for a step input is less than 0.02, while achieving a desired phase margin of at least 45° . (20 points)

C.2.3 Final Exam 3 (2025 G.C.)

Projector Screen Positioning System

Consider a projector screen positioning system in Figure 1.18, using the operational parameters of the DC motor detailed in Table C.2, which includes specifications and the voltage gain of the non-inverting operational amplifier configuration.

1. For the Projector Screen Positioning System shown in Figure C.4; (30 points)
 - (a) Derive the transfer function of the system from the input (source voltage $V_s(s)$) to the output (motor speed $\omega_m(s)$) in zero-pole-gain form (zpk form). (8 points)
 - (b) Approximate the system to second order using dominant poles approximation. (4 points)
 - (c) Determine the DC gain of the system (1a) and the reduced order system (1b). (2 points)
 - (d) Determine the steady-state error (e_{ss}) of both systems for a step input. (2 points)
2. For the reduced order system 1b, (18 points)
 - (a) Determine the time constants for the electrical and mechanical subsystems. What conclusions can you draw from your results about part 1b? (5 points)
 - (b) Determine the damping factor (ξ) and natural frequency (ω_n) of the reduced order system. (5 points)
 - (c) Is the reduced order system Stable? Why? (3 points)
 - (d) Determine the settling time, maximum percentage overshoot, and steady-state error of the reduced-order system with unity feedback for a step input. (5 points)
 - (e) For the closed-loop frequency response, determine the resonant frequency, the magnitude of the system at the resonant frequency, and the phase angle at the resonant frequency. What implications does your answer have for the closed-loop time-domain analysis of the reduced-order system? Relate your answer to part 2d. (4 points)
 - (f) Determine the gain crossover frequency (ω_{gc}), phase crossover frequency (ω_{pc}), gain margin (GM), and phase margin (PM) of the open loop transfer function of the reduced order system and interpret the implication of the results. (8 points)
3. Design a lead compensator to ensure that the steady-state error for a step input is less than 0.02, while achieving a desired phase margin of at least 45° for the reduced-order system in (1b). (20 points)

Appendix D

Digital Control Systems

D.1 Digital Control Systems Exams

D.1.1 Final Exam 1

1. For the following analog transfer function

$$G(s) = \frac{s + 2}{s^2 - 5s + 6}$$

- (a) Determine its discrete equivalent for sampling period $T=0.2\text{sec}$. (2 point)
- (b) If we use a microcontroller to implement the system, write C code only for this system part (1 point)
- (c) If the range of the input is $-12V$ to $12V$, determine the maximum quantization error (assume that the microcontroller is capable of handling $\pm 12V$ input) (1 point)
- (d) Derive a discrete-time transfer function using a numerical approximation method. (2 point)
- (e) Realize the discrete transfer function using standard programming. (1 point)

2. Mr. Tomas has a plant having open open-loop forward transfer function of

$$G(s) = \frac{3(s - 2)}{s^2 + s - 2}$$

The plant operation is required to be controlled by a microcontroller. Assume that the microcontroller ADC uses an ideal sampler to sample the analog input, DAC uses zero-order hold to convert the processed digital signal to an Analog signal, and the system has unity feedback gain. The plant should have a rise time of less than 20 seconds and a settling time of less than 50 seconds. Hint (use 10 samples per rise time and choose t_r and t_s that simplify your calculation).

- (a) Determine the discrete equivalent of the closed-loop transfer function using the zero-order hold equivalent technique. (3 marks)
- (b) Determine the natural frequency (ω_n), damping factor (ξ), maximum overshoot, and steady state error of the system for a unit step input. (2 marks)
- (c) Determine whether the system closed-loop transfer function is stable or not using the Jury stability test. (2 marks)

- (d) If the system is not stable the design a controller using the direct Raggazzini method with one pole of the closed loop transfer function at $z = 0.5$. (2 marks)

3. Given a sampling period $T = 0.5$ sec. determine the pulse transfer function for,

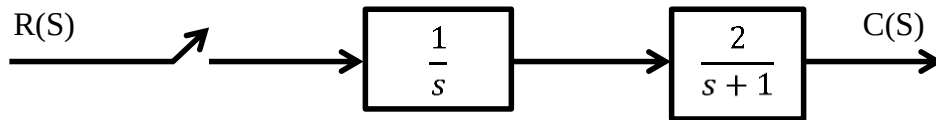


Figure D.1: *a*

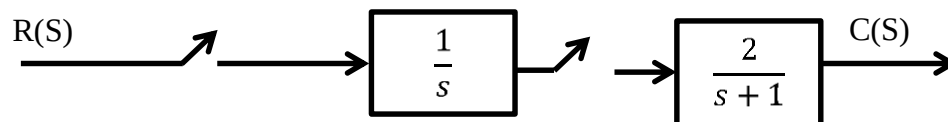


Figure D.2: *b*

Compare and Contrast your result (8 points)

4. Given the discrete-time system represented by the transfer function

$$G(z) = \frac{Kz}{z^2 - 1.5z + 0.5}$$

- Determine the closed-loop transfer function with unity feedback. (4 points)
- Determine the range of values of K that will ensure the stability of the system in equation 4a. (5 points)
- Determine the DC gain of $G(z)$ and the closed-loop system in equation 4a. Compare your results and provide an analysis. (5 points)

D.1.2 Final Exam 2

1. Batter making system has two inputs called water and flour. The system mixes the two inputs and results in one output called batter. The discrete-time equation governing the relation between the batter and flour in the discrete time domain is as follows. $y(k+2) + 5y(k+1) + 6y(k) = u_1(k+1) + 2u_1(k)$
 - (a) Derive a controllable canonical form state space representation for the system (5 points)
 - (b) Derive diagonal canonical form state space representation for the system (5 points)
 - (c) Determine discrete time state space transition matrix. (7 points)
 - (d) For constant flour input $1\text{kg}/\text{min}$ determines the solution for the states vector in the discrete time domain and the output equation in the discrete time domain. (7 points)
2. For system in question 1, if the desired location of the poles is at $z_1 = -0.5$ and $z_2 = 0.4$
 - (a) Determine state feedback gain $\mathbf{K}_f$ using control law. (5 points)
 - (b) Determine prediction estimator gain $\mathbf{L}_p$ for the system. (5 points)
 - (c) Determine current estimator $\mathbf{L}_c$ for the system. (5 points)
 - (d) Compare the results and state your conclusion. (2 points)
3. Derive an expression for diagonal controllable canonical form, assuming that the system has distinct poles $p_1, p_2, \dots, p_n$. (5 points)
4. Derive an expression for the state transition matrix using the z-transform method. (4 points)

D.1.3 Final Exam 3

1. Given a dynamic system,

$$y(k+2) + 5y(k+1) + 6y(k) = u(k+1) + u(k) \quad (\text{D.1.1})$$

- (a) Derive the discrete-time state-space representation of the system. (5 points)
- (b) Is the system stable? Why? (2 points)
- (c) Determine the discrete-time state transition matrix ($\Phi(k)$). (6 points)
- (d) Determine the zero input response of the system ($y_{zinp}(k)$) for initial condition $\mathbf{x}_o = \begin{bmatrix} 1 & 0 \end{bmatrix}^\top$. (6 points)
- (e) Determine the zero initial condition response of the system ($y_{zint}(k)$) for step input. (7 points)

2. For the system derived in [1a](#),
 - (a) Determine the state feedback gain $\mathbf{K}_f$ that locates the poles of the discrete-time system at $s_1 = -3$ and $s_2 = -0.5$. (5 points)
 - (b) Determine the predictive estimator gain $\mathbf{L}_p$ that locates the poles of the estimator at $s_1 = -3$ and $s_2 = -0.5$. (5 points)
 - (c) Determine the current estimator gain $\mathbf{L}_c$ that locates the poles of the estimator at $s_1 = -3$ and $s_2 = -0.5$. (6 points)
 - (d) Compare the results from [2b](#) and [2c](#) and derive a conclusion. (3 points)
3. If a PID controller is used to control the plant derived in [1a](#). Determine the discrete-time equivalent of the controller. (5 points)

"Strive for perfection in everything you do. Take the best that exists
and make it better. When it does not exist, design it."
-Sir Henry Royce, English engineer and car designer.

Remember always! We are yesterday, and you are tomorrow; work hard before you end
up being yesterday!
Good luck and be happy being an engineer!

I am very happy to address or clarify any confusing concepts or anything that feels
wrong. If you feel so, contact me using email: abebe.daniel7@gmail.com
Daniel Abebe Beyene (Science Ambassador, MSc in Control Engineering)
Lecturer at AASTU, Electro-mechanical Engineering Department
Thank you very much!!!

References

- [1] K. Ogata, *Modern control engineering*. Prentice hall, 2010.
- [2] R. C. Dorf and R. H. Bishop, *Modern Control Systems*. Addison-Wesley, 1998.
- [3] C. C. Remsing, “Am3. 2-linear control,” *Lecture Notes, South Africa*, 2006.
- [4] B. Kisacanin and G. C. Agarwal, *Linear Control Systems: with solved problems and MATLAB examples*. Springer Science & Business Media, 2012.
- [5] R. L. Williams, D. A. Lawrence, *et al.*, *Linear state-space control systems*. Wiley Online Library, 2007.
- [6] D. Abebe, “Design and implementation of feedback linearization based adaptive stabilizing controller coupled with fuzzy logic swing-up for pendulum on a cart,” Addis Ababa Institute of Technology, 2019.
- [7] S. Özkan, T. Kara, and M. Arıcı, “Modelling, simulation and control of quadruple tank process,” in *2017 10th International Conference on Electrical and Electronics Engineering (ELECO)*, IEEE, 2017, pp. 866–870.
- [8] P. Paraskevopoulos, *Modern Control Engineering*. Marcel Dekker, Inc., 2004.
- [9] R. E. Kalman, “A new approach to linear filtering and prediction problems,” 1960.
- [10] T. Point. “Time domain specifications.” (2020), [Online]. Available: https://www.tutorialspoint.com/control_systemscontrol_systems_time_domain_specifications.htm.
- [11] K. M. Mahala, P. Gadkari, and A. Deb, “Mathematical models for designing vehicles for ride comfort,” in *ICORD 09: Proceedings of the 2nd International Conference on Research into Design, Bangalore, India*, 2009.
- [12] U. A. Bakshi and M. V. Bakshi, *Modern control theory*. Technical Publications, 2020.
- [13] N. S. Nise, *Control systems engineering*. John Wiley & Sons, 2020.
- [14] J.-J. E. Slotine, W. Li, *et al.*, *Applied nonlinear control*. Prentice hall Englewood Cliffs, NJ, 1991, vol. 199.
- [15] R. Paz, *Analog computing technique*, University of Illinois, 2020.
- [16] C. A. Kluever, *Dynamic systems: modeling, simulation, and control*. John Wiley & Sons, 2020.